

MATSE: Operations Research

Introduction

Lecturer: Barbara Franci

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Office: Paul-Henri Spaaklaan 1, C4.011

Maastricht, the Netherlands

Course Information

When:

- Every Wednesday
- Lectures+ Exercises (3/4 hours every week)
- Till December 4th + January 15th

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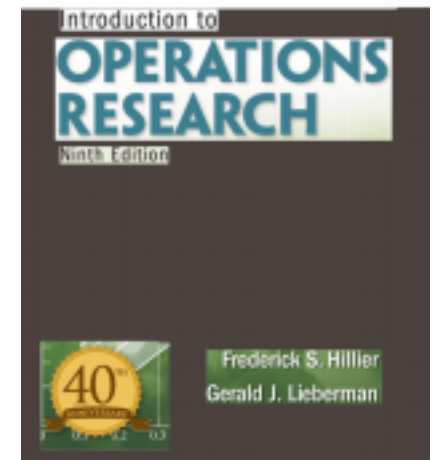
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Textbook

- Introduction to Operations Research by Hillier and Lieberman (9th ed)



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- Understanding the concepts is more important than memorization
- We all tend to forget the precise details, but my aim and hope is for you to learn the concepts that will allow you to relocate the details as and when you need them

Operations Research, a sample

The scope is much too broad for one course, but here we will sample some topics:

Deterministic Models:

Stochastic Models:

Game Theory

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- Linear Programming [HL: §3,4, and more!]
(Linear Algebra but looking for a *best* feasible solution)
- Transportation problem [HL: §8]

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Deterministic Models:

- Linear Programming [HL: §3,4, and more!]
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- Transportation problem [HL: §8]

Stochastic Models:

- Coin flipping and more
- Markov Chains [HL: §16]
- Queuing Theory [HL: §17]

Game Theory

Introduce Yourself!

- Who
- Why
- Word

Introduce Yourself!

- Barbara Franci, Assistant Professor
- Degree in Math, Math is Fun!
- Ich spreche kein Deutsch



MATSE: Operations Research

Lecture 1. (linear) Deterministic Models

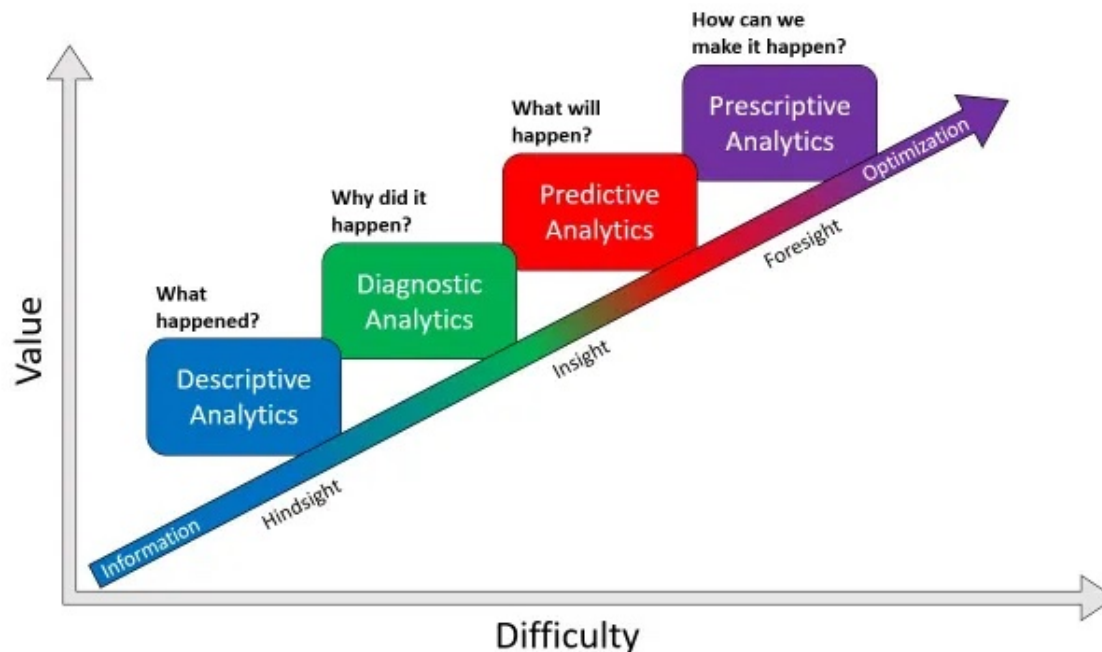
Why OR?

Operation Research is the discipline of applying advanced analytical methods to help **make better decisions**.

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Operation Research is the discipline of applying advanced analytical methods to help **make better decisions**.

Operation Research is **prescriptive**: what should we do in order to optimize performance subject to constraints?



<https://inthenameofai.com/data-analytics-not-data-analytics/>

Maximizing Gain

You are the boss of a small company, producing two products P_1 and P_2 . If you produce x_1 units of P_1 and x_2 units of P_2 , your gain is

$$G(x_1, x_2) = 30x_1 + 50x_2$$

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Your production runs on three machines M_A , M_B and M_C with the following capacities:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$

$$M_B: \quad x_1 + x_2 \leq 150$$

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Which choice of (x_1, x_2) maximizes your gain?

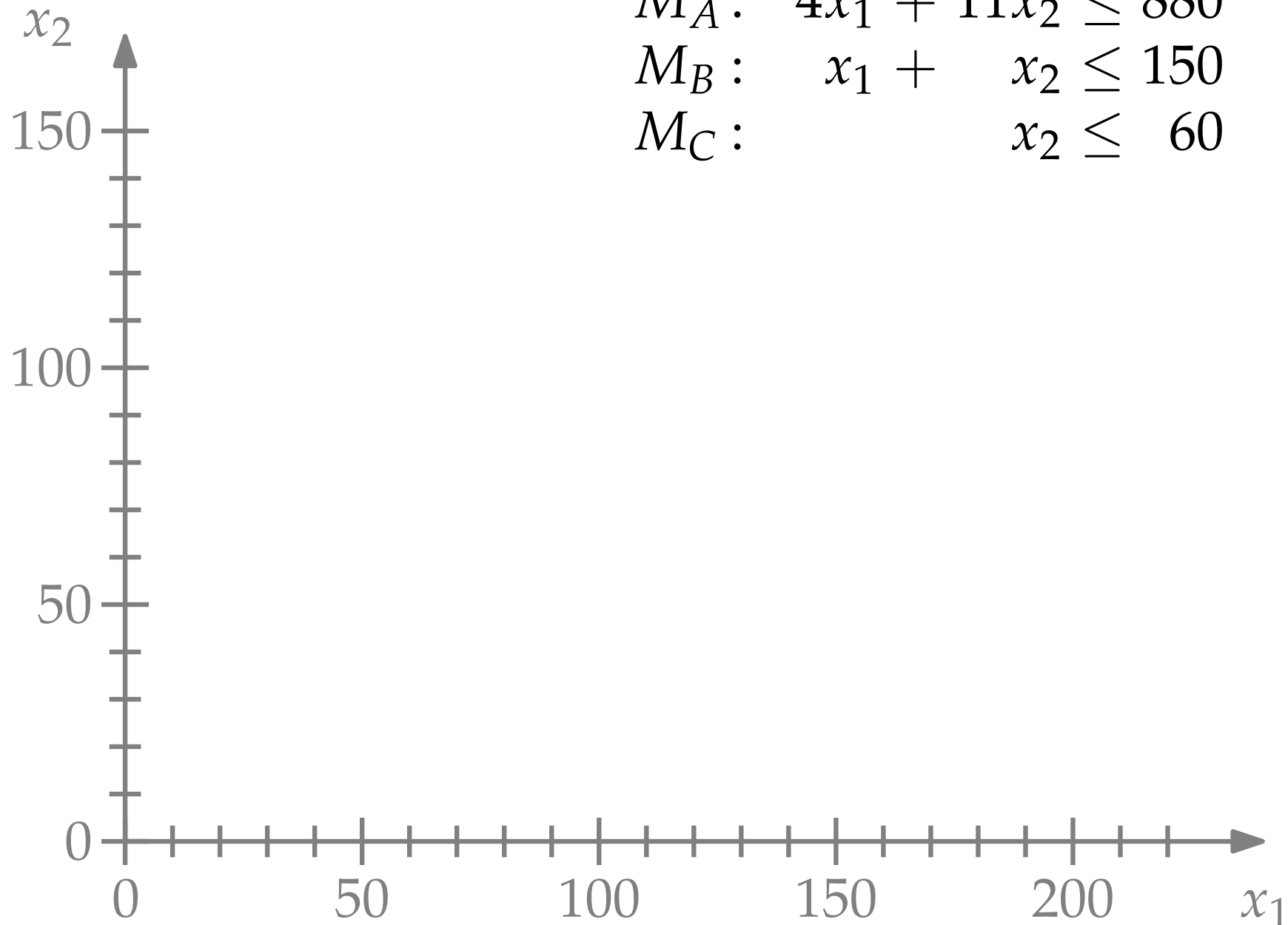
Solving by hand

Linear constraints:

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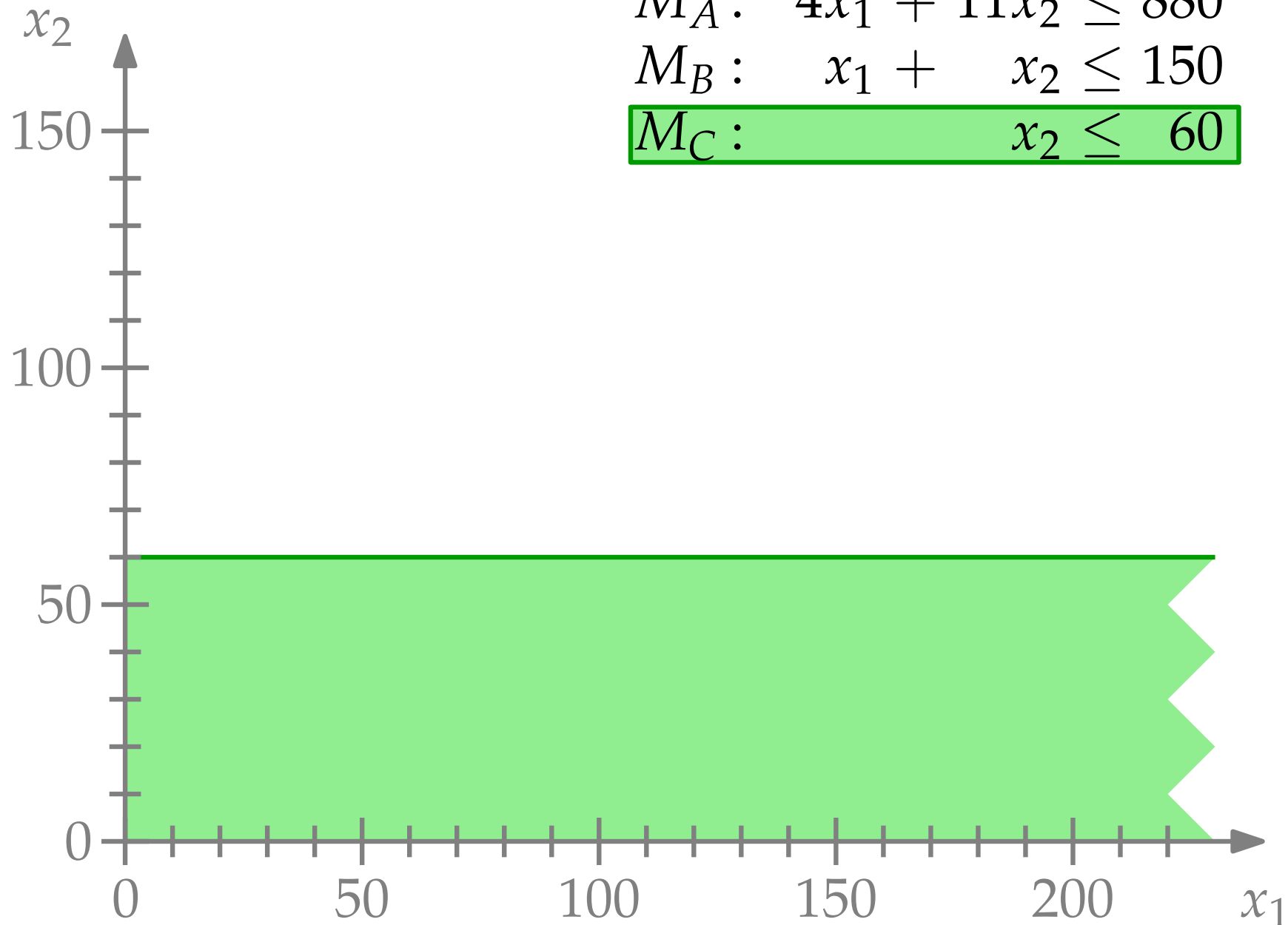
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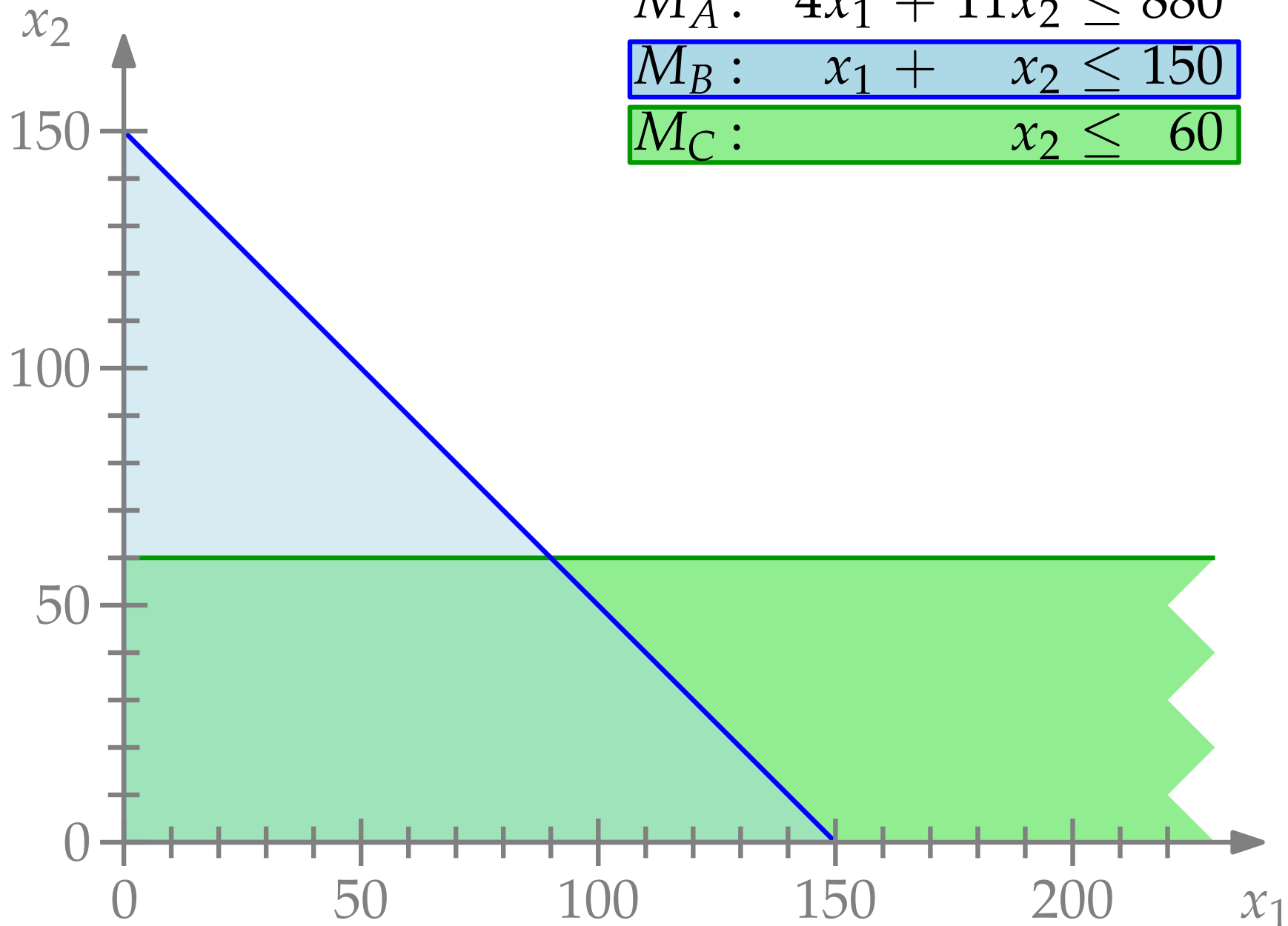
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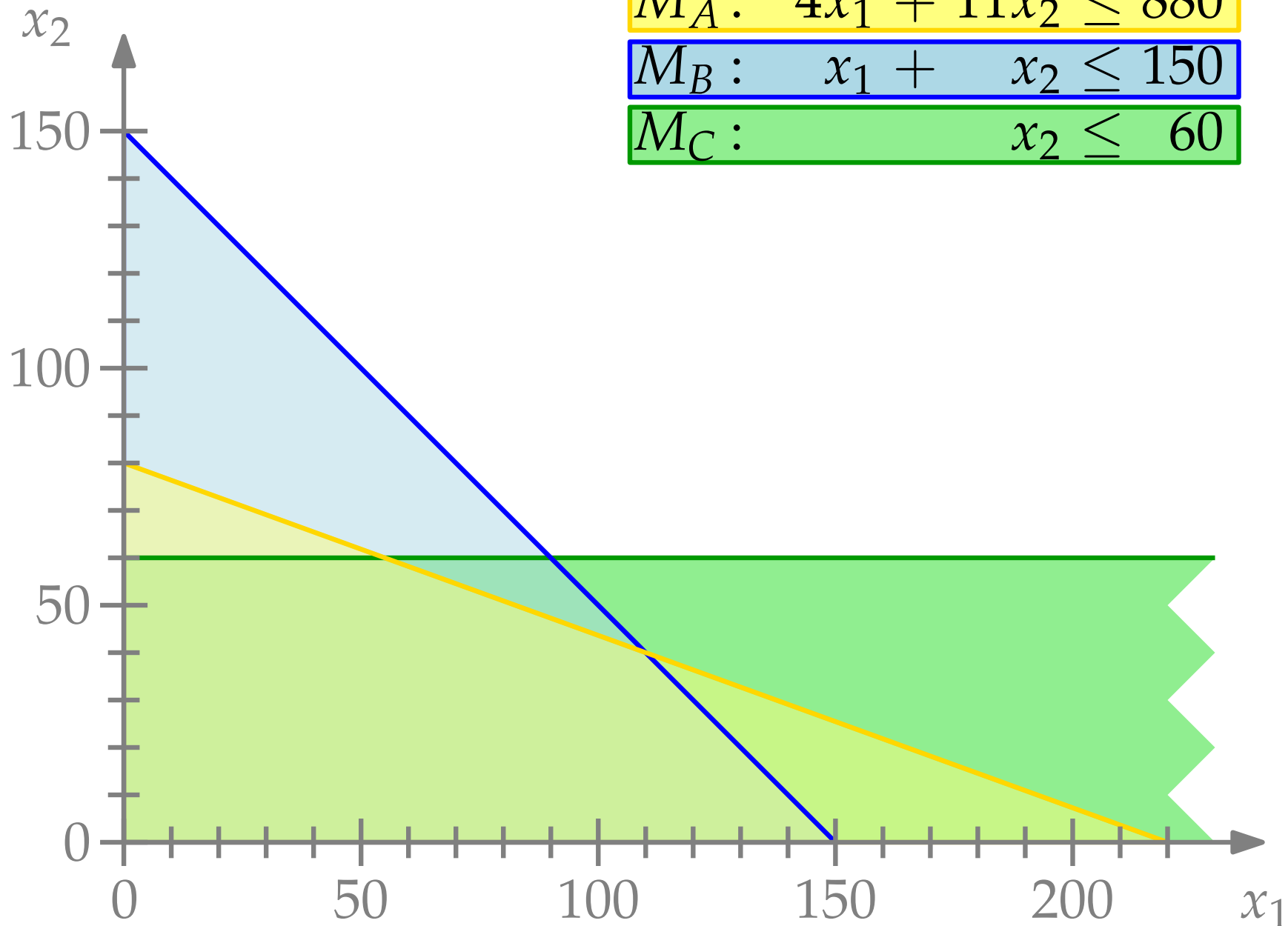
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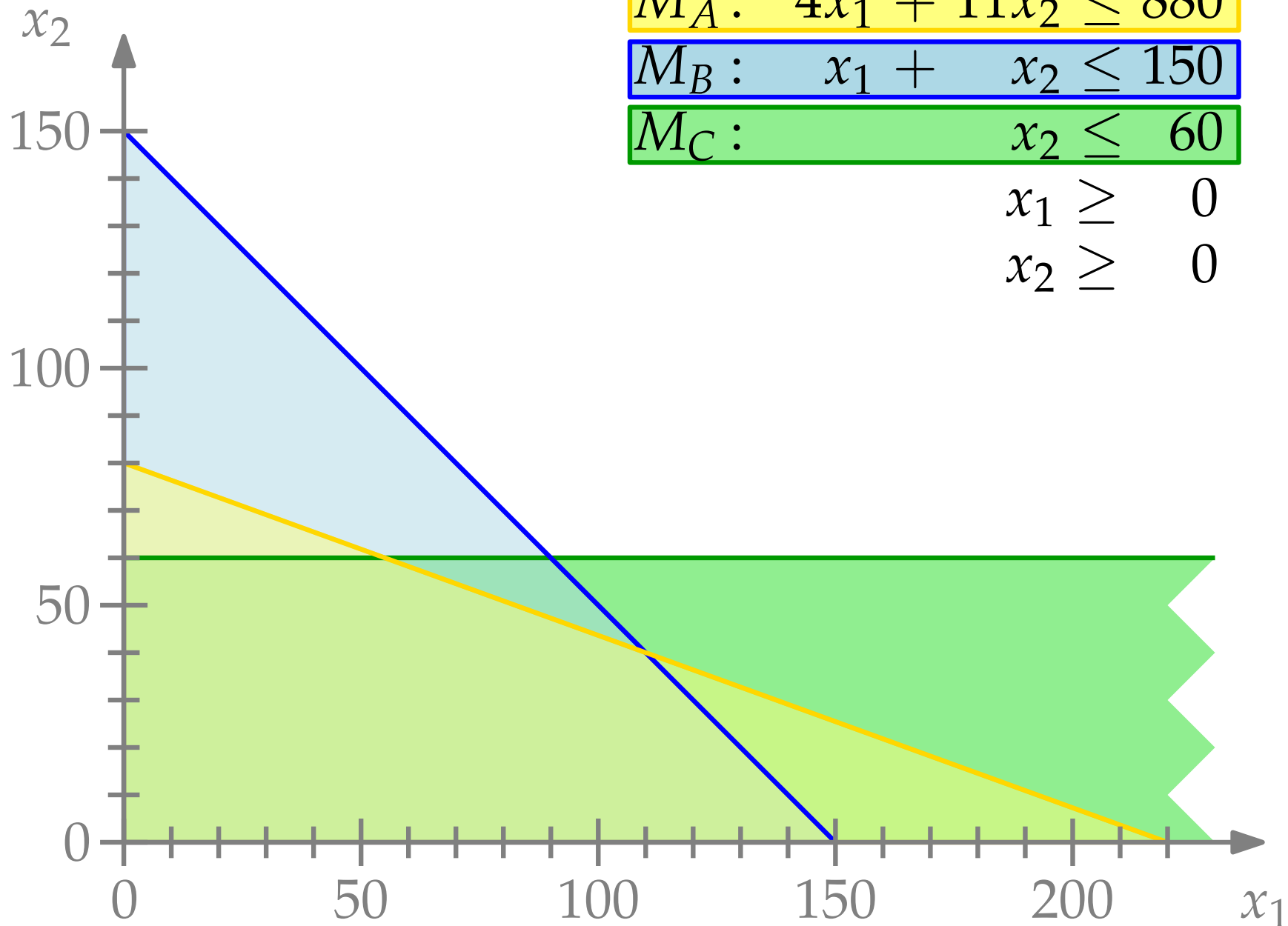
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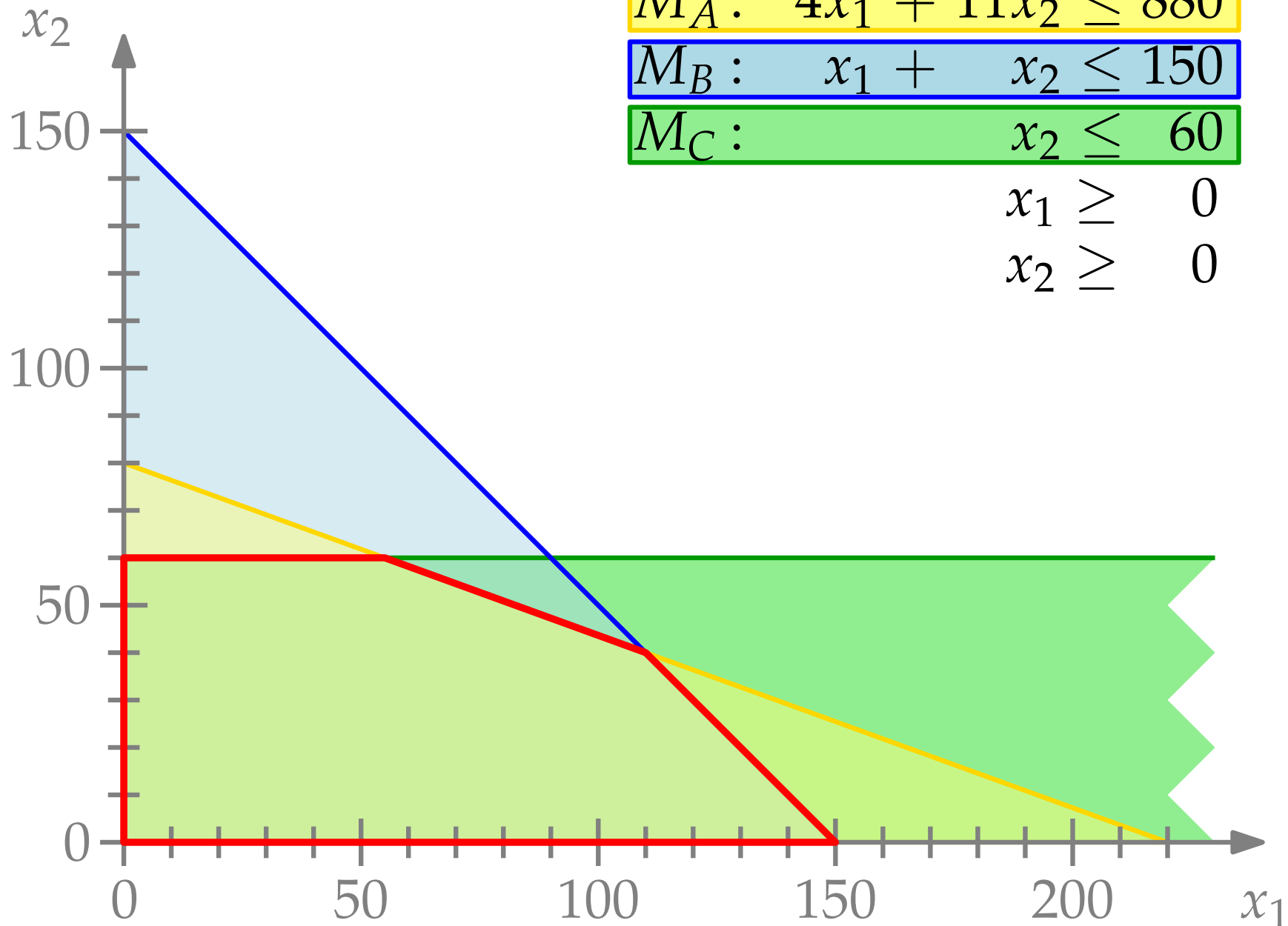
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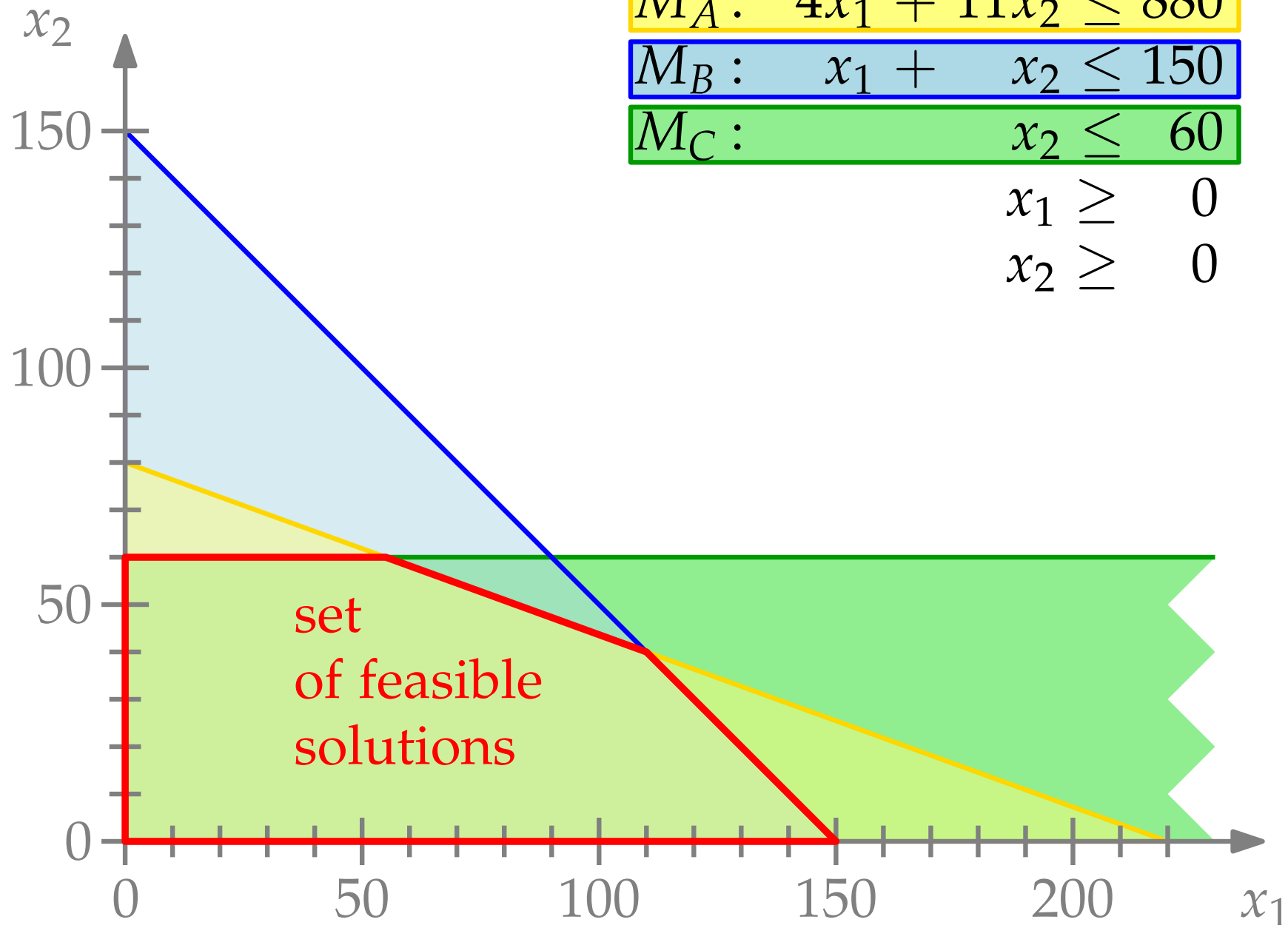
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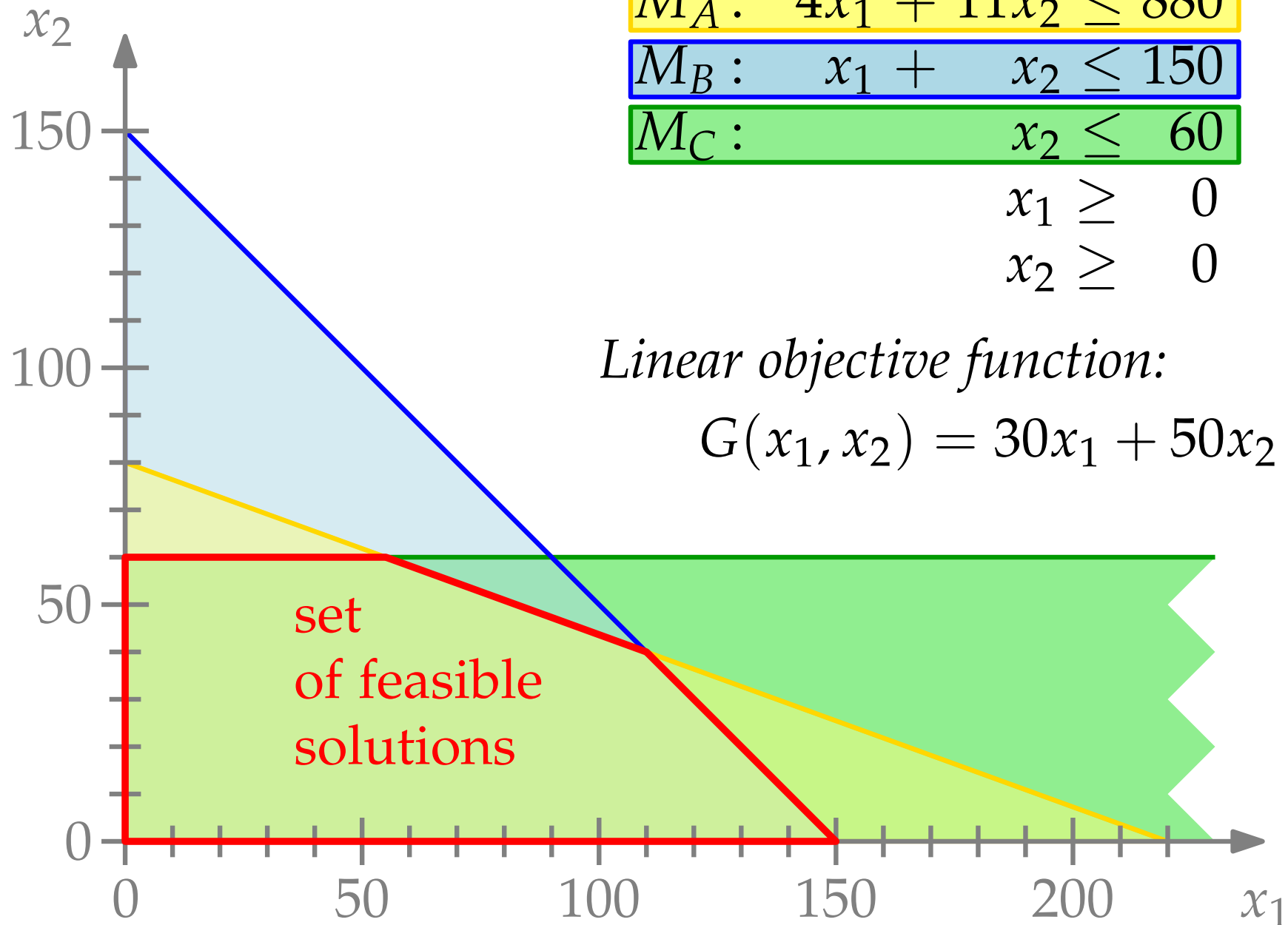
$$M_C: x_2 \leq 60$$

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Linear objective function:

$$G(x_1, x_2) = 30x_1 + 50x_2$$



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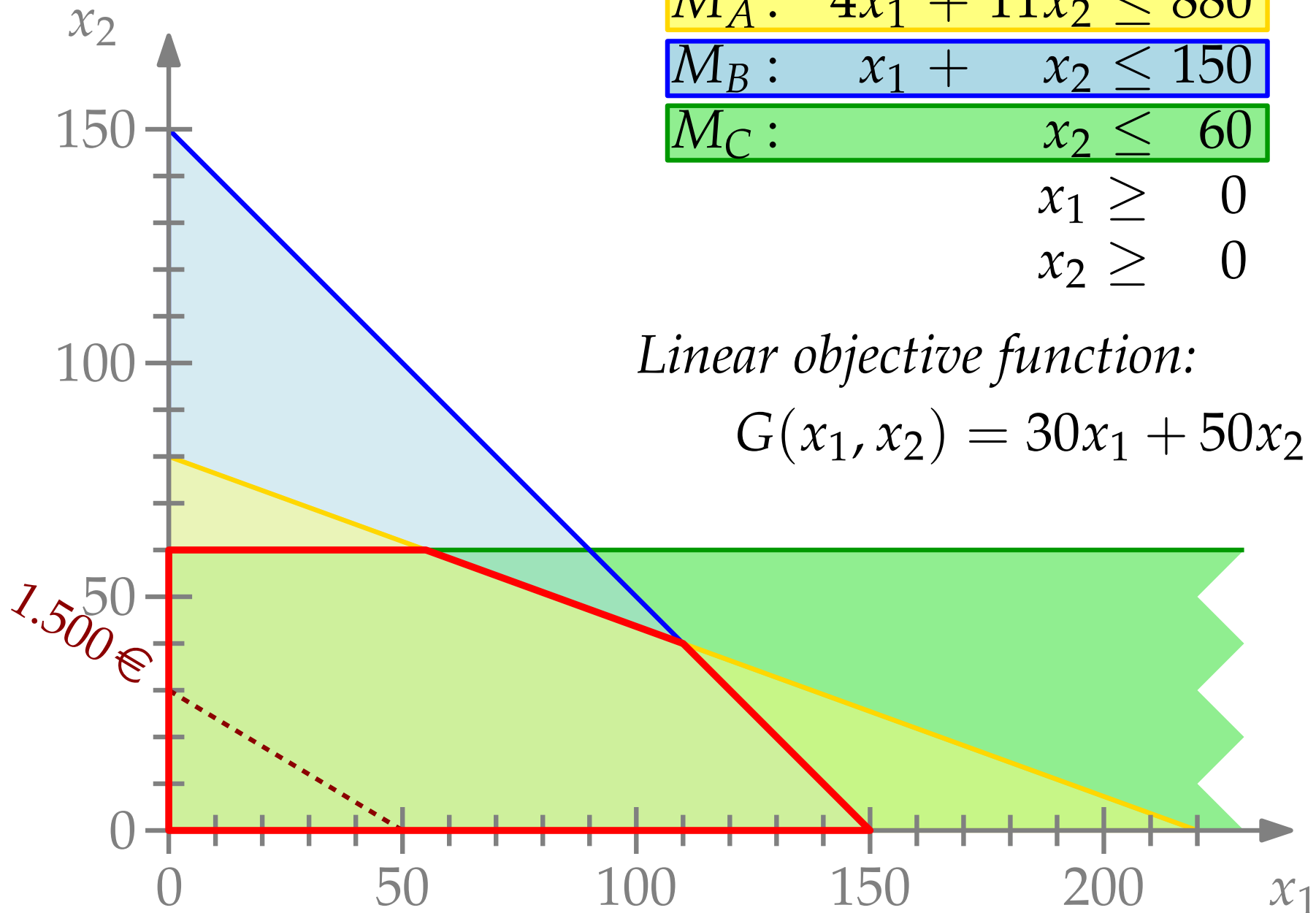
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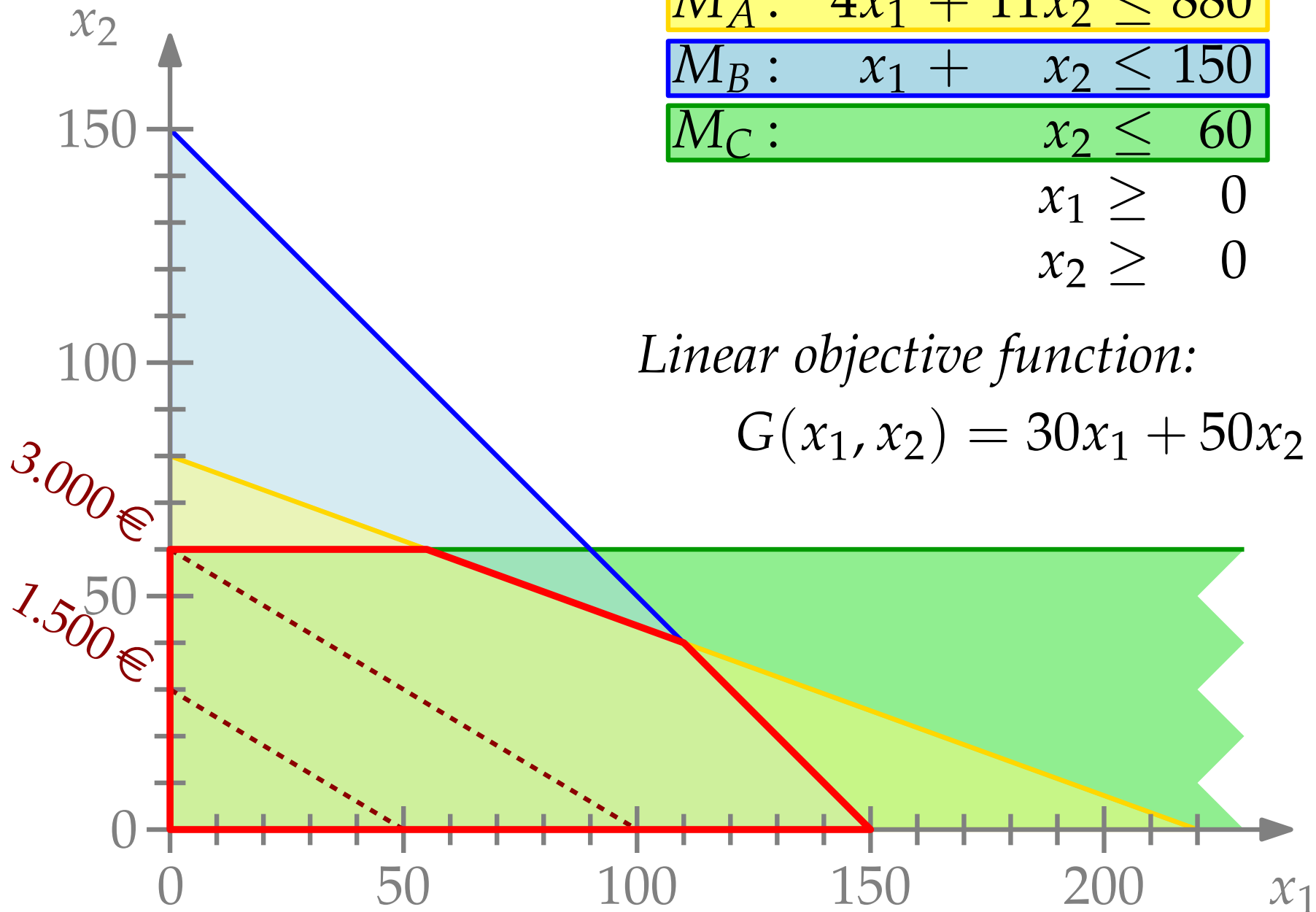
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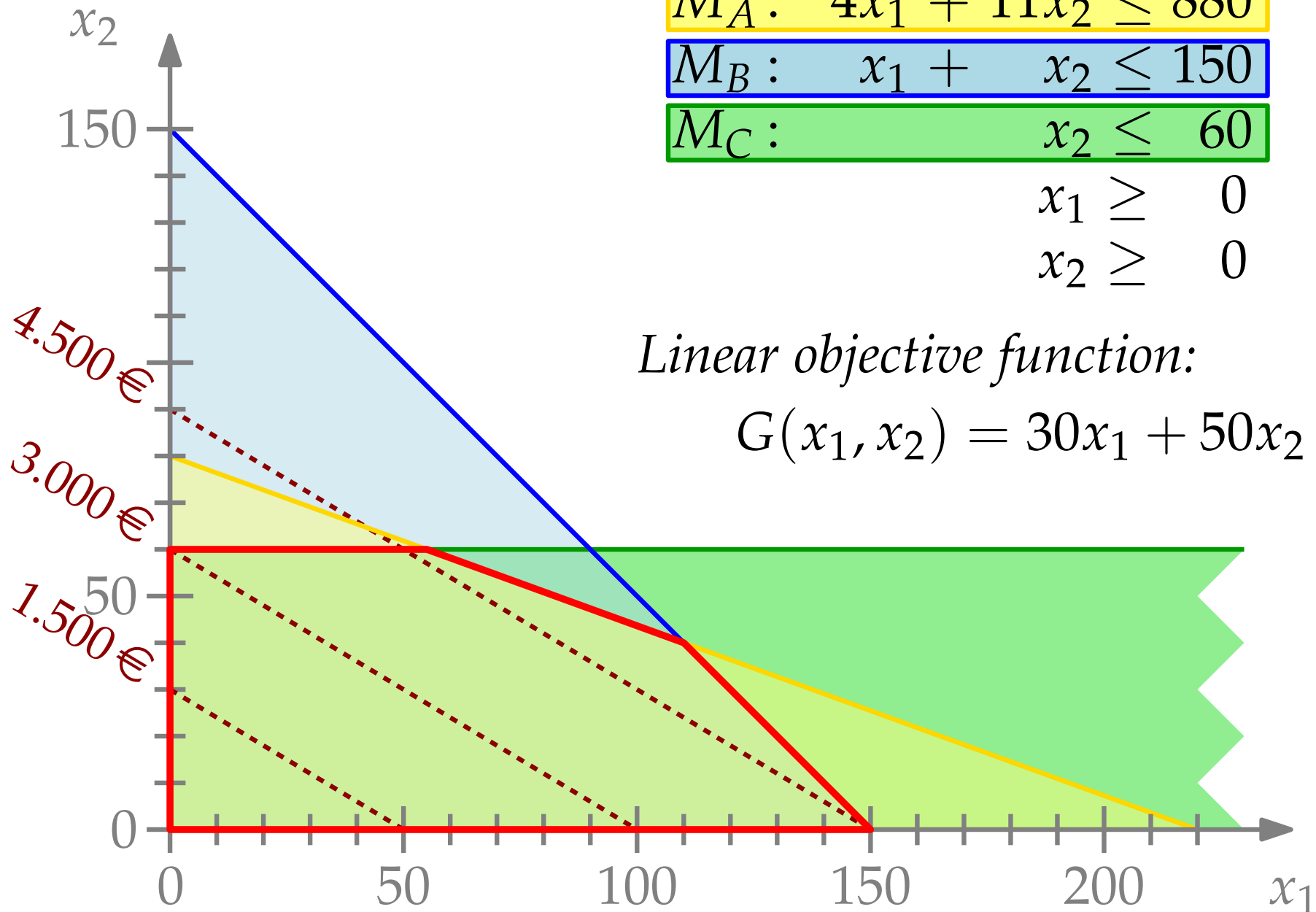
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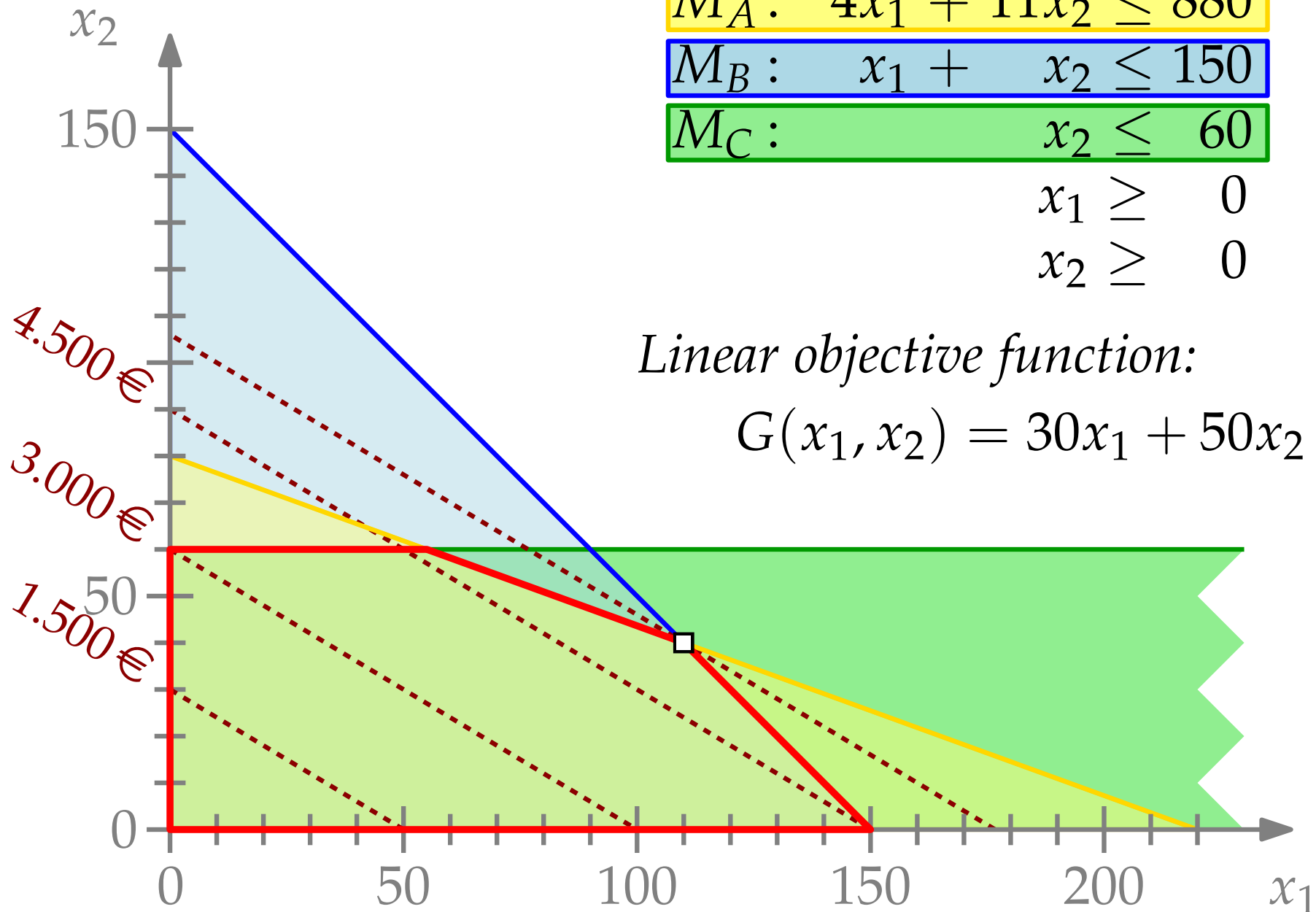
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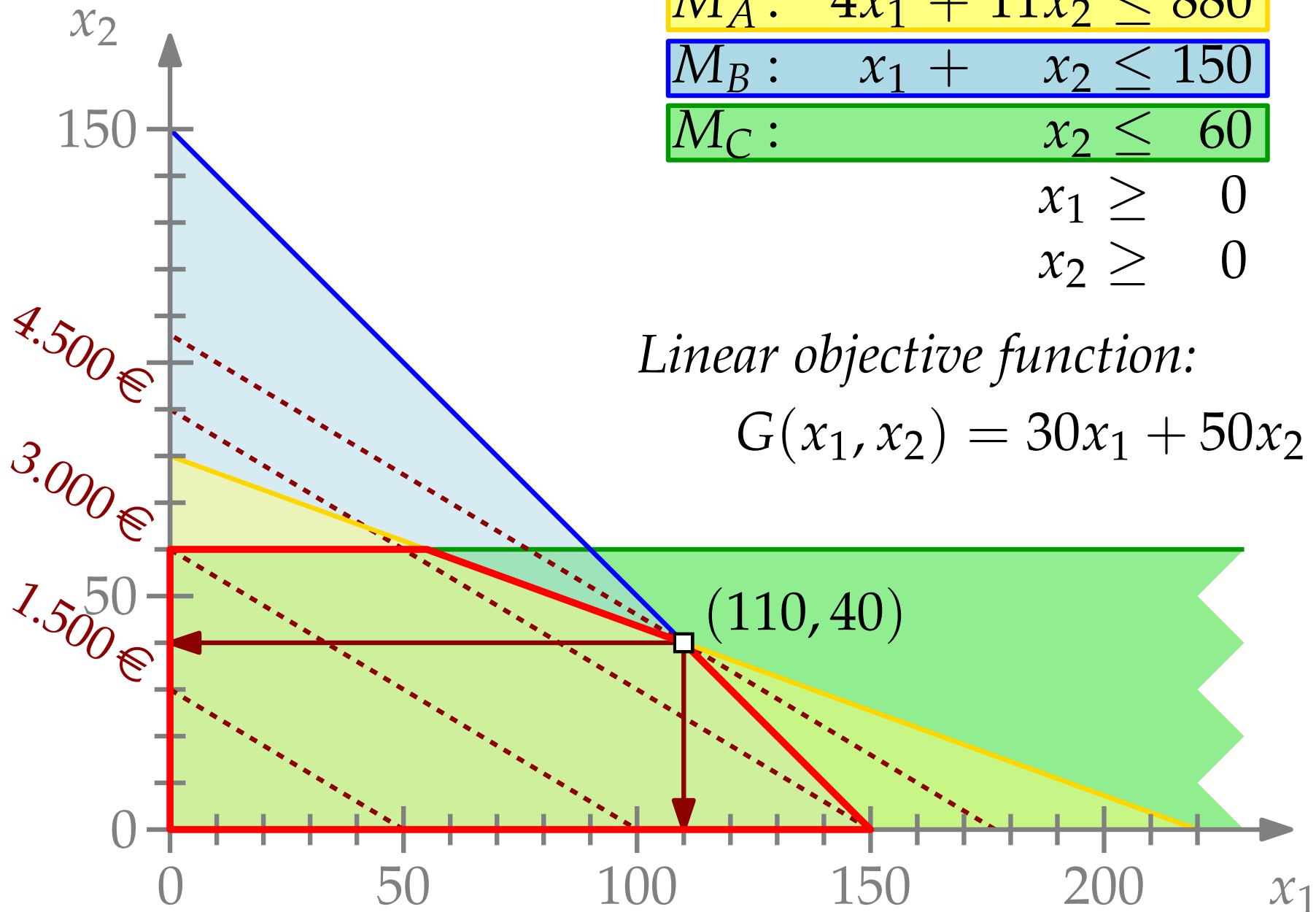
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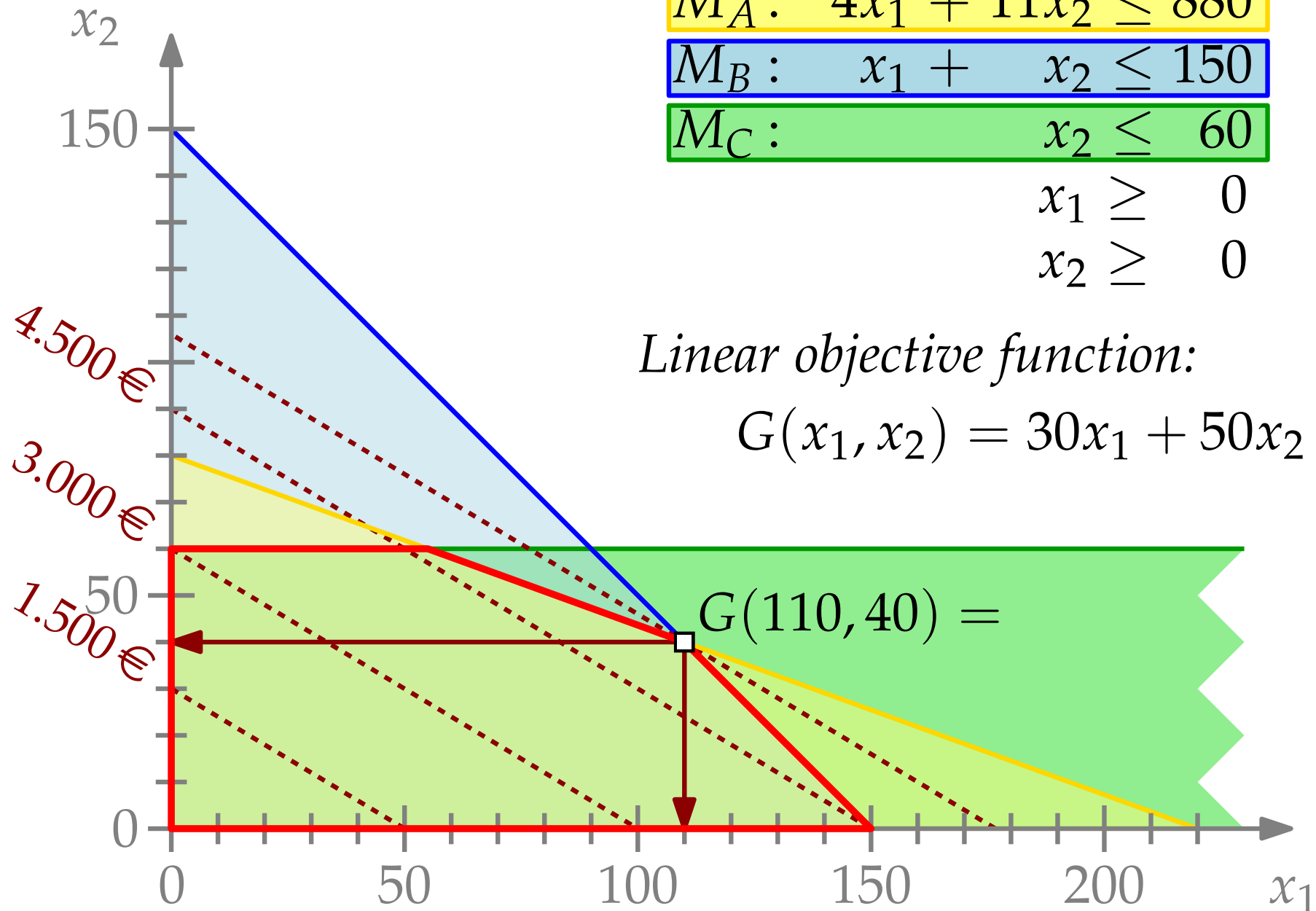
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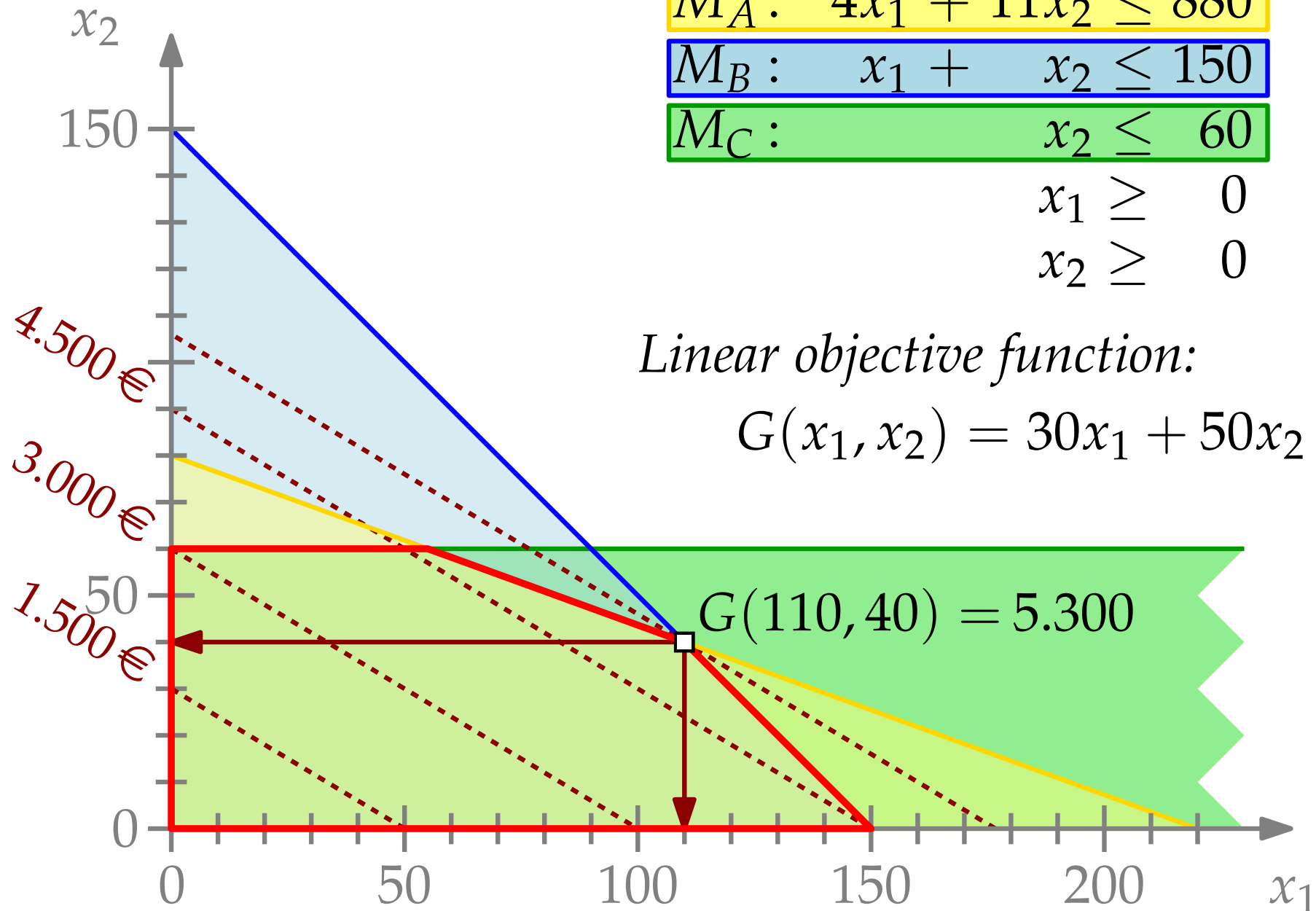
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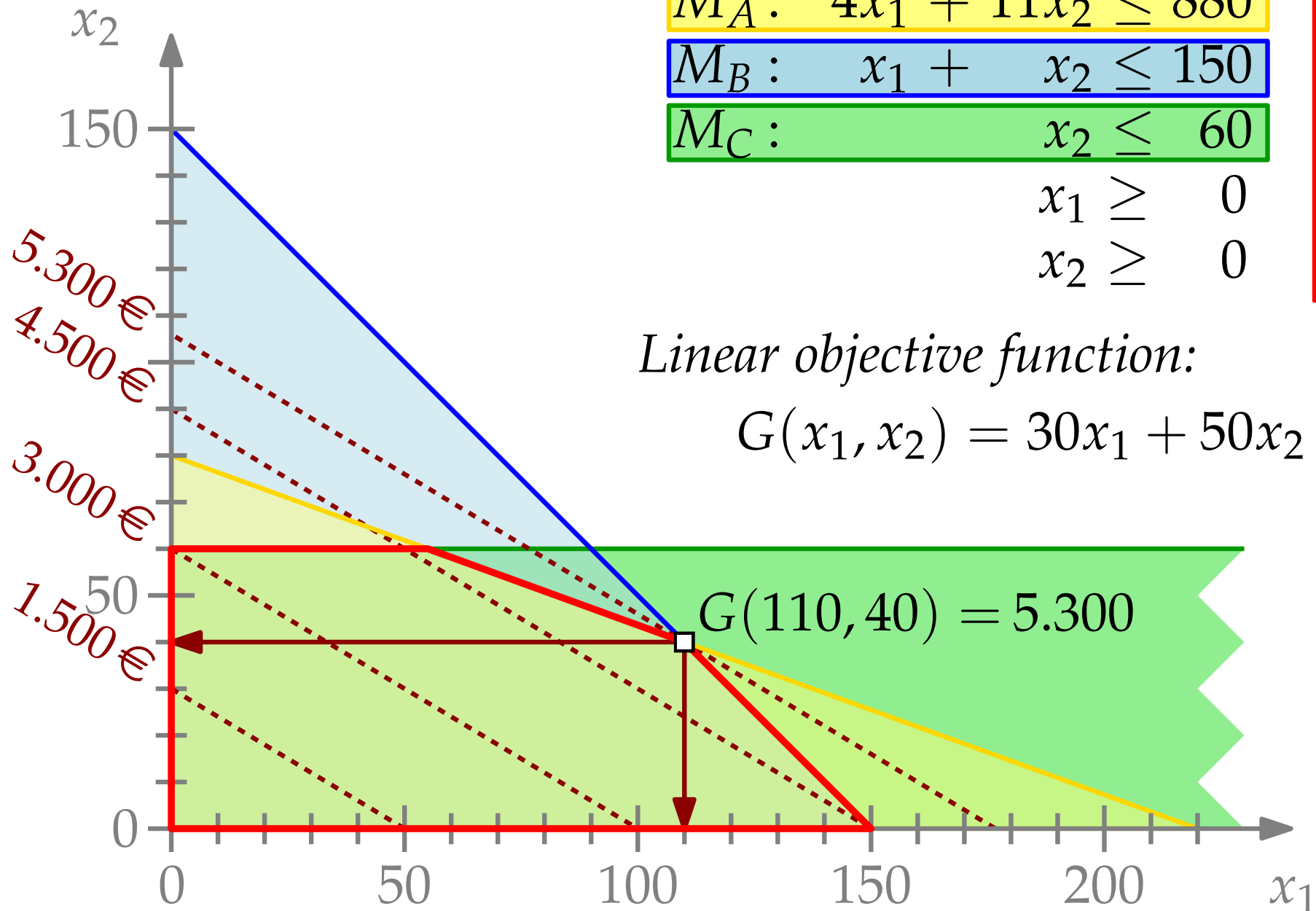
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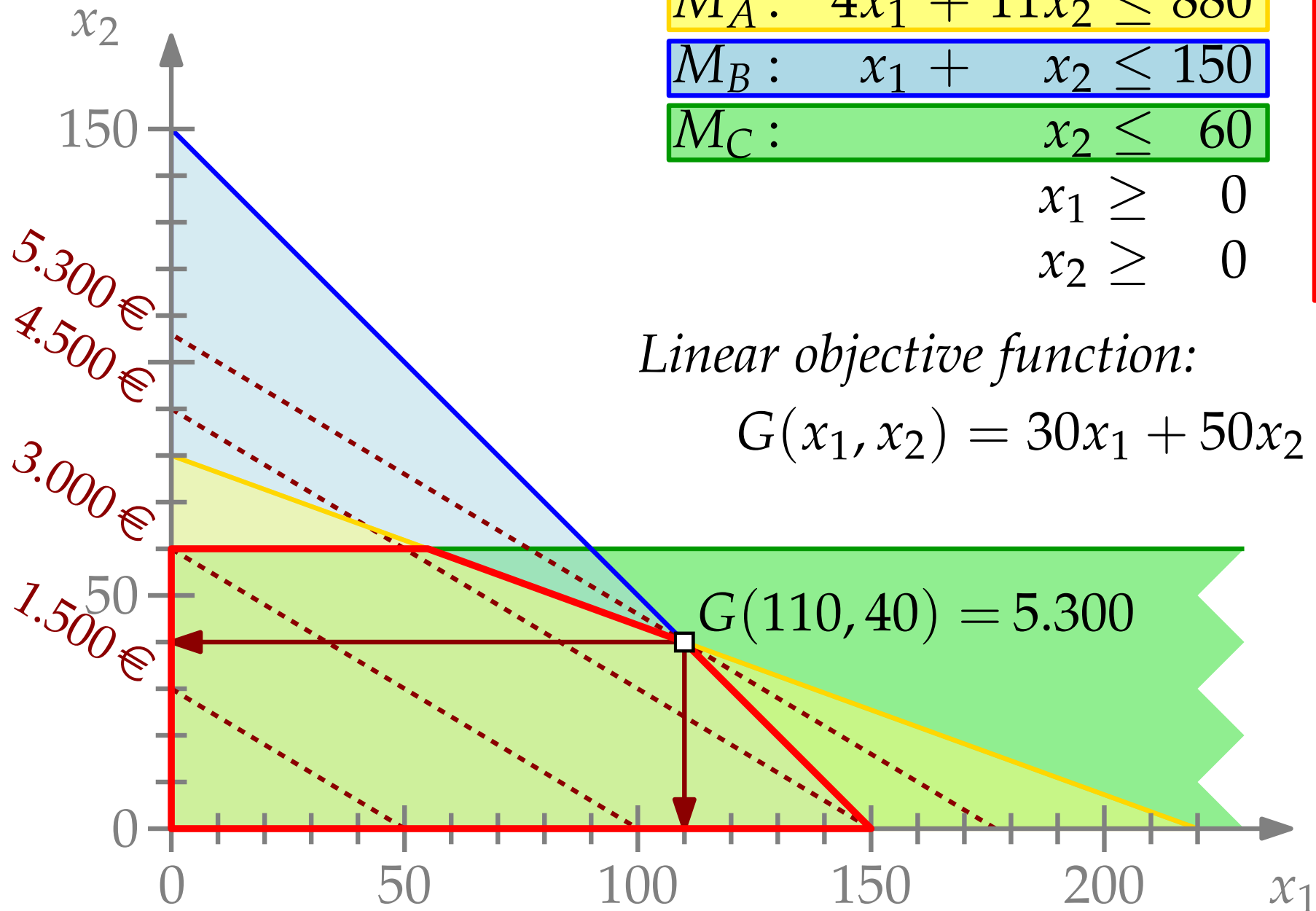
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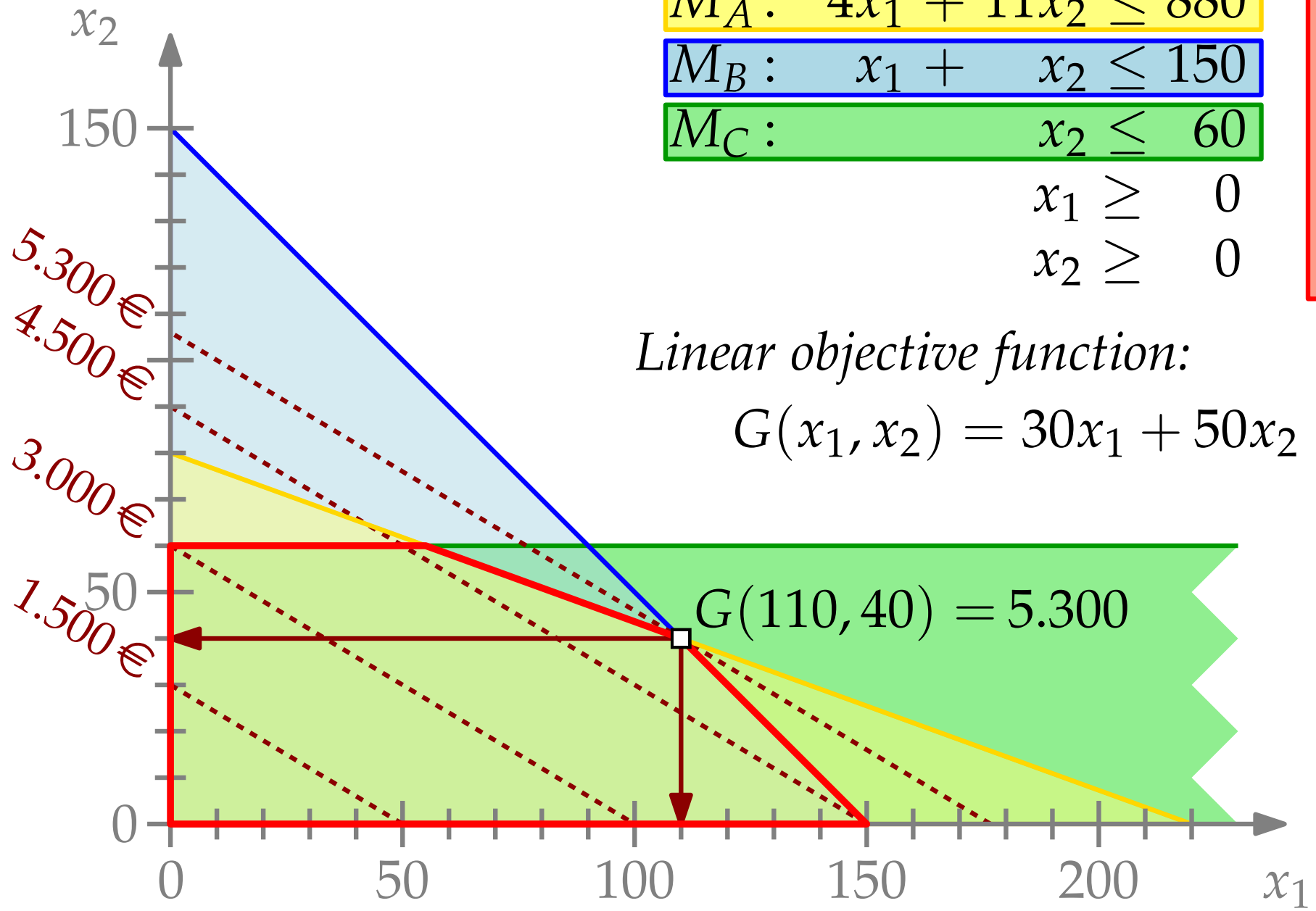
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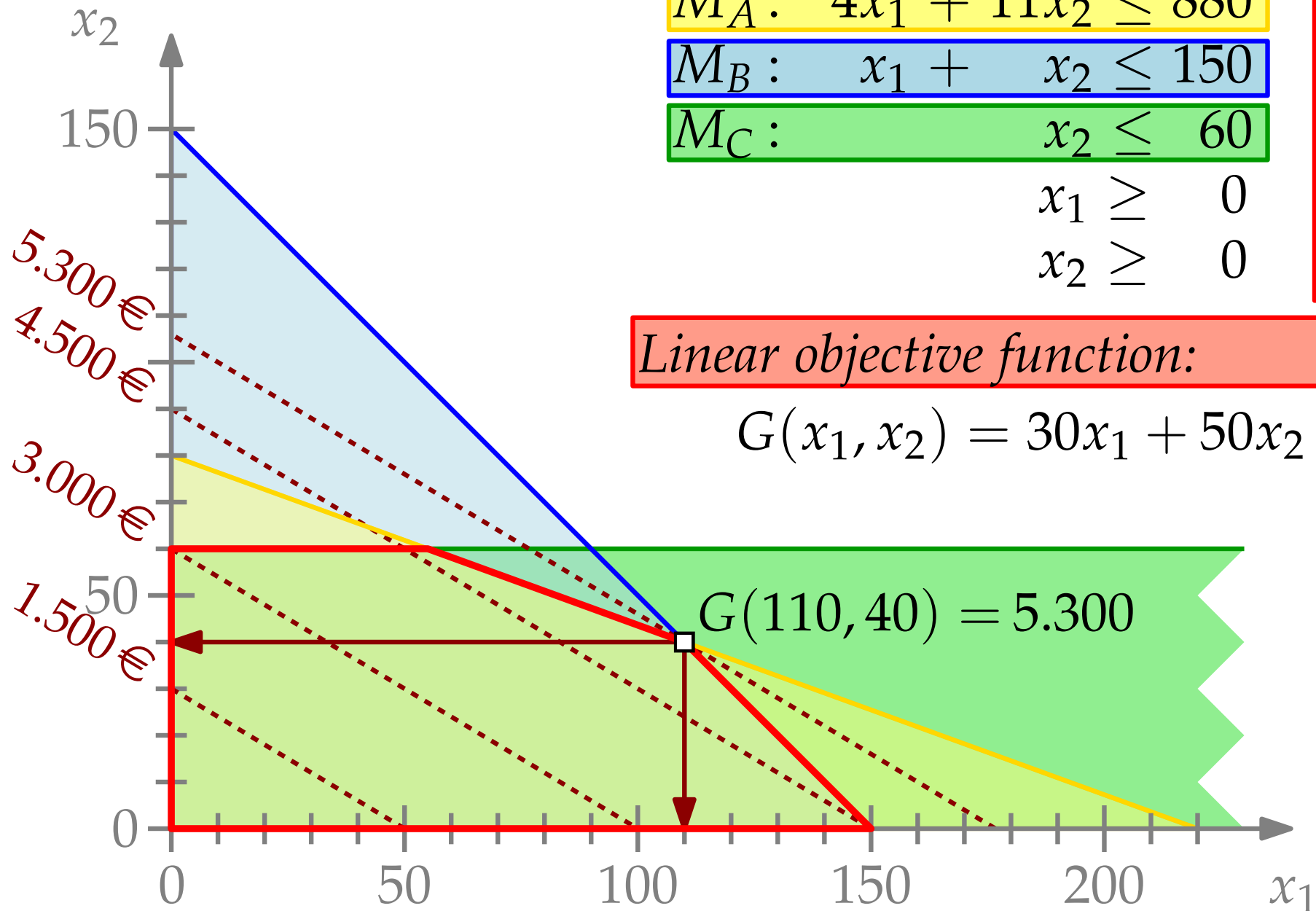
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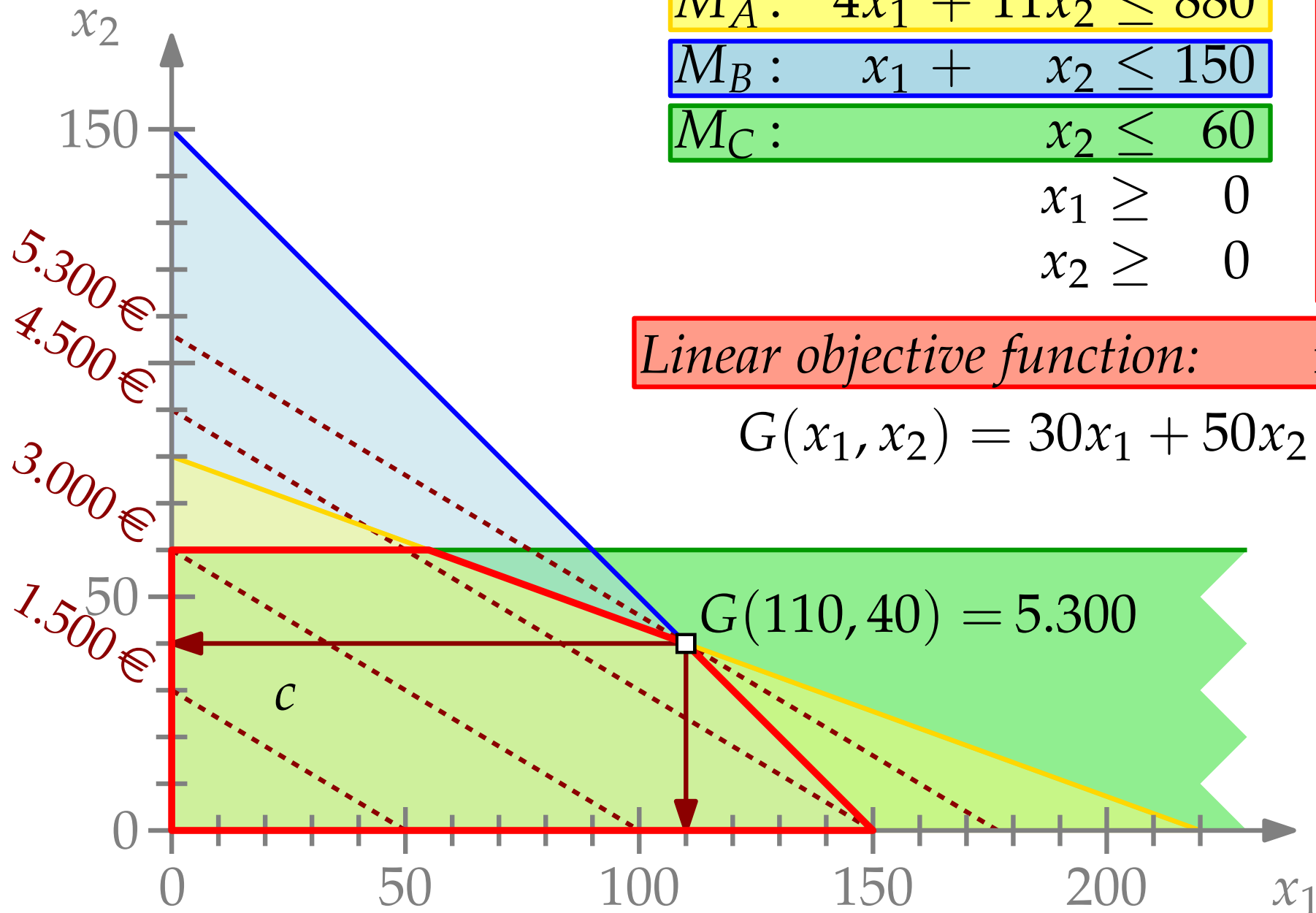
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Linear objective function: $\max c^T x$

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Linear Programming?

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- $x_1 - 4.33x_2 + 0x_4$

Non-Linear:

- $3x_1^2 + 5x_2$
- $10x_1x_2 - 5x_3 + 23 \sin(x_3)$

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- Not programming as in writing code in (say) Java or C++. But it is programming in the sense that we write linear programs (=objective function & constraints) to model and solve real-world problems.

Linear Program (LP)

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$$\begin{aligned} G(x_1, x_2) &= 30x_1 + 50x_2 \\ &= (30, 50) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \end{aligned}$$

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- Sample of hits: “A linear programming approach” ...

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A linear programming approach to the cutting-stock problem

PC Gilmore, RE Gomory - Operations research, 1961 - pubsonline.informs.org

... an approximate solution is being sought by **linear programming**. In this paper, a technique is described for overcoming the difficulty in the **linear programming** formulation of the problem. ...

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A linear programming approach to production and employment scheduling

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[LIBRO] Control of uncertain systems: a linear programming approach

MA Dahleh, IJ Diaz-Bobillo - 1994 - dl.acm.org

We propose a sampling-based **approach** to learn Lyapunov functions for a class of discrete-time autonomous hybrid systems that admit a mixed-integer representation. Such systems ...

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A linear programming approach to max-sum problem: A review

T Werner - IEEE transactions on pattern analysis and machine ..., 2007 - ieeeexplore.ieee.org

... of trees is closest to the **approach** by Schlesinger et al. The **linear programming** relaxation considered by ... The **approach** based on convex combination of trees is more general than the ...

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A linear programming approach to novelty detection

C Campbell, K Bennett - Advances in neural information ..., 2000 - proceedings.neurips.cc

... involves solution of a quadratic **programming** problem. In this paper we propose a simpler kernel method for estimating the support based on **linear programming**. The method is easy to ...

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The linear programming approach to approximate dynamic programming

DP De Farias, B Van Roy - Operations research, 2003 - pubsonline.informs.org

... **linear programming** for approximating solutions to such problems. The **approach** "fits" a **linear** combination of pre-selected basis functions to the dynamic **programming** cost-to-go ...

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An approach to linear programming with 0-1 variables

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... The approximate **approach** using effective gradients for solving problems of **linear programming** with 0-1 variables discussed here will give satisfactory solutions in most cases in a short ...

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Linear programming fundamentals

N Munier - Strategic **Approach** in Multi-Criteria Decision Making: A ..., 2024 - Springer

SIMUS is explained in Chap. 7, and it is the method used in this book to illustrate how most of the aspects related to real-world scenarios can be incorporated in the modelling and ...

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Optimization Study of Cropping Pattern in the Klakah Irrigation Area, Lumajang Regency, Using Linear Programming

W Sejati, TT Akbar - ADI Journal on Recent Innovation, 2024 - adi-journal.org

... The optimization technique used in this crop pattern optimization study employs **linear programming** through the use of the POM QM application. This study plans for 3 alternatives ...

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[PDF] Neutrosophic Fuzzy Regression: A Linear Programming Approach

M Darehmira - Iranian Journal of Operations Research, 2024 - iors.ir

... This work introduces a **linear programming approach** for constructing regression models for ... deviation **approach** to transform the regression issue into a **linear programming** problem. In ...

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Linear program for testing nonclassicality and an open-source implementation

JH Selby, E Wolfe, D Schmid, AB Sainz, VP Rossi - Physical Review Letters, 2024 - APS

... Indeed, the **linear program** we derive is simply a test of whether ... -source code implements the **linear program** we develop herein. ... For now, we simply note that the **linear program** (and ...

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[HTML] Wave energy site location optimizing in Chile: A fuzzy serial linear programming decision-making approach

CN Wang, NL Nhieu, HV Tran - Environment, Development and ..., 2024 - Springer

... To narrow this practice research gap, this study proposes hybrid **approaches** between DEA, FBWM and Bonferroni FCoCoSo to determine suitable locations for WESS. ...

☆ Salva 99 Cita Citato da 5 Articoli correlati Web of Science: 4

Improving estimates accuracy of voter transitions. Two new algorithms for ecological inference based on linear programming

JM Pavia, R Romero - Sociological Methods & Research, 2024 - journals.sagepub.com

... First, we suggest a novel procedure, based on **linear programming** and grounded in the homogeneity hypothesis, to estimate the inner-cells values at the local level. We call this ...

☆ Salva 99 Cita Citato da 16 Articoli correlati Tutte e 8 le versioni Web of Science: 4

Hybrid quantum-classical computing via Dantzig-Wolfe decomposition for integer linear programming

X Wei, J Liu, L Fan, Y Guo, Z Han - ..., 2024 - ieeeexplore.ieee.org

... 1, in HyDWD, the master problem constitutes a **linear programming** problem solvable on a ... In Section III, we briefly review the classical integer **linear programming** problem and its ...

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Linear Program (LP)

Given: matrix $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$

Find: $x^* \in \mathbb{R}^n$ where $x^* = \arg \max \{c^T x \mid Ax \leq b\}$

- Software that solves linear programs – *linear programming solvers* – also generate lots of important auxiliary information (as well as the optimum).
- Sensitivity analysis, shadow prices, alternative optima (all very useful for understanding the meaning, stability and significance of the optimum solutions generated by linear programming solvers; important in industry.)

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- ok, ok, but how do we solve LPs ?

Why study the Math of LPs?

- Many people can do basic modelling using linear programming. In many cases this is good enough.
- But the separation of what and how is, of course, still not complete.
- ...so your job is to also understand what is going on “in the engine room”
- This will help you get maximum performance from your LP solver and deal with phenomena such as degeneracy.

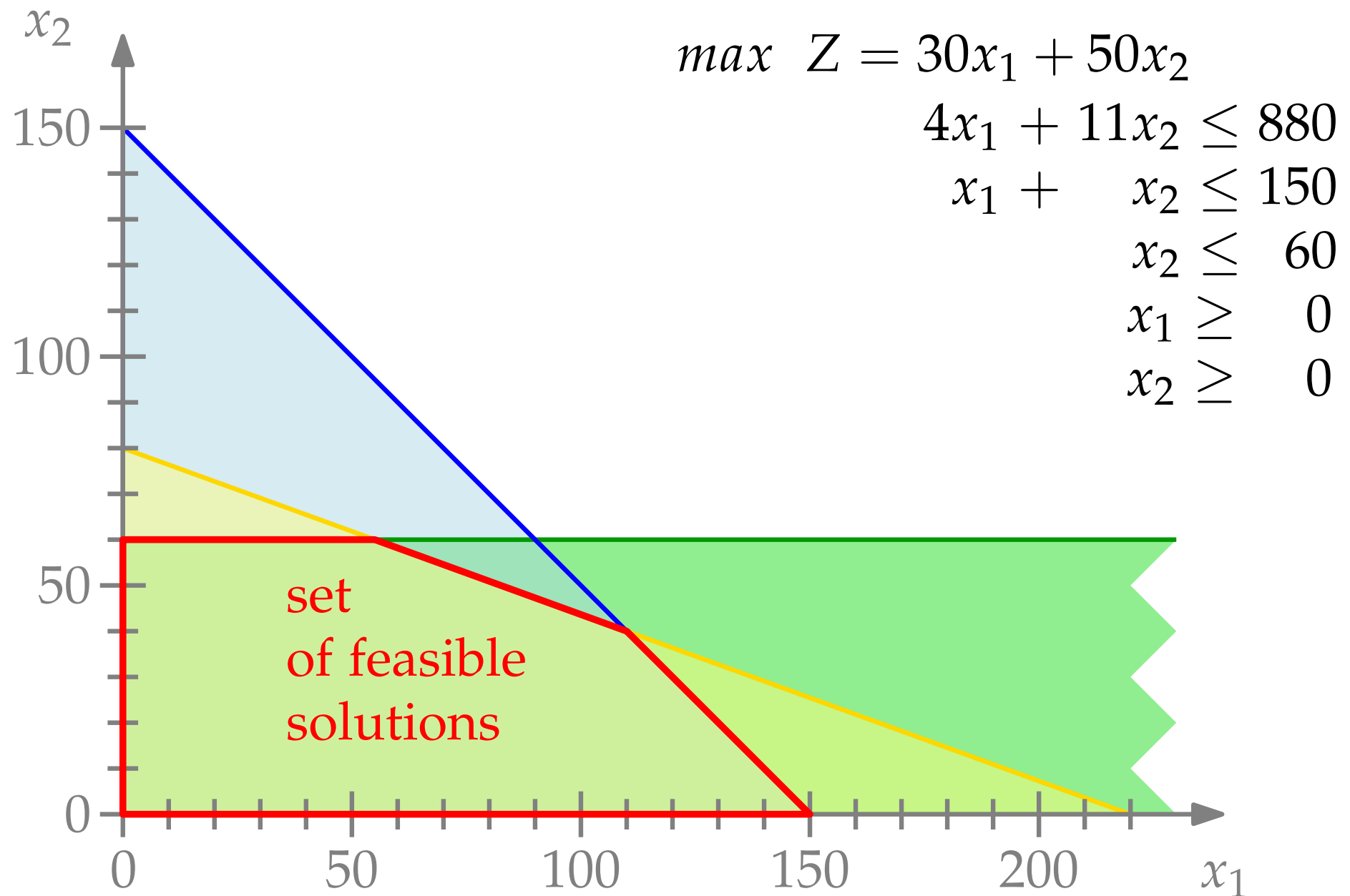
Why study the Math of LPs?

- LP can be viewed as the first step in a hierarchy of increasingly complex forms of mathematical programming (non-linear programming, especially convex programming; integer programming)
- Understanding the mathematics of LP is an important step towards understanding these mathematically more complex paradigms
- The abstract mathematical theory behind linear programming turns out to be extremely important for certain theoretical mathematical fields, e.g., approximation algorithms built around the scaffolding of LP- duality.
- Many of these scientists will never write or solve linear programs, but the theory of linear programming will be ever-present in their work.

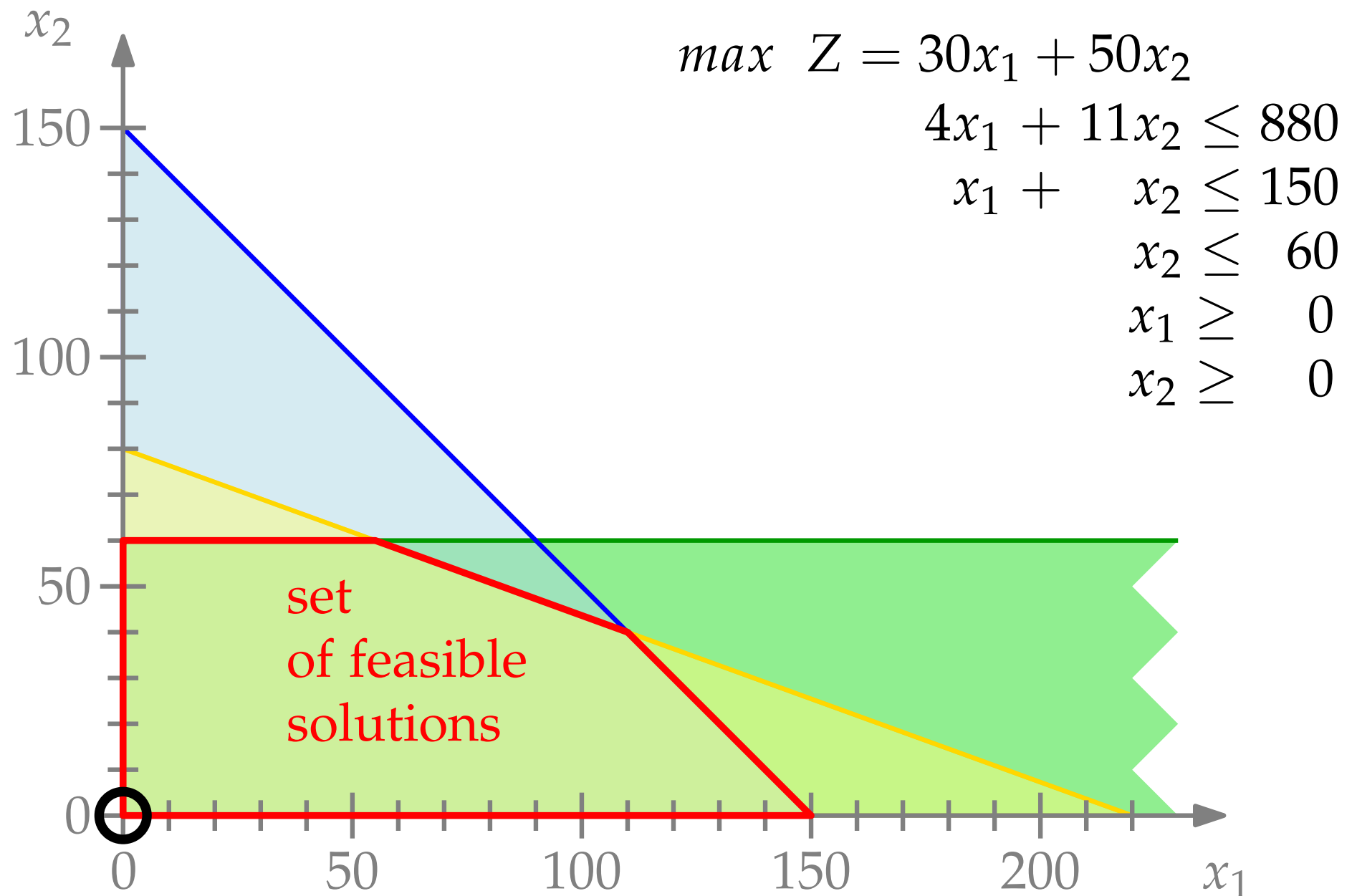
Solving LPs

- Most linear programming solvers are based on either or both of the following two techniques:
 - The Simplex method
 - Interior-point methods
- It is not trivial to solve really big linear programs. All kinds of extra mathematical tricks are needed as the size of the program grows.

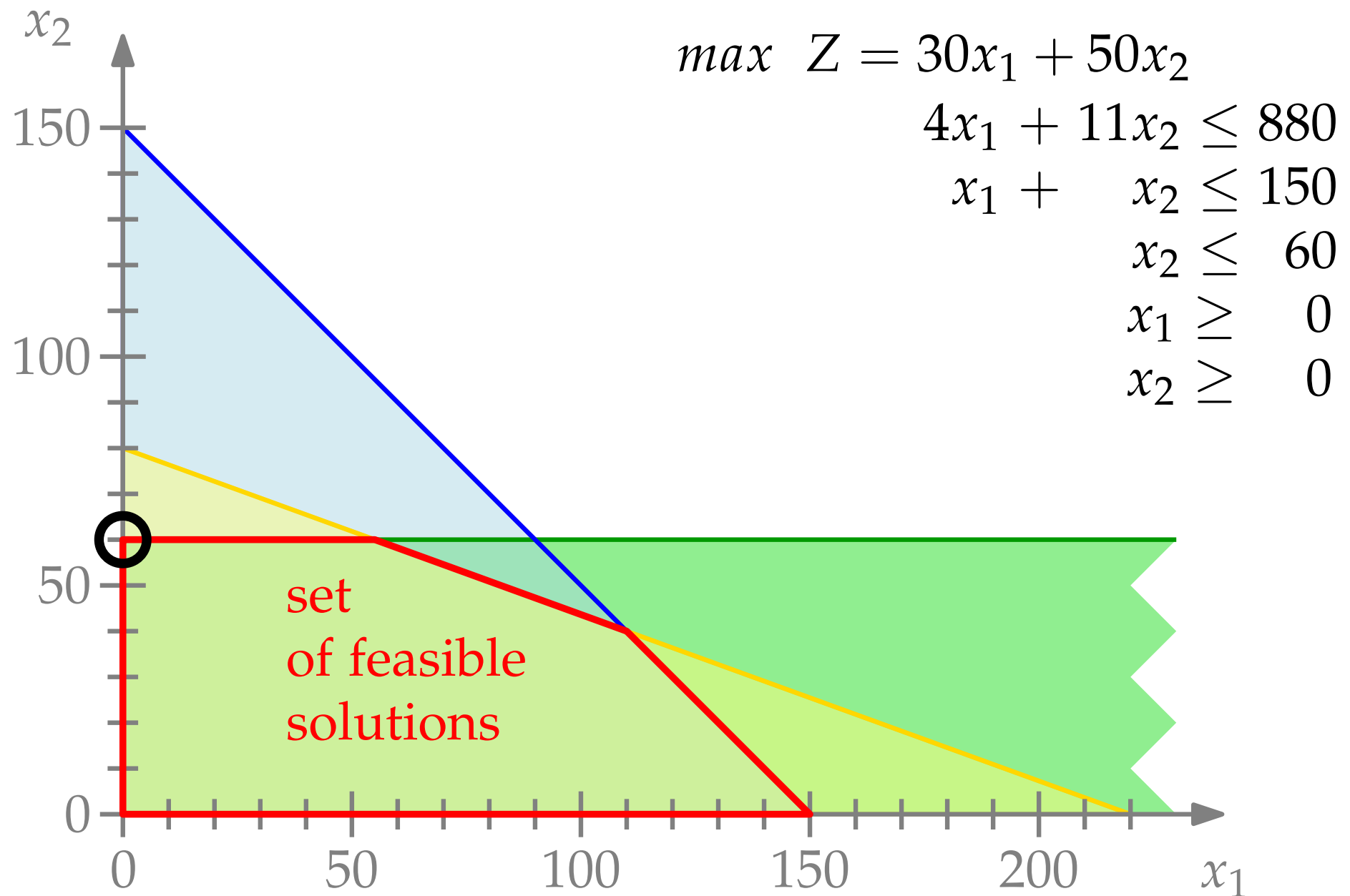
Exterior Point Methods



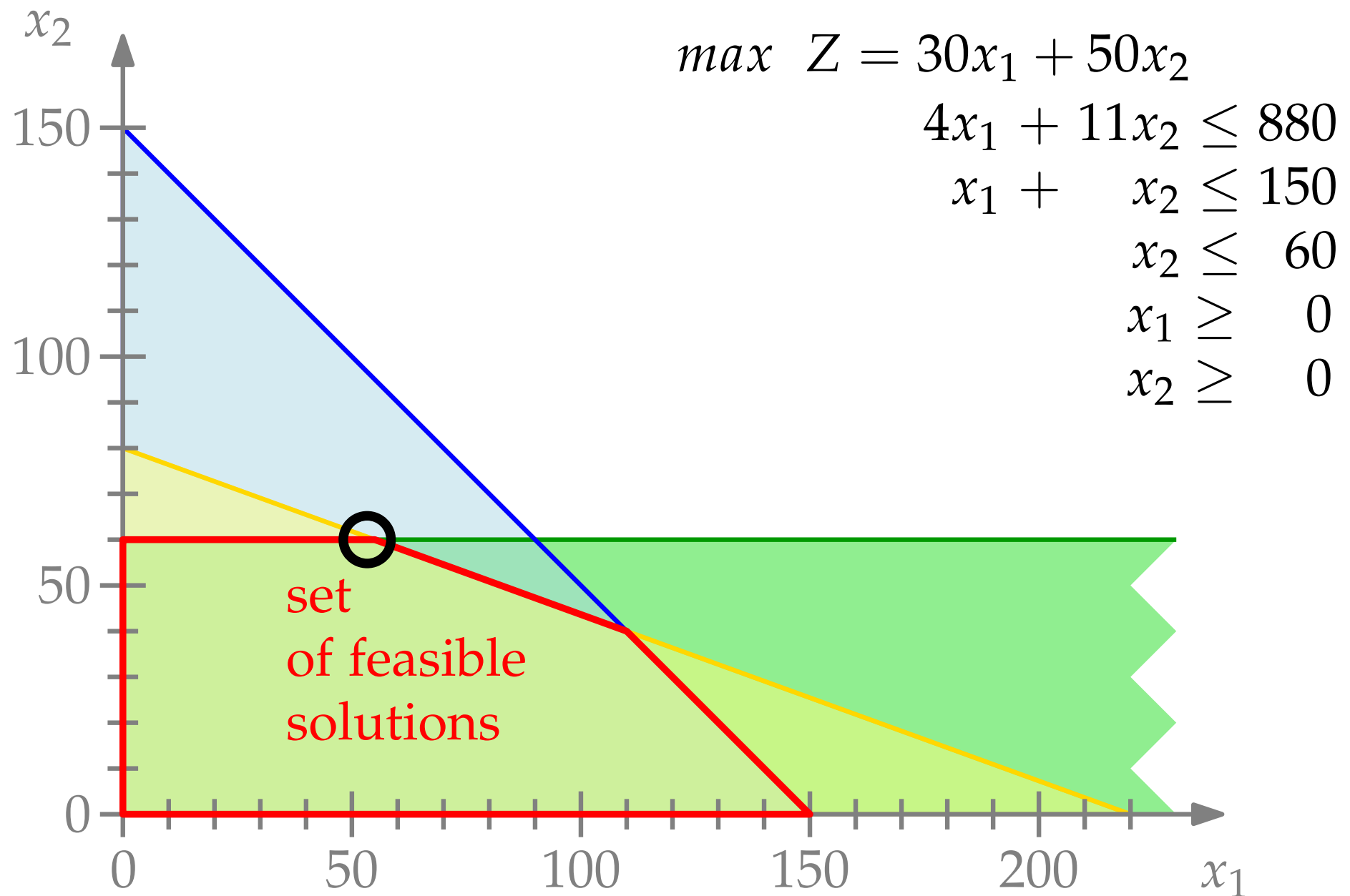
Exterior Point Methods



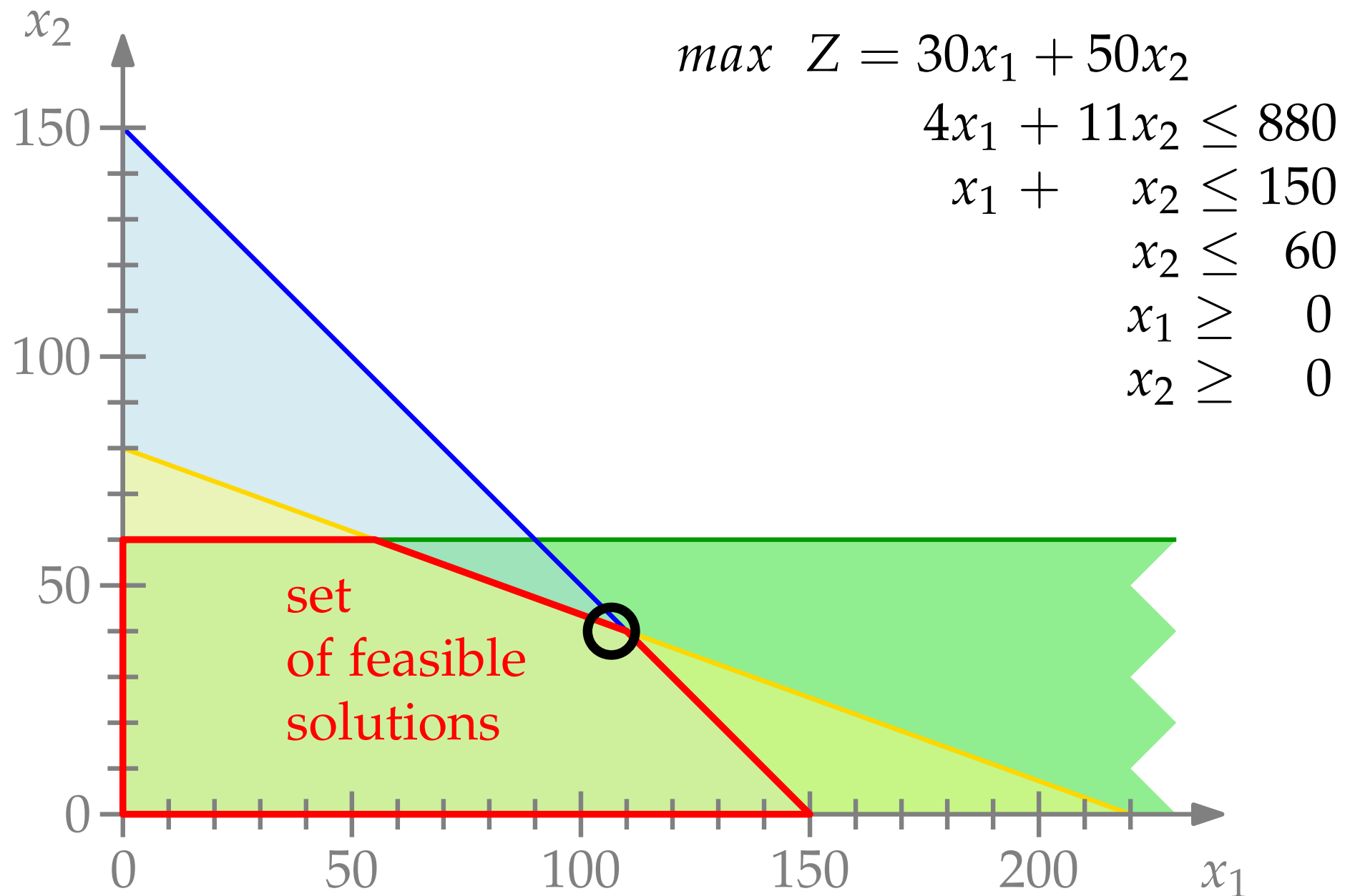
Exterior Point Methods



Exterior Point Methods



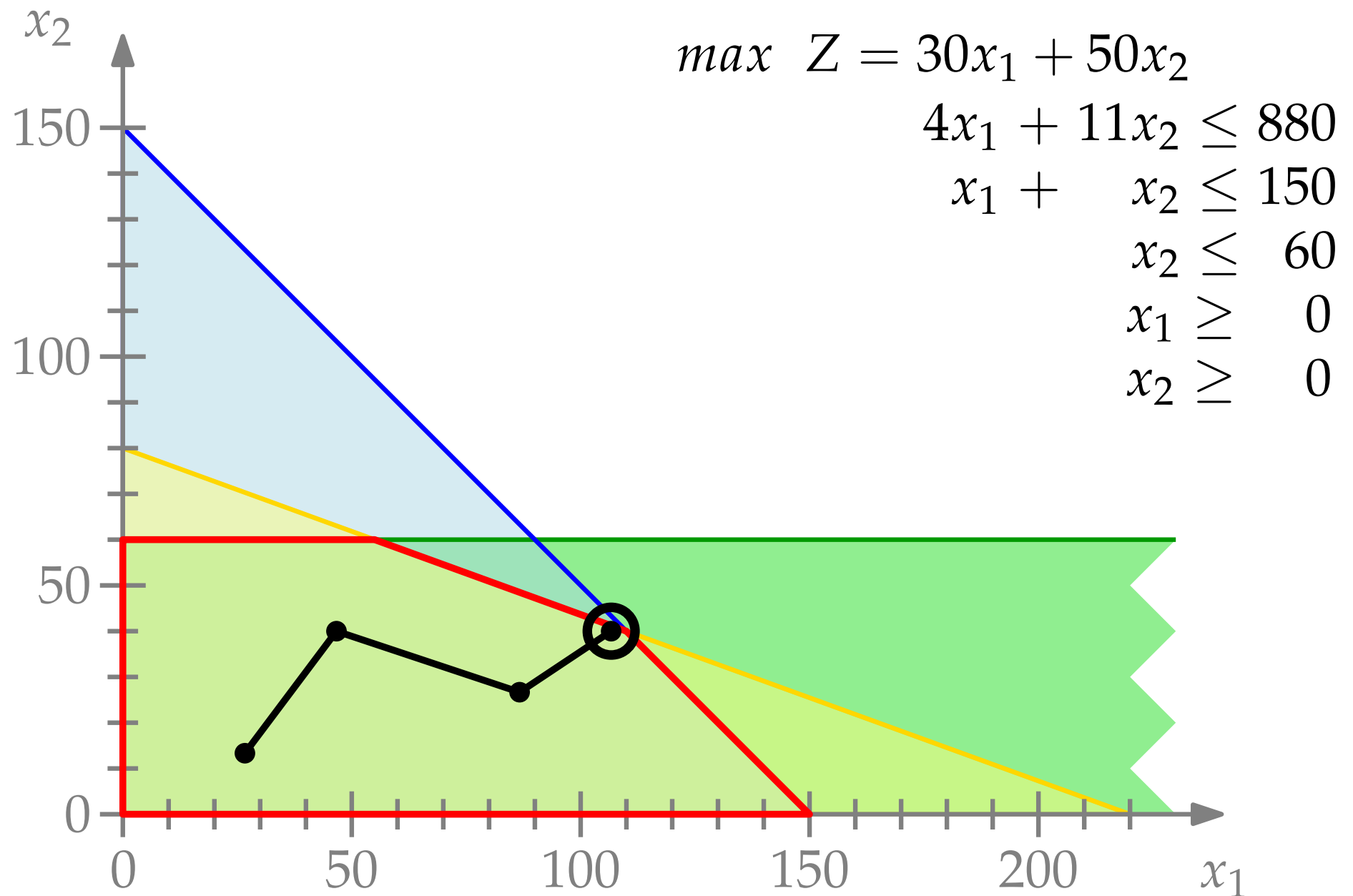
Exterior Point Methods



The Simplex method

- The most frequently used algorithm to solve linear programs
- The core algorithm is quite simple (...)
- Exterior-point method: it goes around the perimeter of the feasible region, one cornerpoint at a time
- Very useful for extended analysis: re- optimisation, shadow prices, sensitivity analysis etc.
- “Bad cases” for the algorithm hardly ever occur in practice

Interior Point Methods



Interior Points Method

- More recent than the Simplex method
- These methods go through the middle of the feasible region, not around the outside
- Per iteration, much slower than the Simplex method
But the number of iterations required increases much more slowly than with the Simplex method
- Thus: superior/competitive for extremely large instances

Terminology

Barbara Franci

The Wyndor Glass Example

■ **TABLE 3.1** Data for the Wyndor Glass Co. problem

Plant	Production Time per Batch, Hours		Production Time Available per Week, Hours
	Product		
	1	2	
1	1	0	4
2	0	2	12
3	3	2	18
Profit per batch	\$3,000	\$5,000	

The Wyndor Glass Example

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$$\begin{aligned}
 &\text{Maximize} && Z = && 3x_1 &+& 5x_2 \\
 &\text{subject to} && && x_1 && \leq 4 \\
 &&& && && x_2 \leq 12 \\
 &&& && 3x_1 &+& 2x_2 \leq 18 \\
 &\text{and} && x_1 \geq 0, && x_2 \geq 0
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*Objective
Function*

$$\begin{aligned}
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Objective Function

Maximize $Z = 3x_1 + 5x_2$

subject to

$$\begin{aligned} x_1 &\leq 4 \\ x_2 &\leq 12 \\ 3x_1 + 2x_2 &\leq 18 \end{aligned}$$

and

$$x_1 \geq 0, \quad x_2 \geq 0$$

x_1 and x_2 :
Decision Variables

The Wyndor Glass Example

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Objective Function *Functional Constraints*

Maximize $Z = 3x_1 + 5x_2$

subject to

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x_1 and x_2 : Decision Variables and $x_1 \geq 0, x_2 \geq 0$

The Wyndor Glass Example

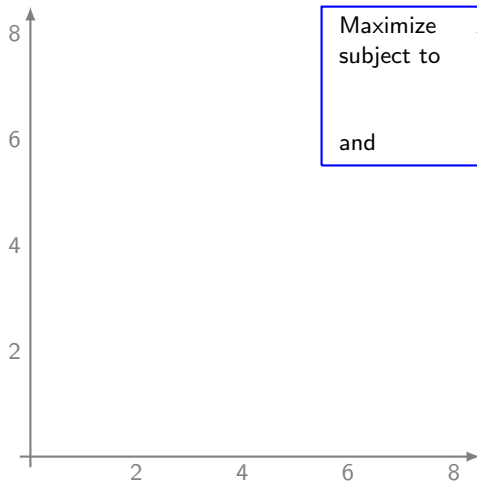
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Objective Function *Functional Constraints*
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 subject to
 $x_1 \leq 4$
 $x_2 \leq 12$
 $3x_1 + 2x_2 \leq 18$
 and $x_1 \geq 0, x_2 \geq 0$ *Nonnegativity Constraints*
Decision Variables

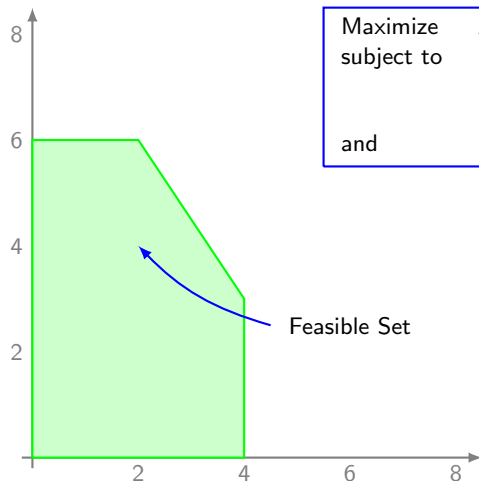
x_1 and x_2 :

The Wyndor Glass Example



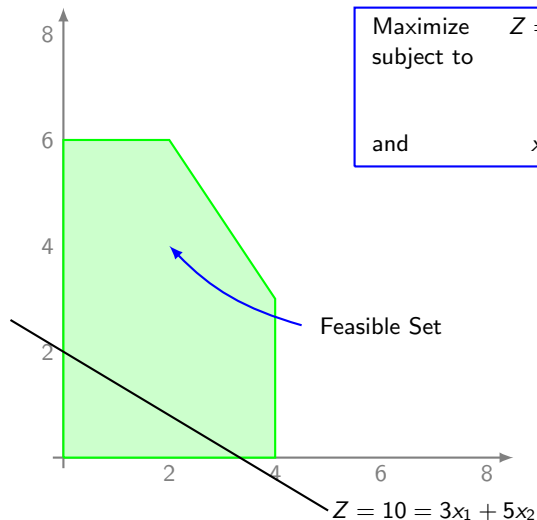
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The Wyndor Glass Example



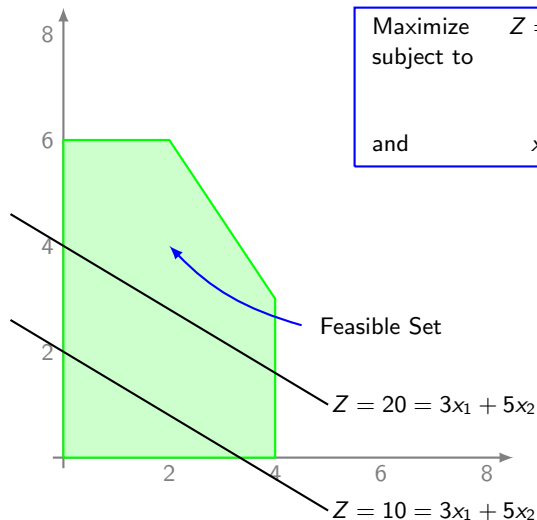
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The Wyndor Glass Example



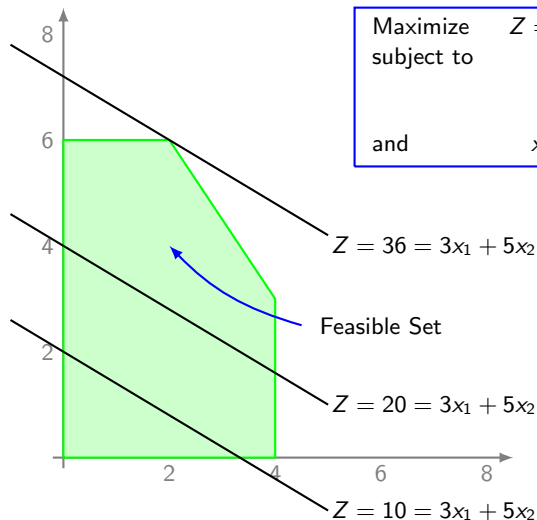
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The Wyndor Glass Example



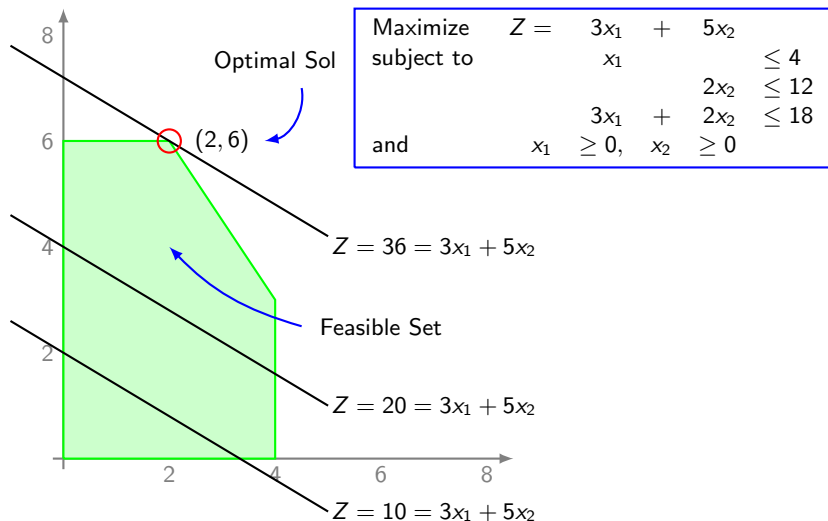
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The Wyndor Glass Example



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The Wyndor Glass Example



The Wyndor Glass Example

- ▶ In this example, the feasible region is **non-empty**, and **bounded**.

The Wyndor Glass Example

- ▶ In this example, the feasible region is **non-empty**, and **bounded**.
- ▶ However, it can happen that the feasible region is **empty**, i.e., there are no solutions possible. We then say that the linear program is **infeasible**.

The Wyndor Glass Example

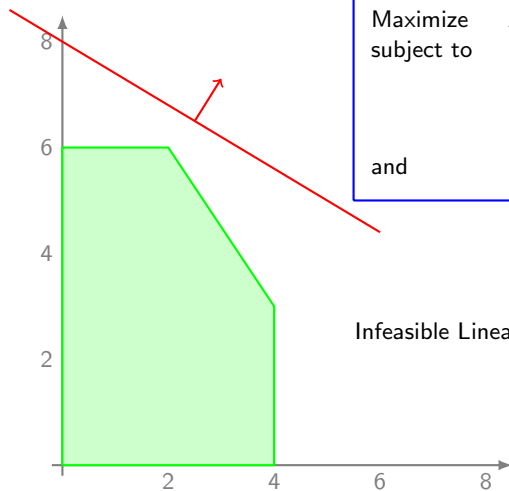
- ▶ In this example, the feasible region is **non-empty**, and **bounded**.
- ▶ However, it can happen that the feasible region is **empty**, i.e., there are no solutions possible. We then say that the linear program is **infeasible**.
- ▶ It can also happen that the feasible region stretches off to infinity in some direction, in which case we say that the feasible region is **unbounded**. In this case the optimal solution might also be unbounded.

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- ▶ There can also be **multiple optima**.

The Wyndor Glass Example

Infeasible

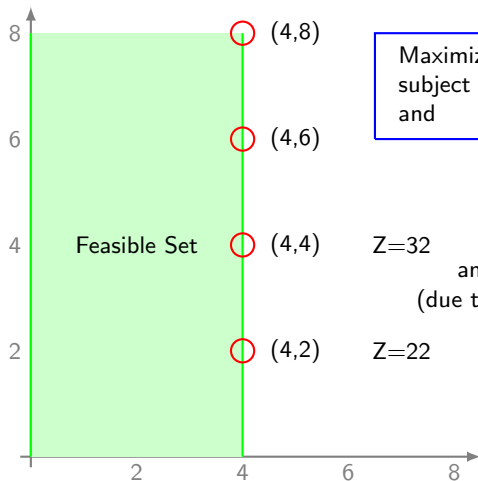


$$\begin{array}{llll} \text{Maximize} & Z = & 3x_1 & + & 5x_2 \\ \text{subject to} & & x_1 & & \leq 4 \\ & & & & x_2 \leq 12 \\ & & 3x_1 & + & 2x_2 \leq 18 \\ & & 3x_1 & + & 5x_2 \geq 40 \\ \text{and} & & x_1 & \geq 0, & x_2 \geq 0 \end{array}$$

Infeasible Linear Program

The Wyndor Glass Example

Unbounded



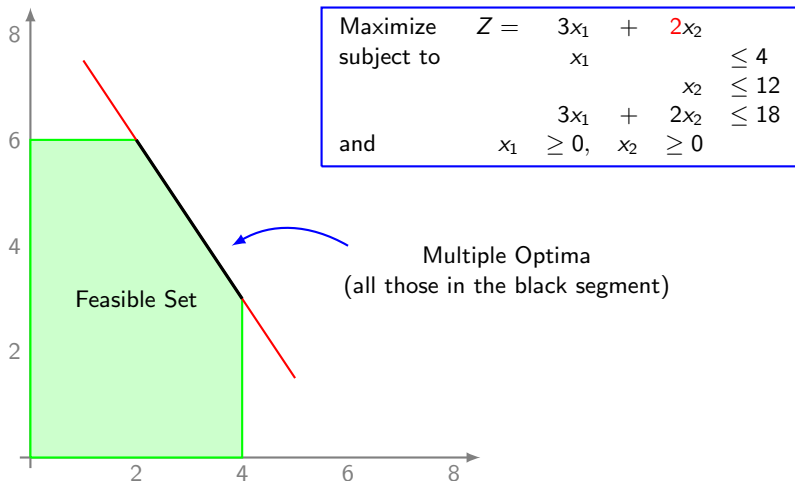
$$\begin{array}{ll} \text{Maximize} & Z = 3x_1 + 5x_2 \\ \text{subject to} & x_1 \leq 4 \\ \text{and} & x_1 \geq 0, \quad x_2 \geq 0 \end{array}$$

$Z=32$ Unbounded Feasible Set
and Unbounded Objective Function
(due to the angle of the objective function)

$Z=22$

The Wyndor Glass Example

Multiple Optima



MATSE: Operations Research

Lecture 2. The Simplex Method for Linear Programming

[based on slides by: Steven Kelk, Steve Chaplick]

Linear Programs: Basic Terminology

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■ **TABLE 3.1** Data for the Wyndor Glass Co. problem (goal: maximize profit)

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decision
variables
 x_1, x_2

Maximize $Z = 3x_1 + 5x_2$,
subject to

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

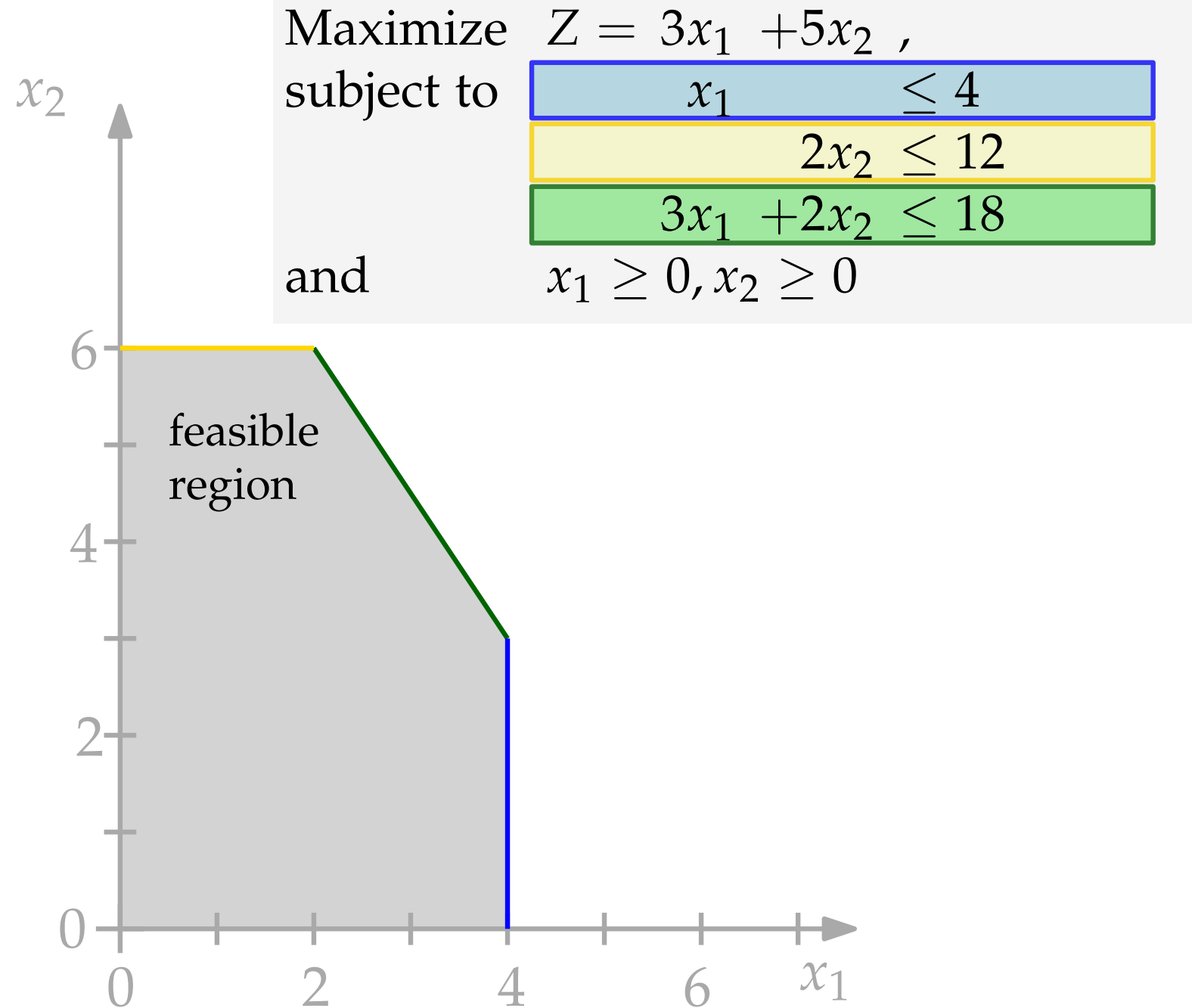
and

$$x_1 \geq 0, x_2 \geq 0$$

non-negativity constraints

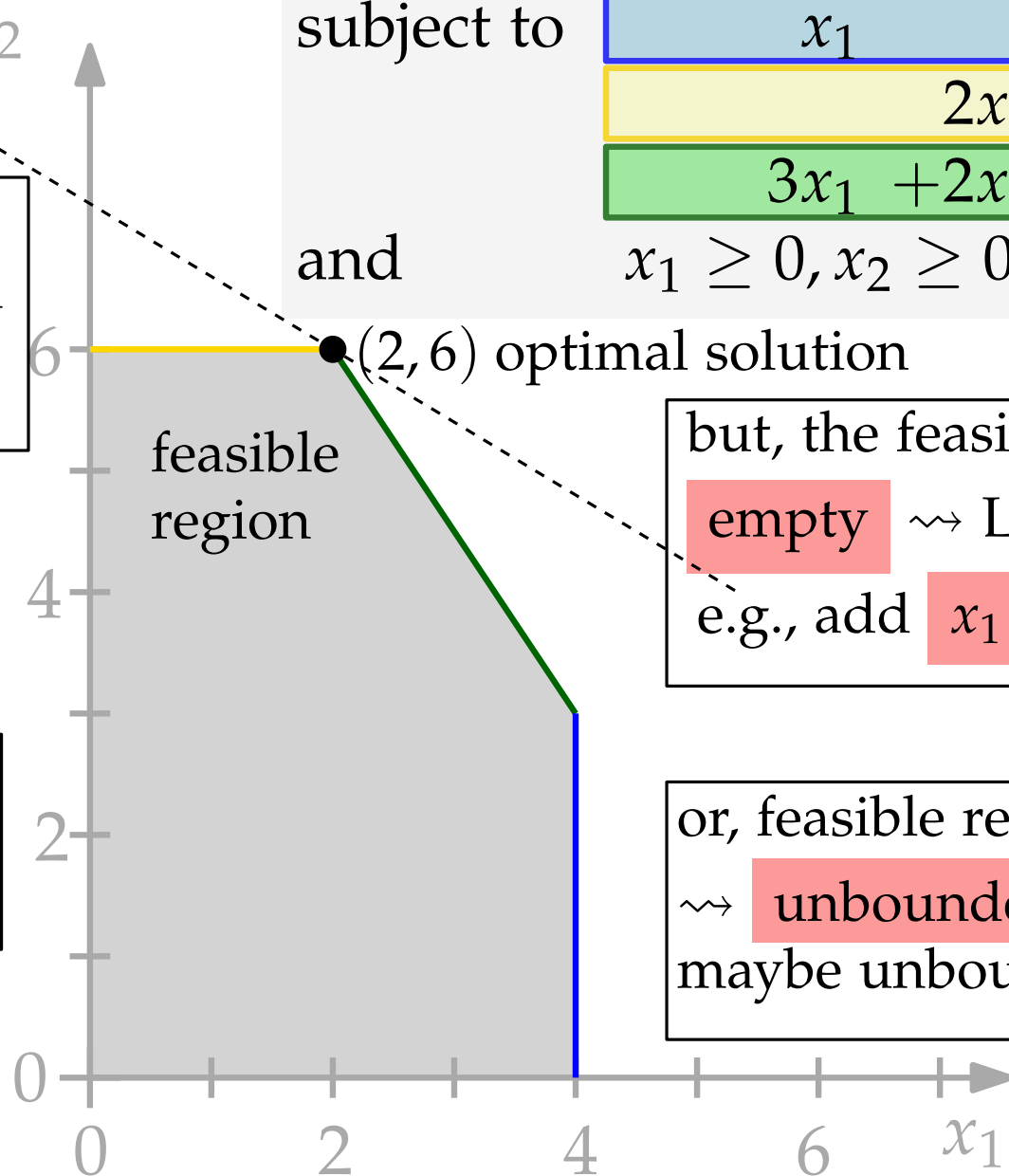
objective
function

functional
constraints



$$Z = 36 = 3x_1 + 5x_2$$

This feasible region is **non-empty**, and **bounded**



Maximize $Z = 3x_1 + 5x_2$,
subject to

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

$$x_1 \geq 0, x_2 \geq 0$$

and

(2, 6) optimal solution

feasible
region

but, the feasible region can be
empty $\leadsto$ LP is **infeasible**
e.g., add $x_1 - x_2 \leq 4.5$

There can also be
many optimal
solutions!

or, feasible region can go to ∞
 $\leadsto$ **unbounded** LP, and
maybe unbounded obj. value

Forms of LP

Given: $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$ [standard form]

where, for each entry b_i in b , $b_i \geq 0$, i.e., $b \geq 0$

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Find: $x^* \in \mathbb{R}^n$ where $x^* = \arg \max \{c^T x \mid Ax \leq b, x \geq 0\}$

This allows us to model more than you might expect,

- $\geq$ constraints $\rightarrow g(x) \geq b \rightarrow -g(x) \leq -b$
- $=$ constraints $\rightarrow g(x) = b \rightarrow g(x) \leq b$ and $-g(x) \leq -b$
- minimization $\rightarrow \text{Minimize } f(x) \rightarrow \text{Maximize } -f(x)$

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Why the standard form?

- it does not sacrifice any generality!
- simplifies the explanation and analysis of the Simplex method

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when modelling a problem as an LP, use the standard form!

Some Linear Algebra

- Consider a set of m linear equalities in m variables.
- We express it as $Cx = d$, where C is an $m \times m$ matrix, x is a length- m column vector and d is a length- m column vector.
- What are necessary and sufficient conditions for this system of equalities to have a **unique solution**?

Some Linear Algebra

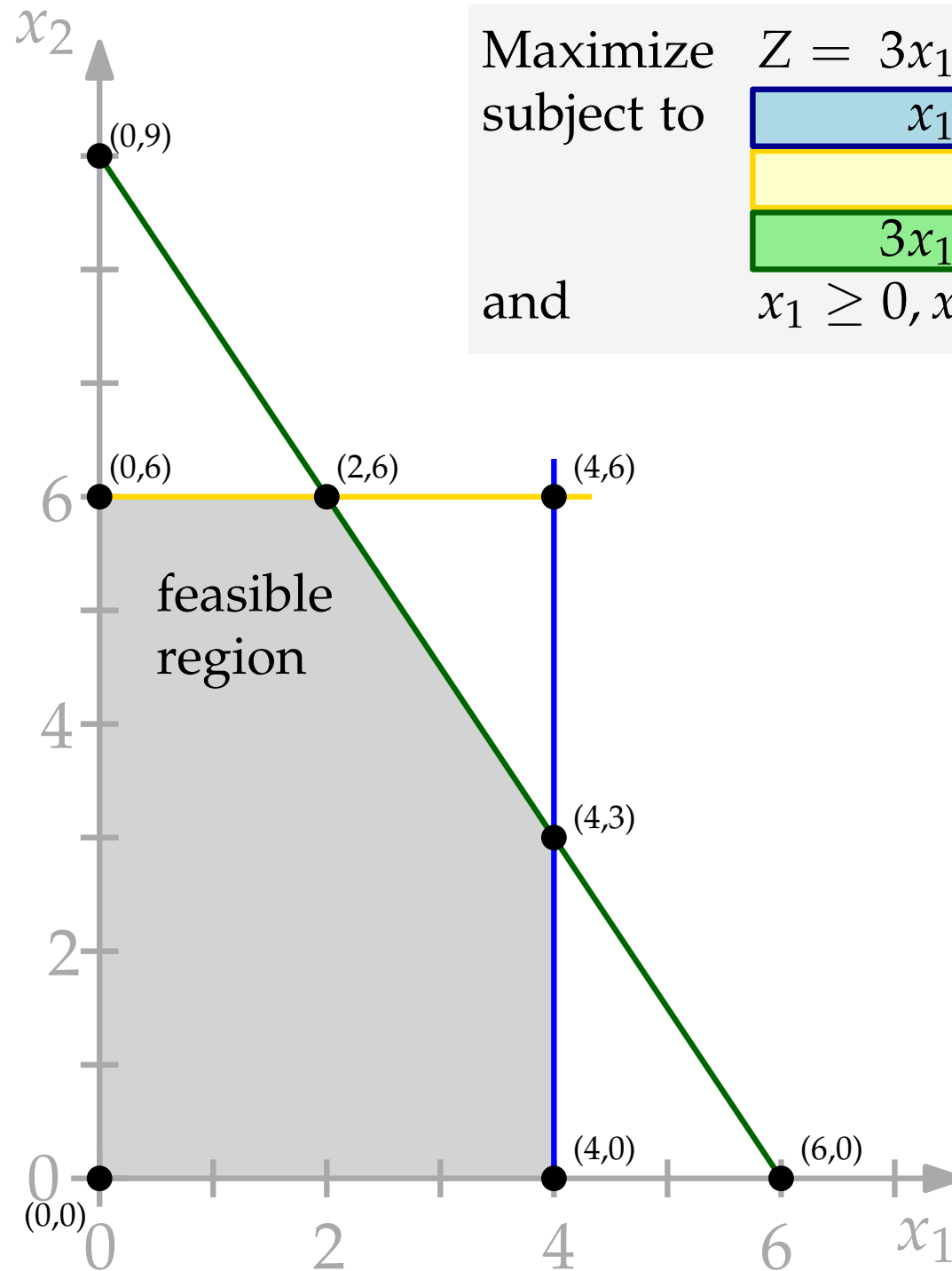
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- What are necessary and sufficient conditions for this system of equalities to have a **unique solution**?
 - row-rank(C) = column-rank(C) = m , equivalently
 - row set of C is **linearly independent**, equivalently,
 - cols. set of C is **linearly independent**, equivalently,
 - C is **invertible**, i.e., C^{-1} exists, i.e., $\det(C) \neq 0$.

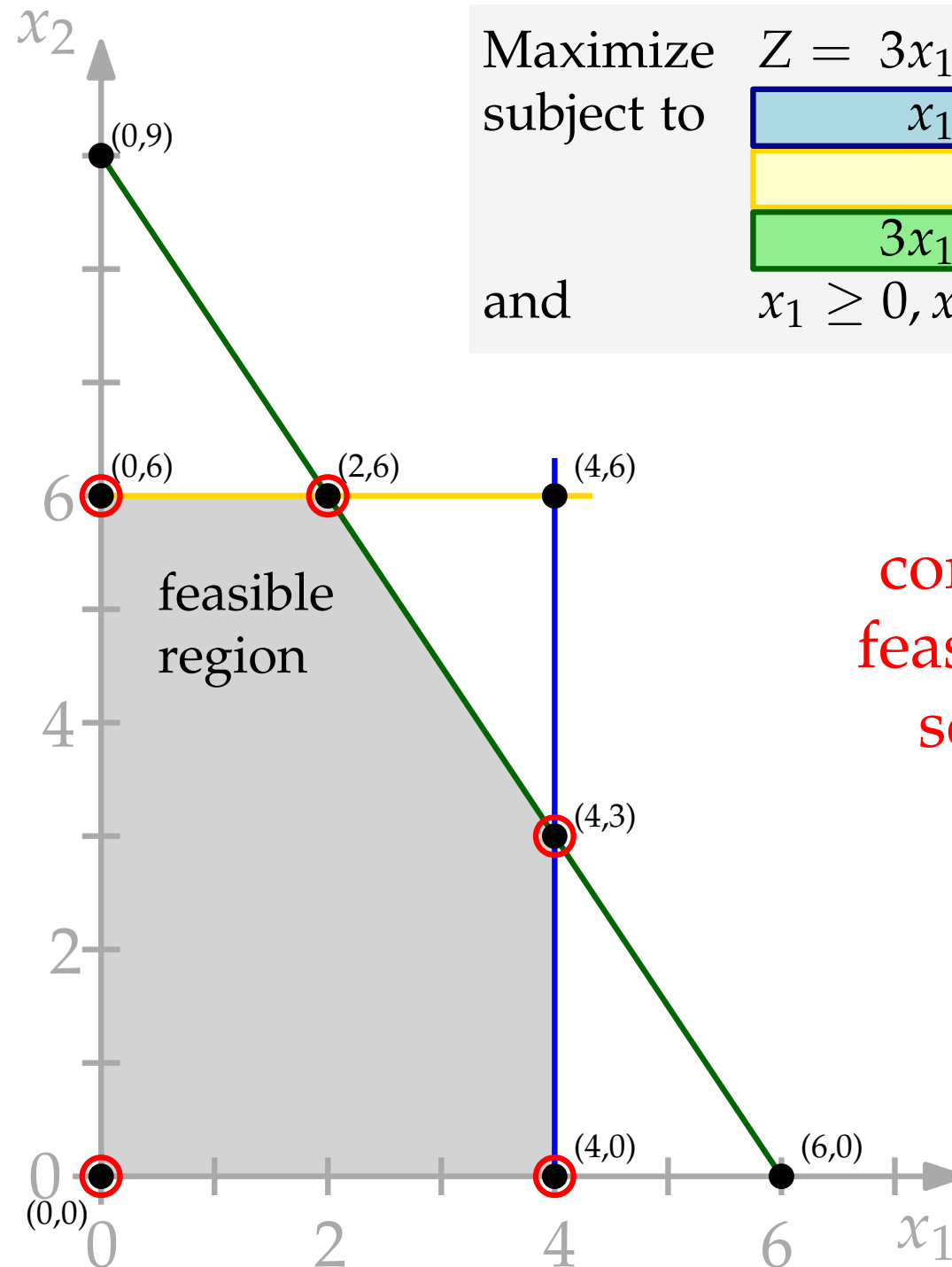
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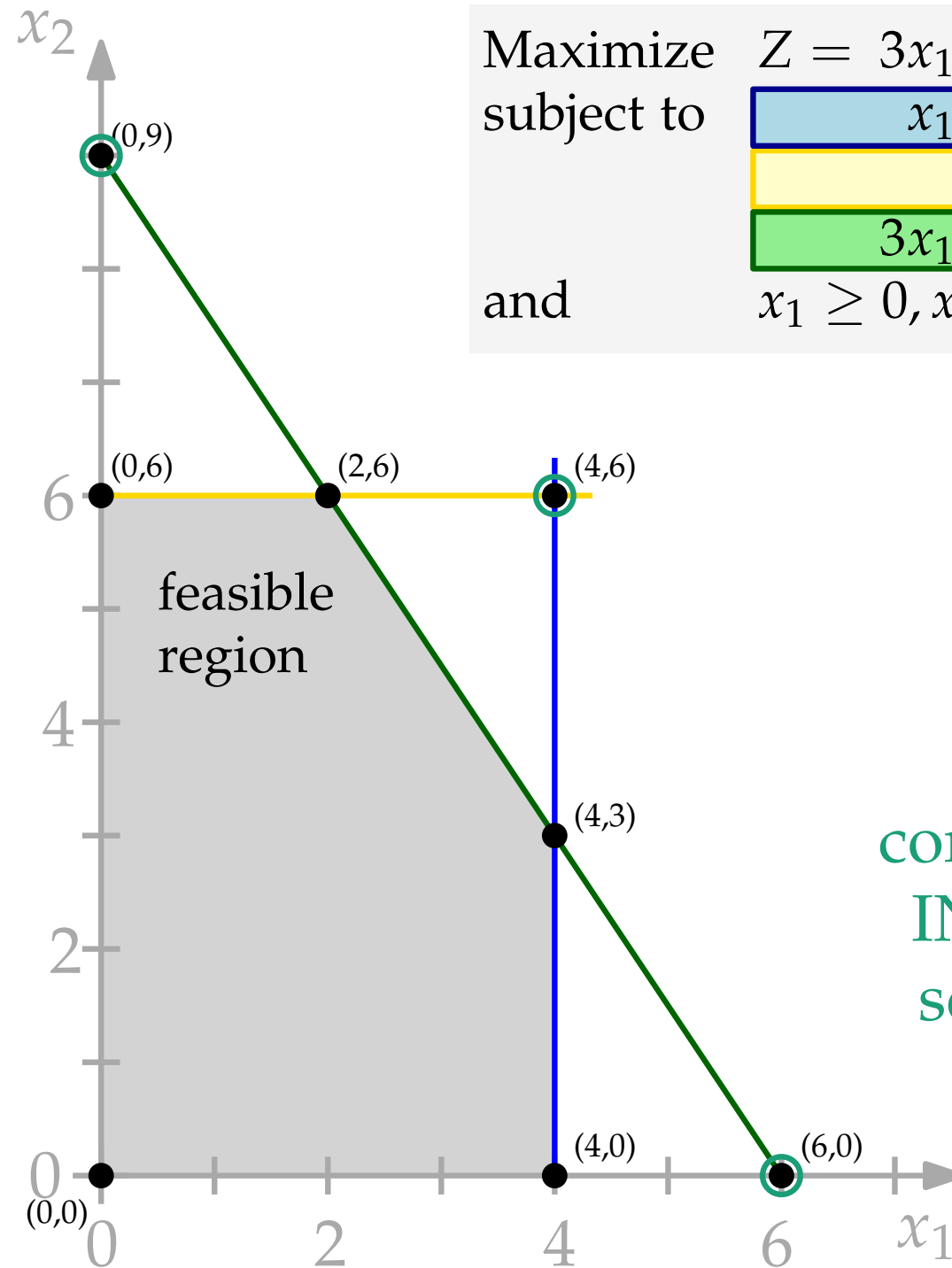
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- If these conditions fail, the system is either infeasible or infinitely many solutions exist.
- Simplex method iterates **corner-point feasible solutions**: certain unique solutions to *carefully* constructed systems of equalities (that are feasible for the linear program).





corner point
feasible (CPF)
solutions!



Maximize
subject to

$$Z = 3x_1 + 5x_2 ,$$

$$x_1 \leq 4$$

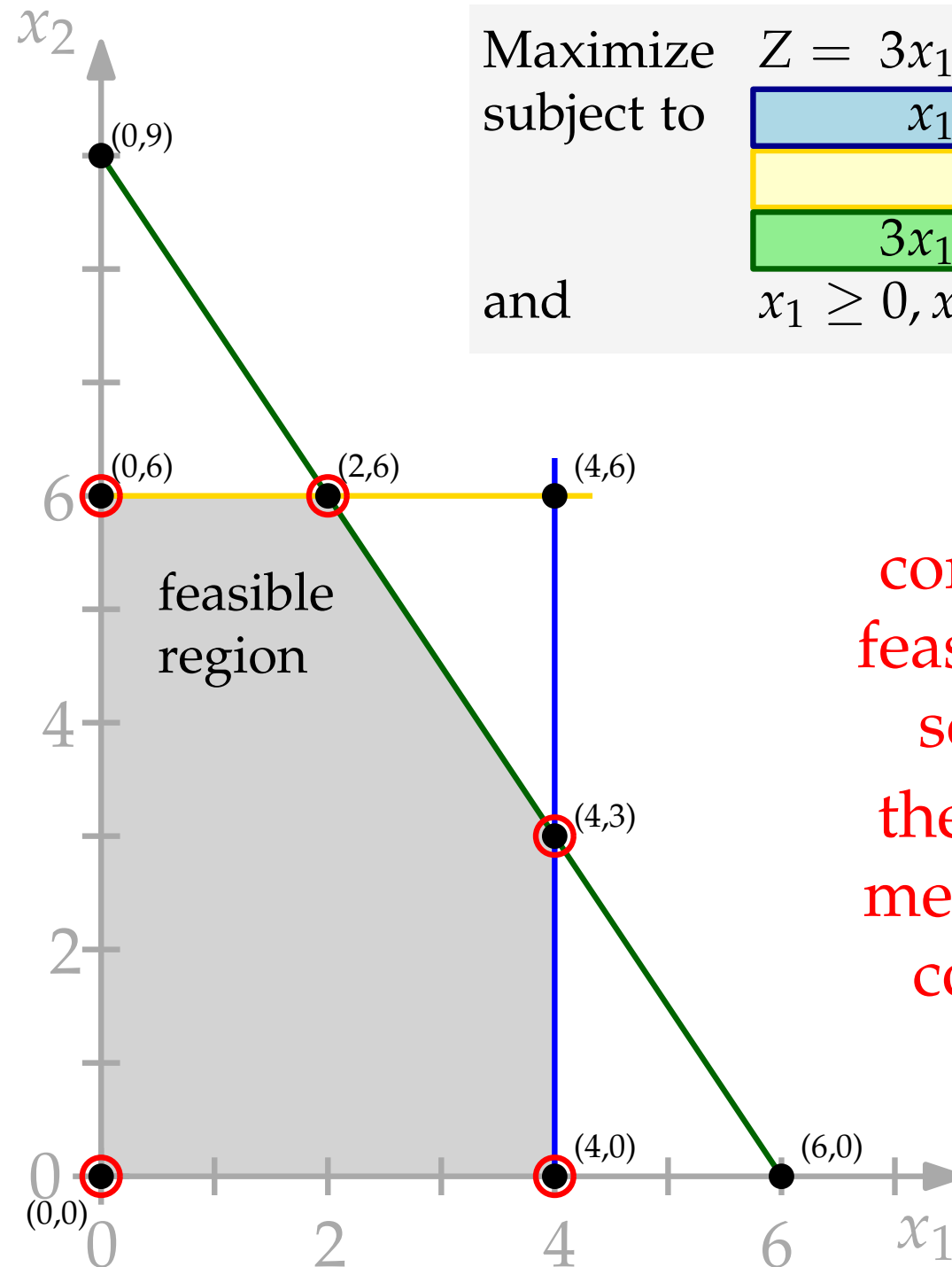
$$2x_2 \leq 12$$

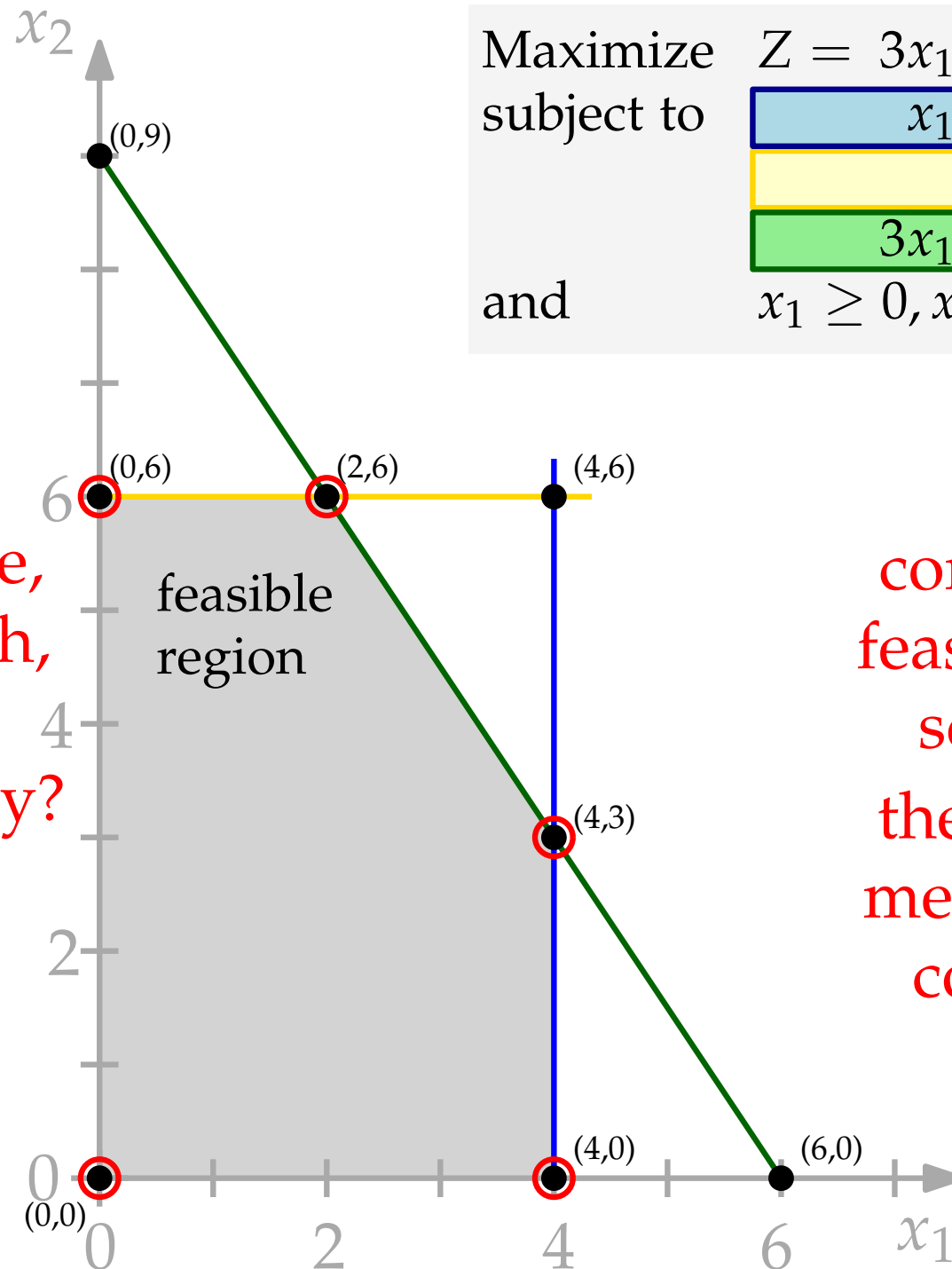
$$3x_1 + 2x_2 \leq 18$$

and

$$x_1 \geq 0, x_2 \geq 0$$

corner-point
INfeasible
solutions!

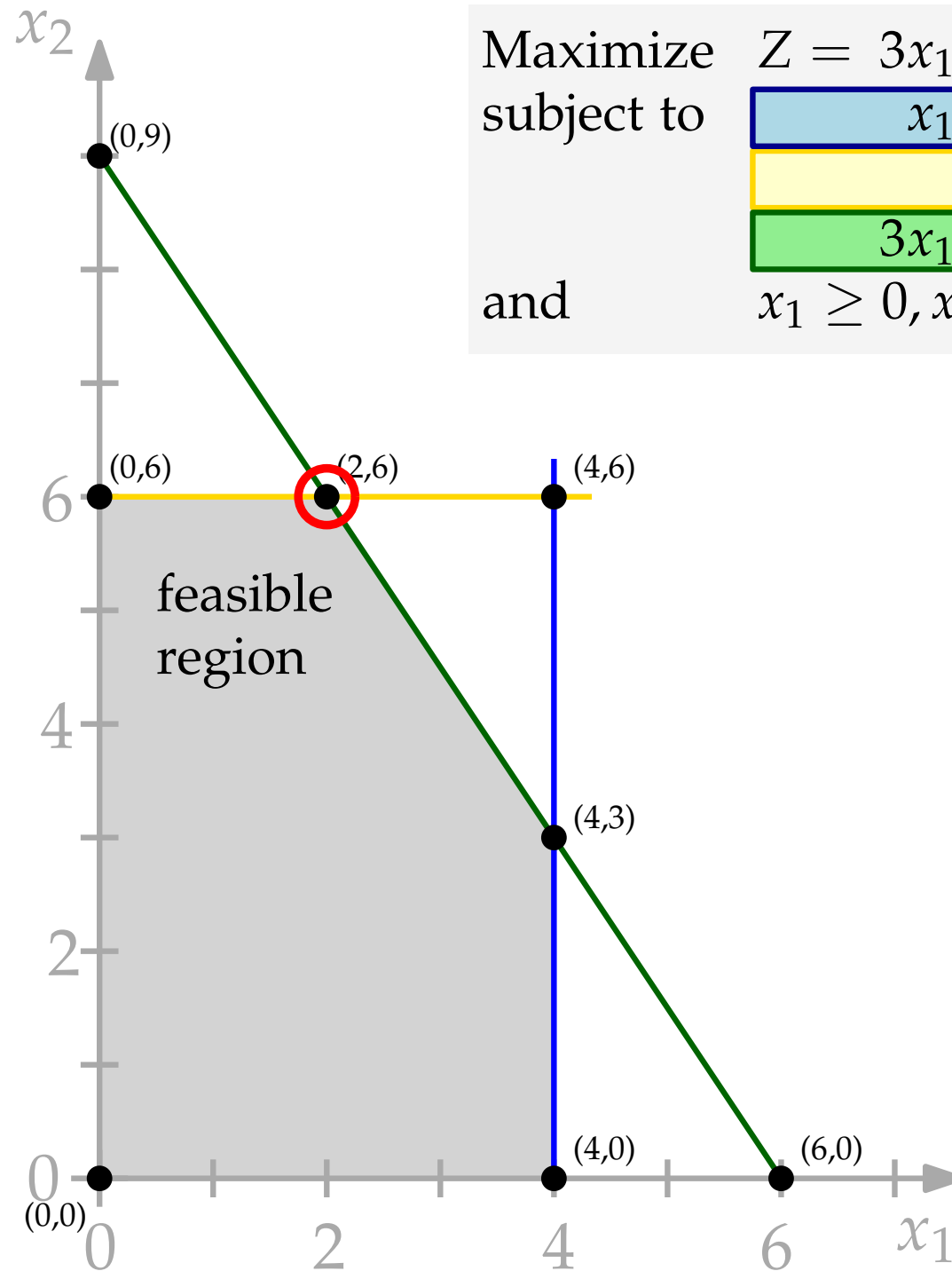




How to define,
and work with,
these CPF
mathematically?

corner point
feasible (CPF)
solutions!
the Simplex
method only
considers
these!

Consider the
CPF (2, 6)



Maximize
subject to

$$Z = 3x_1 + 5x_2 ,$$

$$x_1 \leq 4$$

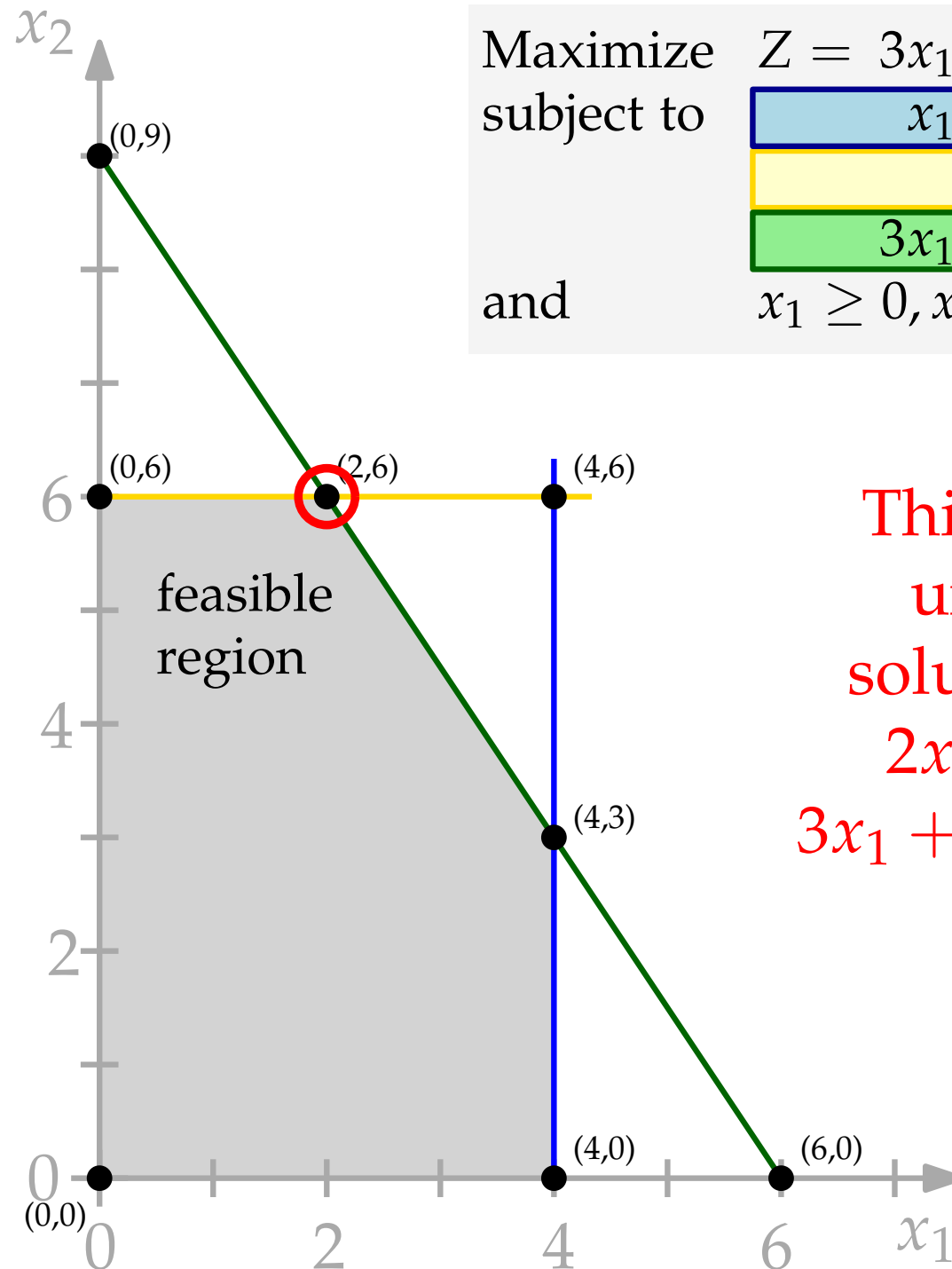
$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

and

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Consider the
CPF (2, 6)



This is the
unique
solution to:
 $2x_2 = 12$
 $3x_1 + 2x_2 = 18$

What's happening here?

- Cornerpoint feasible solutions have something to do with unique solutions to systems of **equalities** induced by subsets of the inequalities in the linear program (and the non-negativity constraints).

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What's happening here?

- Cornerpoint feasible solutions have something to do with unique solutions to systems of **equalities** induced by subsets of the inequalities in the linear program (and the non-negativity constraints).
- But linear programs are much more complex than systems of equalities. They are systems of **inequalities** (and we are looking for *optimal* solutions).
- Solutions to systems of **inequalities** can be *projected* to solutions of systems of **equalities** in higher dimension (i.e. with more variables), and this is exactly what the Simplex method does.

Projecting “higher”

Consider the first constraint from our example:

$$x_1 \leq 4$$

where $x_1 \geq 0$

Projecting “higher”

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equivalent to

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slack variable: x_3
encodes the gap between
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What about the other constraints?

Projecting “higher”

original LP:

Maximize $Z = 3x_1 + 5x_2$

subject to

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

and $x_1 \geq 0, x_2 \geq 0$

Projecting “higher”

original LP:

Maximize $Z = 3x_1 + 5x_2$
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$$\begin{aligned} x_1 &\leq 4 \\ 2x_2 &\leq 12 \\ 3x_1 + 2x_2 &\leq 18 \end{aligned}$$

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augmented form of LP:

Maximize $Z = 3x_1 + 5x_2$
subject to

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If the LP is in the form $Ax \leq b$ (and x is constrained to be ≥ 0), where A has m rows and n columns, then:

the augmented form will have m equalities in $m + n$ variables.

Matrix A : rank, bases

$$A = \begin{matrix} & x_1 & x_2 & x_3 & x_4 & x_5 \\ \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} & b = & \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix} \end{matrix}$$

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- Thus, there exist subsets of 3 columns which are linearly independent (because col-rank = row-rank).
- These subsets are in **many-to-1** correspondence with corner-point solutions!

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Ex. subset: $\{x_2, x_3, x_5\}$.

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The solution is **unique** (the induced 3×3 matrix is **invertible**)

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The solution is **unique** (the induced 3×3 matrix is **invertible**)

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Maximize
subject to

$$Z = 3x_1 + 5x_2 ,$$

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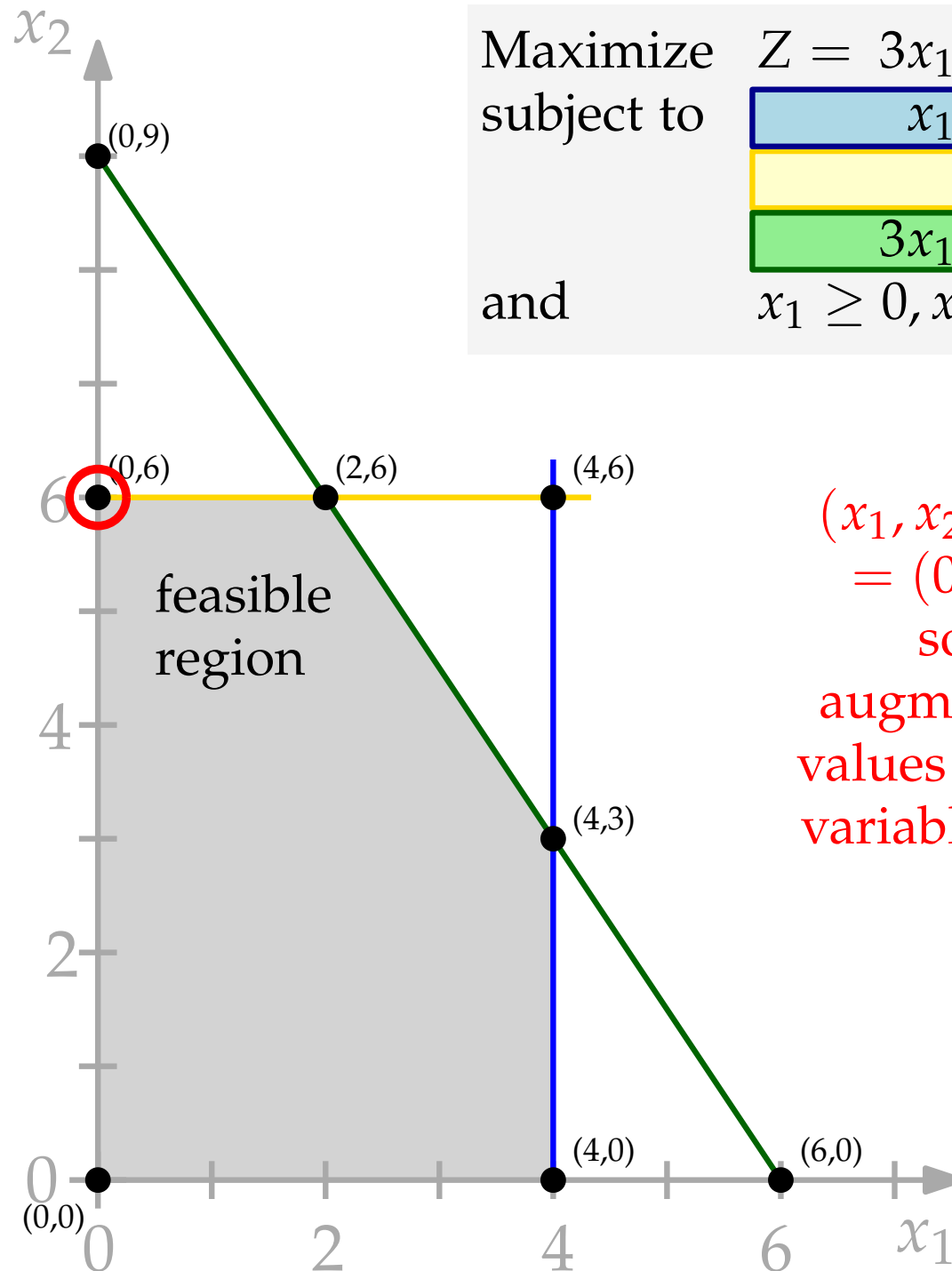
$$2x_2 \leq 12$$

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 $= (0, 6, 4, 0, 6)$
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- The variables x_1 and x_4 are **nonbasic variables**.

Basic (B) vs. Non-Basic (NB)

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- Every **linearly independent subset of 3 columns is a basis**, and generates a **basic solution**, but not all will be feasible.

Basic (B) vs. Non-Basic (NB)

$$\begin{array}{ccccc}
 \text{NB} & \text{B} & \text{B} & \text{NB} & \text{B} \\
 x_1 & x_2 & x_3 & x_4 & x_5 \\
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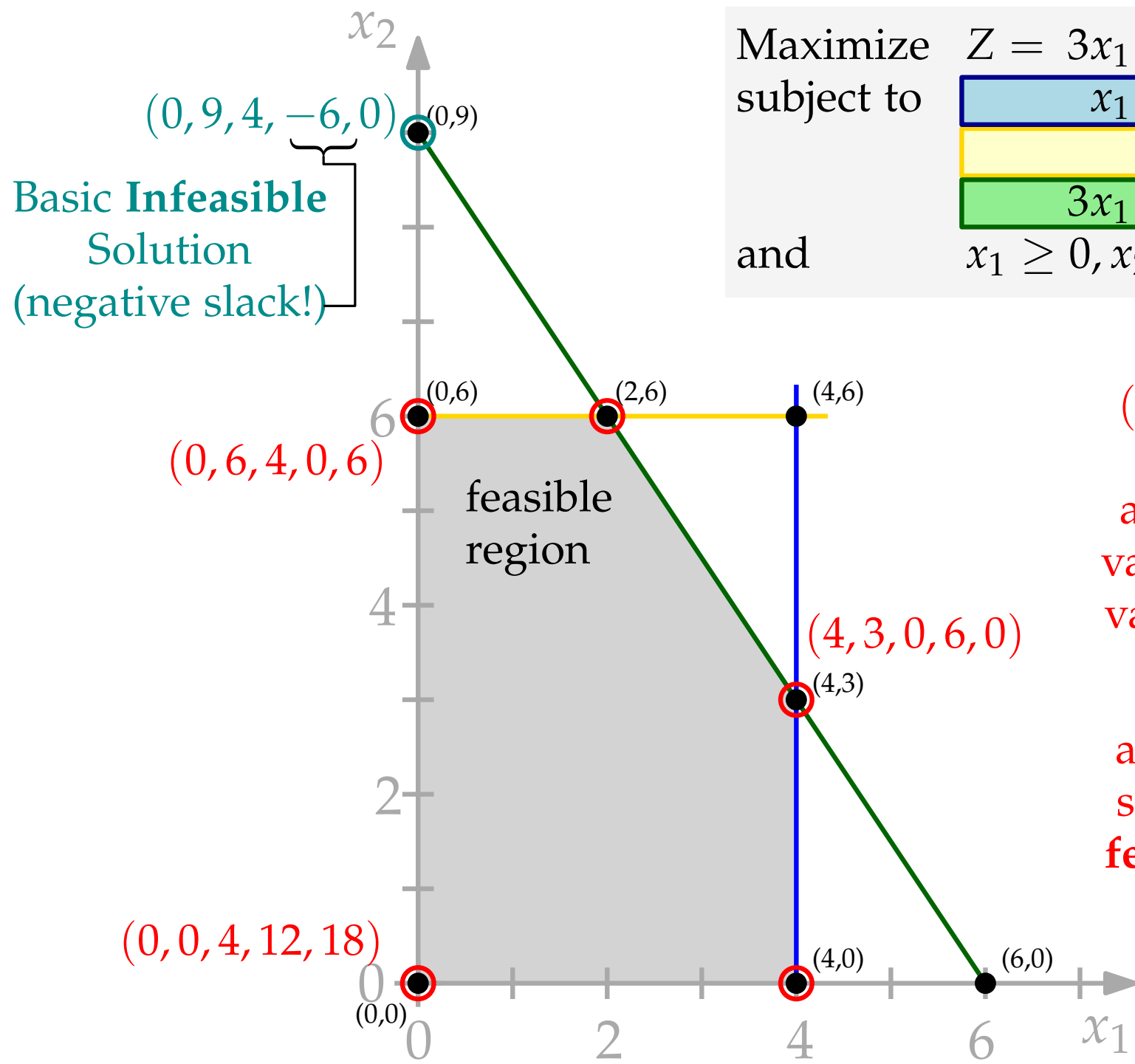
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- Every **linearly independent subset of 3 columns is a basis**, and generates a **basic solution**, but not all will be feasible.
- A basic solution is **feasible**, if and only if all the decision variables and slack variables are **non-negative**.

Maximize $Z = 3x_1 + 5x_2$,
 subject to $x_1 \leq 4$
 $2x_2 \leq 12$
 $3x_1 + 2x_2 \leq 18$
 and $x_1 \geq 0, x_2 \geq 0$



$(x_1, x_2, x_3, x_4, x_5)$
 = solution
 augmented with
 values of the slack
 variables x_3, x_4, x_5

augmented CPF
 solution = **basic
 feasible solution**
 (BFS)

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- LP with n decision variables and m functional constraints $\rightsquigarrow$ augmented form of the LP is a set of m equalities in $(n + m)$ variables i.e. an **underdetermined system of equalities**.

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- To obtain a unique solution to this system of equalities, we **force** n of the variables to zero (such that the remaining columns are linearly independent)

Summary so far

- Geometric interpretation:
this corresponds to the choice of n constraint boundaries which we ask to be simultaneously satisfied with equality.
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Check Table 4.2 from the book!

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- The remaining m variables are the **basic** variables; and the set of m basic variables, is called the **basis**.
- **The idea** : *choose* n variables to be **nonbasic**, the basic variables (i.e. the basis) is the set of m variables that remain, and their values are the **unique solution** to the system of equalities obtained after driving the nonbasic variables to zero.

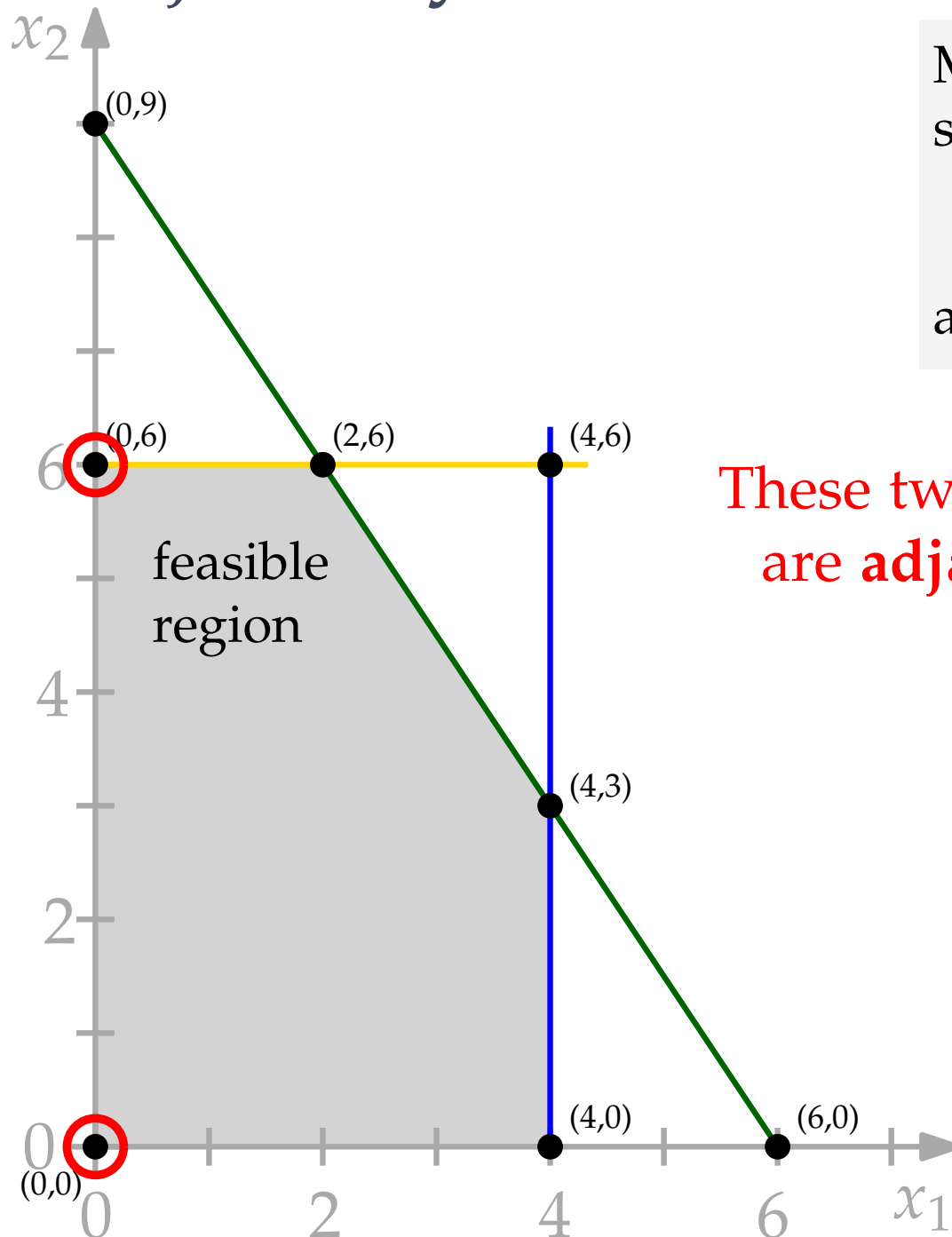
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- So, the **nonbasic variables correspond 1-to-1 with the subset of functional constraints, that we want to be met with equality (“tight”)** in order to define a corner point feasible solution.
- The remaining m variables are the **basic** variables; and the set of m basic variables, is called the **basis**.
- **The idea** : *choose* n variables to be **nonbasic**, the basic variables (i.e. the basis) is the set of m variables that remain, and their values are the **unique solution** to the system of equalities obtained after driving the nonbasic variables to zero.

↪ (1) nonbasic means **zero**, but (2) basic **can** also be **zero**

Adjacency

Adjacency



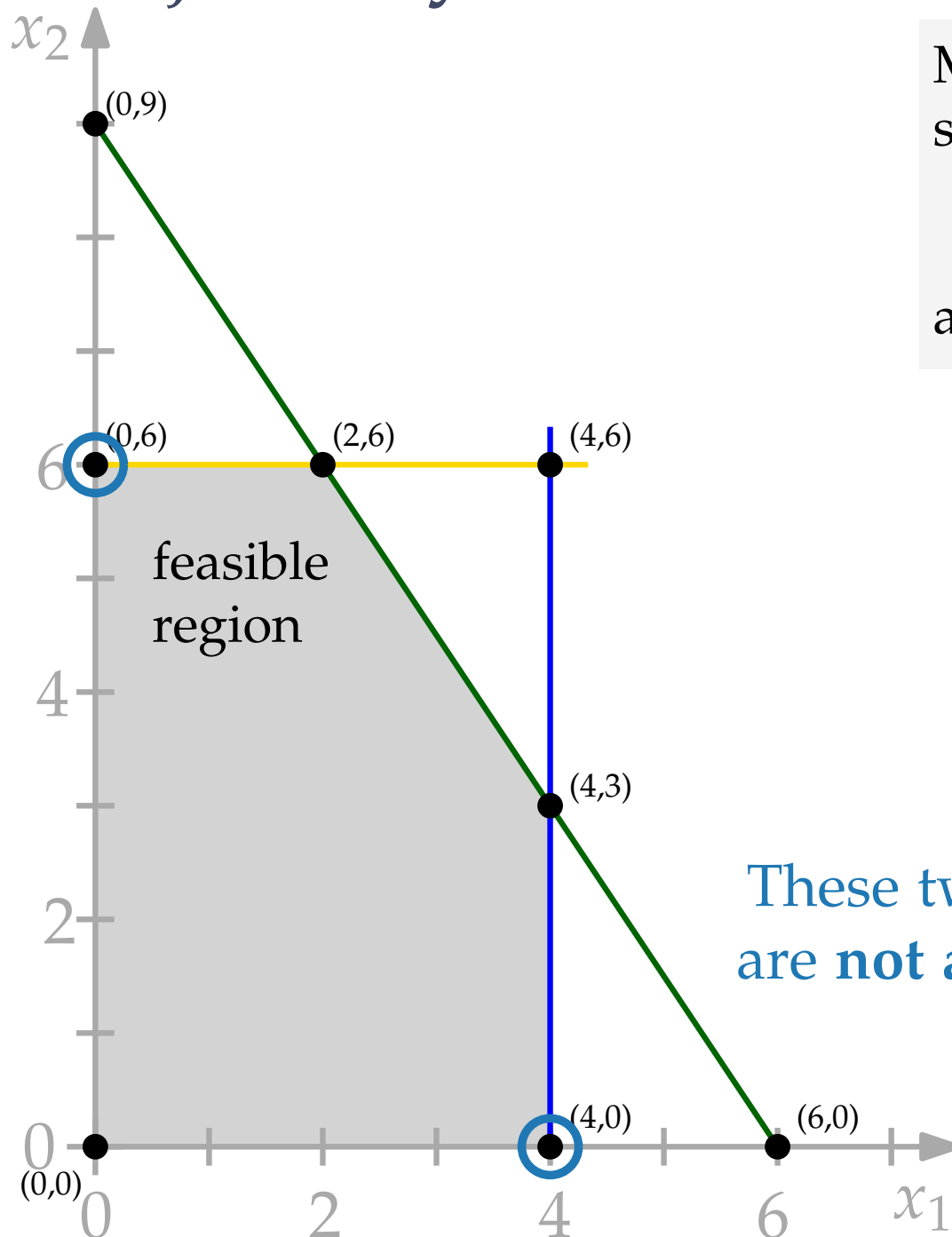
Maximize $Z = 3x_1 + 5x_2$,
 subject to

$x_1 \leq 4$
$2x_2 \leq 12$
$3x_1 + 2x_2 \leq 18$

and $x_1 \geq 0, x_2 \geq 0$

These two BFSs
are adjacent

Adjacency



Maximize
subject to

$$Z = 3x_1 + 5x_2 ,$$

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

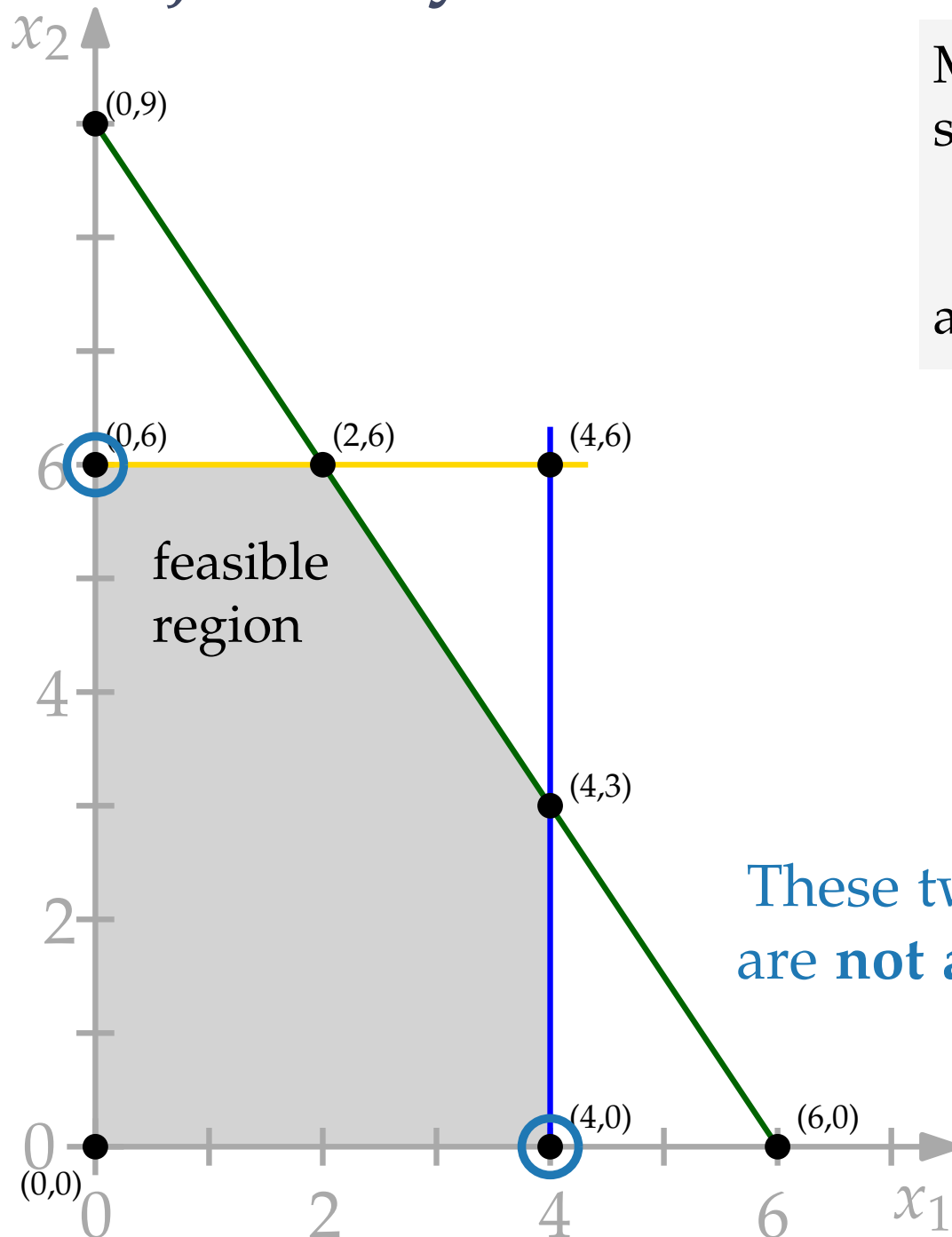
$$3x_1 + 2x_2 \leq 18$$

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These two BFSs
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Adjacency



Maximize
subject to

$$Z = 3x_1 + 5x_2 ,$$

$$x_1 \leq 4$$

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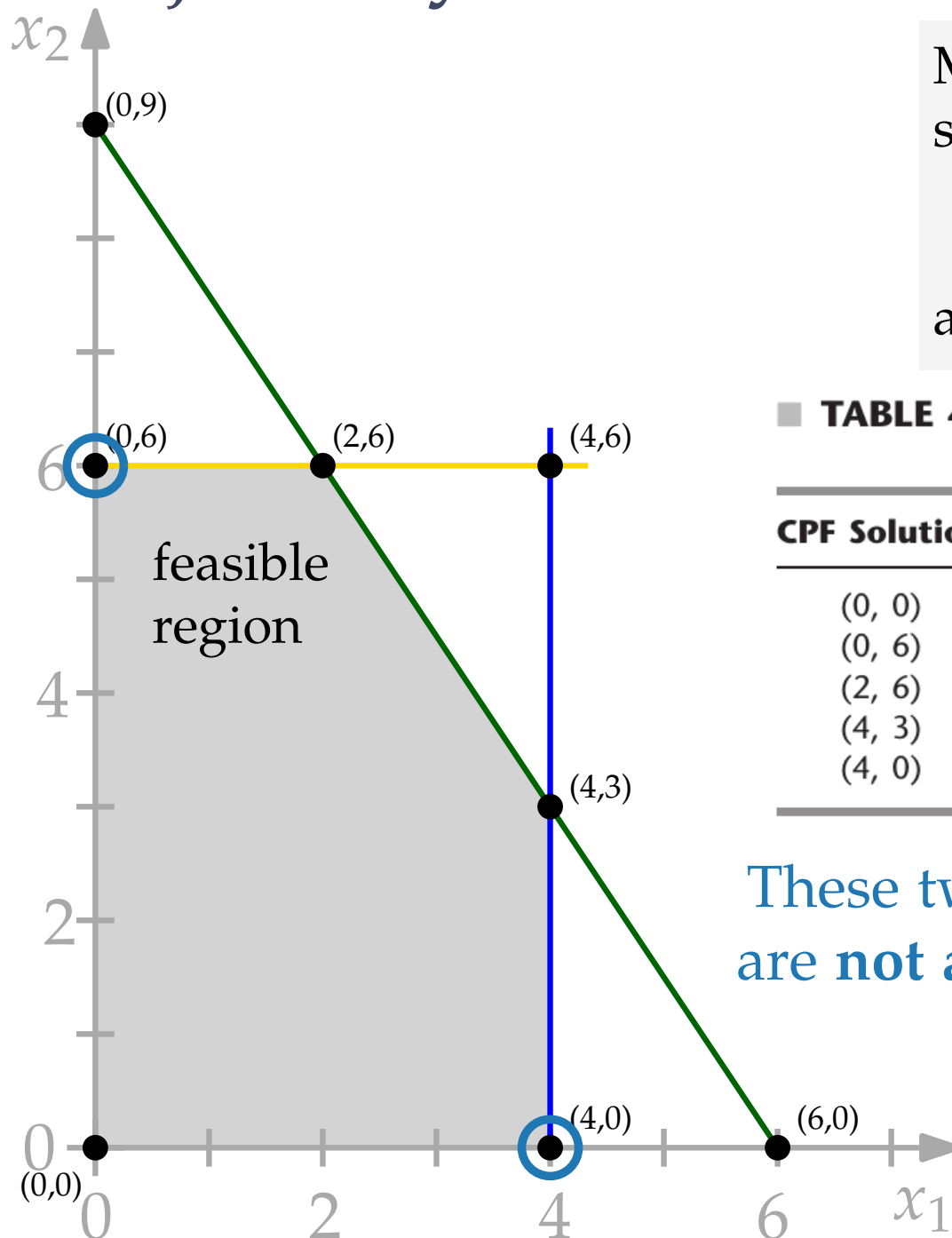
$$3x_1 + 2x_2 \leq 18$$

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Why? How do we formalize adjacency?

Adjacency



Maximize $Z = 3x_1 + 5x_2$,

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$$x_1 \leq 4$$

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and

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■ **TABLE 4.1** Adjacent CPF solutions for each CPF solution of the Wyndor Glass Co. problem

CPF Solution	Its Adjacent CPF Solutions
$(0, 0)$	$(0, 6)$ and $(4, 0)$
$(0, 6)$	$(2, 6)$ and $(0, 0)$
$(2, 6)$	$(4, 3)$ and $(0, 6)$
$(4, 3)$	$(4, 0)$ and $(2, 6)$
$(4, 0)$	$(0, 0)$ and $(4, 3)$

These two BFSs
are not adjacent

Adjacency (special case)

In n dimensions, suppose at each CPF solution exactly n constraint boundaries meet

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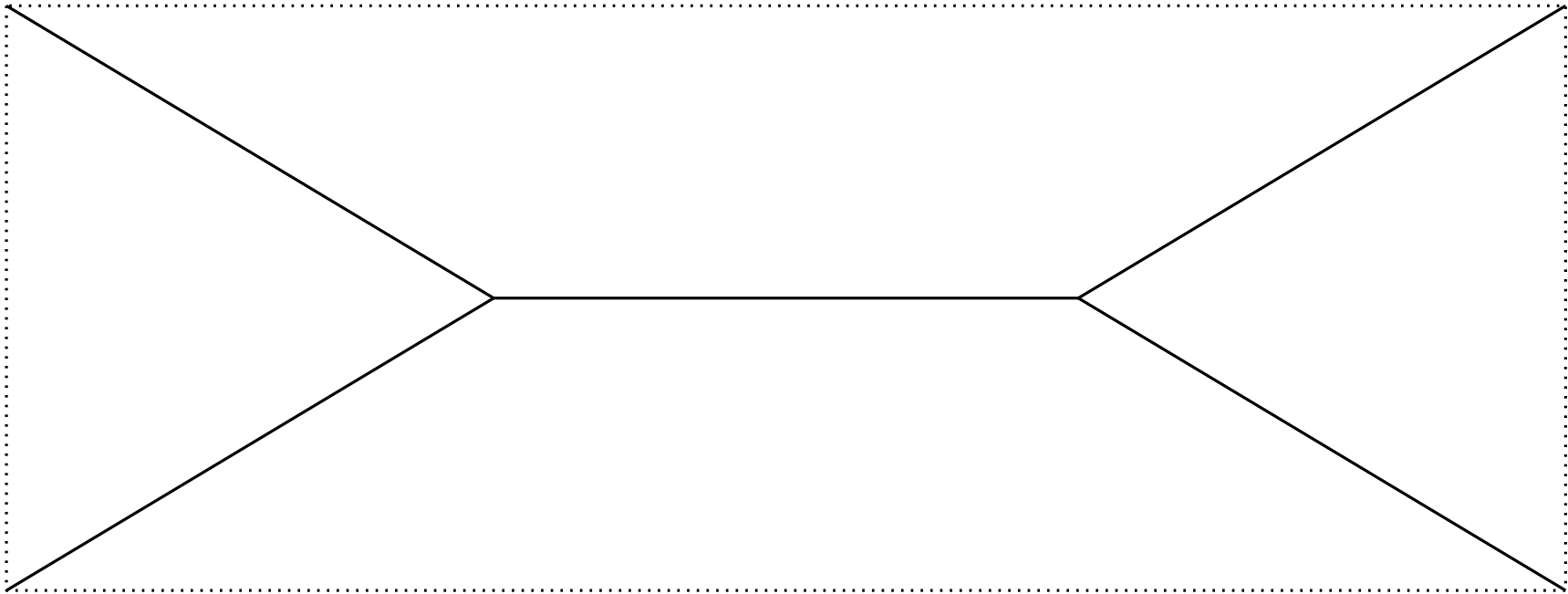
$\Rightarrow$ each CPF/BFS is defined by exactly one basis, rather than many

Adjacency (special case)

In n dimensions, suppose at each CPF solution exactly n constraint boundaries meet

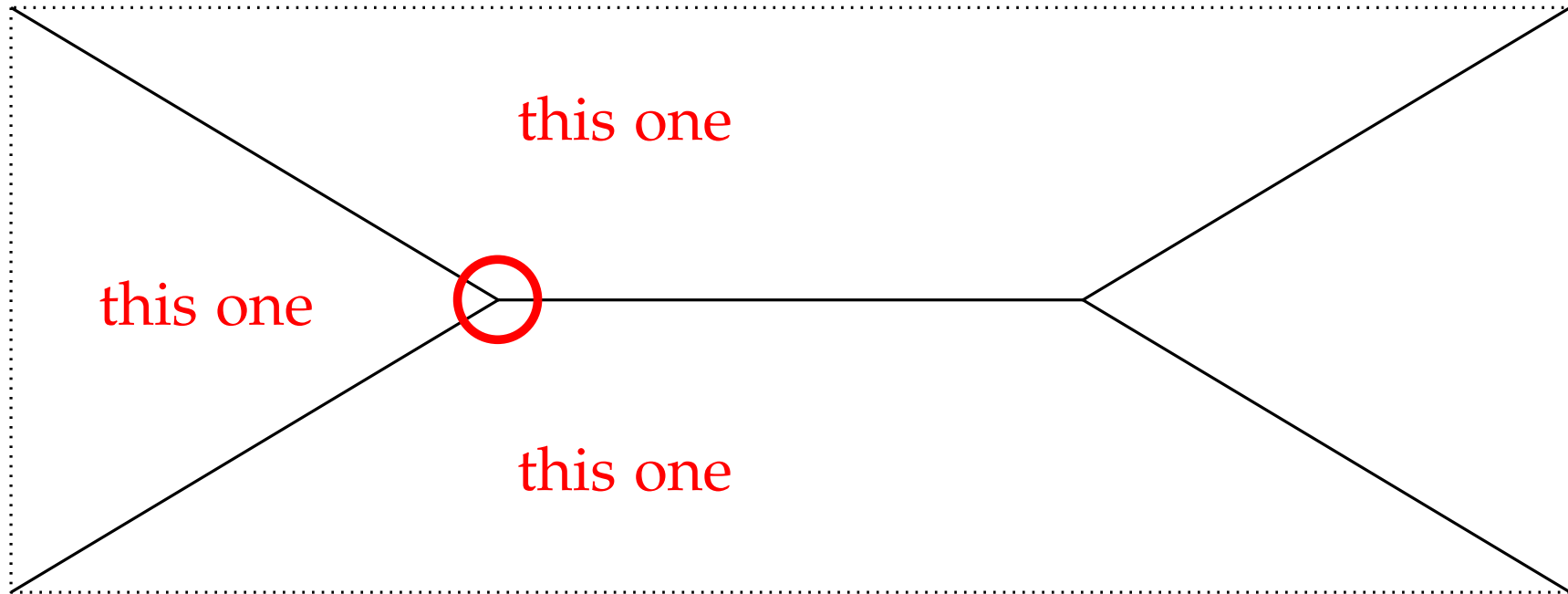
- $\Rightarrow$ each CPF/BFS is defined by exactly one basis, rather than many
- $\Rightarrow$ CPFs with $(n - 1)$ common constraint boundaries are adjacent, i.e., the two sets of nonbasic variables differ in **exactly one** variable

Adjacency (special case)



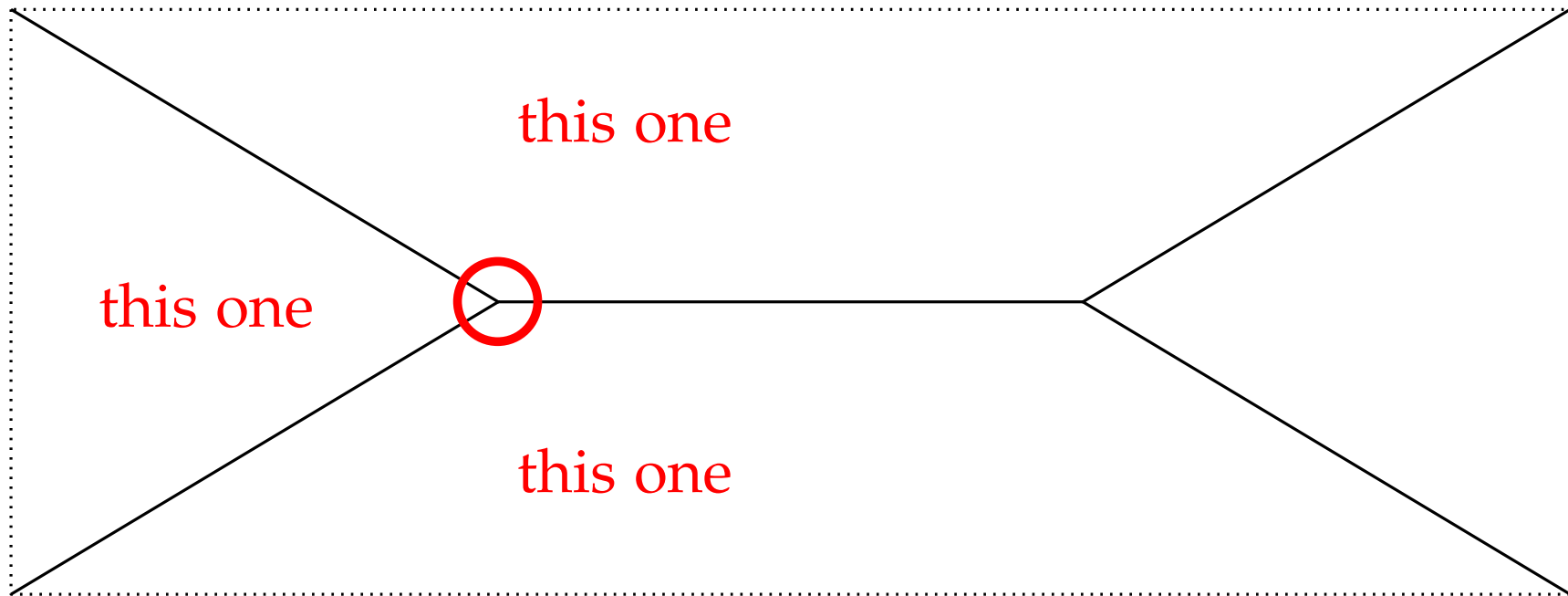
Imagine this is a 3-dimensional feasible region where the horizontal line in the middle is sticking furthest out of the page (i.e. is closest to you)

Adjacency (special case)



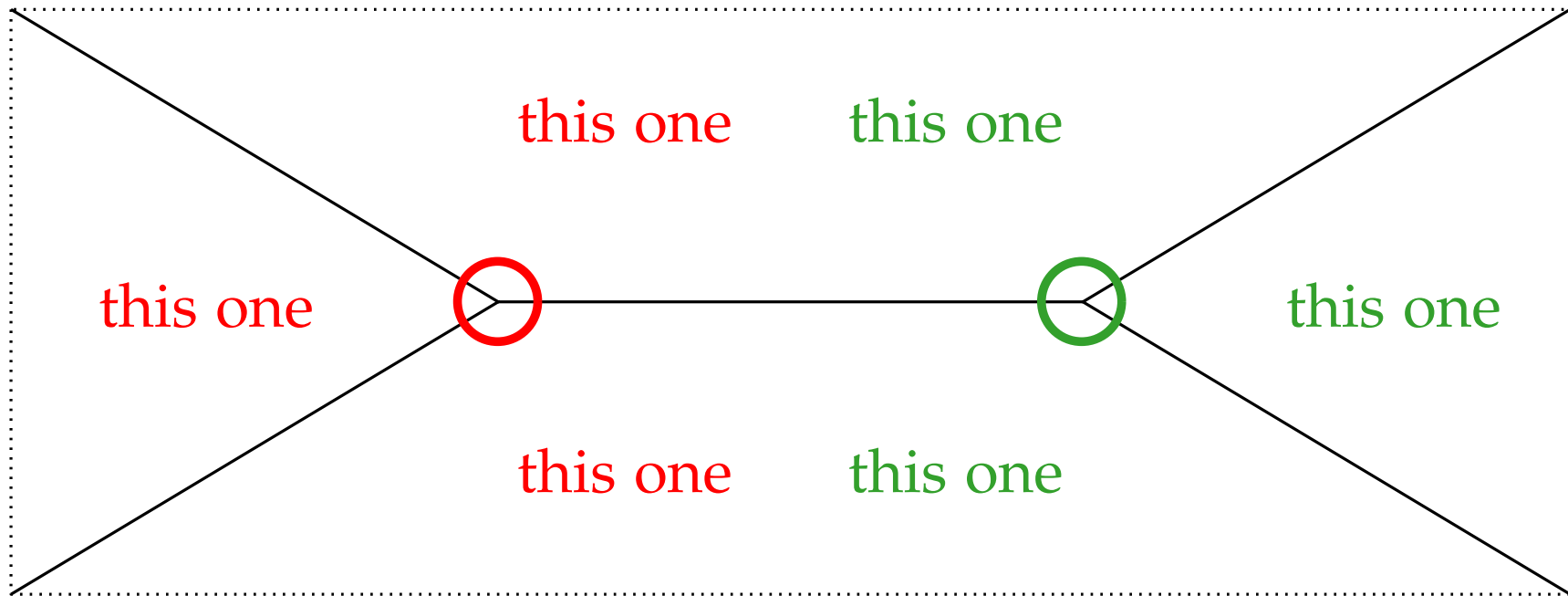
The circled CPF/BFS is defined by three constraint boundaries

Adjacency (special case)



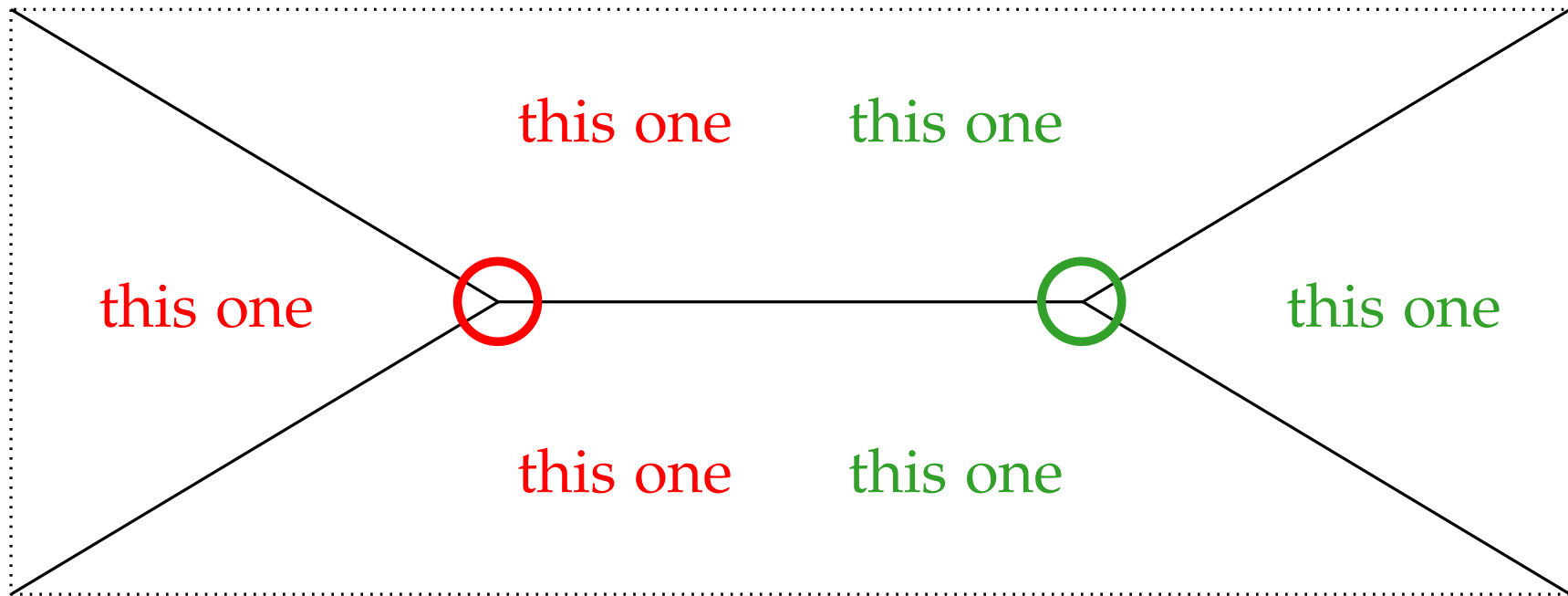
The circled CPF/BFS is defined by three constraint boundaries
their slack variables can be considered **nonbasic**

Adjacency (special case)



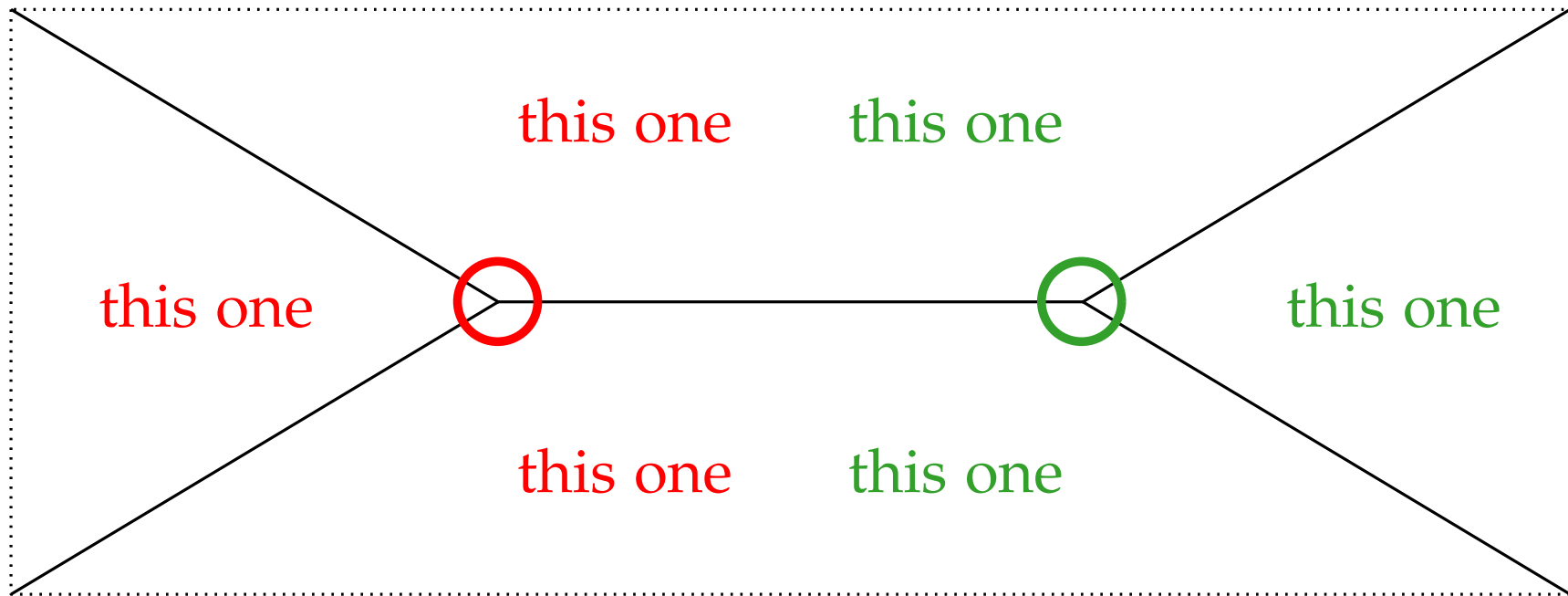
Consider the new CPF/BFS, shown in green and its constraints.

Adjacency (special case)



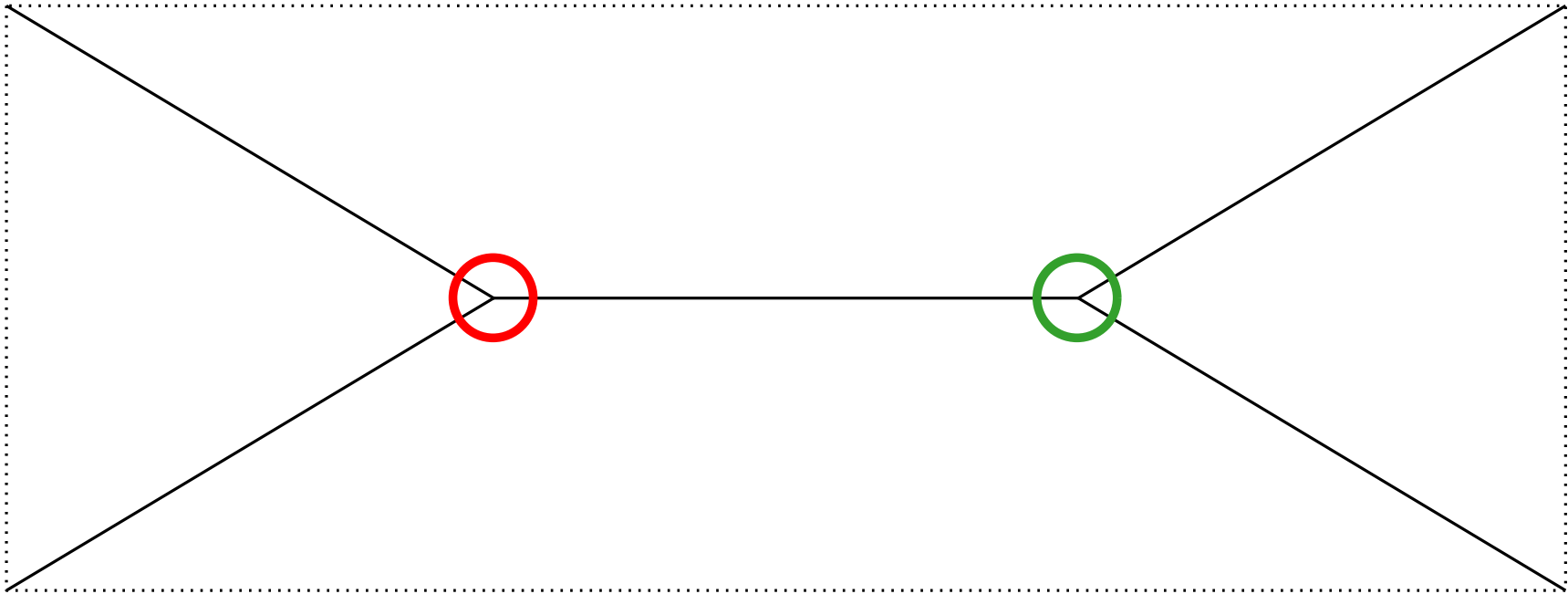
Consider the new CPF/BFS, shown in green and its constraints. The slack variables of its constraints can be considered **nonbasic**

Adjacency (special case)



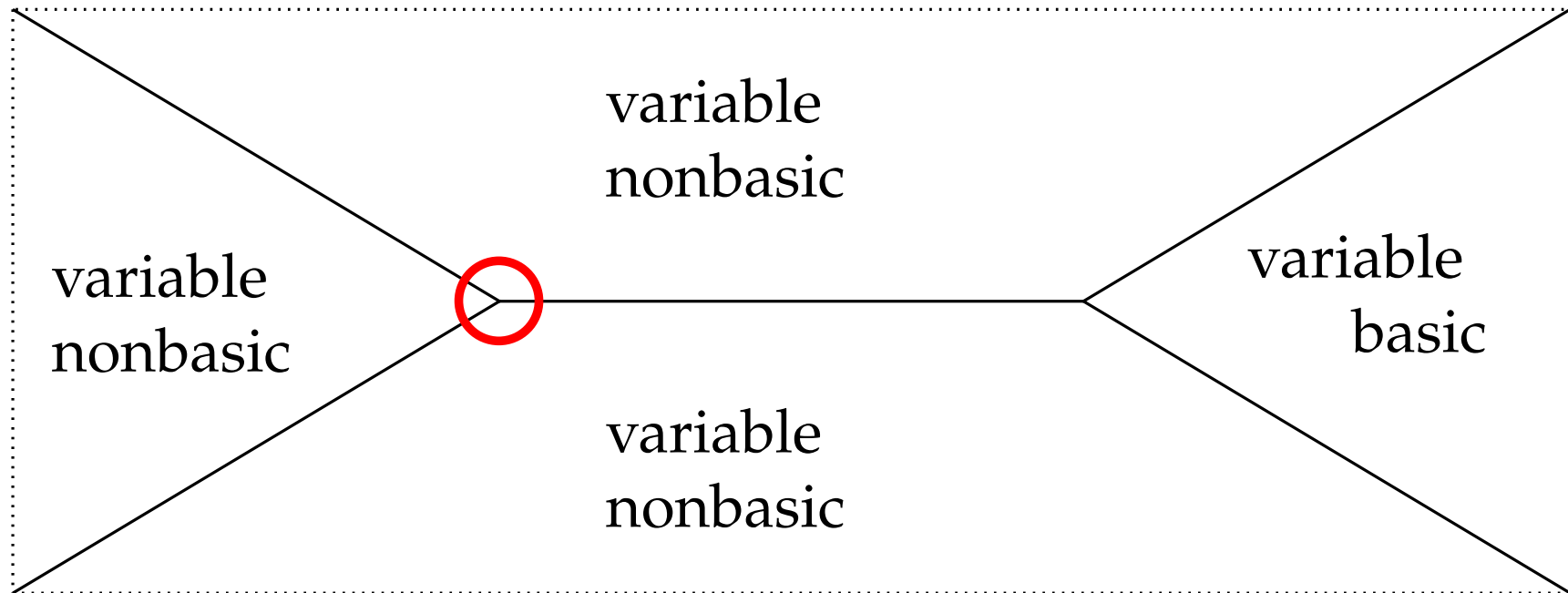
They share $n - 1 = 2$ constraint boundaries and are adjacent!
 $n - 1$ common boundaries $\Rightarrow$ the two adjacent CPFs/BFSs are connected by a line (an *edge* of the feasible region) i.e. a 1-dimensional structure

Adjacency (special case)



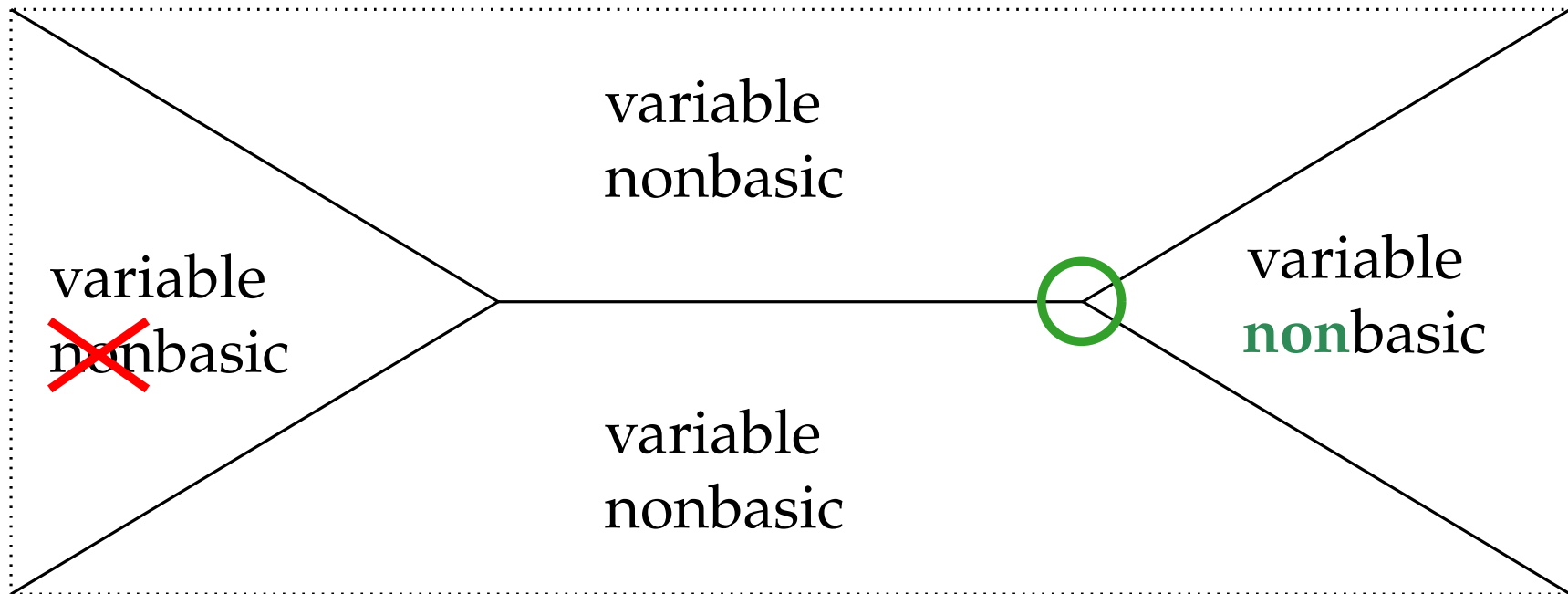
To go from red to green we:

Adjacency (special case)



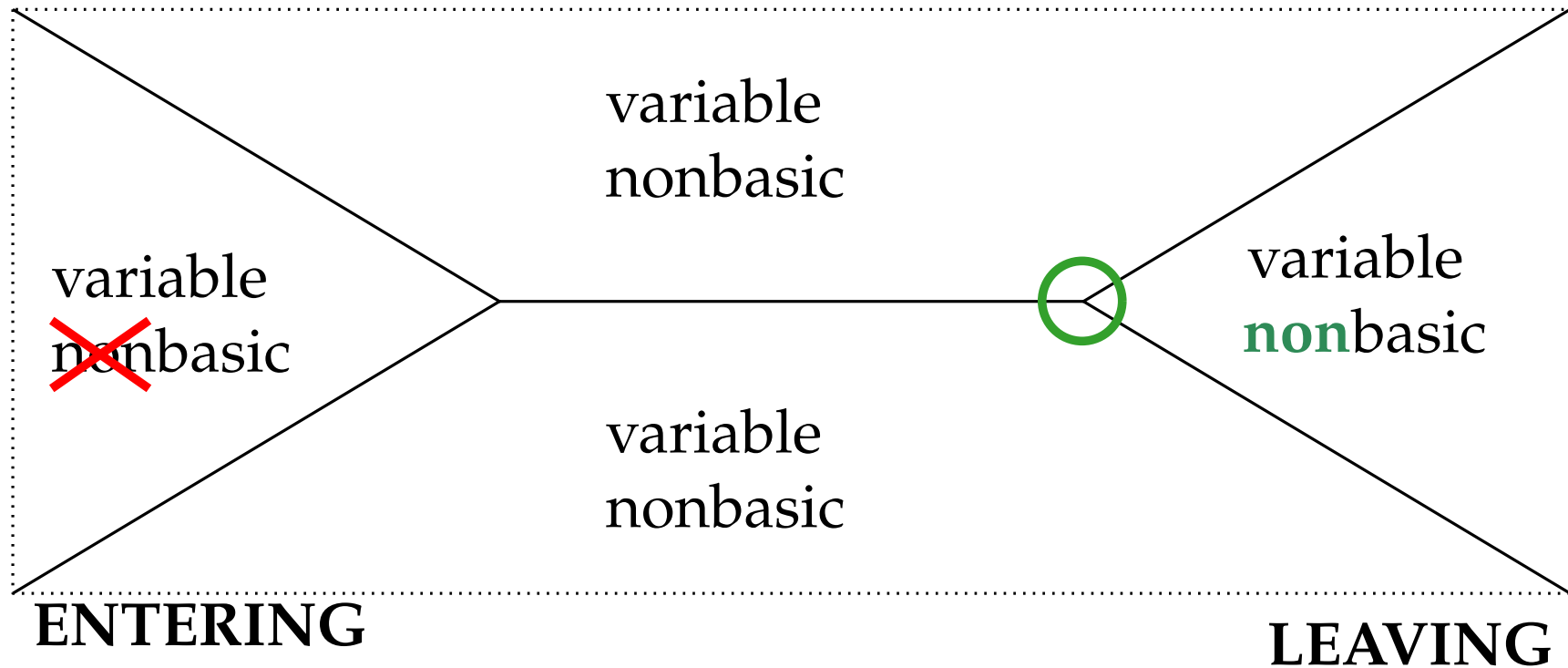
To go from red to green we:
start from this

Adjacency (special case)



To go from red to green we:
start from this and go to this

Adjacency (special case)



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start from this and go to this

The Simplex Method

- Start from a CPF/BFS, use slack variables!

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if it can, it starts at the origin. In the running example, this is the BFS for the basis $\{x_3, x_4, x_5\}$ i.e. the BFS $(0, 0, 4, 12, 18)$.

The Simplex Method

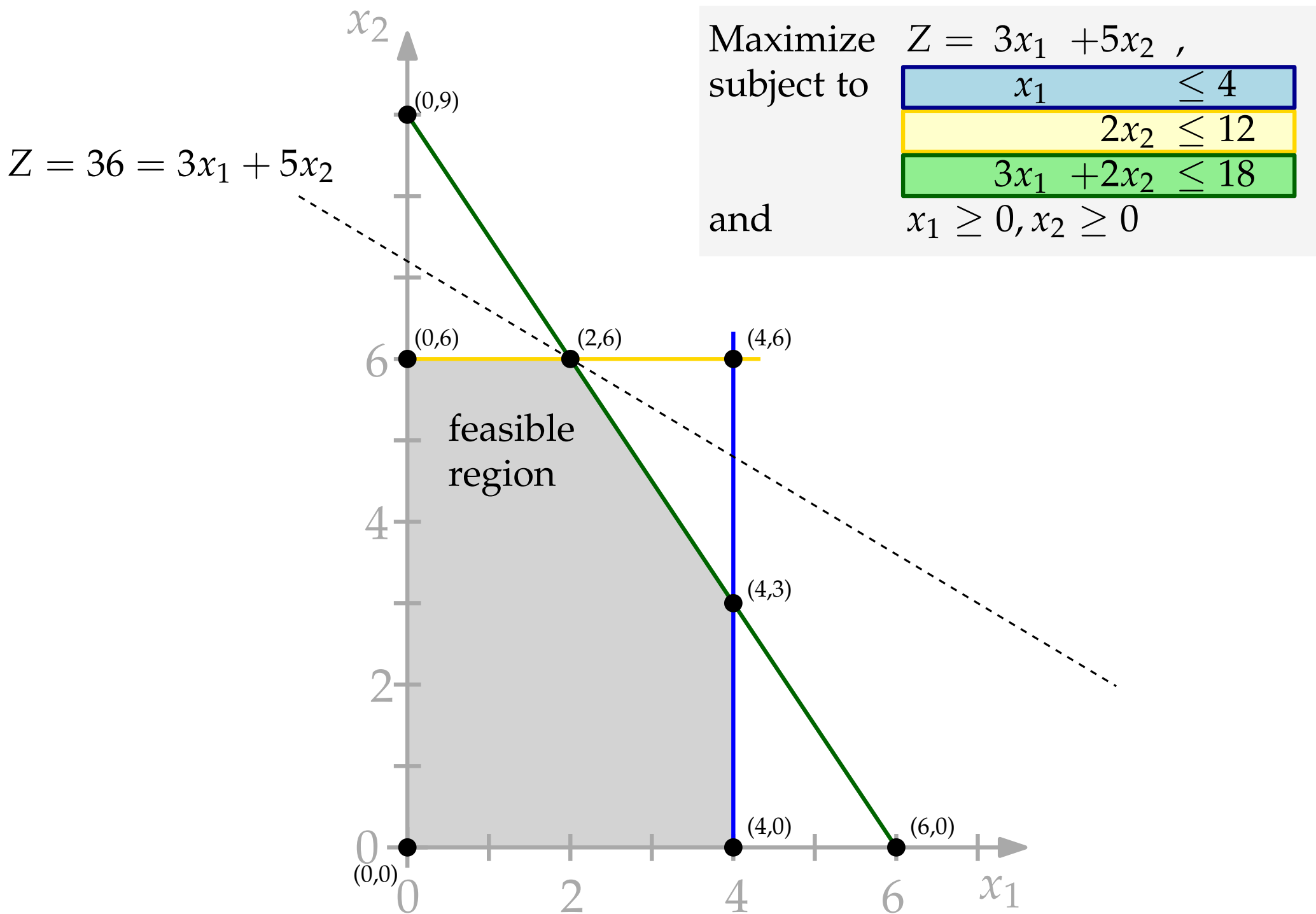
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- Iteratively move from a CPF/BFS to an adjacent CPF/BFS.

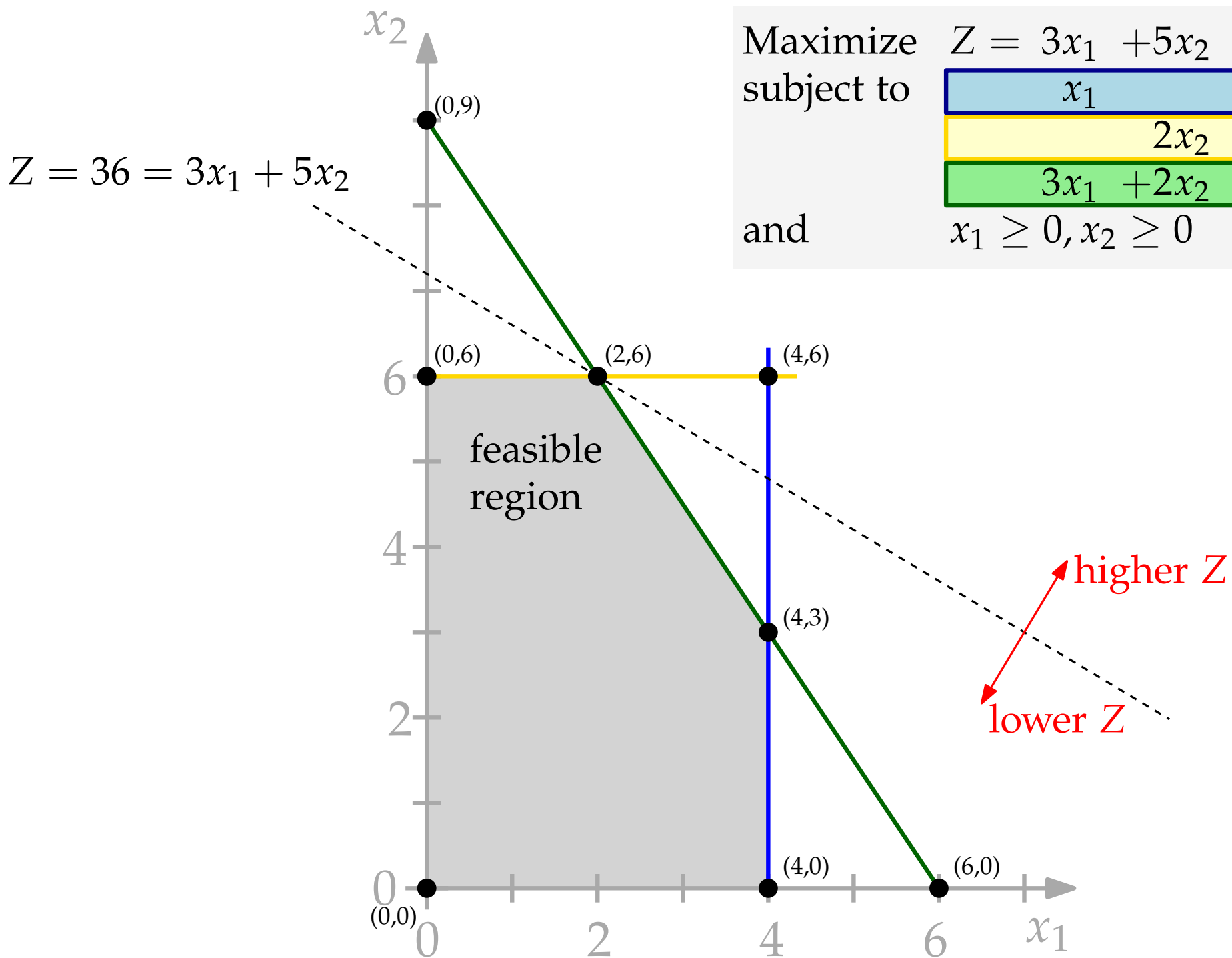
The Simplex Method

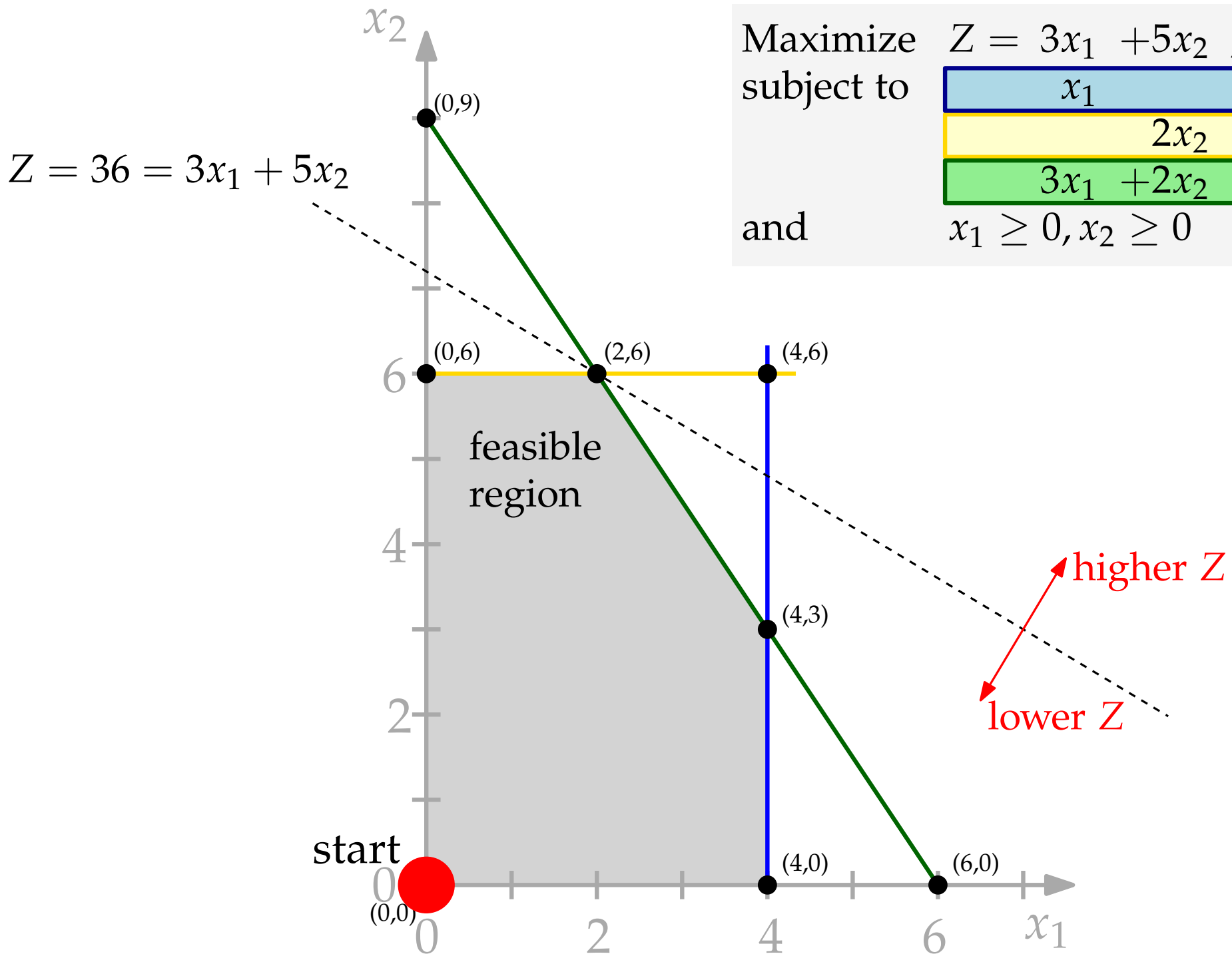
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- Iteratively move from a CPF/BFS to an adjacent CPF/BFS.
- It is *greedy*: moves to a neighbouring CPF/BFS with the most potential to improve the objective function. (It never chooses neighbouring CPF/BFS that worsens the objective function).

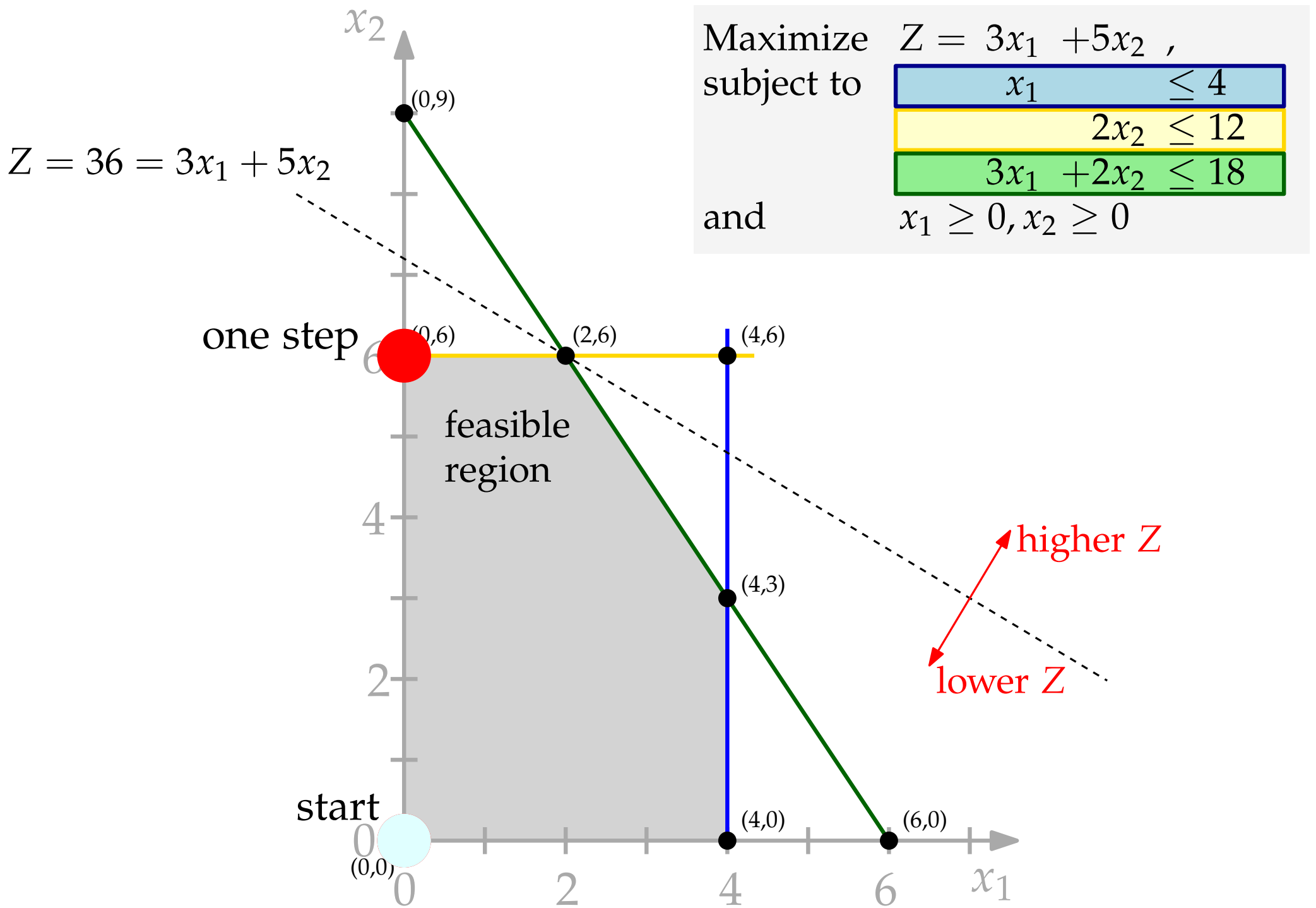
The Simplex Method

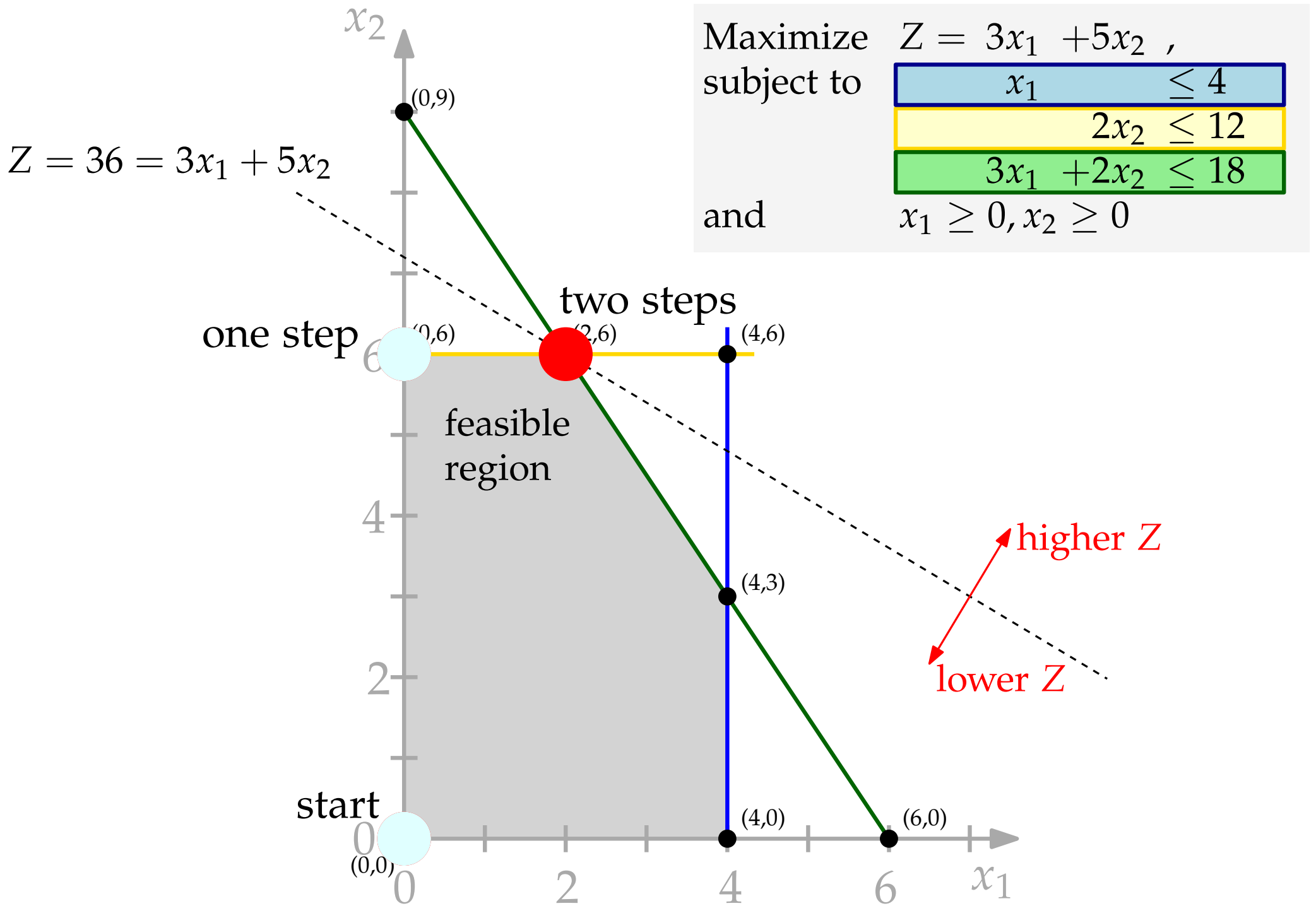
- Start from a CPF/BFS, use slack variables!
if it can, it starts at the origin. In the running example, this is the BFS for the basis $\{x_3, x_4, x_5\}$ i.e. the BFS $(0, 0, 4, 12, 18)$.
- Iteratively move from a CPF/BFS to an adjacent CPF/BFS.
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- Stop when no improvement is available on adjacent solutions.

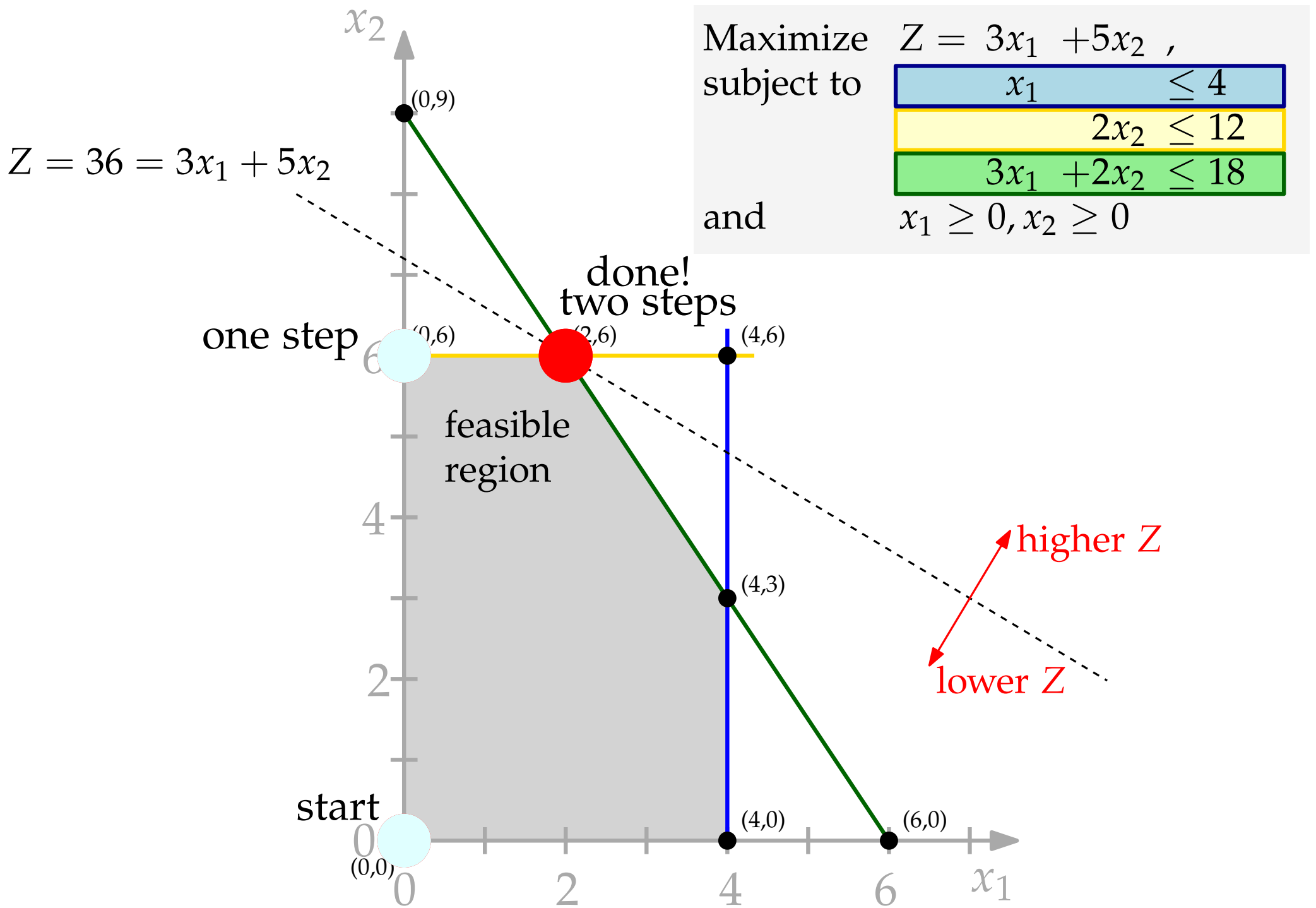












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How do you know when you've reached optimality?

- if the LP is feasible and has a bounded optimum, there exists at least one optimal CPF/BFS.
- if none of the neighbours of the current CPF/BFS can improve the objective function, we have reached optimality.
 $\Rightarrow$ suffices to look at adjacent CPF/BFS. This is an immensely powerful feature of the Simplex algorithm.

The Simplex Method

Can we reach an optimal CPF/BFS from any starting point?

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How to pick the entering/leaving variables?

- **Pivoting Rule**
- Move in a direction with the most **per unit** increase in the objective function.

Geometrically: the **steepest angle** of approach to the objective function.

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How do we calculate which currently basic variable leaves?

How do we ensure that our Basis really corresponds to linear independent columns?

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- Machinery of the Simplex Method

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{llll}
 (0) & Z - 3x_1 - 5x_2 & & = 0 \\
 (1) & x_1 & + x_3 & = 4 \\
 (2) & & 2x_2 + x_4 & = 12 \\
 (3) & 3x_1 + 2x_2 & & + x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

For convenience, we treat the current value of the objective function (Z) as a variable

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For convenience, we treat the current value of the objective function (Z) as a variable
 $\Rightarrow$ at every iteration, the objective value occurs on the RHS, along with the values of the basic variables.

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Maximize Z , subject to

$$\begin{array}{rclcl}
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 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- Start with $\{x_3, x_4, x_5\}$ as the basis, all other variables as nonbasic, i.e., $x_1 = x_2 = 0$. Thus, $x_3 = 4$, $x_4 = 12$, $x_5 = 18$, and $Z = 0$.

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$\Rightarrow$ if we allow a **single nonbasic** variable to become basic, we can directly see the improvement in Z **per unit** increase of that variable

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E.g. Change in Z per unit increase x_1 : 3
 Change in Z per unit increase x_2 : 5

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$$\begin{array}{rclcl}
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Increase x_2 (decrease basic variables) until one hits zero.
(note: all other nonbasics stay at 0)

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and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Increase x_2 (decrease basic variables) until one hits zero.
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$$\begin{array}{l}
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 x_3 = 4 \\
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 \end{array}$$

- Note that x_3 is unchanged by an increase in x_2 , because the constraint boundary it represents, the vertical line at $x_3 = 4$, is **parallel** to the direction of increase.

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Maximize Z , subject to

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- Note: disregard rows with non-positive (≤ 0) coefficients in the pivot column

Simplex Tableau

Maximize Z , subject to

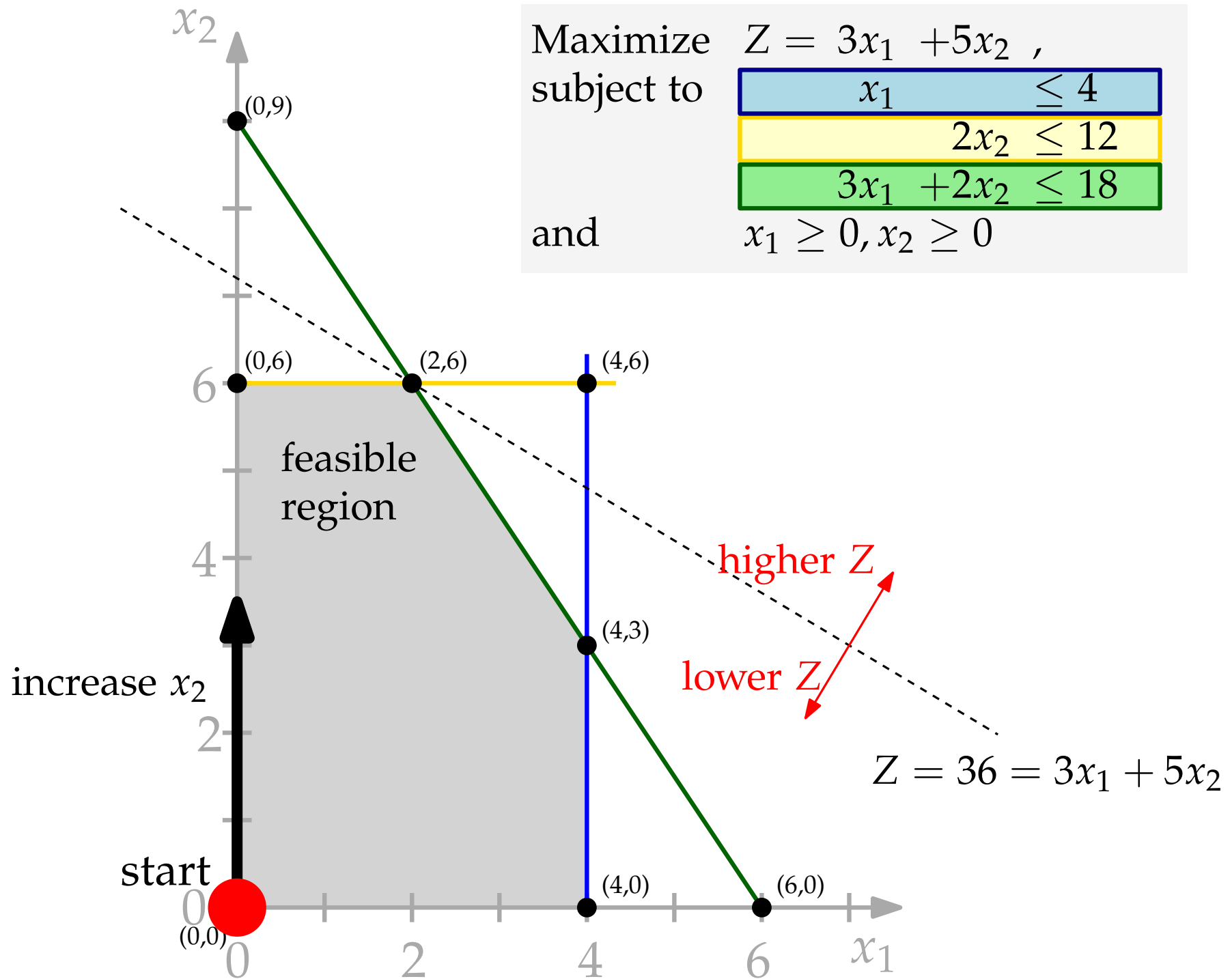
$$\begin{array}{rclcl}
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 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Increase x_2 (decrease basic variables) until one hits zero.
(note: all other nonbasics stay at 0)

$$\begin{array}{l}
 x_3 = 4 \\
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- Note: disregard rows with non-positive (≤ 0) coefficients in the pivot column
- Since $12/2 < 18/2$, the first basic variable that hits 0 is x_4 !
This is the **MINIMUM RATIO TEST**.



Maximize $Z = 3x_1 + 5x_2$,
 subject to

$$x_1 \leq 4$$

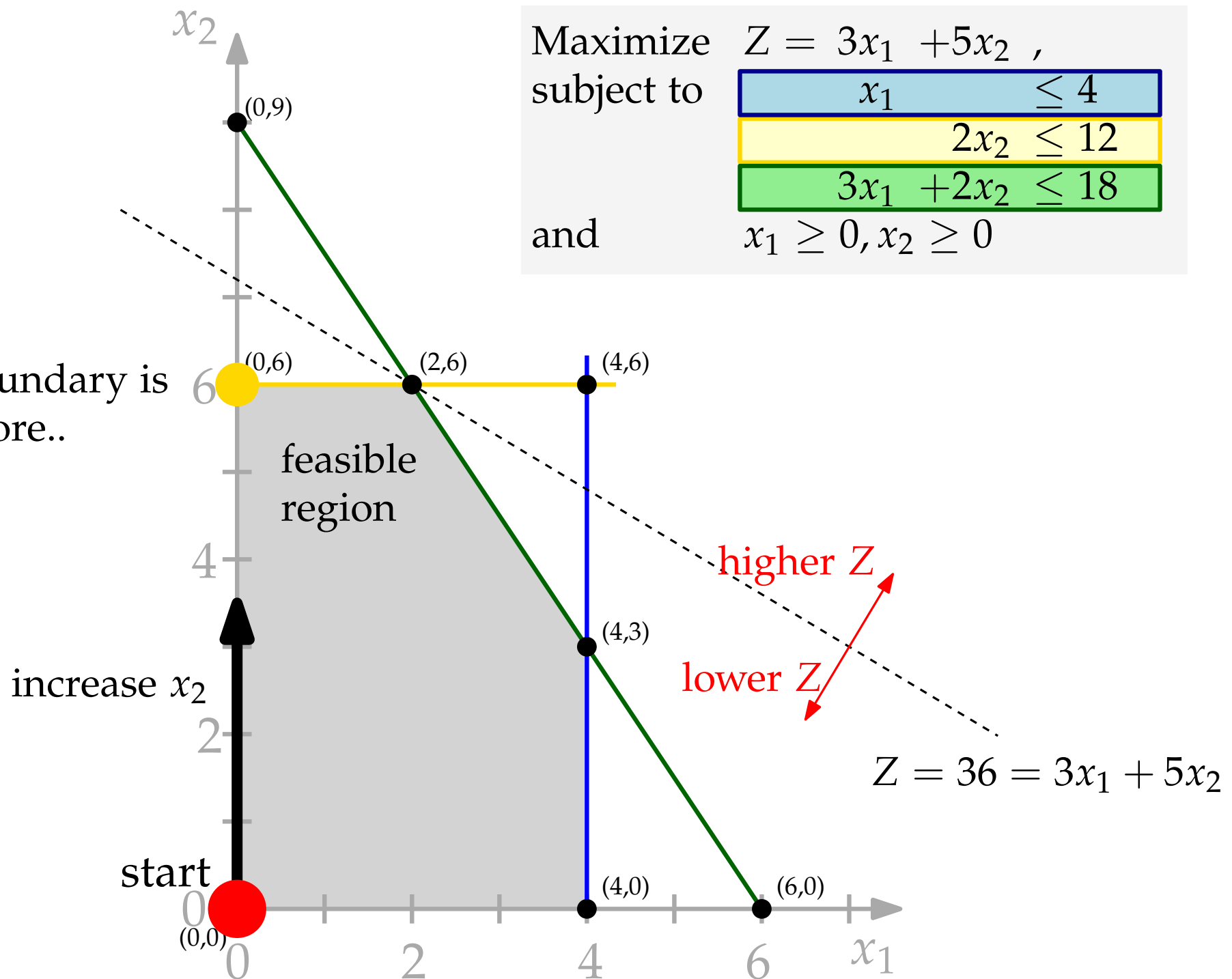
$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

and

$$x_1 \geq 0, x_2 \geq 0$$

this boundary is
hit before..



Maximize $Z = 3x_1 + 5x_2$,
 subject to

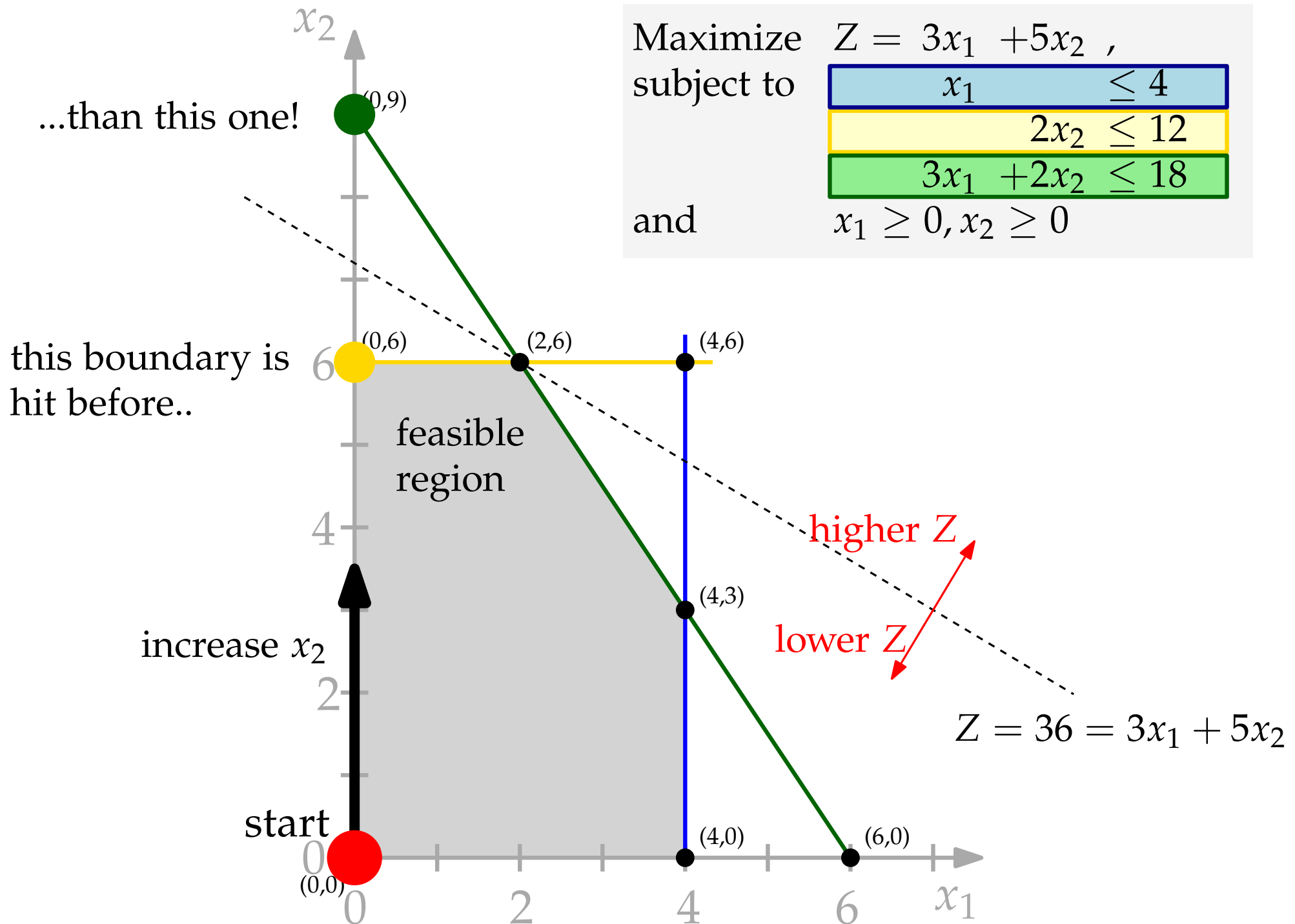
$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

and

$$x_1 \geq 0, x_2 \geq 0$$



Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclcl}
 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
 (1) & x_1 & & +x_3 & = 4 \\
 (2) & & 2x_2 & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

(note: all other nonbasics stay at 0)

Increase x_2 (lowering basic variables) until one hits zero.

$$\begin{array}{l}
 x_3 = 4 \\
 x_4 = 12 - 2x_2 \\
 x_5 = 18 - 2x_2
 \end{array}$$

- Note that x_3 is unchanged by an increase in x_2 , because the constraint boundary it represents, the vertical line at $x_1 = 4$, is **parallel** to the direction of increase.
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This is the **MINIMUM RATIO TEST**.

This ensures we stay within the feasible region.

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 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

To sum up:

1. Pick the **entering** variable by the coefficient in the objective (here: x_2)
2. Pick who **leaves** by the Minimum Ratio Test. (here: x_4)

Simplex Tableau

Maximize Z , subject to

$$(0) \quad Z - 3x_1 - 5x_2 = 0$$

$$(1) \quad x_1 + x_3 = 4$$

$$(2) \quad 2x_2 + x_4 = 12$$

$$(3) \quad 3x_1 + 2x_2 + x_5 = 18$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Now update the Tableau according to this change!

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclcl}
 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
 (1) & x_1 & & +x_3 & = 4 \\
 (2) & & \boxed{2x_2} & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Now update the Tableau according to this change!

- update x_2 's column so that it has all 0s except for one **1** in the row of “owned” by the leaving variable x_4 .

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclclcl}
 (0) & Z - 3x_1 & -5x_2 & & & = 0 \\
 (1) & x_1 & & +x_3 & & = 4 \\
 (2) & & 2x_2 & & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Now update the Tableau according to this change!

- update x_2 's column so that it has all 0s except for one **1** in the row of “owned” by the leaving variable x_4 .
- **eliminate** x_2 from the objective function (and other rows)

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclcl}
 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
 (1) & x_1 & & +x_3 & = 4 \\
 (2) & & 2x_2 & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Now update the Tableau according to this change!

- update x_2 's column so that it has all 0s except for one **1** in the row of "owned" by the leaving variable x_4 .
- **eliminate** x_2 from the objective function (and other rows)
- use Gaussian Elimination (i.e., elementary row operations) to make these changes!

Simplex Tableau

Maximize Z , subject to

$$(0) \quad Z - 3x_1 \quad \boxed{} \quad + \frac{5}{2}x_4 \quad = 30$$

$$(1) \quad x_1 \quad \boxed{} \quad + x_3 \quad = 4$$

$$(2) \quad \quad x_2 \quad \boxed{} \quad + \frac{1}{2}x_4 \quad = 6$$

$$(3) \quad 3x_1 \quad \boxed{} \quad - x_4 \quad + x_5 \quad = 6$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

$$\text{row}(0) \leftarrow \text{row}(0) + 5 \text{ row}(2)$$

$$\text{row}(2) \leftarrow \frac{1}{2} \text{ row}(2), \text{ then}$$

$$\text{row}(3) \leftarrow \text{row}(3) - 2 \text{ row}(2)$$

Now update the Tableau according to this change!

- update x_2 's column so that it has all 0s except for one **1** in the row of "owned" by the leaving variable x_4 .
- **eliminate** x_2 from the objective function (and other rows)
- use Gaussian Elimination (i.e., elementary row operations) to make these changes!

Simplex Tableau

Maximize Z , subject to

$$(0) \quad Z - 3x_1 \quad \quad \quad + \frac{5}{2}x_4 \quad \quad = 30$$

$$(1) \quad \quad x_1 \quad \quad + x_3 \quad \quad = 4$$

$$(2) \quad \quad \quad x_2 \quad \quad + \frac{1}{2}x_4 \quad \quad = 6$$

$$(3) \quad \quad 3x_1 \quad \quad - x_4 \quad + x_5 = 6$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclclclcl}
 (0) & Z - 3x_1 & & & + \frac{5}{2}x_4 & & = & 30 \\
 (1) & & x_1 & & & + x_3 & & = 4 \\
 (2) & & & x_2 & & & + \frac{1}{2}x_4 & = 6 \\
 (3) & & 3x_1 & & & & - x_4 & + x_5 = 6
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.
- New value of objective function is 30.

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rcllclcl}
 (0) & Z - 3x_1 & & + \frac{5}{2}x_4 & & = & 30 \\
 (1) & x_1 & & + x_3 & & = & 4 \\
 (2) & & x_2 & & + \frac{1}{2}x_4 & = & 6 \\
 (3) & 3x_1 & & - x_4 & + x_5 & = & 6
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.
- New value of objective function is 30.

Are we finished?

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclclclclcl}
 (0) & Z & -3x_1 & & & +\frac{5}{2}x_4 & & = & 30 \\
 (1) & & x_1 & & +x_3 & & & = & 4 \\
 (2) & & & x_2 & & +\frac{1}{2}x_4 & & = & 6 \\
 (3) & & 3x_1 & & & -x_4 & +x_5 & = & 6
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.
- New value of objective function is 30.

Are we finished?

- No! Nonbasic variable x_1 still has potential to improve the objective function. (Note: coefficient of x_4 in the top row is non-negative means that increasing x_4 does not improve the objective function.)

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclclclcl}
 (0) & Z & -3x_1 & & +\frac{5}{2}x_4 & & = & 30 \\
 (1) & & x_1 & & +x_3 & & = & 4 \\
 (2) & & & x_2 & & +\frac{1}{2}x_4 & = & 6 \\
 (3) & & 3x_1 & & & -x_4 & +x_5 & = & 6
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.
- New value of objective function is 30.

Are we finished?

- No! Nonbasic variable x_1 still has potential to improve the objective function. (Note: coefficient of x_4 in the top row is non-negative means that increasing x_4 does not improve the objective function.)
- So, x_1 enters ... who leaves?

Simplex Tableau

Maximize Z , subject to

$$(0) \quad Z - 3x_1 \quad \quad \quad + \frac{5}{2}x_4 \quad \quad = 30$$

$$(1) \quad \quad x_1 \quad \quad + x_3 \quad \quad = 4$$

$$(2) \quad \quad \quad x_2 \quad \quad + \frac{1}{2}x_4 \quad \quad = 6$$

$$(3) \quad \quad 3x_1 \quad \quad - x_4 \quad + x_5 = 6$$

$$x_3 = 4 - x_1 \rightarrow \text{zero at } x_1 = (4/1) = 4$$

x_2 : unaffected by x_1

$$x_5 = 6 - 3x_1 \rightarrow \text{zero at } x_1 = (6/3) = 2$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.
- New value of objective function is 30.

Are we finished?

- No! Nonbasic variable x_1 still has potential to improve the objective function. (Note: coefficient of x_4 in the top row is non-negative means that increasing x_4 does not improve the objective function.)
- So, x_1 enters ... who leaves?

Simplex Tableau

Maximize Z , subject to

$$(0) \quad Z - 3x_1 \quad \quad \quad + \frac{5}{2}x_4 \quad \quad = 30$$

$$(1) \quad \quad x_1 \quad \quad + x_3 \quad \quad = 4$$

$$(2) \quad \quad \quad x_2 \quad \quad + \frac{1}{2}x_4 \quad \quad = 6$$

$$(3) \quad \quad 3x_1 \quad \quad - x_4 \quad + x_5 = 6$$

$$x_3 = 4 - x_1 \rightarrow \text{zero at } x_1 = (4/1) = 4$$

x_2 : unaffected by x_1

$$x_5 = 6 - 3x_1 \rightarrow \text{zero at } x_1 = (6/3) = 2$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.
- New value of objective function is 30.

Are we finished?

- No! Nonbasic variable x_1 still has potential to improve the objective function. (Note: coefficient of x_4 in the top row is non-negative means that increasing x_4 does not improve the objective function.)
- So, x_1 enters ... who leaves? x_5 will go!

Simplex Tableau

Maximize Z , subject to

(0)	Z			$+\frac{3}{2}x_4$	$+x_5$	$= 36$
(1)			x_3	$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= 4$
(2)		x_2		$+\frac{1}{2}x_4$		$= 6$
(3)		x_1		$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= 2$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_1, x_2, x_3\}$ and $(x_1, x_2, x_3, x_4, x_5) = (2, 6, 2, 0, 0)$.
New value of objective function is 36.

Simplex Tableau

Maximize Z , subject to

(0)	Z			$+\frac{3}{2}x_4$	$+x_5$	$= 36$
(1)			x_3	$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= 4$
(2)		x_2		$+\frac{1}{2}x_4$		$= 6$
(3)		x_1		$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= 2$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_1, x_2, x_3\}$ and $(x_1, x_2, x_3, x_4, x_5) = (2, 6, 2, 0, 0)$.
New value of objective function is 36.

Are we done?

Simplex Tableau

Maximize Z , subject to

(0)	Z			$+\frac{3}{2}x_4$	$+x_5$	$= 36$
(1)			x_3	$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= 4$
(2)		x_2		$+\frac{1}{2}x_4$		$= 6$
(3)		x_1		$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= 2$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_1, x_2, x_3\}$ and $(x_1, x_2, x_3, x_4, x_5) = (2, 6, 2, 0, 0)$.
New value of objective function is 36.

Are we done?

- YES! Neither of the two nonbasic variables can improve the objective function. So we know we have reached optimality!

The Simplex Method

1. Find a BFS to start (just use the slack variables as the basis).
2. Can a currently nonbasic variable help to improve the objective function? (**just check coefficients in row(0)**)
No? We are done! We have an optimal solution!
3. Pick a nonbasic variable with the greatest potential to improve the objective function, **the entering variable**
4. Use the **minimum ratio test** to identify the **leaving variable**
5. Update the system for the new BFS using Gaussian elimination
6. Go to 2.



Maastricht University

Department of Advanced Computing Sciences

MATSE: Operations Research

Lecture 3. More Simplex

[based on slides by: Steven Kelk and Steven Chaplick]

Barbara Franci

Standard LP ... or is it?

Given: $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$ [standard form]

where, for each entry b_i in b , $b_i \geq 0$, i.e., $b \geq 0$

Find: $x^* \in \mathbb{R}^n$ where $x^* = \arg \max \{c^T x \mid Ax \leq b, x \geq 0\}$

- Maximization
- All decision variables: non-negative (≥ 0)
- All constraints: $\leq$

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Find: $x^* \in \mathbb{R}^n$ where $x^* = \arg \max \{c^T x \mid Ax \leq b, x \geq 0\}$

- Maximization
 - All decision variables: non-negative (≥ 0)
 - All constraints: $\leq$
- } all of these are **safe**,
without loss of generality
regarding LP

Standard LP ... or is it?

Given: $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$ [standard form]

where, for each entry b_i in b , $b_i \geq 0$, i.e., $b \geq 0$

Find: $x^* \in \mathbb{R}^n$ where $x^* = \arg \max \{c^T x \mid Ax \leq b, x \geq 0\}$

- Maximization
 - All decision variables: non-negative (≥ 0)
 - All constraints: $\leq$
- } all of these are **safe**,
without loss of generality
regarding LP

With this assumption, an initial BFS is easy to find:

- Set the decision variables as non-basic (set = 0), and the slack variables as basic.

The Simplex Method

1. Find a BFS to start (just use the slack variables as the basis).
2. Can a currently nonbasic variable help to improve the objective function? (**just check coefficients in row(0)**)
No? We are done! We have an optimal solution!
3. Pick a nonbasic variable with the greatest potential to improve the objective function, **the entering variable**
4. Use the **minimum ratio test** to identify the **leaving variable**
5. Update the system for the new BFS using Gaussian elimination
6. GOTO 2.

Algebraic vs. Tabular Form

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■ **TABLE 4.3** Initial system of equations for the Wyndor Glass Co. problem

(a) Algebraic Form			(b) Tabular Form									
			Basic Variable	Eq.	Coefficient of:					Right Side		
					Z	x ₁	x ₂	x ₃	x ₄		x ₅	
(0)	Z	− 3x ₁ − 5x ₂	= 0	Z	(0)	1	−3	−5	0	0	0	0
(1)	x ₁	+ x ₃	= 4	x ₃	(1)	0	1	0	1	0	0	4
(2)	2x ₂	+ x ₄	= 12	x ₄	(2)	0	0	2	0	1	0	12
(3)	3x ₁ + 2x ₂	+ x ₅	= 18	x ₅	(3)	0	3	2	0	0	1	18

Algebraic vs. Tabular Form

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■ **TABLE 4.3** Initial system of equations for the Wyndor Glass Co. problem

(a) Algebraic Form			(b) Tabular Form									
			Basic Variable	Eq.	Coefficient of:					Right Side		
					Z	x ₁	x ₂	x ₃	x ₄		x ₅	
(0)	Z	− 3x ₁ − 5x ₂	= 0	Z	(0)	1	−3	−5	0	0	0	0
(1)	x ₁	+ x ₃	= 4	x ₃	(1)	0	1	0	1	0	0	4
(2)	2x ₂	+ x ₄	= 12	x ₄	(2)	0	0	2	0	1	0	12
(3)	3x ₁ + 2x ₂	+ x ₅	= 18	x ₅	(3)	0	3	2	0	0	1	18

Starting BFS: (0, 0, 4, 12, 18)

■ **TABLE 4.4** Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of:						Right Side	Ratio
		Z	x_1	x_2	x_3	x_4	x_5		
Z	(0)	1	-3	-5	0	0	0	0	
x_3	(1)	0	1	0	1	0	0	4	
x_4	(2)	0	0	2	0	1	0	12	
x_5	(3)	0	3	2	0	0	1	18	

- Negative value in row(0), so not yet optimal

■ **TABLE 4.4** Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of:						Right Side	Ratio
		Z	x_1	x_2	x_3	x_4	x_5		
Z	(0)	1	-3	-5	0	0	0	0	
x_3	(1)	0	1	0	1	0	0	4	
x_4	(2)	0	0	2	0	1	0	12	
x_5	(3)	0	3	2	0	0	1	18	

- Negative value in row(0), so not yet optimal
- Pick entering variable (nonbasic $\rightarrow$ basic):

■ **TABLE 4.4** Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of:						Right Side	Ratio
		Z	x_1	x_2	x_3	x_4	x_5		
Z	(0)	1	-3	-5	0	0	0	0	
x_3	(1)	0	1	0	1	0	0	4	
x_4	(2)	0	0	2	0	1	0	12	
x_5	(3)	0	3	2	0	0	1	18	

- Negative value in row(0), so not yet optimal
- Pick entering variable (nonbasic $\rightarrow$ basic):

-5 is most negative in row(0) $\leadsto$ pick x_2

■ **TABLE 4.4** Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of:						Right Side	Ratio
		Z	x_1	x_2	x_3	x_4	x_5		
Z	(0)	1	-3	-5	0	0	0	0	
x_3	(1)	0	1	0	1	0	0	4	
x_4	(2)	0	0	2	0	1	0	12	
x_5	(3)	0	3	2	0	0	1	18	

- Negative value in row(0), so not yet optimal
- Pick entering variable (nonbasic $\rightarrow$ basic):

-5 is most negative in row(0) $\leadsto$ pick x_2
- Find leaving variable (basic $\rightarrow$ nonbasic): Minimum Ratio Test

■ **TABLE 4.4** Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of:						Right Side	Ratio
		Z	x_1	x_2	x_3	x_4	x_5		
Z	(0)	1	-3	-5	0	0	0	0	
x_3	(1)	0	1	0	1	0	0	4	
x_4	(2)	0	0	2	0	1	0	12	
x_5	(3)	0	3	2	0	0	1	18	

- Negative value in row(0), so not yet optimal
- Pick entering variable (nonbasic $\rightarrow$ basic):

-5 is most negative in row(0) $\leadsto$ pick x_2
- Find leaving variable (basic $\rightarrow$ nonbasic): **Minimum Ratio Test**
 for each row, find the value of entering var (x_2) such that, basic var. = 0
 pick the row (and basic variable) with **minimum** value

■ **TABLE 4.4** Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of:						Right Side	Ratio
		Z	x_1	x_2	x_3	x_4	x_5		
Z	(0)	1	-3	-5	0	0	0	0	
x_3	(1)	0	1	0	1	0	0	4	
x_4	(2)	0	0	2	0	1	0	$12 \rightarrow \frac{12}{2} = 6$	$\leftarrow$ minimum
x_5	(3)	0	3	2	0	0	1	$18 \rightarrow \frac{18}{2} = 9$	

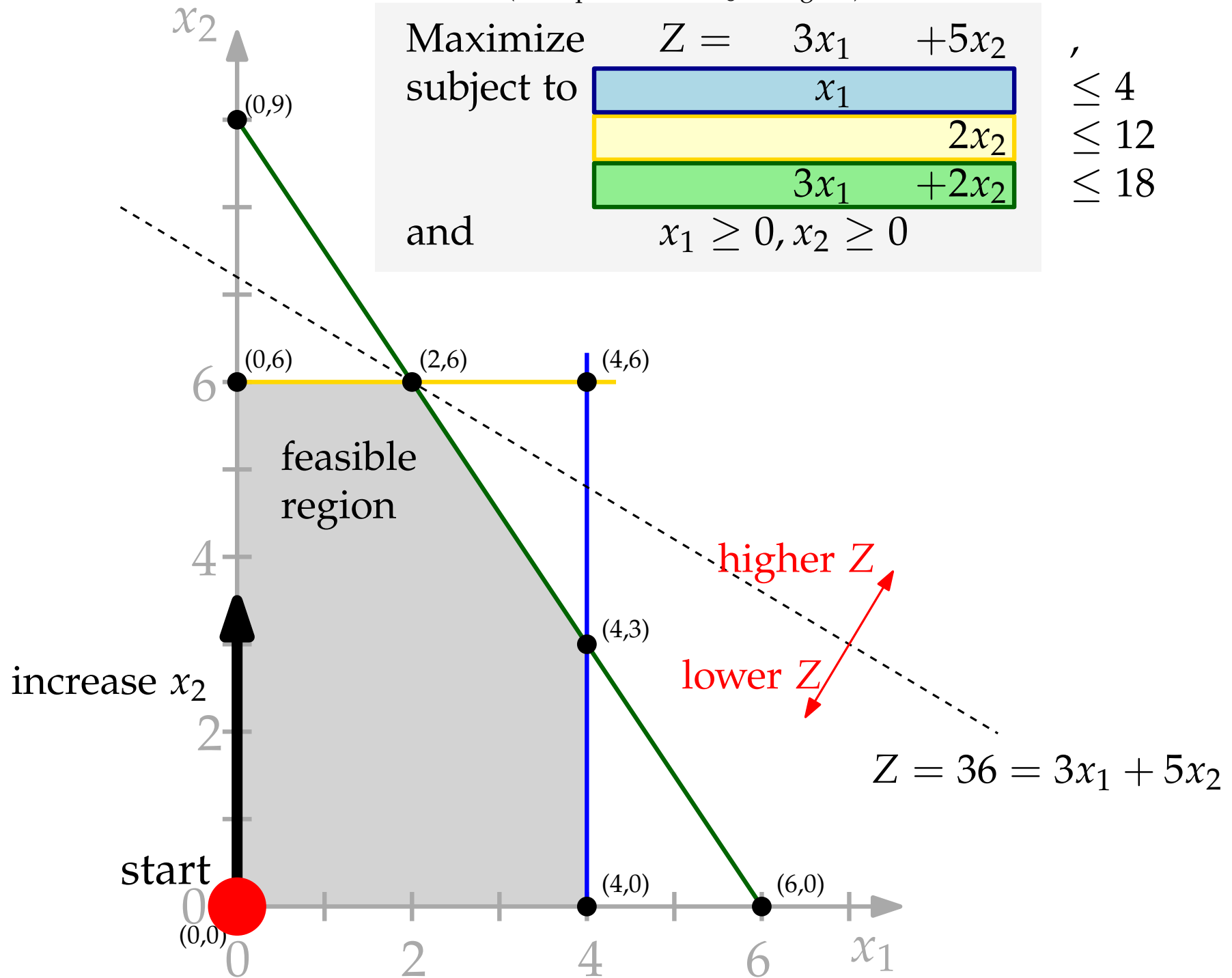
- Negative value in row(0), so not yet optimal
- Pick entering variable (nonbasic $\rightarrow$ basic):
 -5 is most negative in row(0) $\leadsto$ pick x_2
- Find leaving variable (basic $\rightarrow$ nonbasic): **Minimum Ratio Test**
 for each row, find the value of entering var (x_2) such that, basic var. = 0
 pick the row (and basic variable) with **minimum** value

■ **TABLE 4.4** Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of:						Right Side	Ratio
		Z	x_1	x_2	x_3	x_4	x_5		
Z	(0)	1	-3	-5	0	0	0	0	
x_3	(1)	0	1	0	1	0	0	4	
x_4	(2)	0	0	2	0	1	0	$12 \rightarrow \frac{12}{2} = 6$	← minimum
x_5	(3)	0	3	2	0	0	1	$18 \rightarrow \frac{18}{2} = 9$	

- Negative value in row(0), so not yet optimal
- Pick entering variable (nonbasic $\rightarrow$ basic):
 -5 is most negative in row(0) $\rightsquigarrow$ pick x_2
- Find leaving variable (basic $\rightarrow$ nonbasic): **Minimum Ratio Test**
 for each row, find the value of entering var (x_2) such that, basic var. = 0
 pick the row (and basic variable) with **minimum** value
 row(2) gives minimum ratio $\rightsquigarrow x_4$ leaves

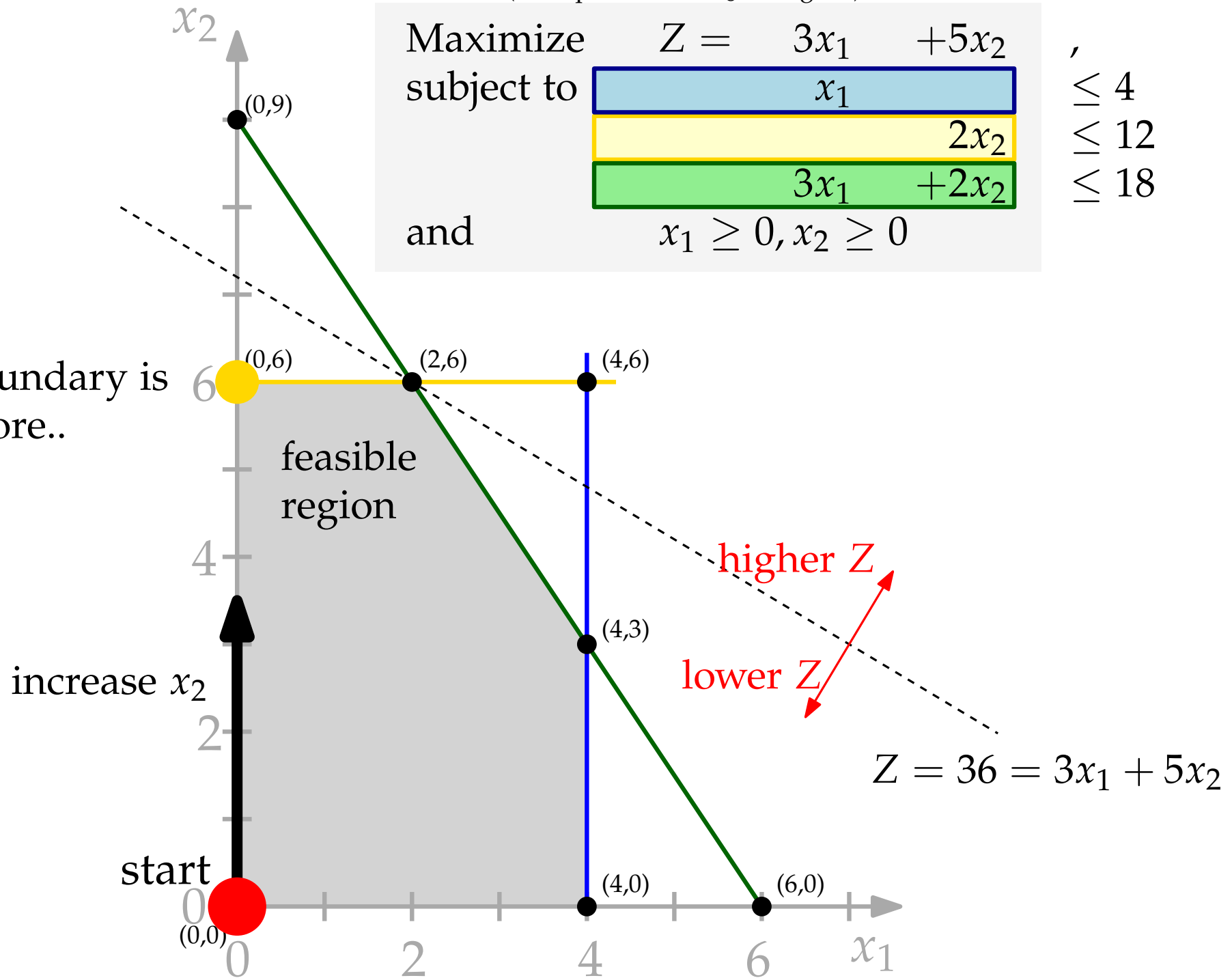
(example as in HL §4.1, fig 4.1)



(example as in HL §4.1, fig 4.1)

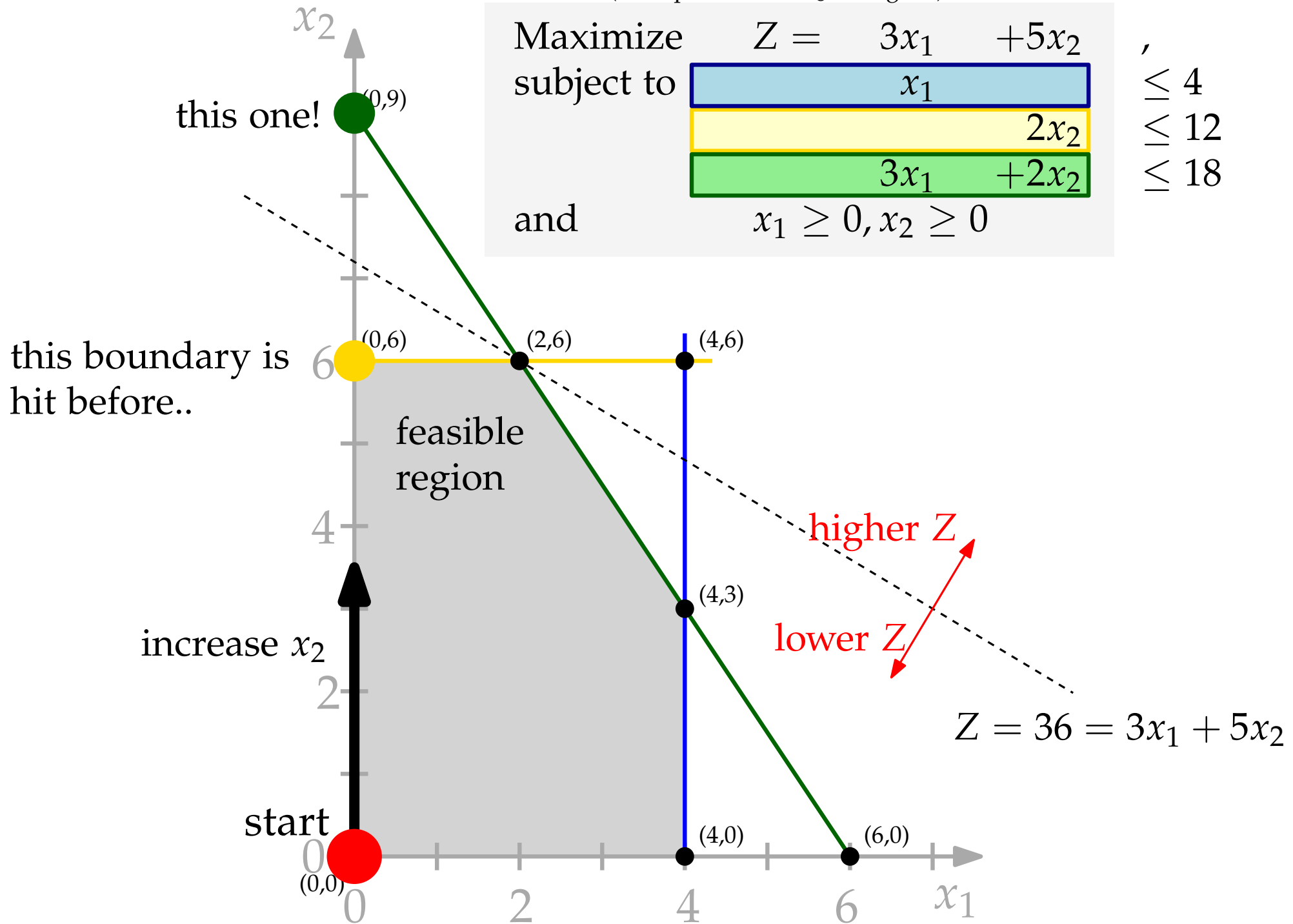
Maximize $Z = 3x_1 + 5x_2$,
 subject to $x_1 \leq 4$,
 $2x_2 \leq 12$,
 $3x_1 + 2x_2 \leq 18$
 and $x_1 \geq 0, x_2 \geq 0$

this boundary is
hit before..



(example as in HL §4.1, fig 4.1)

Maximize $Z = 3x_1 + 5x_2$,
 subject to $x_1 \leq 4$,
 $2x_2 \leq 12$,
 $3x_1 + 2x_2 \leq 18$
 and $x_1 \geq 0, x_2 \geq 0$



■ **TABLE 4.4** Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of:						Right Side	Ratio
		Z	x_1	x_2	x_3	x_4	x_5		
Z	(0)	1	-3	-5	0	0	0	0	
x_3	(1)	0	1	0	1	0	0	4	
x_4	(2)	0	0	2	0	1	0	$12 \rightarrow \frac{12}{2} = 6$	$\leftarrow$ minimum
x_5	(3)	0	3	2	0	0	1	$18 \rightarrow \frac{18}{2} = 9$	

- Negative value in row(0), so not yet optimal
- Pick entering variable (nonbasic $\rightarrow$ basic):
 -5 is most negative in row(0) $\leadsto$ pick x_2
- Find leaving variable (basic $\rightarrow$ nonbasic): **Minimum Ratio Test**
 for each row, find the value of entering var (x_2) such that, basic var. = 0
 pick the row (and basic variable) with **minimum** value
 $\text{row}(2)$ gives minimum ratio $\leadsto x_4$ leaves

■ **TABLE 4.4** Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of:						Right Side	Ratio
		Z	x_1	x_2	x_3	x_4	x_5		
Z	(0)	1	-3	-5	0	0	0	0	
x_3	(1)	0	1	0	1	0	0	4	
x_4	(2)	0	0	2	0	1	0	$12 \rightarrow \frac{12}{2} = 6 \leftarrow$ minimum	
x_5	(3)	0	3	2	0	0	1	$18 \rightarrow \frac{18}{2} = 9$	

pivot
row

pivot column

- Negative value in row(0), so not yet optimal
- Pick entering variable (nonbasic $\rightarrow$ basic):
 -5 is most negative in row(0) $\leadsto$ pick x_2
- Find leaving variable (basic $\rightarrow$ nonbasic): **Minimum Ratio Test**
 for each row, find the value of entering var (x_2) such that, basic var. = 0
 pick the row (and basic variable) with **minimum** value
 row(2) gives minimum ratio $\leadsto x_4$ leaves

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■ **TABLE 4.4** Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of:						Right Side	Ratio
		Z	x_1	x_2	x_3	x_4	x_5		
Z	(0)	1	-3	-5	0	0	0	0	
x_3	(1)	0	1	0	1	0	0	4	
x_4	(2)	0	0	2	0	1	0	$12 \rightarrow \frac{12}{2} = 6 \leftarrow \text{minimum}$	
x_5	(3)	0	3	2	0	0	1	$18 \rightarrow \frac{18}{2} = 9$	

pivot
row

pivot column

- Next Step: Establish identity 0/1 pattern in the pivot column (1 in pivot row, 0 otherwise).

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Basic Variable	Eq.	Coefficient of:						Right Side	Ratio
		Z	x_1	x_2	x_3	x_4	x_5		
Z	(0)	1		0					
x_3	(1)	0		0					
x_4	(2)	0		1					
x_5	(3)	0		0					

- Next Step: Establish identity 0/1 pattern in the pivot column (1 in pivot row, 0 otherwise).

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■ **TABLE 4.4** Applying the minimum ratio test to determine the first leaving basic variable for the Wyndor Glass Co. problem

Basic Variable	Eq.	Coefficient of:						Right Side	Ratio
		Z	x_1	x_2	x_3	x_4	x_5		
Z	(0)	1		0					
x_3	(1)	0		0					
x_4	(2)	0		1					
x_5	(3)	0		0					

- Next Step: Establish identity 0/1 pattern in the pivot column (1 in pivot row, 0 otherwise).
- Do this via elementary row operations (this will of course change ... many things!)

■ **TABLE 4.5** Simplex tableaux for the Wyndor Glass Co. problem after the first pivot row is divided by the first pivot number

Iteration	Basic Variable	Eq.	Coefficient of:						Right Side
			Z	x_1	x_2	x_3	x_4	x_5	
0	Z	(0)	1	-3	-5	0	0	0	0
	x_3	(1)	0	1	0	1	0	0	4
	x_4	(2)	0	0	2	0	1	0	12
	x_5	(3)	0	3	2	0	0	1	18
1	Z	(0)	1						
	x_3	(1)	0						
	x_2	(2)	0						
	x_5	(3)	0						

■ **TABLE 4.5** Simplex tableaux for the Wyndor Glass Co. problem after the first pivot row is divided by the first pivot number

Iteration	Basic Variable	Eq.	Coefficient of:						Right Side
			Z	x_1	x_2	x_3	x_4	x_5	
0	Z	(0)	1	-3	-5	0	0	0	0
	x_3	(1)	0	1	0	1	0	0	4
	x_4	(2)	0	0	2	0	1	0	12
	x_5	(3)	0	3	2	0	0	1	18
1	Z	(0)	1						
	x_3	(1)	0						
	x_2	(2)	0	0	1	0	$\frac{1}{2}$	0	6
	x_5	(3)	0						

■ **TABLE 4.6** First two simplex tableaux for the Wyndor Glass Co. problem

Iteration	Basic Variable	Eq.	Coefficient of:						Right Side
			Z	x_1	x_2	x_3	x_4	x_5	
0	Z	(0)	1	-3	-5	0	0	0	0
	x_3	(1)	0	1	0	1	0	0	4
	x_4	(2)	0	0	2	0	1	0	12
	x_5	(3)	0	3	2	0	0	1	18
1	Z	(0)	1	-3	0	0	$\frac{5}{2}$	0	30
	x_3	(1)	0	1	0	1	0	0	4
	x_2	(2)	0	0	1	0	$\frac{1}{2}$	0	6
	x_5	(3)	0	3	0	0	-1	1	6

■ **TABLE 4.6** First two simplex tableaux for the Wyndor Glass Co. problem

Iteration	Basic Variable	Eq.	Coefficient of:						Right Side
			Z	x_1	x_2	x_3	x_4	x_5	
0	Z	(0)	1	-3	-5	0	0	0	0
	x_3	(1)	0	1	0	1	0	0	4
	x_4	(2)	0	0	2	0	1	0	12
	x_5	(3)	0	3	2	0	0	1	18
1	Z	(0)	1	-3	0	0	$\frac{5}{2}$	0	30
	x_3	(1)	0	1	0	1	0	0	4
	x_2	(2)	0	0	1	0	$\frac{1}{2}$	0	6
	x_5	(3)	0	3	0	0	-1	1	6

- start the next round: pick an entering variable, then find a leaving variable!

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■ **TABLE 4.7** Steps 1 and 2 of iteration 2 for the Wyndor Glass Co. problem

Iteration	Basic Variable	Eq.	Coefficient of:					Right Side	Ratio
			Z	x_1	x_2	x_3	x_4	x_5	
1	Z	(0)	1	-3	0	0	$\frac{5}{2}$	0	30
	x_3	(1)	0	1	0	1	0	0	4 $\frac{4}{1} = 4$
	x_2	(2)	0	0	1	0	$\frac{1}{2}$	0	6
	x_5	(3)	0	3	0	0	-1	1	6 $\frac{6}{3} = 2 \leftarrow \text{minimum}$

- start the next round: pick an entering variable, then find a leaving variable!

■ **TABLE 4.8** Complete set of simplex tableaux for the Wyndor Glass Co. problem

Iteration	Basic Variable	Eq.	Coefficient of:						Right Side
			Z	x ₁	x ₂	x ₃	x ₄	x ₅	
0	Z	(0)	1	−3	−5	0	0	0	0
	x ₃	(1)	0	1	0	1	0	0	4
	x ₄	(2)	0	0	2	0	1	0	12
	x ₅	(3)	0	3	2	0	0	1	18
1	Z	(0)	1	−3	0	0	$\frac{5}{2}$	0	30
	x ₃	(1)	0	1	0	1	0	0	4
	x ₂	(2)	0	0	1	0	$\frac{1}{2}$	0	6
	x ₅	(3)	0	3	0	0	−1	1	6
2	Z	(0)	1	0	0	0	$\frac{3}{2}$	1	36
	x ₃	(1)	0	0	0	1	$\frac{1}{3}$	− $\frac{1}{3}$	2
	x ₂	(2)	0	0	1	0	$\frac{1}{2}$	0	6
	x ₁	(3)	0	1	0	0	− $\frac{1}{3}$	$\frac{1}{3}$	2

Complications (§4.5)

Complications

What if there are unbounded/multiple optima?

Complications

What if there are unbounded/multiple optima?

- Tie for Entering Basic Variable:
multiple nonbasic variables with the same **per unit increase**
(most negative coefficient in row(0))
- Tie for Leaving Basic Variable:
multiple basic variables with the same **Minimum Ratio**

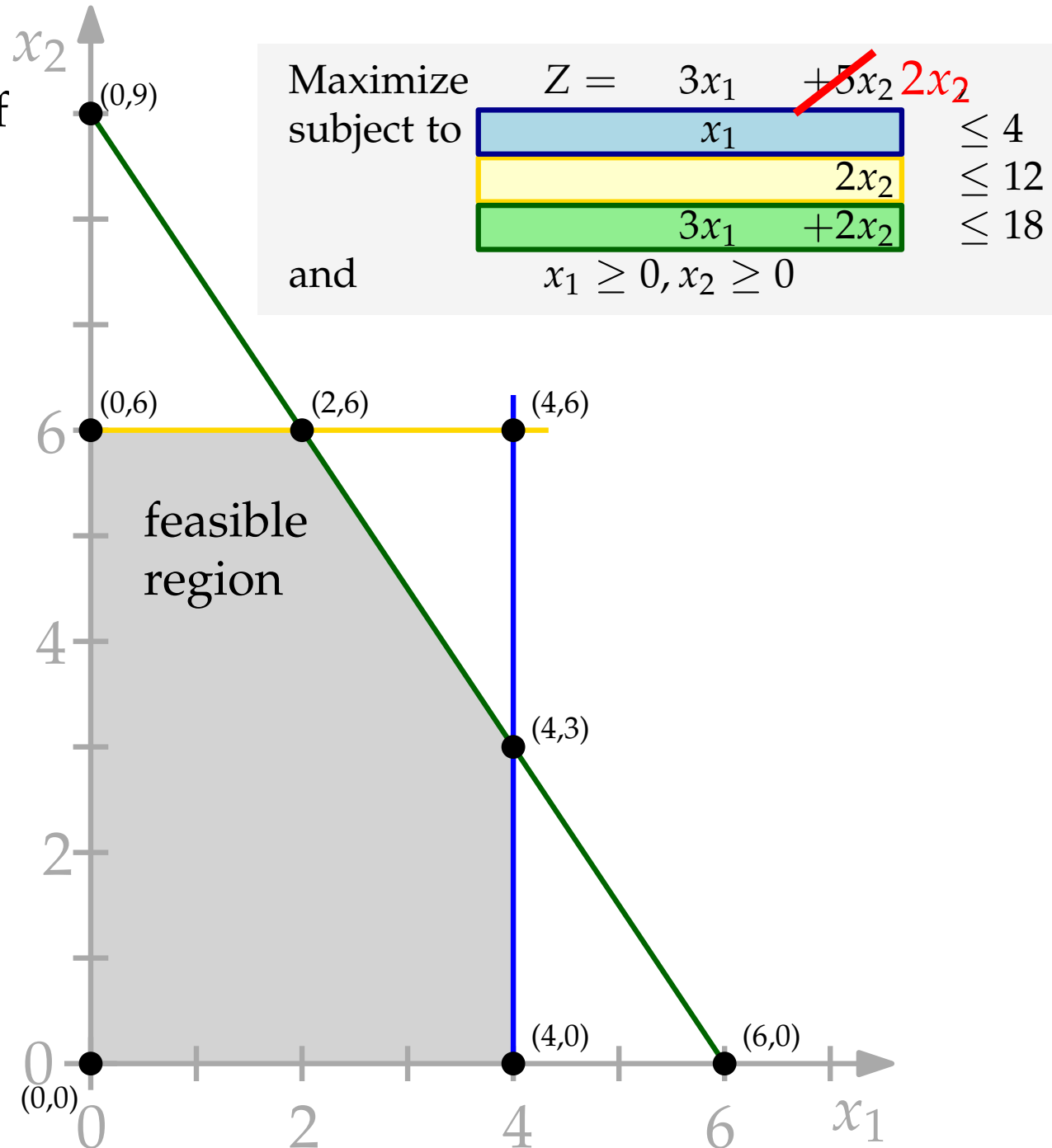
Complications

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- Tie for Entering Basic Variable:
multiple nonbasic variables with the same **per unit increase**
(most negative coefficient in row(0))
 - Tie for Leaving Basic Variable:
multiple basic variables with the same **Minimum Ratio**
- ⇒ **Degeneracy**: not just theoretical, it happens!

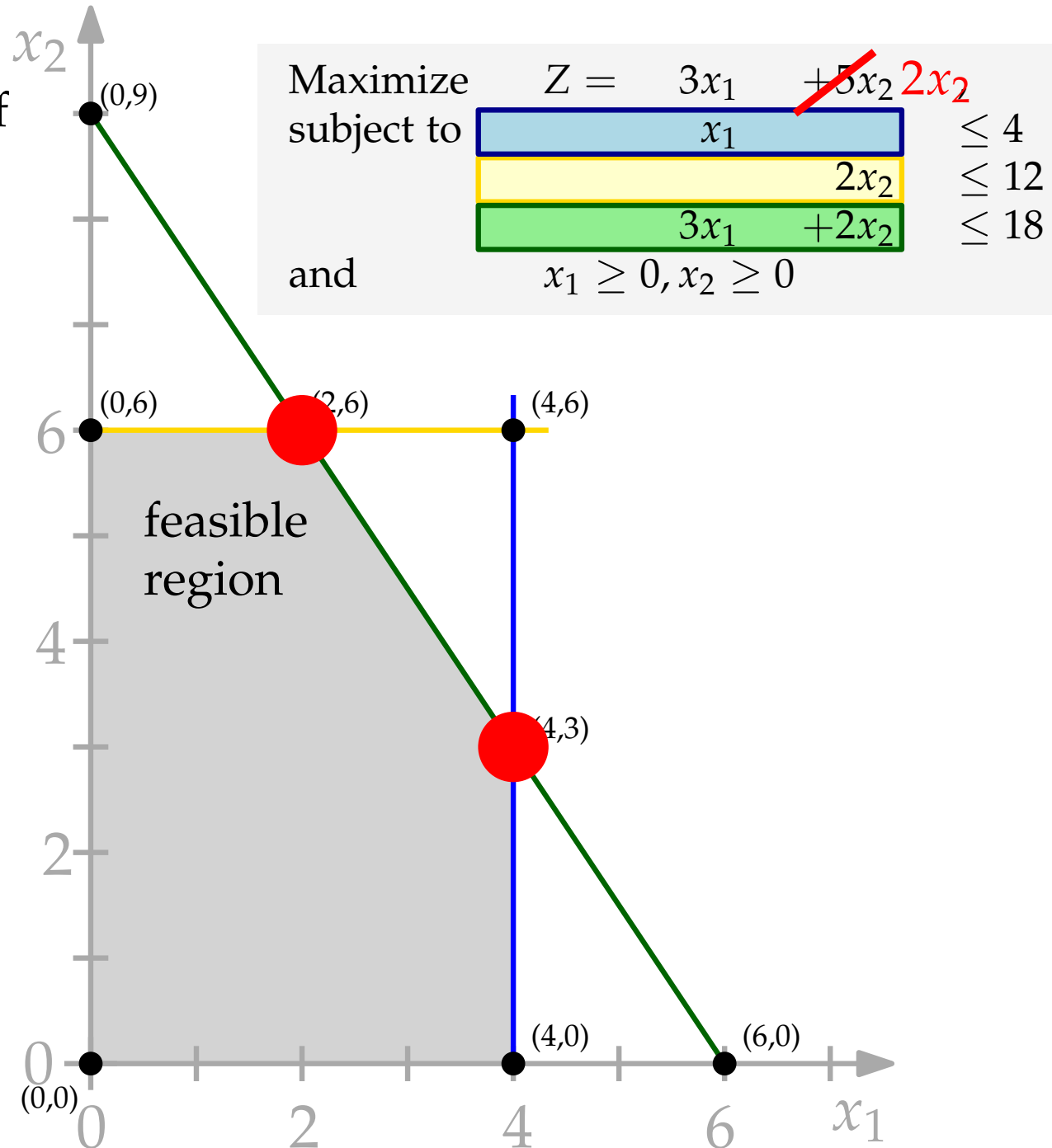
Multiple Optima

- change the coefficient of x_2 to 2 instead of 5.



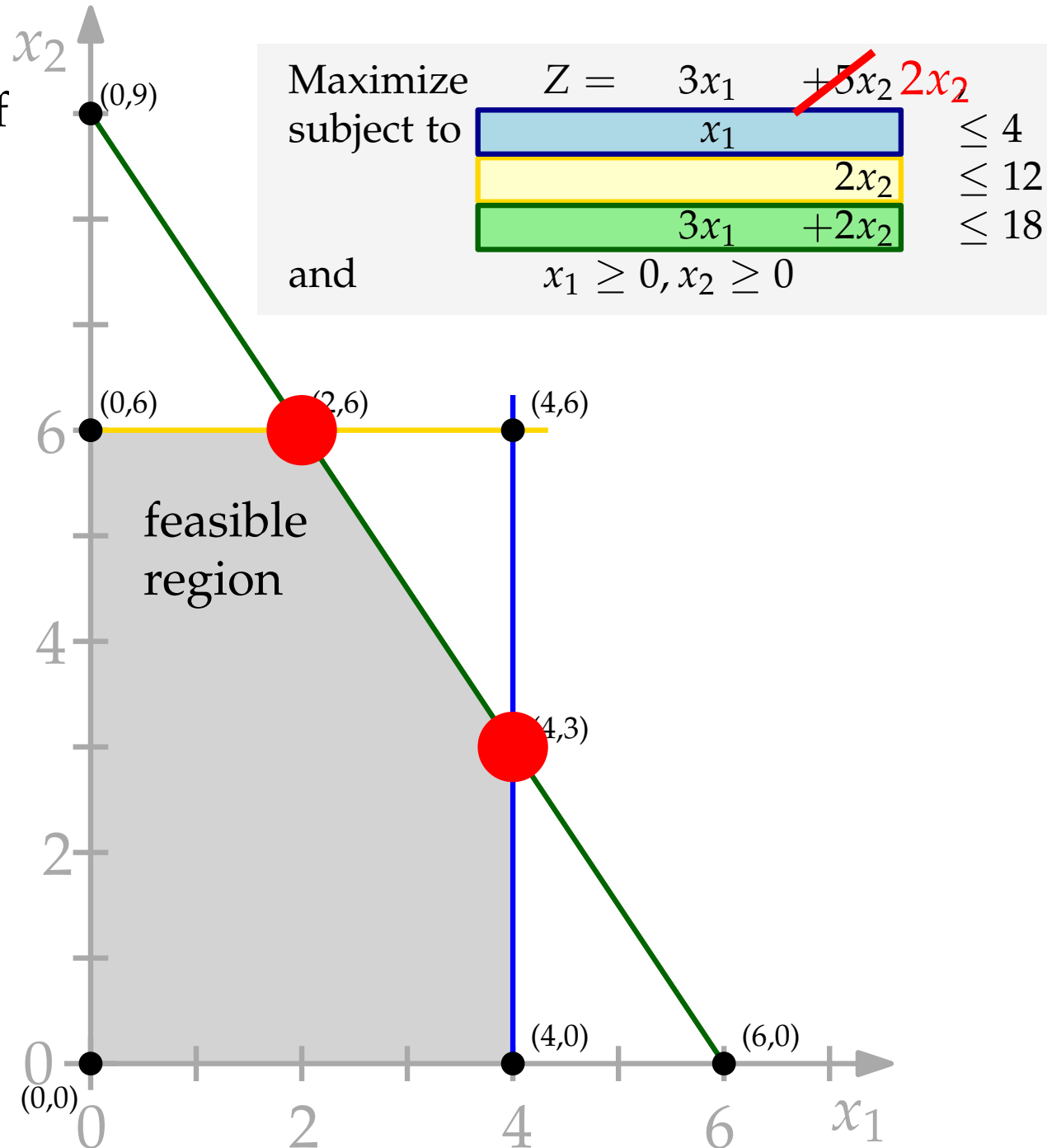
Multiple Optima

- change the coefficient of x_2 to 2 instead of 5.
- Now, two optimal CPF solutions.
(objective function parallel to green functional constraint)



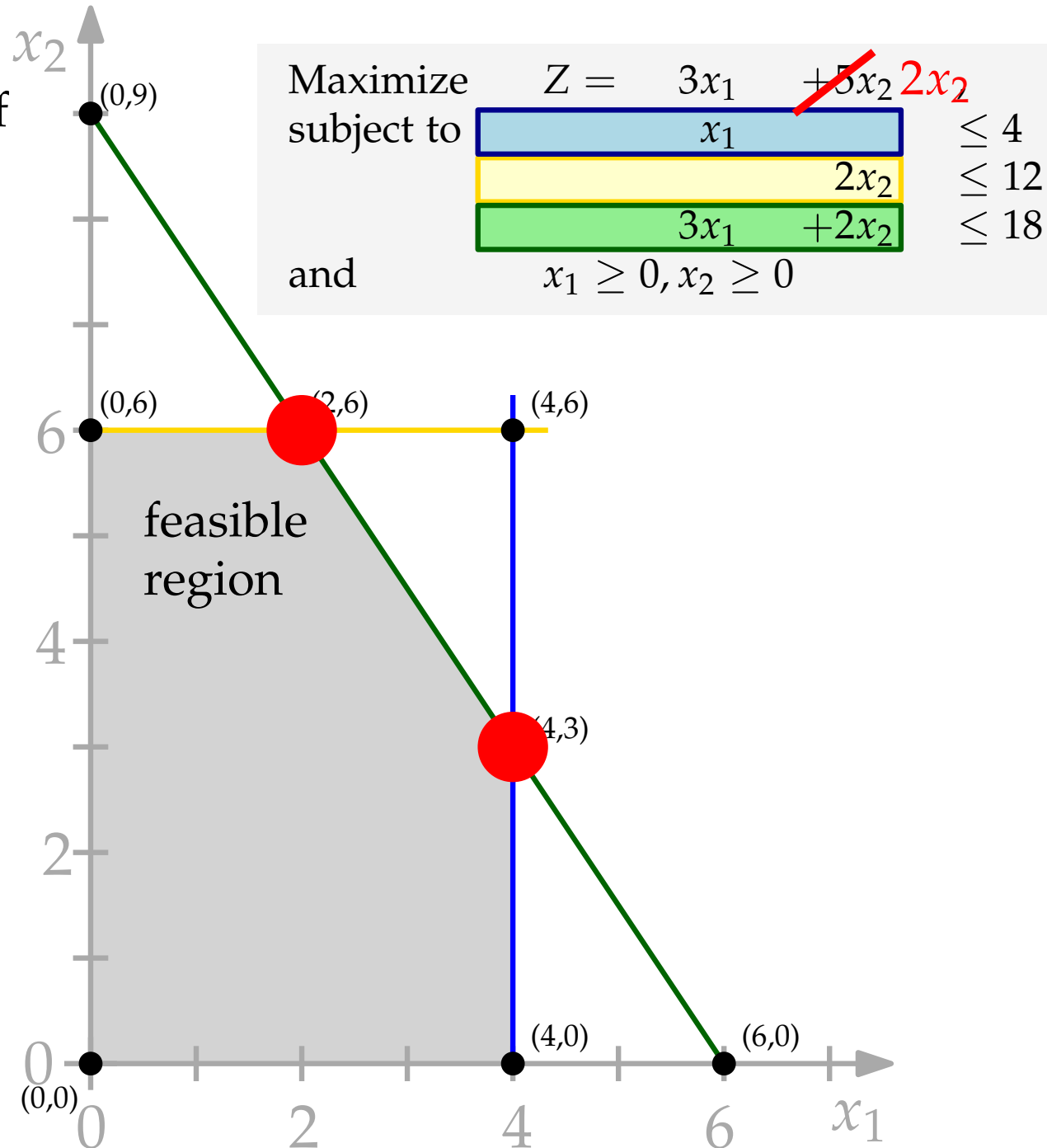
Multiple Optima

- change the coefficient of x_2 to 2 instead of 5.
- Now, two optimal CPF solutions.
(objective function parallel to green functional constraint)
- To explore these alternative optima, continue to pivot, choosing a nonbasic variable with a 0 in the top row as the entering basic variable



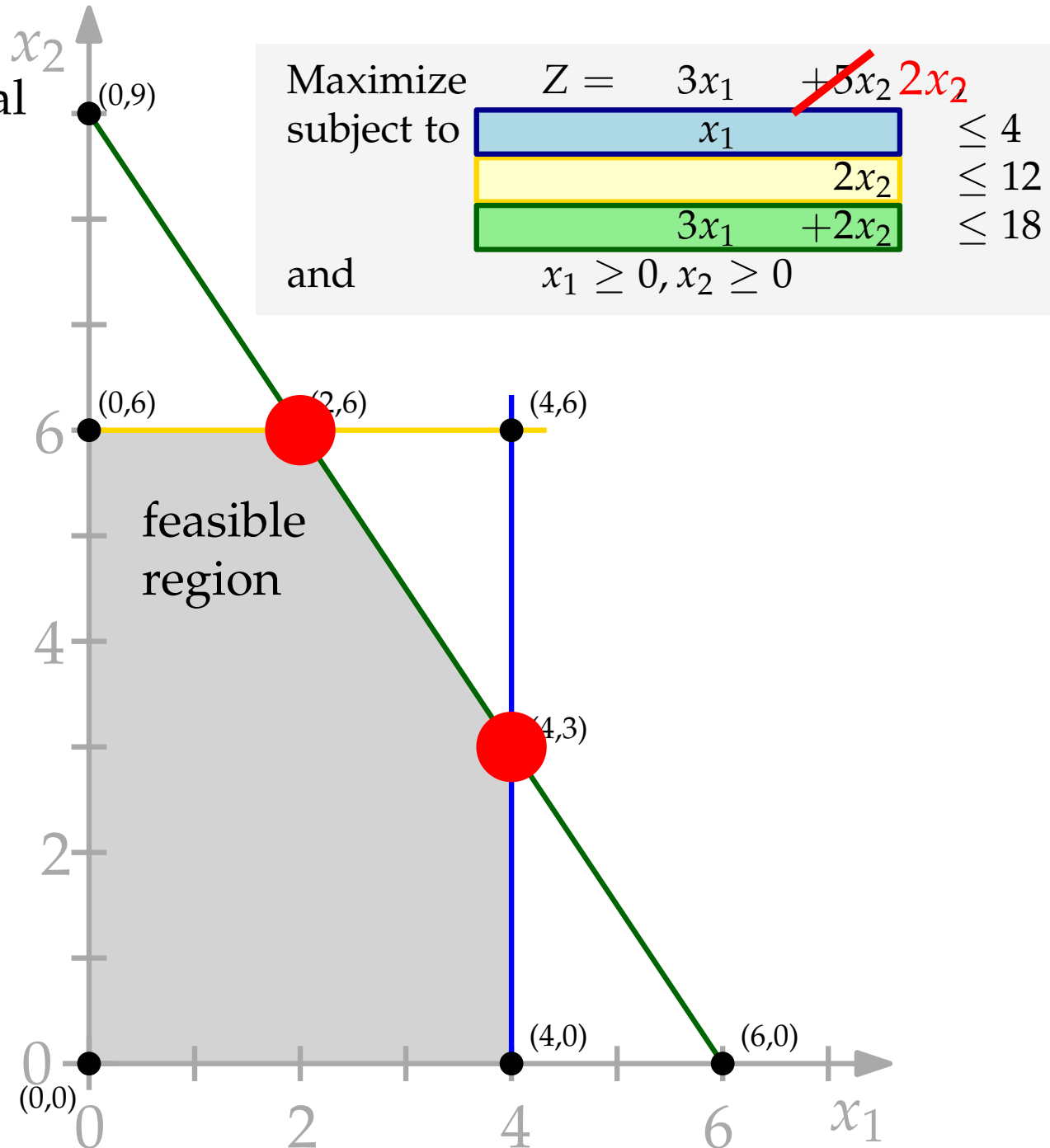
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- change the coefficient of x_2 to 2 instead of 5.
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Multiple Optima

What are all possible optimal solutions?



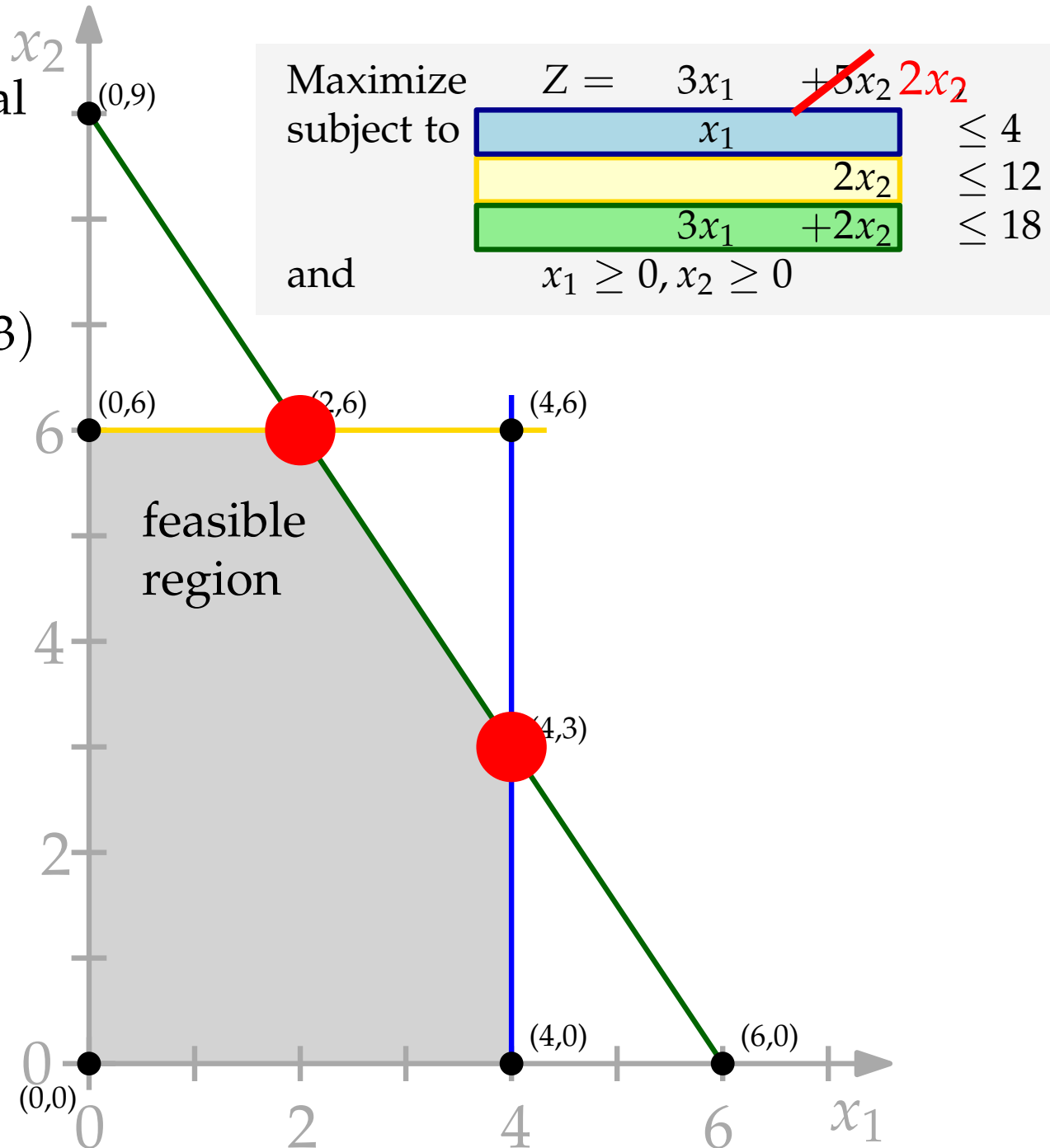
Multiple Optima

What are all possible optimal solutions?

$$(x_1, x_2) = w_1(2, 6) + w_2(4, 3)$$

$$w_1 + w_2 = 1, w_1, w_2 \geq 0$$

$\Rightarrow$ **Convex combinations** of solutions



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■ **TABLE 4.10** Complete set of simplex tableaux to obtain all optimal BF solutions for the Wyndor Glass Co. problem with $c_2 = 2$

Iteration	Basic Variable	Eq.	Coefficient of:						Right Side	Solution Optimal?
			Z	x_1	x_2	x_3	x_4	x_5		
0	Z	(0)	1	-3	-2	0	0	0	0	No
	x_3	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	2	0	1	0	12	
	x_5	(3)	0	3	2	0	0	1	18	
1	Z	(0)	1	0	-2	3	0	0	12	No
	x_1	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	2	0	1	0	12	
	x_5	(3)	0	0	2	-3	0	1	6	
2	Z	(0)	1	0	0	0	0	1	18	Yes
	x_1	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	0	3	1	-1	6	
	x_2	(3)	0	0	1	$-\frac{3}{2}$	0	$\frac{1}{2}$	3	
Extra	Z	(0)	1	0	0	0	0	1	18	Yes
	x_1	(1)	0	1	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	2	
	x_3	(2)	0	0	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2	
	x_2	(3)	0	0	1	0	$\frac{1}{2}$	0	6	

Multiple Optima

- Modern LP solvers can often provide these 

Multiple Optima

- Modern LP solvers can often provide these 😊
- How to efficiently list/enumerate all the optimal CPF/BFS of an LP is actually a topic of research in its own right!

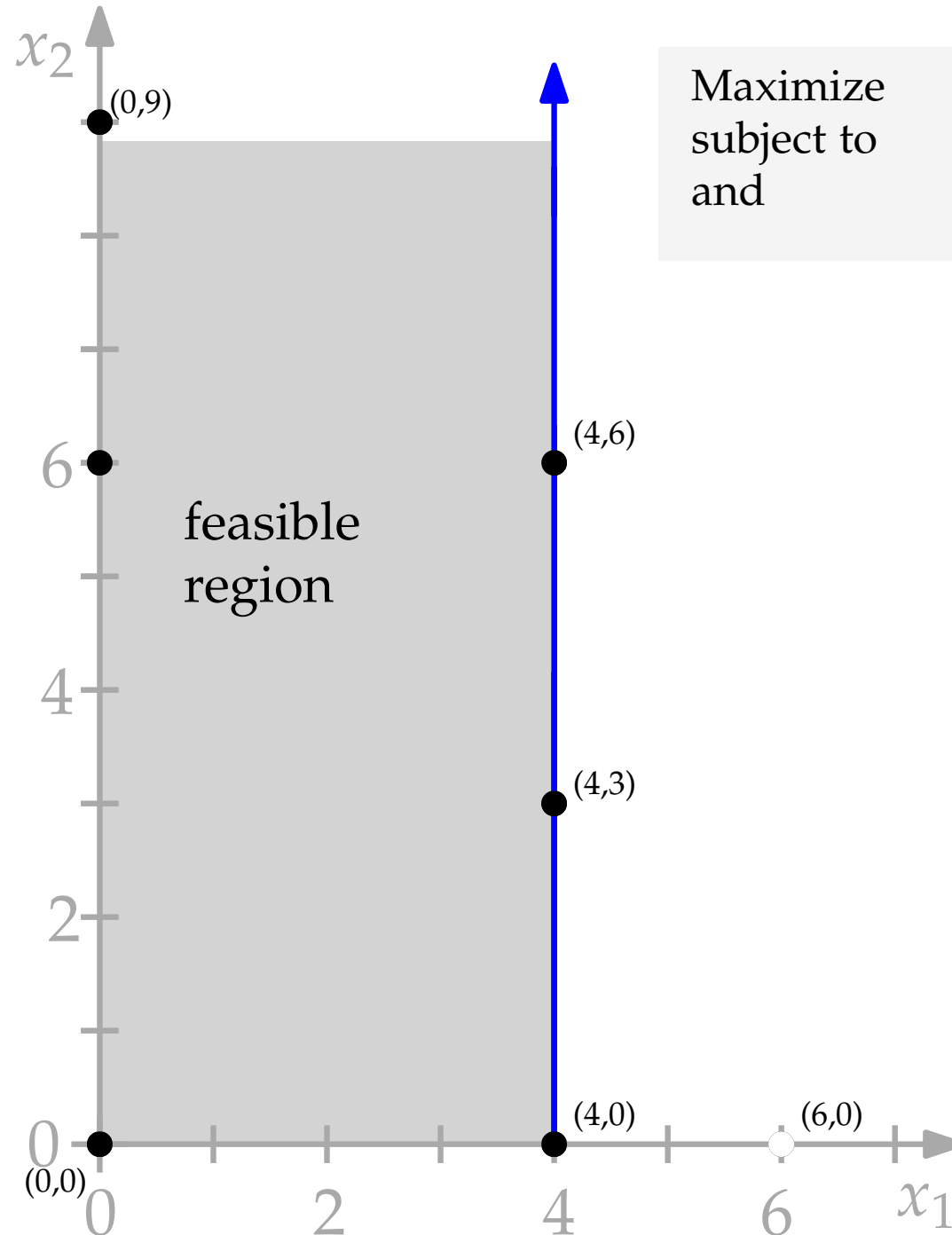
Multiple Optima

- Modern LP solvers can often provide these 😊
- How to efficiently list/enumerate all the optimal CPF/BFS of an LP is actually a topic of research in its own right!
- Finding all the optimal CPF/BFS of an LP is mathematically very similar to the problem of finding all the (not necessarily optimal) CPF/BFS of the feasible region.

Multiple Optima

- Modern LP solvers can often provide these ☺
- How to efficiently list/enumerate all the optimal CPF/BFS of an LP is actually a topic of research in its own right!
- Finding all the optimal CPF/BFS of an LP is mathematically very similar to the problem of finding all the (not necessarily optimal) CPF/BFS of the feasible region.
- Such exhaustive lists of CPF/BFS can be very useful for characterizing the **entire space** of (optimal) solutions

Unbounded solution



Maximize
subject to
and

$$Z = 3x_1 + 5x_2,$$

$$x_1 \leq 4$$
$$x_1 \geq 0, x_2 \geq 0$$

Undefined Minimum Ratio?

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■ **TABLE 4.9** Initial simplex tableau for the Wyndor Glass Co. problem without the last two functional constraints

Basic Variable	Eq.	Coefficient of:				Right Side	Ratio
		Z	x_1	x_2	x_3		
Z	(0)	1	-3	-5	0	0	
x_3	(1)	0	1	0	1	4	None

With $x_1 = 0$ and x_2 increasing,
 $x_3 = 4 - 1x_1 - 0x_2 = 4 > 0$.

Undefined Minimum Ratio?

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With $x_1 = 0$ and x_2 increasing,
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- But, don't we ignore rows with ≤ 0 in the pivot column?

Undefined Minimum Ratio?

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- Here, we can increase x_2 arbitrarily (while staying feasible), and as such increase Z arbitrarily!

Undefined Minimum Ratio?

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With $x_1 = 0$ and x_2 increasing,
 $x_3 = 4 - 1x_1 - 0x_2 = 4 > 0$.

- But, don't we ignore rows with ≤ 0 in the pivot column?
- The minimum ratio is ∞ (or undefined) ????
- Here, we can increase x_2 arbitrarily (while staying feasible), and as such increase Z arbitrarily!
- Thus, unbounded optimal solution
- Yes, we will find this, but first we need to be **adjacent** to such unbounded *face* of the feasible region.

Degeneracy

- (1) Tie among entering variables, i.e., multiple nonbasic variables with the same **potential**
- (2) Tie among leaving variables: basic variables with the same **minimum ratio**

Degeneracy

(1) Tie among entering variables, i.e., multiple nonbasic variables with the same **potential**

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What to do?

Degeneracy

- (1) Tie among entering variables, i.e., multiple nonbasic variables with the same **potential**
- (2) Tie among leaving variables: basic variables with the same **minimum ratio**

What to do?

don't worry it will be fine, arbitrarily break ties!
(actually, works well in practice...)

Degeneracy

- (1) Tie among entering variables, i.e., multiple nonbasic variables with the same **potential**
- (2) Tie among leaving variables: basic variables with the same **minimum ratio**

What to do?

don't worry it will be fine, arbitrarily break ties!
(actually, works well in practice...)

- But, seriously, while it is often claimed to be *rarely* problematic, it is important to be aware of it and have an idea that there are ways to **properly** handle it.

Degeneracy

- A BFS is **degenerate** if one or more of its basic variables is zero, i.e., has $\text{RHS} = 0$. Such basic variables are called **degenerate** basic variables.

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- Related to the (geometric) concept of adjacency, see lecture 2.

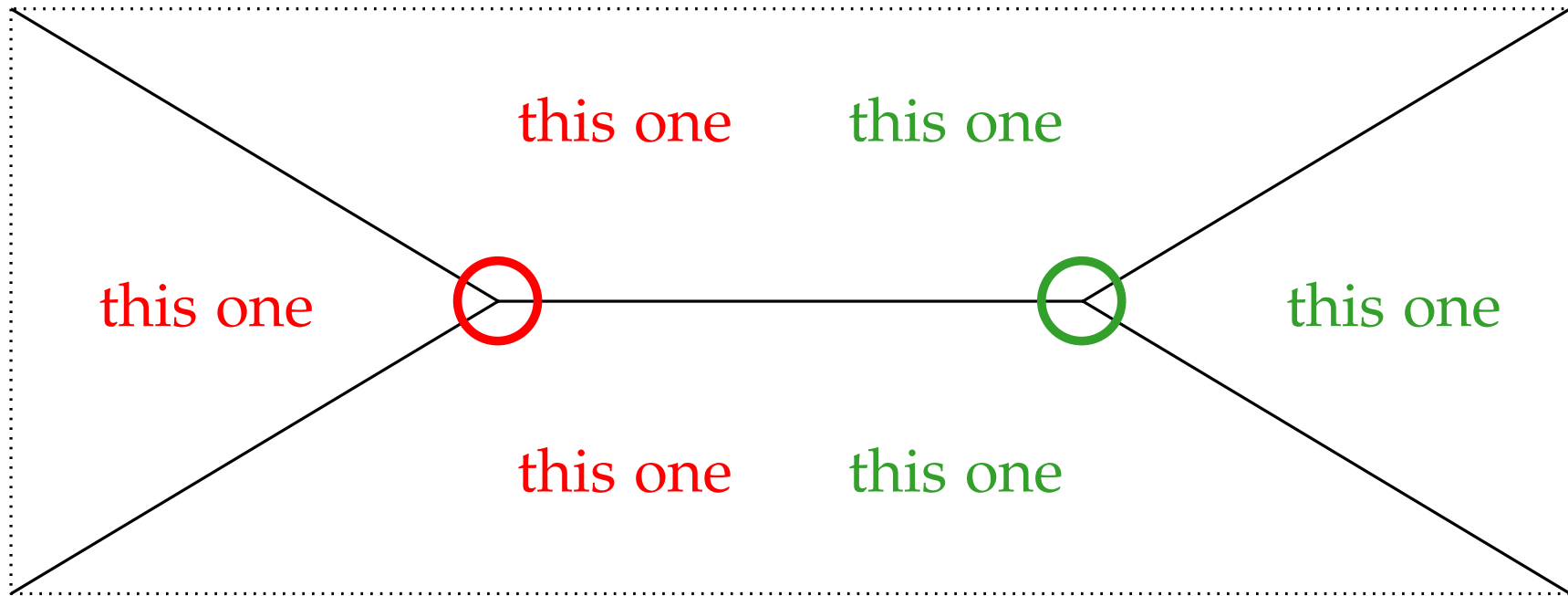
Degeneracy

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- Related to the (geometric) concept of adjacency, see lecture 2.
- The book [HL] (deliberately) over-simplifies its definition of **adjacency**, it says that a CPF solution can have at most n adjacent CPF solutions

Degeneracy

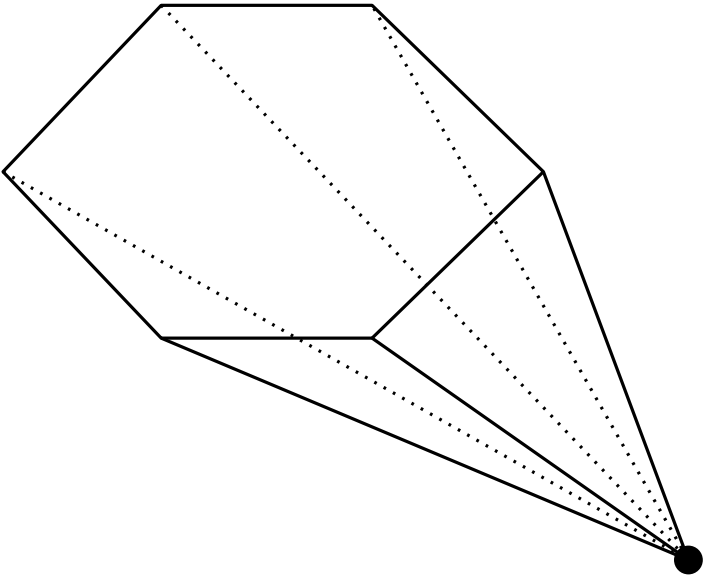
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- The book [HL] (deliberately) over-simplifies its definition of **adjacency**, it says that a CPF solution can have at most n adjacent CPF solutions
(based on the idea that in n -dimensional space a CPF solution requires n constraints to be tight, i.e. met with equality, at that point. Sure, but...)

Degeneracy

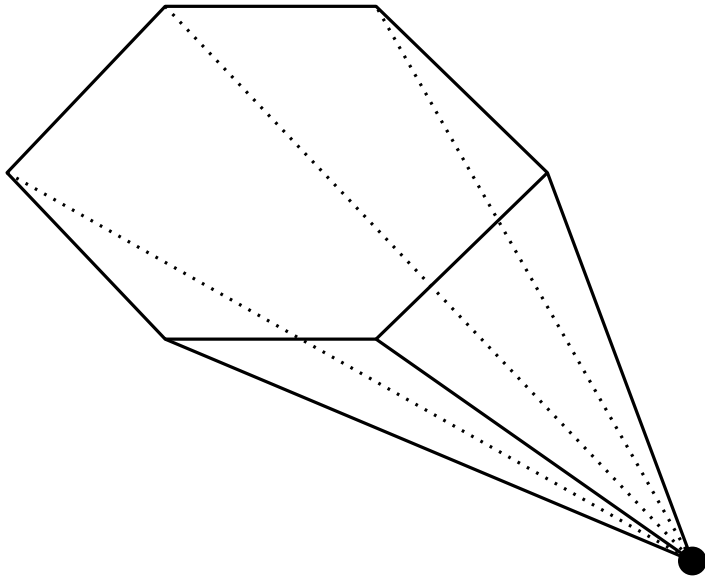


But, what if...

Degeneracy

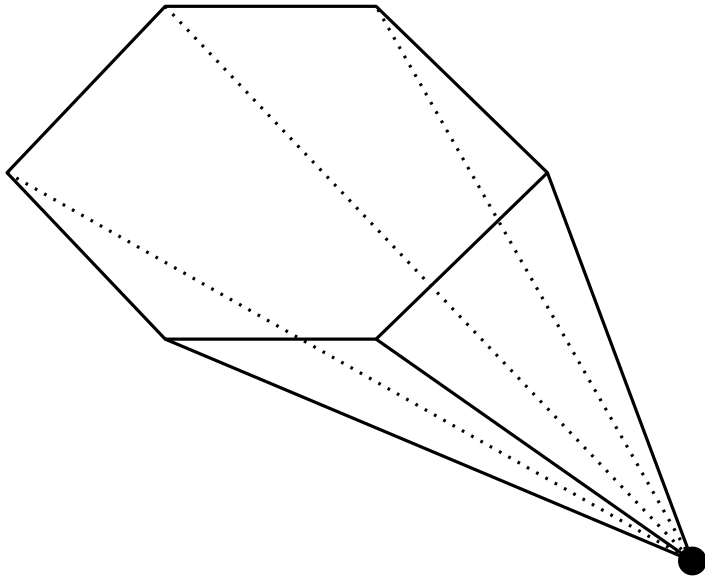


Degeneracy



Any **three** of the **six** long *faces* define the corner point!

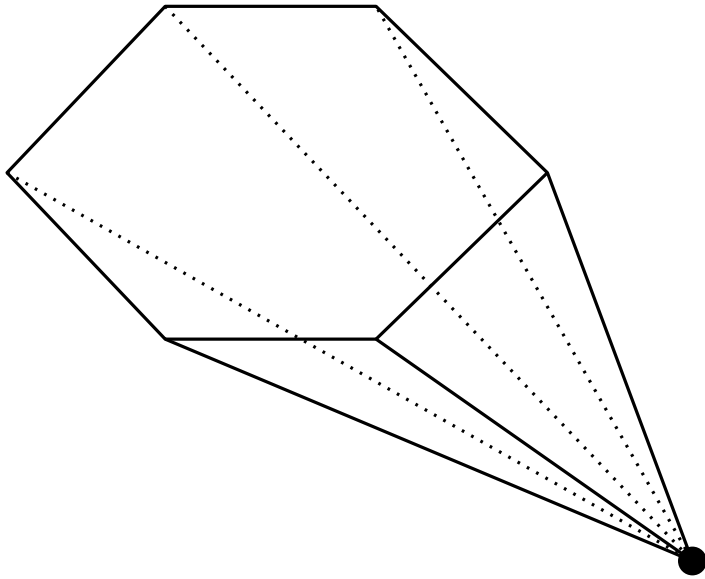
Degeneracy



Any **three** of the **six** long *faces* define the corner point!

but the **six** slack variables (of these faces) are zero at this point!

Degeneracy

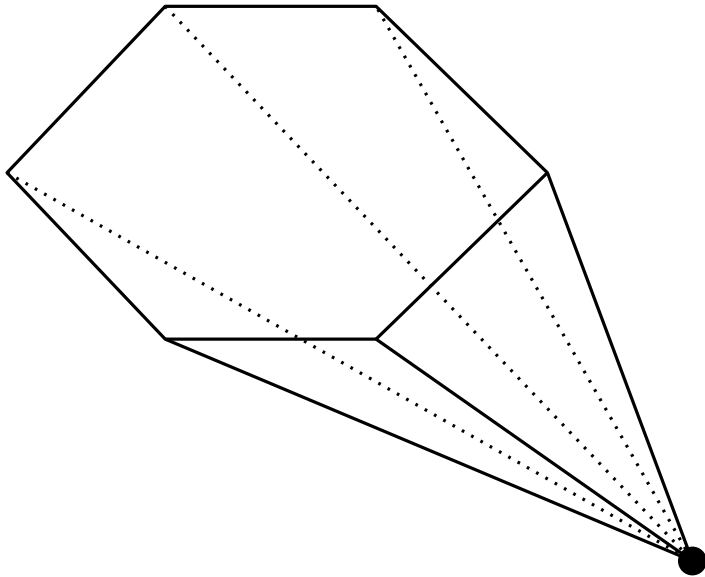


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$\rightsquigarrow$ degenerate CPF (> 3 tight constraints)

Degeneracy



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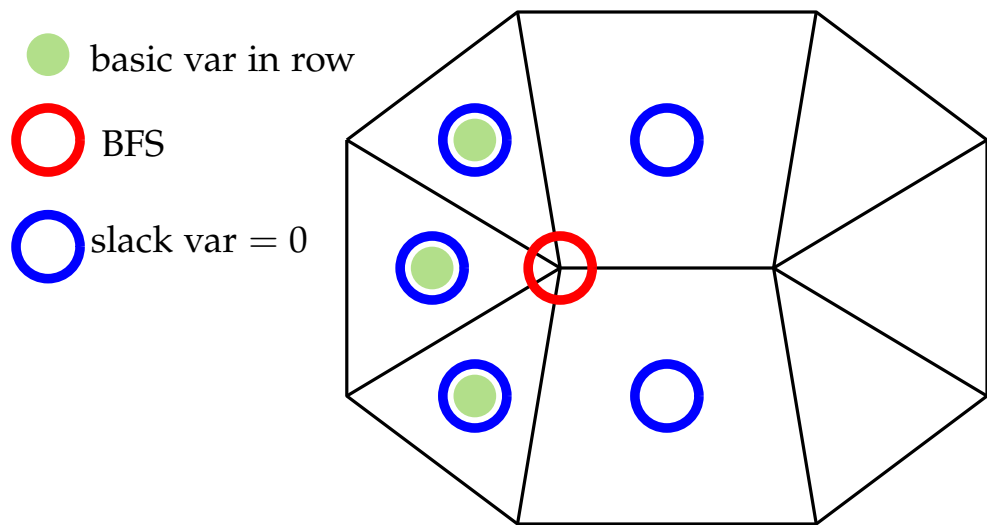
~> degenerate CPF (> 3 tight constraints)

This can make the simplex method go to the next basis **without** increasing the objective value (i.e. because it is stuck inside the same basic feasible solution)

~> **stalling**, or even worse **cycling**

Degeneracy: Stalling

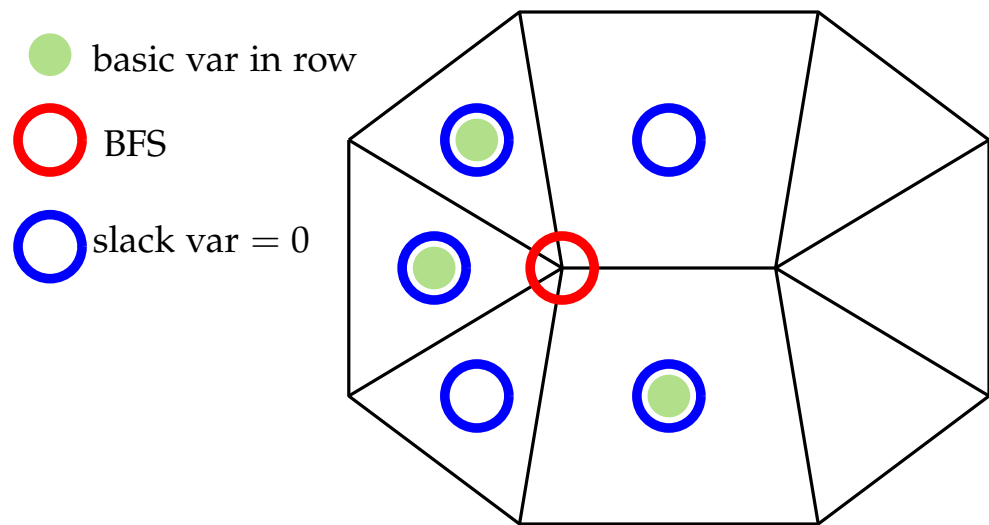
- **Stalling** (taking several iterations to improve the objective function) is sometimes inevitable. Depending on the LP you are trying to solve it can be more or less of a problem.



view from above
with the horizontal
line “sticking out”
direction of objective
function →

Degeneracy: Stalling

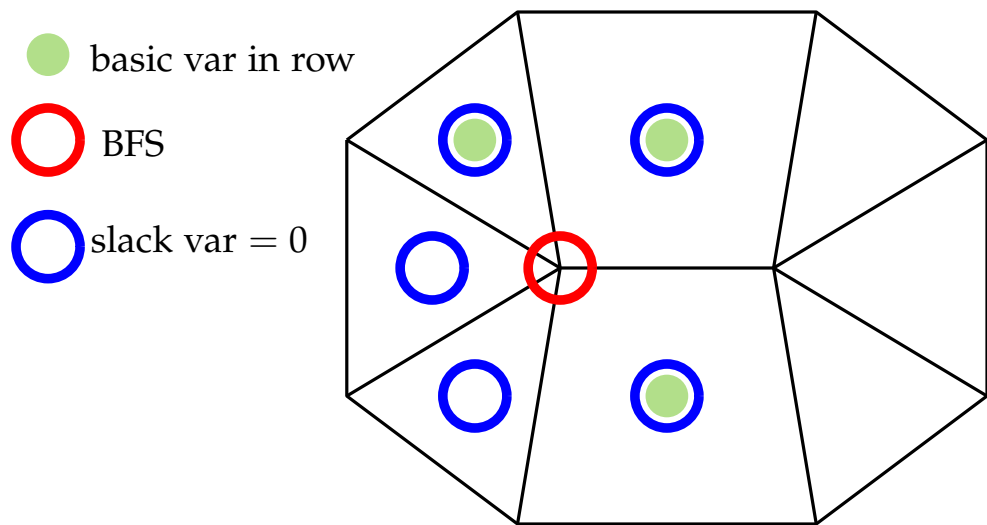
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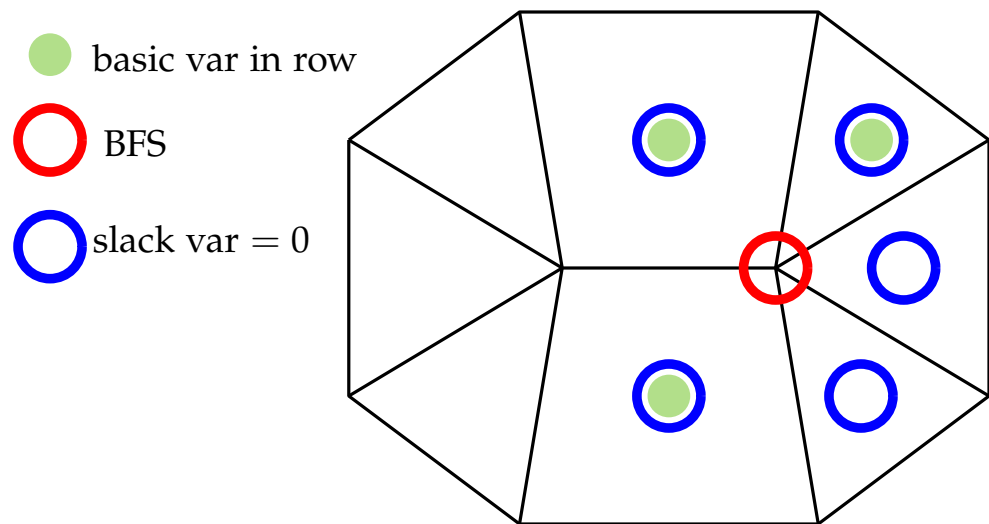
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Degeneracy: Stalling

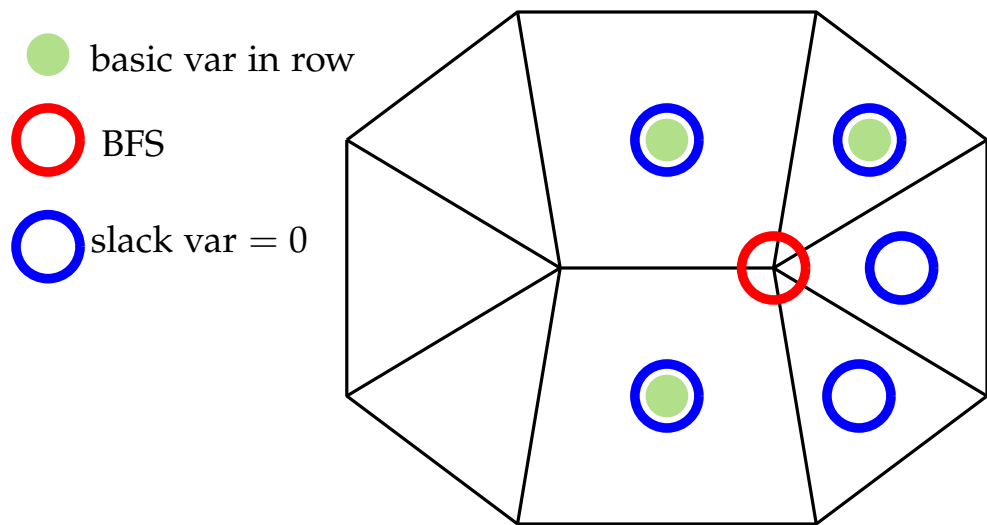
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Degeneracy: Stalling

- **Stalling** (taking several iterations to improve the objective function) is sometimes inevitable. Depending on the LP you are trying to solve it can be more or less of a problem.



view from above
with the horizontal
line “sticking out”
direction of objective
function →

- The objective function only improved after the third pivot!

Degeneracy: Algebraic View

- you can “see degeneracy coming” in the simplex tableau.
- Suppose we have a tie for the leaving variable

<i>basic</i>	<i>Z</i>	x_1	x_2	x_3	x_4	x_5	x_6	<i>RHS</i>
			-7			0	0	.
.						0	0	.
x_3			2			1	0	6
.						0	0	.
x_4			4			0	1	12
.						0	0	.

Degeneracy: Algebraic View

- you can “see degeneracy coming” in the simplex tableau.
- Suppose we have a tie for the leaving variable

<i>basic</i>	<i>Z</i>	x_1	x_2	x_3	x_4	x_5	x_6	<i>RHS</i>	Ratio
			-7			0	0	.	
.						0	0	.	
x_3			2			1	0	6	3
.						0	0	.	
x_4			4			0	1	12	3
.						0	0	.	

Degeneracy: Algebraic View

- you can “see degeneracy coming” in the simplex tableau.
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.						0	0	.	
x_3			2			1	0	6	3
.						0	0	.	
x_4			4			0	1	12	3
.						0	0	.	

tie, but pick x_4

Degeneracy: Algebraic View

- you can “see degeneracy coming” in the simplex tableau.
- Suppose we have a tie for the leaving variable

<i>basic</i>	<i>Z</i>	x_1	x_2	x_3	x_4	x_5	x_6	<i>RHS</i>
			0			0	0	.
.						0	0	.
x_3			0			1	$-1/2$	0
.						0	0	.
x_2			1			0	$1/4$	3
.						0	0	.

x_3 is now in the basis with value 0

Degeneracy: Algebraic View

- you can “see degeneracy coming” in the simplex tableau.
- Suppose we have a tie for the leaving variable

<i>basic</i>	<i>Z</i>	x_1	x_2	x_3	x_4	x_5	x_6	<i>RHS</i>
	-4	0				0	0	.
.						0	0	.
x_3	6	0				1	-1/2	0
.						0	0	.
x_2	2	1				0	1/4	3
.						0	0	.

- suppose x_1 enters next ...

Degeneracy: Algebraic View

- you can “see degeneracy coming” in the simplex tableau.
- Suppose we have a tie for the leaving variable

<i>basic</i>	<i>Z</i>	x_1	x_2	x_3	x_4	x_5	x_6	<i>RHS</i>
	-4	0				0	0	.
.						0	0	.
x_3	6	0				1	-1/2	0
.						0	0	.
x_2	2	1				0	1/4	3
.						0	0	.

- suppose x_1 enters next ...
- row of x_3 : ratio of $\frac{0}{6}=0 \rightsquigarrow x_3$ **wins** the minimum ratio test

Degeneracy: Algebraic View

- you can “see degeneracy coming” in the simplex tableau.
- Suppose we have a tie for the leaving variable

<i>basic</i>	<i>Z</i>	x_1	x_2	x_3	x_4	x_5	x_6	<i>RHS</i>
	-4	0				0	0	.
.						0	0	.
x_3	6	0				1	-1/2	0
.						0	0	.
x_2	2	1				0	1/4	3
.						0	0	.

- suppose x_1 enters next ...
- row of x_3 : ratio of $\frac{0}{6}=0 \rightsquigarrow x_3$ **wins** the minimum ratio test
- when x_3 leaves the basis, the “0” means that the objective function does not change!

Degeneracy: Comments

- **Stalling** can be handled, but it does happen!

Degeneracy: Comments

- **Stalling** can be handled, but it does happen!
- Usually degeneracy is not a problem . . .

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- . . . it depends, e.g., the structure of your LP might be vulnerable to degeneracy (many constraints tight at once)

Degeneracy: Comments

- **Stalling** can be handled, but it does happen!
- Usually degeneracy is not a problem . . .
- . . . it depends, e.g., the structure of your LP might be vulnerable to degeneracy (many constraints tight at once)
- If a Simplex LP solver is stalling regularly and badly, this can be the right moment to switch to an interior-point solver!

Degeneracy: Comments

- **Stalling** can be handled, but it does happen!
- Usually degeneracy is not a problem . . .
- . . . it depends, e.g., the structure of your LP might be vulnerable to degeneracy (many constraints tight at once)
- If a Simplex LP solver is stalling regularly and badly, this can be the right moment to switch to an interior-point solver!
- We can't prove that the Simplex algorithm is an efficient (i.e. polynomial-time) algorithm (even in practice it is very fast).
All pivot rules (so far) have pathologically bad examples

Degeneracy: Cycling??

- Possible in theory to loop around BFSs ...
- The problem is that, in a sequence of degenerate pivots, the objective function does not increase, so there is a risk that you return to a basis that you have already seen before

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- The good news, rules can avoid this!

Degeneracy: Cycling??

- Possible in theory to loop around BFSs ...
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- The good news, rules can avoid this!
 $\rightsquigarrow$ **Bland's rule**

Pick the entering variable of smallest index, and if there are several leaving variable options, also take the one of the smallest index.

Degeneracy: Cycling??

- Possible in theory to loop around BFSs ...
- The problem is that, in a sequence of degenerate pivots, the objective function does not increase, so there is a risk that you return to a basis that you have already seen before
- The good news, rules can avoid this!
 $\rightsquigarrow$ **Bland's rule**

Pick the entering variable of smallest index, and if there are several leaving variable options, also take the one of the smallest index.

(such cycle-avoiding rules tend to be slow in practice ...)

Recap and Properties

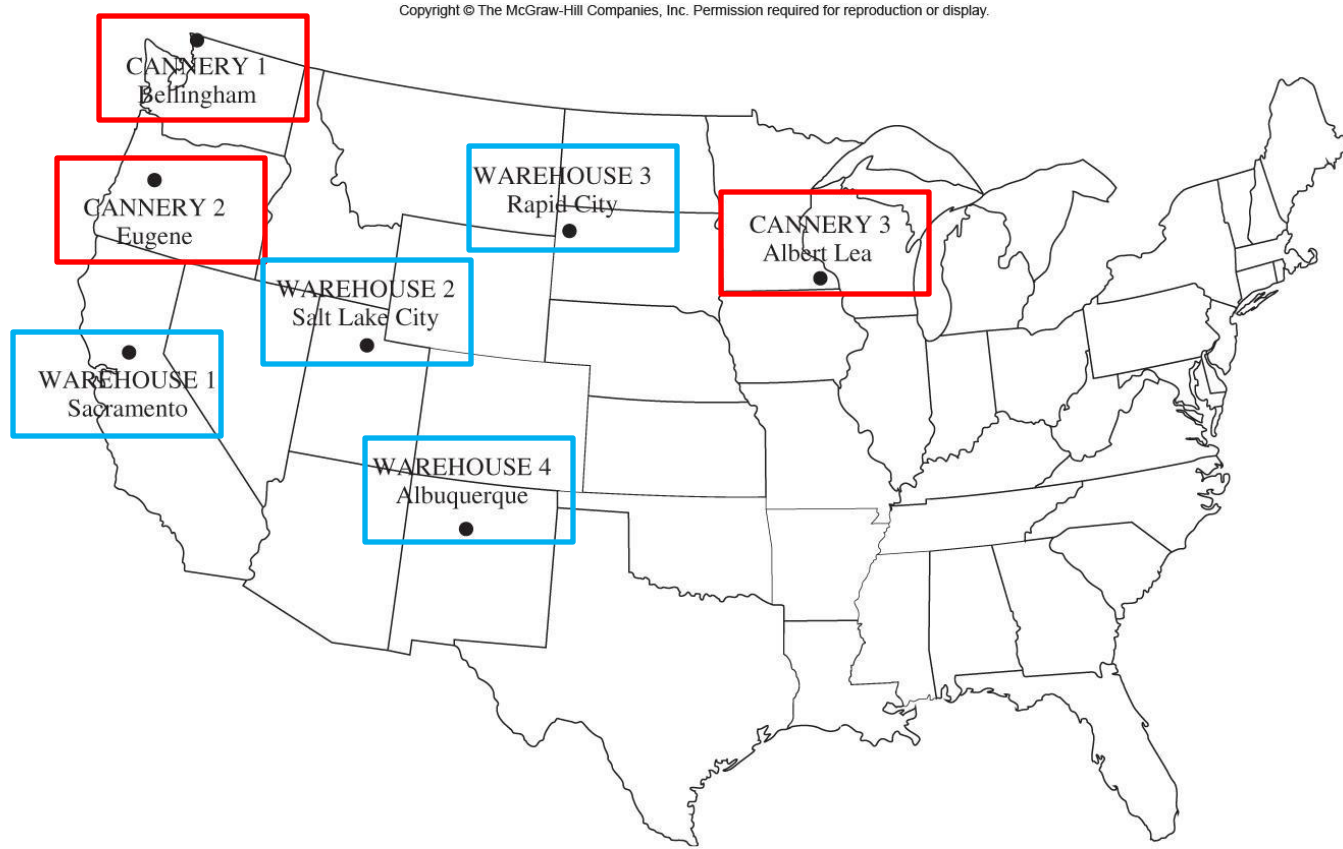
1. If there is **exactly one** optimal solution, then it must be a CPF solution.
2. If there are multiple optimal solutions (and a bounded feasible region), then at least two must be **adjacent CPF** solutions.
3. There are only a **finite number** of CPF solutions.
4. If a CPF solution has no adjacent CPF solutions that are better (as measured by Z), then there are **no better CPF** solutions anywhere.



Lecture 4: The Transportation Problem

[based on slides by: Steven Kelk]

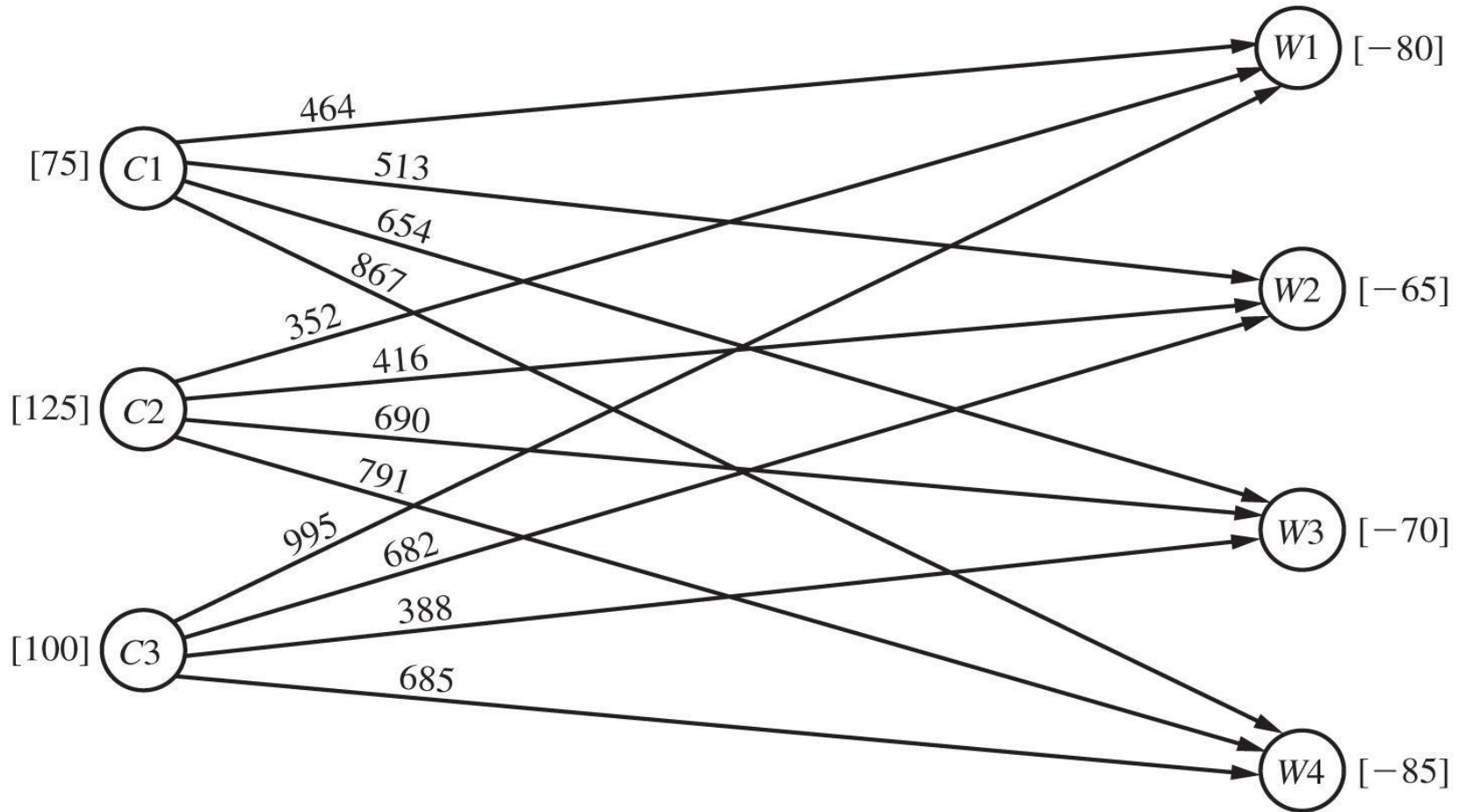
P&T's canned peas



- There are 300 truckloads to be shipped
- We have to plan how to assign the shipments to the canneries-warehouses combinations while minimizing the shipment costs

P&T's canned peas

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P&T's canned peas

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■ **TABLE 8.2** Shipping data for P & T Co.

		Shipping Cost (\$) per Truckload				
		Warehouse				
		1	2	3	4	
Cannery	1	464	513	654	867	75
	2	352	416	690	791	125
	3	995	682	388	685	100
Allocation		80	65	70	85	

Transportation costs

Needs of warehouses

Production
of canneries
=300

How would you write the LP problem?

P&T's canned peas

$$\begin{aligned} \text{Minimize } Z = & 464x_{11} + 513x_{12} + 654x_{13} + 867x_{14} + 352x_{21} + 416x_{22} \\ & + 690x_{23} + 791x_{24} + 995x_{31} + 682x_{32} + 388x_{33} + 685x_{34}, \end{aligned}$$

subject to the constraints

$$\begin{array}{rcccccccc} x_{11} + x_{12} + x_{13} + x_{14} & & & & & & & = 75 \\ & & & x_{21} + x_{22} + x_{23} + x_{24} & & & & = 125 \\ & & & & & x_{31} + x_{32} + x_{33} + x_{34} & & = 100 \\ x_{11} & & & + x_{21} & & + x_{31} & & = 80 \\ & x_{12} & & + x_{22} & & + x_{32} & & = 65 \\ & & x_{13} & & + x_{23} & & + x_{33} & = 70 \\ & & & x_{14} & & + x_{24} & & + x_{34} = 85 \end{array}$$

and

$$x_{ij} \geq 0 \quad (i = 1, 2, 3; j = 1, 2, 3, 4).$$

Do you see something interesting here?

P&T's canned peas

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■ **TABLE 8.3** Constraint coefficients for P & T Co.

Coefficient of:												
	x_{11}	x_{12}	x_{13}	x_{14}	x_{21}	x_{22}	x_{23}	x_{24}	x_{31}	x_{32}	x_{33}	x_{34}
A =	1 1 1 1				1 1 1 1				1 1 1 1			
	1 1 1 1				1 1 1 1				1 1 1 1			

1 1 1 1

1 1 1 1

1 1 1 1

1 1 1 1

1 1 1 1

1 1 1 1

}

Cannery constraints

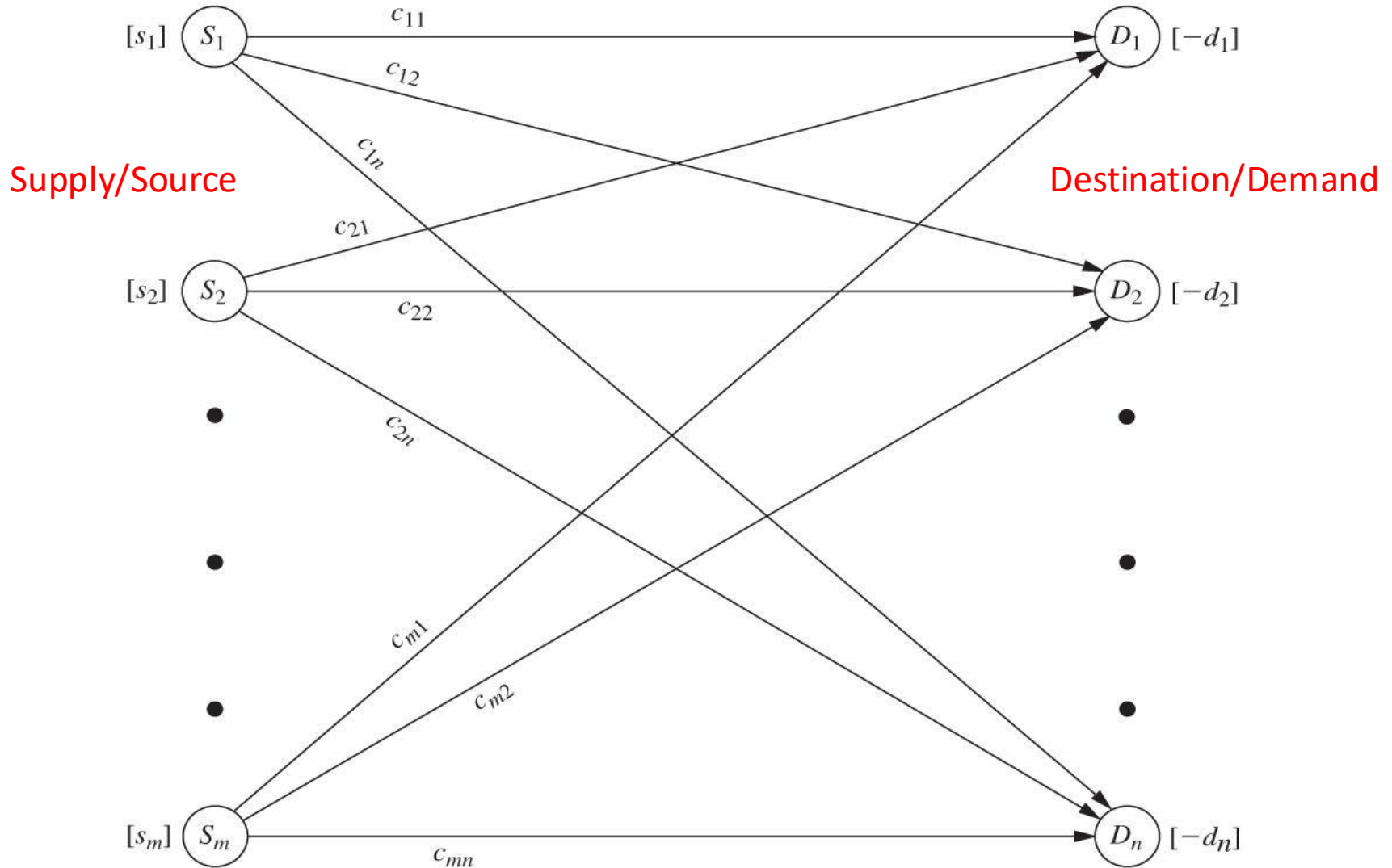
}

Warehouse constraints

The model can be generalized to every source-destination problem

General case

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General case

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■ **TABLE 8.5** Parameter table for the transportation problem

		Cost per Unit Distributed				
		Destination				
		1	2	...	n	
Source	1	c_{11}	c_{12}	...	c_{1n}	s_1
	2	c_{21}	c_{22}	...	c_{2n}	s_2
	$\vdots$					$\vdots$
	m	c_{m1}	c_{m2}	...	c_{mn}	s_m
Demand		d_1	d_2	...	d_n	

Sum of demands must equal sum of supplies!

General case

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■ **TABLE 8.6** Constraint coefficients for the transportation problem

Coefficient of:													
	x_{11}	x_{12}	...	x_{1n}	x_{21}	x_{22}	...	x_{2n}	...	x_{m1}	x_{m2}	...	x_{mn}
A =	1	1	...	1									
					1	1	...	1					
									...				
										1	1	...	1
	1				1					1			
		1				1					1		
			...				...					...	
				1				1					1
									...				

Supply constraints

Demand constraints

(m+n) constraints corresponding to m sources (i.e. supply points) and n destinations (i.e. demand points)

General case

Minimize $Z = \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij},$

subject to

Source constraints $\sum_{j=1}^n x_{ij} = s_i \quad \text{for } i = 1, 2, \dots, m,$

Demand constraints $\sum_{i=1}^m x_{ij} = d_j \quad \text{for } j = 1, 2, \dots, n,$

and

$$x_{ij} \geq 0, \quad \text{for all } i \text{ and } j.$$

Relaxation

The sum of supplies = sum of demands restriction can be softened

➡ Use *dummy destinations* and/or *dummy sources*.

This increases the modelling power of the Transportation Problem considerably.

Dummy Destination

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■ **TABLE 8.7** Production scheduling data for Northern Airplane Co.

Month	Scheduled Installations	Maximum Production	Unit Cost* of Production	Unit Cost* of Storage
1	10	25	1.08	0.015
2	15	35	1.11	0.015
3	25	30	1.10	0.015
4	20	10	1.13	

Cumulative number of engines produced in a month

There might be an advantage in producing and storing, i.e., to save money you are allowed to produce engines in month x that are only going to be fitted in month $y > x$, but each such engine will have to be stored for $(y-x)$ months.

How to model this as a Transportation Problem?
(And how to prevent fractional engine production...?)

Dummy Destination

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■ **TABLE 8.8** Incomplete parameter table for Northern Airplane Co.

	Cost per Unit Distributed				
	Destination				
	1	2	3	4	
Source	1	2	3	4	Supply
Demand					

Dummy Destination

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■ **TABLE 8.8** Incomplete parameter table for Northern Airplane Co.

		Cost per Unit Distributed				
		Destination (month)				
		1	2	3	4	
Source (month)	1	1.080	1.095	1.110	1.125	?
	2	?	1.110	1.125	1.140	?
	3	?	?	1.100	1.115	?
	4	?	?	?	1.130	?
Demand		10	15	25	20	

- Let $x_{i,j}$ be the number of engines produced in month i for installation in month j
- Fill the costs: $c_{1,3} = 1.110 = 1.080 + ((3-1)*0.015)$.
- But what to do about the positions marked with a “?” symbol?

Filling the blanks

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■ **TABLE 8.9** Complete parameter table for Northern Airplane Co.

	Cost per Unit Distributed					
	Destination					
	1	2	3	4	5(D)	
Source	1	1.080	1.095	1.110	1.125	
	2	M	1.110	1.125	1.140	
	3	M	M	1.100	1.115	
	4	M	M	M	1.130	
Demand	10	15	25	20		

- We cannot produce after the scheduled intervention
- **Big M** technique to make it too expensive to produce for the past
- How do we fix the supply?

Filling the blanks

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■ **TABLE 8.9** Complete parameter table for Northern Airplane Co.

	Cost per Unit Distributed					
	Destination					
	1	2	3	4	5(D)	
1	1.080	1.095	1.110	1.125	0	25
2	M	1.110	1.125	1.140	0	35
3	M	M	1.100	1.115	0	30
4	M	M	M	1.130	0	10
Demand	10	15	25	20		

- Set month supply to maximum month production
- Use a **dummy destination** to absorb “fictional” production

Filling the blanks

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■ **TABLE 8.9** Complete parameter table for Northern Airplane Co.

	Cost per Unit Distributed					Supply	
	Destination						
	1	2	3	4	5(D)		
Source	1	1.080	1.095	1.110	1.125	0	25
	2	M	1.110	1.125	1.140	0	35
	3	M	M	1.100	1.115	0	30
	4	M	M	M	1.130	0	10
Demand	10	15	25	20	30		

- How did we determine the demand of the dummy destination?
- Total unused capacity = sum of supply – sum of demand
 $(25+35+30+10)-(10+15+25+20)=30$

Big M technique

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■ **TABLE 8.9** Complete parameter table for Northern Airplane Co.

	Cost per Unit Distributed					Supply	
	Destination						
	1	2	3	4	5(D)		
Source	1	1.080	1.095	1.110	1.125	0	25
	2	M	1.110	1.125	1.140	0	35
	3	M	M	1.100	1.115	0	30
	4	M	M	M	1.130	0	10
Demand	10	15	25	20	30		

- How do we prevent fractional production of energy?
- **Magical property of the Transportation Problem:** if the sources and the demands are all integers, there will exist an integer optimal solution.
- **Every** Basic Feasible Solution will be integer!

Dummy sources

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■ **TABLE 8.10** Water resources data for Metro Water District

	Cost (Tens of Dollars) per Acre Foot				Supply
	Berdoo	Los Devils	San Go	Hollyglass	
Colombo River	16	13	22	17	50
Sacron River	14	13	19	15	60
Calorie River	19	20	23	—	50
Minimum needed	30	70	0	10	(in units of 1 million acre feet)
Requested	50	70	30	∞	

- We want to supply the 4 cities with water from the 3 rivers.
- Hollyglass cannot be supplied from Calorie River.
- Cities have minimum water requirements (“*minimum needed*”).
- Cities have maximum water requirements (“*requested*”).

Dummy sources

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■ **TABLE 8.10** Water resources data for Metro Water District

	Cost (Tens of Dollars) per Acre Foot				Supply
	Berdoo	Los Devils	San Jo	Hollyglass	
Colombo River	16	13	22	17	50
Sacron River	14	13	19	15	60
Calorie River	19	20	23	—	50
Minimum needed	30	70	0	10	(in units of 1 million acre feet)
Requested	50	70	30	∞	

- Note that all the water in the rivers has to be shipped
- Note that Hollyglass will take as much as it can get!

Dummy sources

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■ **TABLE 8.10** Water resources data for Metro Water District

	Cost (Tens of Dollars) per Acre Foot				Supply
	Berdoo	Los Devils	San Jo	Hollyglass	
Colombo River	16	13	22	17	50
Sacramento River	14	13	19	15	60
Calaveras River	19	20	23	—	50
Minimum needed	30	70	0	10	(in units of 1 million acre feet)
Requested	50	70	30	60	

- Note that all the water in the rivers has to be shipped
 - Note that Hollyglass will take as much as it can get!
- Requested demand for Hollyglass: exceeded total supply – minimum need of other cities
- $$(50+60+50)-(30+70+0)=60$$

Dummy sources

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■ **TABLE 8.11** Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	M	50
Demand		50	70	30	60	

Calorie River cannot supply
Hollyglass

Dummy sources

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■ **TABLE 8.11** Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	M	50
Demand		50	70	30	60	

- We have an ECCESS DEMAND CAPACITY!
- Supply = sum of the requested amount – total real supply
 $(50+70+30+60)-(50+60+50)=50$

Dummy sources

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■ **TABLE 8.11** Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	M	50
	Dummy	0	0	0	0	50
Demand		50	70	30	60	

- We have an ECCESS DEMAND CAPACITY!
- Supply = sum of the requested amount – total real supply
 $(50+70+30+60)-(50+60+50)=50$
- Hence, dummy source!
 No costs (it is not a real source)

Dummy sources

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■ **TABLE 8.11** Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	M	50
	Dummy	0	0	0	0	50
Demand		50	70	30	60	

- The table shown above is a *partial* solution to the problem. It correctly models the upper bound on consumption by a city (the “requirement”), allows cities to receive less than their requirement, and prevents Hollyglass from consuming Calorie River water.
- What about the lower bound...?

Dummy sources

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■ **TABLE 8.11** Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	M	50
	Dummy	0	0	0	0	50
Demand		50	70	30	60	

Min Needed: 30 70 0 10

- This is a fascinating example because for each city we can use a different technique for modelling its lower bound.

■ **TABLE 8.11** Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	M	50
	Dummy	0	0	0	0	50
Demand		50	70	30	60	

Min Needed:

30

70

0



10

- San Go: Easy! San Go has no minimum requirement.

■ **TABLE 8.11** Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	M	50
	Dummy	0	0	0	0	50
Demand		50	70	30	60	

Min Needed:

30

70



0



10

- San Go: Easy! San Go has no minimum requirement.
- Los Devils: Use Big M to forbid *any* dummy consumption

■ **TABLE 8.11** Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	M	50
	Dummy	0	0	0	0	50
Demand		50	70	30	60	

Min Needed:

30

70



0



10



- San Go: Easy! San Go has no minimum requirement.
- Los Devils: Use Big M to forbid *any* dummy consumption
- Hollyglass: We don't need to change anything.
(receives 50 from dummy source hence it will get at least 10 from real ones)

■ **TABLE 8.11** Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	M	50
	Dummy	0	0	0	0	50
Demand		50	70	30	60	

Min Needed:

30

70



0



10



- San Go: Easy! San Go has no minimum requirement.
- Los Devils: Use Big M to forbid *any* dummy consumption
- Hollyglass: We don't need to change anything.
- Berdoo: we need a slightly more sophisticated technique...

■ **TABLE 8.12** Parameter table for Metro Water District

			Cost (Tens of Millions of Dollars) per Unit Distributed					Supply
			Destination					
			Berdoo (min.) 1	Berdoo (extra) 2	Los Devils 3	San Go 4	Hollyglass 5	
Source	Colombo River	1	16	16	13	22	17	50
	Sacron River	2	14	14	13	19	15	60
	Calorie River	3	19	19	20	23	M	50
	Dummy	4(D)	M	0	M	0	0	50
Demand			30	20	70	30	60	

- We split Berdoo into **two** destinations.
- The first part demands the minimum requirement (30 units) and refuses to accept “dummy” water, like Los Devils.
- The second part accepts any surplus above the minimum, up to a maximum of 20 (because $30+20=50$), like San Go.

Why the TSM?

- The Transportation Problem can easily be modelled by a normal LP, and we know how to solve LPs already (Simplex Method), so why do we bother with a special formulation? And why do we bother trying to squeeze complex models into a format suitable for the Transportation Problem?
- **Speed.** We can solve it **much faster** than using a standard LP solver. Especially important for very big problems!
- **Insight.** We know a huge amount about Transportation Problems, (magical **integer solutions property**,) so no need for Integer Linear Programming. (It also has various nice theoretical properties which are not known to hold for general Linear Programs).

Transportation Simplex Method

Conceptually, the Transportation Simplex Method is very similar to the general Simplex Method:

1. Construct an **initial** Basic Feasible Solution (BFS).
2. Check whether the current BFS is **optimal**.
YES: we are done!
3. NO: **Identify** a *nonbasic* variable with the most potential to improve the objective function. This is the *entering* variable.
Identify a variable in the current basis that will *leave* when the entering variable enters
4. This gives us a new BFS; go back to 2.

Transportation Simplex Method

Fortunately, the Transportation Simplex Method (TSM) has some shortcuts:

1. Construct an **initial** Basic Feasible Solution (BFS).
 - For the general Simplex Method this can be hard work, because we have to introduce artificial variables and pivot multiple times to drive them to zero (i.e. to enter the feasible region).
 - With the TSM we can find an initial BFS very easily (there are rules)

Transportation Simplex Method

Fortunately, the Transportation Simplex Method (TSM) has some shortcuts:

1. Construct an **initial** Basic Feasible Solution (BFS).
2. Check whether the current BFS is **optimal**.

YES: we are done!

- In the general Simplex Method, we can determine this by looking at the top row of the Simplex Tableaux (no negative numbers in the top row).
- Exactly the same idea works with TSM, the difference is simplicity: we can quickly and dynamically generate the top row by deriving a very simple set of linear equations from the current BFS and solving them.

Transportation Simplex Method

- Fortunately, the Transportation Simplex Method (TSM) has some shortcuts:

1. Construct an **initial** Basic Feasible Solution (BFS).
2. Check whether the current BFS is **optimal**.
YES: we are done!
3. NO: **Identify** entering basic variable and leaving basic variable

Just like the normal Simplex method, we choose the nonbasic variable with the most negative “top-row entry” as the entering basic variable.

Transportation Simplex Method

Fortunately, the Transportation Simplex Method (TSM) has some shortcuts:

1. Construct an **initial** Basic Feasible Solution (BFS).
2. Check whether the current BFS is **optimal**.
YES: we are done!
3. NO: **Identify** entering basic variable and leaving basic variable
 - With the general Simplex Method, we identify the leaving basic variable with the Minimum Ratio Test, and then pivot.
 - With TSM we instead look for “closed loops” in a specially modified Transportation tableaux to find the leaving basic variable. The subsequent pivot can then be performed more quickly than in a general Simplex tableaux.

Table for the TSM

■ **TABLE 8.13** Original simplex tableau before simplex method is applied to transportation problem

Basic Variable	Eq.	Coefficient of:							Right side
		Z	...	x_{ij}	...	z_i	...	z_{m+j}	
Z	(0)	-1		c_{ij}		M		M	0
	(1)								
	⋮								
z_i	(i)	0		1		1			s_i Supply
	⋮								
z_{m+j}	(m + j)	0		1				1	d_j Demand
	⋮								
	(m + n)								

Table for the TSM

■ **TABLE 8.14** Row 0 of simplex tableau when simplex method is applied to transportation problem

Basic Variable	Eq.	Coefficient of:							Right Side
		Z	...	x_{ij}	...	z_i	...	z_{m+j}	
Z	(0)	-1		$c_{ij} - u_i - v_j$		$M - u_i$		$M - v_j$	$-\sum_{i=1}^m s_i u_i - \sum_{j=1}^n d_j v_j$

u_i = multiple of original row(i) that has been subtracted from the original row(0) during the iterations up to this one

v_j = multiple of original row(m+j) that has been subtracted from the original row(0) during the iterations up to this one

■ **TABLE 8.15** Format of a transportation simplex tableau

		Destination				Supply	u_i
		1	2	...	n		
Source	1	c_{11}	c_{12}	...	c_{1n}	s_1	
	2	c_{21}	c_{22}	...	c_{2n}	s_2	
	$\vdots$	...	...	...	...	$\vdots$	
	m	c_{m1}	c_{m2}	...	c_{mn}	s_m	
Demand		d_1	d_2	...	d_n	$Z =$	
v_j							

■ **TABLE 8.15** Format of a transportation simplex tableau

		Destination				Supply	u_i
		1	2	...	n		
Source	1	c_{11}	c_{12}	...	c_{1n}	s_1	
	2	c_{21}	c_{22}	...	c_{2n}	s_2	
	⋮	...	...	...	...	⋮	
	m	c_{m1}	c_{m2}	...	c_{mn}	s_m	
Demand		d_1	d_2	...	d_n	$Z =$	
v_j							

Additional information to be added to each cell:

If x_{ij} is a basic variable

c_{ij}	
	x_{ij}

value of basic variable

If x_{ij} is a nonbasic variable

c_{ij}	
	$c_{ij} - u_i - v_j$

“top-row” entry of corresponding nonbasic variable.

■ **TABLE 8.15** Format of a transportation simplex tableau

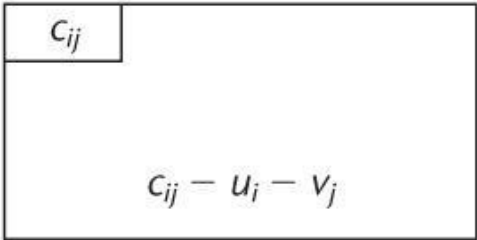
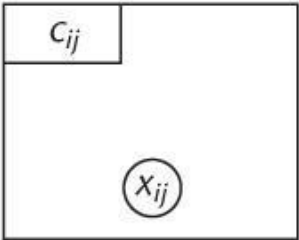
		Destination				Supply	u_i
		1	2	...	n		
Source	1	c_{11}	c_{12}	...	c_{1n}	s_1	
	2	c_{21}	c_{22}	...	c_{2n}	s_2	
	⋮	...	...	...	...	⋮	
	m	c_{m1}	c_{m2}	...	c_{mn}	s_m	
Demand		d_1	d_2	...	d_n	$Z =$	
v_j							

Additional information to be added to each cell:

*If x_{ij} is a
basic variable*

*If x_{ij} is a
nonbasic variable*

*There will always
be **(m+n-1)** basic
variables.*



Transportation Simplex Method

1. Construct the transportation table

Metro Water District

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■ **TABLE 8.12** Parameter table for Metro Water District

			Cost (Tens of Millions of Dollars) per Unit Distributed					Supply
			Destination					
			Berdoo (min.) 1	Berdoo (extra) 2	Los Devils 3	San Go 4	Hollyglass 5	
Source	Colombo River	1	16	16	13	22	17	50
	Sacron River	2	14	14	13	19	15	60
	Calorie River	3	19	19	20	23	M	50
	Dummy	4(D)	M	0	M	0	0	50
Demand			30	20	70	30	60	

Metro Water District

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■ **TABLE 8.16** Initial BF solution from the Northwest Corner Rule

	Destination					Supply	u_i
	1	2	3	4	5		
Source	1	16	16	13	22	17	50
	2	14	14	13	19	15	60
	3	19	19	20	23	M	50
	4(D)	M	0	M	0	0	50
Demand	30	20	70	30	60		
v_j							

Transportation Simplex Method

1. Construct the table
2. Construct an initial Basic Feasible Solution (BFS).
 - North west corner rule
 - Make basic variable large enough to meet supply or demand.

Transportation Simplex Method

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■ **TABLE 8.16** Initial BF solution from the Northwest Corner Rule

	Destination					Supply	u_i
	1	2	3	4	5		
Source	1	16	16	13	22	17	50
	2	14	14	13	19	15	60
	3	19	19	20	23	M	50
	4(D)	M	0	M	0	0	50
Demand	30	20	70	30	60		
v_j							

Transportation Simplex Method

1. Construct the table
2. Construct an initial Basic Feasible Solution (BFS).
 - North west corner rule
 - Make basic variable large enough to meet supply or demand.
 - Select remaining variables to be basic with the remaining feasible allocations

Transportation Simplex Method

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■ **TABLE 8.16** Initial BF solution from the Northwest Corner Rule

	Destination					Supply	u_i
	1	2	3	4	5		
Source 1 2 3 4(D)	16	16	13	22	17	50	
	14	14	13	19	15	60	
	19	19	20	23	M	50	
	M	0	M	0	0	50	
Demand	30	20	70	30	60		
v_j							

Transportation Simplex Method

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■ **TABLE 8.16** Initial BF solution from the Northwest Corner Rule

		Destination					Supply	u_i
		1	2	3	4	5		
Source	1	16 (30)	16 (20)	13	22	17	50	
	2	14	14 (0)	13 (60)	19	15	60	
	3	19	19	20 (10)	23 (30)	M (10)	50	
	4(D)	M	0	M	0	0 (50)	50	
Demand		30	20	70	30	60	$Z = 2,470 + 10M$	
v_j								

$$16 \cdot 30 + 16 \cdot 20 + 14 \cdot 0 + 13 \cdot 60 + 20 \cdot 10 + 23 \cdot 30 + M \cdot 10$$

Transportation Simplex Method

1. Construct the table
2. Construct an initial Basic Feasible Solution (BFS).
 - North west corner rule
 - Make basic variable large enough to meet supply or demand.
 - Select remaining variables to be basic with the remaining feasible allocations
3. Determine whether we have reached optimality

Transportation Simplex Method

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■ **TABLE 8.16** Initial BF solution from the Northwest Corner Rule

		Destination					Supply	u_i
		1	2	3	4	5		
Source	1	16 (30)	16 (20)	13 ?	22 ?	17 ?	50	
	2	14 ?	14 (0)	13 (60)	19 ?	15 ?	60	
	3	19 ?	19 ?	20 (10)	23 (30)	M (10)	50	
	4(D)	M ?	0 ?	M ?	0 ?	0 (50)	50	
Demand		30	20	70	30	60	$Z = 2,470 + 10M$	
v_j								

Fill in the red question marks for all the nonbasic variables.

If none of the values we fill in are negative, we have reached optimality.

Transportation Simplex Method

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■ **TABLE 8.16** Initial BF solution from the Northwest Corner Rule

		Destination					Supply	u_i
		1	2	3	4	5		
Source	1	16 30	16 20	13 ?	22 ?	17 ?	50	u_1
	2	14 ?	14 0	13 60	19 ?	15 ?	60	
	3	19 ?	19 ?	20 10	23 30	M 10	50	
	4(D)	M ?	0 ?	M ?	0 ?	0 50	50	
Demand		30	20	70	30	60	$Z = 2,470 + 10M$	
v_j		v_4						

For each cell they are given by the equation $c_{ij} - u_i - v_j$ For example, we would replace the “?” in cell (1,4) with $22 - u_1 - v_4$

Transportation Simplex Method

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■ **TABLE 8.16** Initial BF solution from the Northwest Corner Rule

		Destination					Supply	u_i
		1	2	3	4	5		
Source	1	16 30	16 20	13 ?	22 ?	17 ?	50	
	2	14 ?	14 0	13 60	19 ?	15 ?	60	
	3	19 ?	19 ?	20 10	23 30	M 10	50	
	4(D)	M ?	0 ?	M ?	0 ?	0 50	50	
Demand		30	20	70	30	60	$Z = 2,470 + 10M$	
v_j								

So we need to know the values of u_i and v_j !

Fortunately these can be derived from the current BFS (i.e., the circled guys).

Filling the blanks

- The idea is as follows. We need to compute the u_i and v_j values to be able to fill in the missing ? values.
- For the $(m+n-1)$ basic variables, $c_{ij} - u_i - v_j = 0$ holds.
- So, by looking at the basic variables, we obtain $(m+n-1)$ equations in $(m+n)$ unknowns.
- We have one degree of freedom; we can kill this degree of freedom by arbitrarily setting one of the u_i and v_j values to zero. Then we can get a solution for the remaining u_i and v_j values by solving the remaining two-variable equations.

Transportation Simplex Method

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■ **TABLE 8.16** Initial BF solution from the Northwest Corner Rule

		Destination					Supply	u_i
		1	2	3	4	5		
Source	1	16 (30)	16 (20)	13	22	17	50	
	2	14	14 (0)	13 (60)	19	15		
	3	19	19	20 (10)	23 (30)	M (10)		
	4(D)	M	0	M	0	0 (50)		
Demand		30	20	70	30	60	$Z = 2,470 + 10M$	
v_j								

Transportation Simplex Method

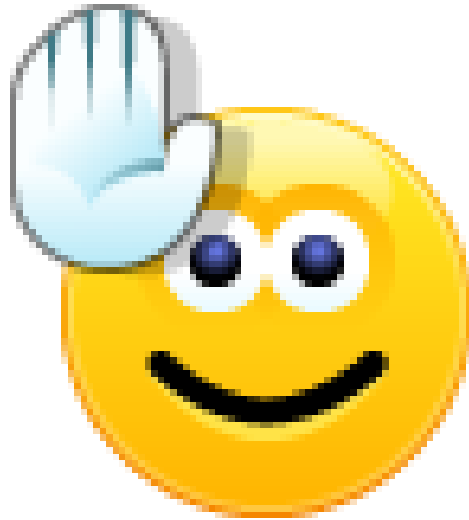
1. Construct the table
2. Construct an initial Basic Feasible Solution (BFS).
 - North west corner rule
 - Make basic variable large enough to meet supply or demand.
 - Select remaining variables to be basic with the remaining feasible allocations
3. Determine whether we have reached optimality
4. Identify entering basic variable and leaving basic variable, then pivot.

Find EBV and LBV

- We choose the **most negative** value (for $c_{ij} - u_i - v_j$).
This is just like choosing the column with the most negative top-row value in the normal Simplex Tableaux.
- This will become the **entering** variable.
At the moment it is nonbasic, (i.e. equal to zero).
We will make it basic (i.e. potentially larger than zero) and this will force a currently basic variable to **leave** and become nonbasic.

	D1	D2	D3	D4	D5	Supply	$u(i)$
S1	16 30	16 20	13 -2	22 4	17 -978	50	-5
S2	14 0	14 0	13 60	19 3	15 -978	60	-7
S3	19 -2	19 -2	20 10	23 30	1000 10	50	0 ←
S4	1000 1979	0 979	1000 1980	0 977	0 50	50	-1000
Demand	30	20	70	30	60		
$v(j)$	21	21	20	23	1000		
$Z =$							12470

- Set $u_3=0$, then the rest of the u and v variables are uniquely determined.
- After this we can fill in the “top row” entries of the nonbasic variables (in the grey squares).
- There are two “most negative” entries: the two -978 squares. We can choose either as the entering basic variable.



“Highfive”
vintage smiley

A NON-DEGENERATE PIVOT IN
THE TRANSPORTATION SIMPLEX
METHOD. OBJECTIVE FUNCTION
IMPROVES!



“Facepalm”
vintage smiley

A DEGENERATE PIVOT, CAUSED
BY A BASIC VARIABLE CARRYING ZERO
VALUE THAT NEEDS TO DECREASE WHEN
THE ENTERING (CURRENTLY NON-BASIC)
VARIABLE WANTS TO INCREASE ABOVE ZERO.
OBJECTIVE FUNCTION IS NOT GOING TO
IMPROVE

	D1	D2	D3	D4	D5	Supply	$u(i)$
S1	16 30	16 20	13 -2	22 4	17 -978	50	-5
S2	14 0	14 0	13 60	19 3	15 -978	60	-7
S3	19 -2	19 -2	20 10	23 30	1000 10	50	0
S4	1000 1979	0 979	1000 1980	0 977	0 50	50	-1000
Demand	30	20	70	30	60		
$v(j)$	21	21	20	23	1000		

$Z = 12470$

- (S2,D5) basic: it will have a value of 10, which is the Min (basic variables that have to decrease).
- (S3,D5) will leave. The Z value will improve (i.e., get lower).



	D1	D2	D3	D4	D5	Supply	$u(i)$
S1	16 30	16 20	13 -2	22 4	17 0	50	2
S2	14 0	14 0	13 50	19 3	15 10	60	0
S3	19 -2	19 -2	20 20	23 30	1000 978	50	7
S4	1000 1001	0 1	1000 1002	0 -1	0 50	50	-15
Demand	30	20	70	30	60		
$v(j)$	14	14	13	16	15		

$Z = 2690$

- Determine if we have reached optimality

	D1	D2	D3	D4	D5	Supply	$u(i)$
S1	16 30	16 20	13 -2	22 4	17 0	50	2
S2	14 0	14 0	13 50	19 3	15 10	60	0
S3	19 -2	19 -2	20 20	23 30	1000 978	50	7
S4	1000 1001	0 1	1000 1002	0 -1	0 50	50	-15
Demand	30	20	70	30	60		
$v(j)$	14	14	13	16	15		

$Z = 2690$

- (S3,D2) becomes basic.
This is going to be a degenerate pivot, can you see why?
The Z value is not going to change...

	D1	D2	D3	D4	D5	Supply	$u(i)$
S1	16 30	16 20	13 -4	22 2	17 -2	50	-3
S2	14 2	14 2	13 50	19 3	15 10	60	-7
S3	19 0	19 0	20 20	23 30	1000 978	50	0
S4	1000 1003	0 3	1000 1002	0 -1	0 50	50	-22
Demand	30	20	70	30	60		
$v(j)$	19	19	20	23	22		

$Z = 2690$



	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 30	16 20	13 -4	22 2	17 -2	50	-3
S2	14 2	14 2	13 50	19 3	15 10	60	-7
S3	19 0	19 0	20 20	23 30	1000 978	50	0
S4	1000 1003	0 3	1000 1002	0 -1	0 50	50	-22
Demand	30	20	70	30	60		
v(j)	19	19	20	23	22		

$Z = 2690$

	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 30	16 0	13 20	22 2	17 2	50	0
S2	14 -2	14 -2	13 50	19 -1	15 10	60	0
S3	19 0	19 20	20 4	23 30	1000 982	50	3
S4	1000 999	0 -1	1000 1002	0 -5	0 50	50	-15
Demand	30	20	70	30	60		
v(j)	16	16	13	20	15		

Z = 2610



	D1	D2	D3	D4	D5	Supply	$u(i)$
S1	16 30	16 0	13 20	22 2	17 2	50	0
S2	14 -2	14 -2	13 50	19 -1	15 10	60	0
S3	19 0	19 20	20 4	23 30	1000 982	50	3
S4	1000 999	0 -1	1000 1002	0 -5	0 50	50	-15
Demand	30	20	70	30	60		
$v(j)$	16	16	13	20	15		

$Z = 2610$

- (S4,D4) basic. This will again be degenerate. The Z value is not going to change.

	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 30	16 5	13 20	22 7	17 2	50	0
S2	14 -2	14 3	13 50	19 4	15 10	60	0
S3	19 -5	19 20	20 -1	23 30	1000 977	50	8
S4	1000 999	0 4	1000 1002	0 0	0 50	50	-15
Demand	30	20	70	30	60		
v(j)	16	11	13	15	15		

Z = 2610



	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 30	16 5	13 20	22 7	17 2	50	0
S2	14 -2	14 3	13 50	19 4	15 10	60	0
S3	19 -5	19 20	20 -1	23 30	1000 977	50	8
S4	1000 999	0 4	1000 1002	0 0	0 50	50	-15
Demand	30	20	70	30	60		
v(j)	16	11	13	15	15		

Z = 2610

- (S3,D1) basic. We can set its value to 30.
The Z Value will change by $(30 \times -5) = -150$, to 2460.

	D1	D2	D3	D4	D5	Supply	$u(i)$
S1	16 0	16 0	13 50	22 7	17 2	50	0
S2	14 -2	14 -2	13 20	19 4	15 40	60	0
S3	19 30	19 20	20 4	23 5	1000 982	50	3
S4	1000 999	0 -1	1000 1002	0 30	0 20	50	-15
Demand	30	20	70	30	60		
$v(j)$	16	16	13	15	15		

$Z = 2460$



	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 0	- 0	16 50	13 7	22 2	50	0
S2	14 -2	14 -2	13 20	19 4	15 40	60	0
S3	19 30	19 20	20 4	23 5	1000 982	50	3
S4	1000 999	0 -1	1000 1002	0 30	0 20	50	-15
Demand	30	20	70	30	60		
v(j)	16	16	13	15	15		

Z = 2460

	D1	D2	D3	D4	D5	Supply	$u(i)$
S1	16 2	16 2	13 50	22 7	17 2	50	0
S2	14 0	14 0	13 20	19 4	15 40	60	0
S3	19 30	19 20	20 2	23 3	1000 980	50	5
S4	1000 1001	0 1	1000 1002	0 30	0 20	50	-15
Demand	30	20	70	30	60		
$v(j)$	14	14	13	15	15		

$Z = 2460$

- Finished because no more negative entries in the grey squares!
- Optimum is 2460, values of basic variables are in the yellow circles, all other variables have value 0.



Comments

- The Northwest Corner Rule is simple and easy to apply, but you might start off quite far from optimality. The other starting rules mentioned in the book take longer to apply, but tend to give you a much better starting point.

Comments

- The Northwest Corner Rule is simple and easy to apply, but you might start off quite far from optimality. The other starting rules mentioned in the book take longer to apply, but tend to give you a much better starting point.
- We saw here the phenomenon of **degeneracy**.
A BFS is *degenerate* if at least one of the basic variables are zero. (Remember, nonbasic variables are *always* zero, while basic variables *might* be zero). As in the general Simplex Method this can be a cause for concern.

Comments

- A common problem with degeneracy is that it can cause iterations to occur where the Z value does not increase.

Comments

- A common problem with degeneracy is that it can cause iterations to occur where the Z value does not increase.
- Relatedly, we saw that degeneracy can *slightly* confuse the optimality test. Namely, we saw a BFS where there was a ? with negative value, but such that the BFS was actually already at optimality. Don't worry, the golden rule still holds: *IF* there are no negative values *THEN* we are definitely at optimality.

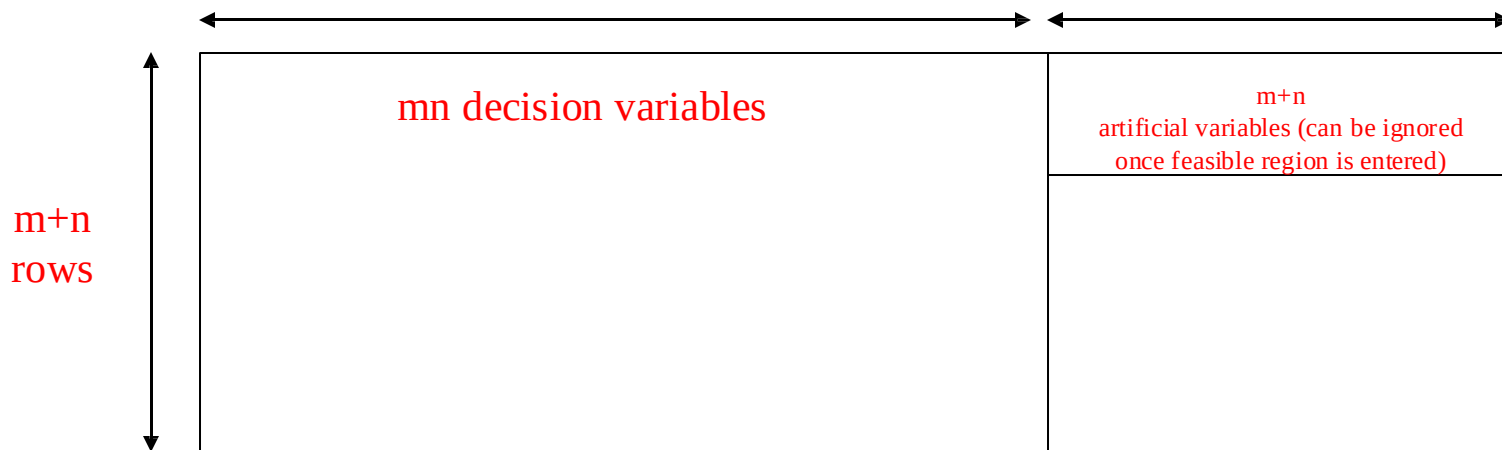
Comments

- A common problem with degeneracy is that it can cause iterations to occur where the Z value does not increase.
- Relatedly, we saw that degeneracy can *slightly* confuse the optimality test. Namely, we saw a BFS where there was a ? with negative value, but such that the BFS was actually already at optimality. Don't worry, the golden rule still holds: *IF* there are no negative values *THEN* we are definitely at optimality.
- In the worst case degeneracy can cause *cycling*...you return to a previous BFS without having improved the Z value, and thus get stuck in an infinite loop. But in practice this rarely happens and you can, if you want, guard against it.

Pros

Where does the TSM save most time, compared to the normal Simplex method?

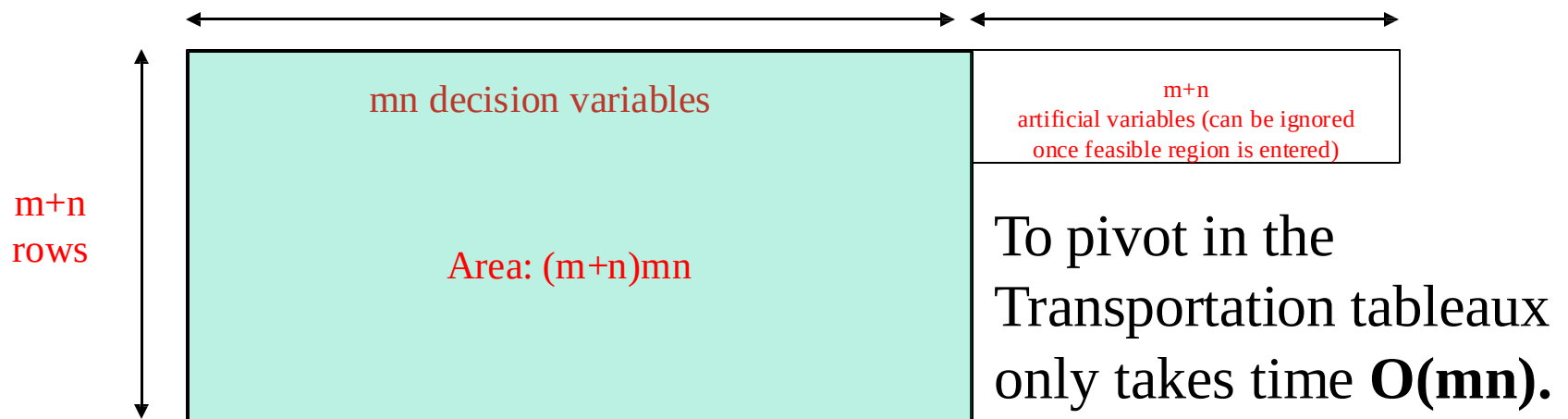
- In the pivoting step (i.e. updating the tableaux after having identified the leaving and entering basic variables).
- If you solve a Transportation problem with the normal simplex method, each pivot will require roughly the following number of arithmetic operations: $O((m+n)mn) = O(\max(m^2n, mn^2))$.



Pros

Where does the TSM save most time, compared to the normal Simplex method?

- In the pivoting step (i.e. updating the tableaux after having identified the leaving and entering basic variables).
- If you solve a Transportation problem with the normal simplex method, each pivot will require roughly the following number of arithmetic operations: $O((m+n)mn) = O(\max(m^2n, mn^2))$.





Maastricht University

Department of Data Science and Knowledge Engineering

MATSE: Operations Research

Lecture 6. Intro Stochastic Models: Flipping Coins

[based on notes by: Frank Thuijsman and Steven Chaplick]

Coin Flipping

Suppose you have a coin with two sides H and T .

Coin Flipping

Suppose you have a coin with two sides H and T .

- Consider a sequence $HHT \cdots HTHT$ of coin flips.

Coin Flipping

Suppose you have a coin with two sides H and T .

- Consider a sequence $HHT \cdots HTHT$ of coin flips.
- What is the **expected** number of coin flips until **first** H ?

Coin Flipping

Suppose you have a coin with two sides H and T .

- Consider a sequence $HHT \cdots HTHT$ of coin flips.
- What is the **expected** number of coin flips until **first** H ?
(i.e., on average over all possible sequences, when does the first H occur?)

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

x	1	2	3	4	5	...	k	
$P(X=x)$								

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

x	1	2	3	4	5	...	k	
$P(X=x)$	$\frac{1}{2}$							

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

x	1	2	3	4	5	...	k	
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$						

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

x	1	2	3	4	5	...	k	
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$					

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

x	1	2	3	4	5	...	k	
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$			

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

x	1	2	3	4	5	...	k
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$		$\frac{1}{2^k}$

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

x	1	2	3	4	5	...	k	
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$		$\frac{1}{2^k}$	

$$\left. \vphantom{\begin{matrix} x \\ P(X=x) \end{matrix}} \right\} \mathbb{E}(X) = ?$$

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

$$\left. \begin{array}{c|c|c|c|c|c|c|c|c} x & 1 & 2 & 3 & 4 & 5 & \dots & k & \\ \hline P(X=x) & \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{16} & \frac{1}{32} & & \frac{1}{2^k} & \end{array} \right\} \mathbb{E}(X) = \sum_{i=1}^{\infty} i \frac{1}{2^i}$$

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

$$\left. \begin{array}{c|c|c|c|c|c|c|c|c} x & 1 & 2 & 3 & 4 & 5 & \dots & k & \\ \hline P(X=x) & \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{16} & \frac{1}{32} & & \frac{1}{2^k} & \end{array} \right\} \mathbb{E}(X) = \sum_{i=1}^{\infty} i \frac{1}{2^i}$$

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Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

$$\left. \begin{array}{c|c|c|c|c|c|c|c|c} x & 1 & 2 & 3 & 4 & 5 & \dots & k & \\ \hline P(X=x) & \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{16} & \frac{1}{32} & & \frac{1}{2^k} & \end{array} \right\} \mathbb{E}(X) = \sum_{i=1}^{\infty} i \frac{1}{2^i}$$

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} i \frac{1}{2^i} = 1 \frac{1}{2} + 2 \frac{1}{4} + 3 \frac{1}{8} + 4 \frac{1}{16} + \dots$$

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

$$\left. \begin{array}{c|c|c|c|c|c|c|c|c} x & 1 & 2 & 3 & 4 & 5 & \dots & k & \\ \hline P(X=x) & \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{16} & \frac{1}{32} & & \frac{1}{2^k} & \end{array} \right\} \mathbb{E}(X) = \sum_{i=1}^{\infty} i \frac{1}{2^i}$$

$$\begin{aligned} \mathbb{E}(X) &= \sum_{i=1}^{\infty} i \frac{1}{2^i} = 1\frac{1}{2} + 2\frac{1}{4} + 3\frac{1}{8} + 4\frac{1}{16} + \dots \\ &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\ &\quad + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\ &\quad \quad + \frac{1}{8} + \frac{1}{16} + \dots \\ &\quad \quad \quad + \frac{1}{16} + \dots \end{aligned}$$

Coin Flipping (Computations)

$$\begin{aligned}\sum_{i=1}^{\infty} i \frac{1}{2^i} &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\ &\quad + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\ &\quad \quad + \frac{1}{8} + \frac{1}{16} + \dots \\ &\quad \quad \quad + \frac{1}{16} + \dots\end{aligned}$$

Coin Flipping (Computations)

$$\begin{aligned}
 \sum_{i=1}^{\infty} i \frac{1}{2^i} &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad \quad + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad \quad \quad + \frac{1}{16} + \dots
 \end{aligned}$$

$$\sum_{i=1}^{\infty} \frac{1}{2^i}$$

Coin Flipping (Computations)

$$\begin{aligned}
 \sum_{i=1}^{\infty} i \frac{1}{2^i} &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad \quad + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad \quad \quad + \frac{1}{16} + \dots
 \end{aligned}$$

$$\begin{aligned}
 &\sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &\frac{1}{2} \sum_{i=1}^{\infty} \frac{1}{2^i}
 \end{aligned}$$

Coin Flipping (Computations)

$$\begin{aligned}
 \sum_{i=1}^{\infty} i \frac{1}{2^i} &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad \quad + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad \quad \quad + \frac{1}{16} + \dots
 \end{aligned}$$

$$\begin{aligned}
 &\sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &\frac{1}{2} \sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &\frac{1}{2^2} \sum_{i=1}^{\infty} \frac{1}{2^i}
 \end{aligned}$$

Coin Flipping (Computations)

$$\begin{aligned}
 \sum_{i=1}^{\infty} i \frac{1}{2^i} &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad \quad + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad \quad \quad + \frac{1}{16} + \dots
 \end{aligned}$$

$$\begin{aligned}
 &\sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &\frac{1}{2} \sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &\frac{1}{2^2} \sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &= \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \left(\frac{1}{2^i} \sum_{j=1}^{\infty} \frac{1}{2^j} \right)
 \end{aligned}$$

Coin Flipping (Computations)

$$\begin{aligned}
 \sum_{i=1}^{\infty} i \frac{1}{2^i} &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad + \frac{1}{16} + \dots
 \end{aligned}$$

$$\begin{aligned}
 &\sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &\frac{1}{2} \sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &\frac{1}{2^2} \sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &= \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \left(\frac{1}{2^i} \sum_{j=1}^{\infty} \frac{1}{2^j} \right)
 \end{aligned}$$

Coin Flipping (Computations)

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\sum_{j=1}^{\infty} \frac{1}{2^j} \right) =$$

Coin Flipping (Computations)

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\sum_{j=1}^{\infty} \frac{1}{2^j} \right) =$$

Geometric Serie: $\sum_{i=0}^{\infty} q^i = \frac{1}{1-q} \quad q \in (0, 1)$

Coin Flipping (Computations)

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Geometric Serie: $\sum_{i=0}^{\infty} q^i = \frac{1}{1-q} \quad q \in (0, 1)$

$$\sum_{i=1}^{\infty} \frac{1}{2^i} =$$

Coin Flipping (Computations)

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\sum_{j=1}^{\infty} \frac{1}{2^j} \right) =$$

Geometric Serie: $\sum_{i=0}^{\infty} q^i = \frac{1}{1-q} \quad q \in (0, 1)$

$$\sum_{i=1}^{\infty} \frac{1}{2^i} = \sum_{i=0}^{\infty} \frac{1}{2^i} - \frac{1}{2^0} =$$

Coin Flipping (Computations)

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\sum_{j=1}^{\infty} \frac{1}{2^j} \right) =$$

Geometric Serie: $\sum_{i=0}^{\infty} q^i = \frac{1}{1-q} \quad q \in (0, 1)$

$$\sum_{i=1}^{\infty} \frac{1}{2^i} = \sum_{i=0}^{\infty} \frac{1}{2^i} - \frac{1}{2^0} = \frac{1}{1 - \frac{1}{2}} - 1 = 1$$

Coin Flipping (Computations)

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\sum_{j=1}^{\infty} \frac{1}{2^j} \right) = \boxed{1} + \sum_{i=1}^{\infty} \frac{1}{2^i} \cdot \boxed{1} = 1 + \boxed{1} = 2$$

Geometric Serie: $\sum_{i=0}^{\infty} q^i = \frac{1}{1-q} \quad q \in (0, 1)$

$$\sum_{i=1}^{\infty} \frac{1}{2^i} = \sum_{i=0}^{\infty} \frac{1}{2^i} - \frac{1}{2^0} = \frac{1}{1 - \frac{1}{2}} - 1 = \boxed{1}$$

Coin Flipping (Computations)

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\sum_{j=1}^{\infty} \frac{1}{2^j} \right) = \boxed{1} + \sum_{i=1}^{\infty} \frac{1}{2^i} \cdot \boxed{1} = 1 + \boxed{1} = 2$$

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$$\sum_{i=1}^{\infty} \frac{1}{2^i} = \sum_{i=0}^{\infty} \frac{1}{2^i} - \frac{1}{2^0} = \frac{1}{1 - \frac{1}{2}} - 1 = \boxed{1}$$

- What is the expected number of coin flips until **first** H ?
2!

Coin Flipping (Computations)

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\sum_{j=1}^{\infty} \frac{1}{2^j} \right) = \boxed{1} + \sum_{i=1}^{\infty} \frac{1}{2^i} \cdot \boxed{1} = 1 + \boxed{1} = 2$$

Geometric Serie: $\sum_{i=0}^{\infty} q^i = \frac{1}{1-q} \quad q \in (0, 1)$

$$\sum_{i=1}^{\infty} \frac{1}{2^i} = \sum_{i=0}^{\infty} \frac{1}{2^i} - \frac{1}{2^0} = \frac{1}{1 - \frac{1}{2}} - 1 = \boxed{1}$$

- What is the expected number of coin flips until **first** H ?
2!
- 2 is also the inverse of $\frac{1}{2}$ which is the probability of flipping H...

Coin Flipping (cont)

Seems complex ...

Let X be the number of coin flips until we see HT .

Take a Guess!

Coin Flipping (cont)

Seems complex ...

Let X be the number of coin flips until we see HT .

x	1	2	3	4	5	...
$P(X=x)$	0	$\frac{1}{4}$				
		HT				

Coin Flipping (cont)

Seems complex ...

Let X be the number of coin flips until we see HT .

x	1	2	3	4	5	...
$P(X=x)$	0	$\frac{1}{4}$	$\frac{2}{8}$			
		HT	zHT			

Coin Flipping (cont)

Seems complex ...

Let X be the number of coin flips until we see HT .

x	1	2	3	4	5	...
$P(X=x)$	0	$\frac{1}{4}$	$\frac{2}{8}$	$\frac{3}{16}$		
		HT	zHT	$zzHT$ $\neg HTHT$		

Coin Flipping (cont)

Seems complex ...

Let X be the number of coin flips until we see HT .

x	1	2	3	4	5	...
$P(X=x)$	0	$\frac{1}{4}$	$\frac{2}{8}$	$\frac{3}{16}$	$\frac{4}{32}$	
		HT	zHT	$zzHT$ $\neg HTHT$		

Coin Flipping (cont)

Seems complex ...

Let X be the number of coin flips until we see HT .

x	1	2	3	4	5	...
$P(X=x)$	0	$\frac{1}{4}$	$\frac{2}{8}$	$\frac{3}{16}$	$\frac{4}{32}$	
		HT	zHT	$zzHT$ $\neg HTHT$		

$$\mathbb{E}(X) = \sum_x x \cdot P(X = x)$$

Coin Flipping (cont)

Seems complex ...

Let X be the number of coin flips until we see HT .

x	1	2	3	4	5	...	$\mathbb{E}(X) = \sum_x x \cdot P(X = x)$ There has to be a better way!
$P(X=x)$	0	$\frac{1}{4}$	$\frac{2}{8}$	$\frac{3}{16}$	$\frac{4}{32}$		
		HT	zHT	$zzHT$ $\neg HTHT$			

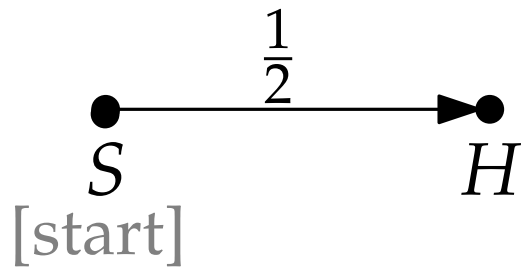
Coin Flipping (cont)

There is a better way!

●
S
[start]

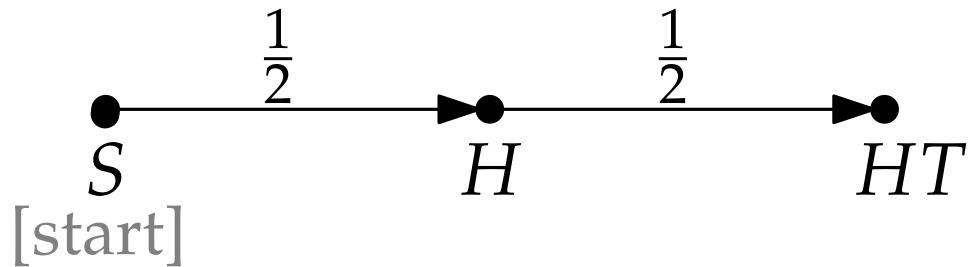
Coin Flipping (cont)

There is a better way!



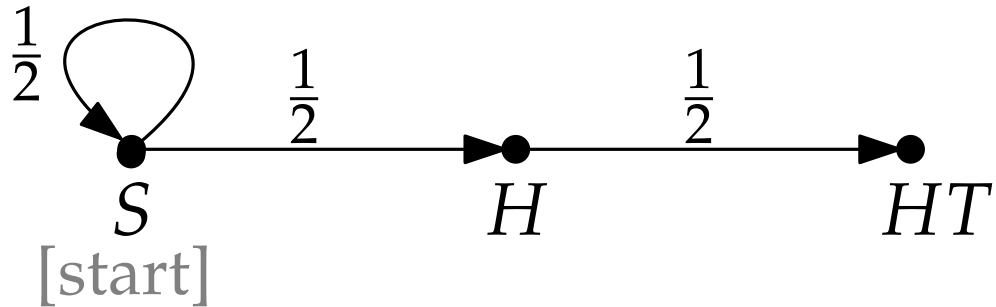
Coin Flipping (cont)

There is a better way!



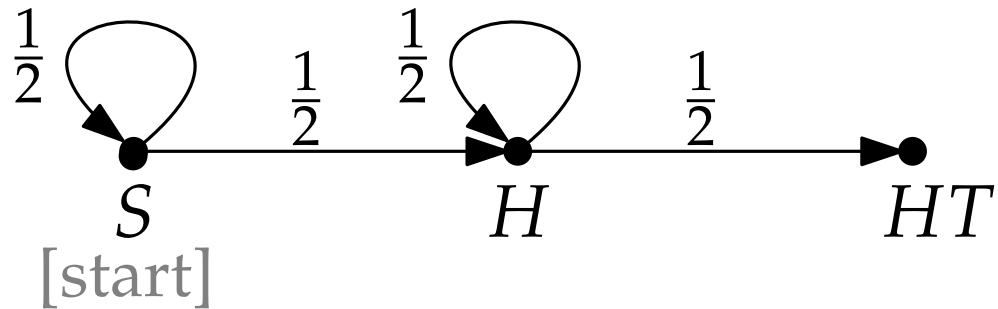
Coin Flipping (cont)

There is a better way!



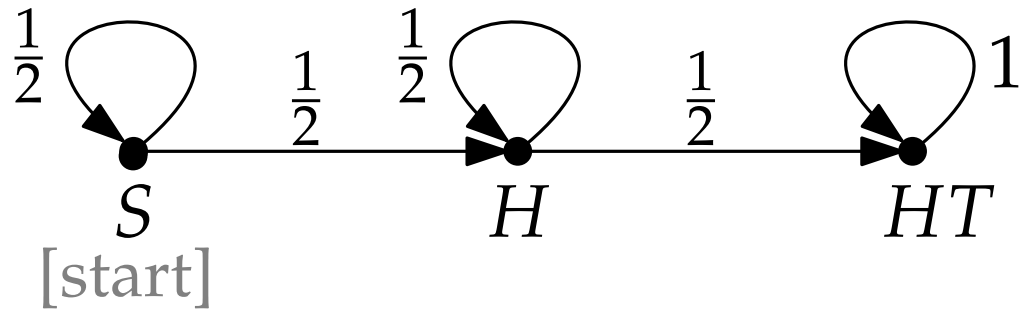
Coin Flipping (cont)

There is a better way!



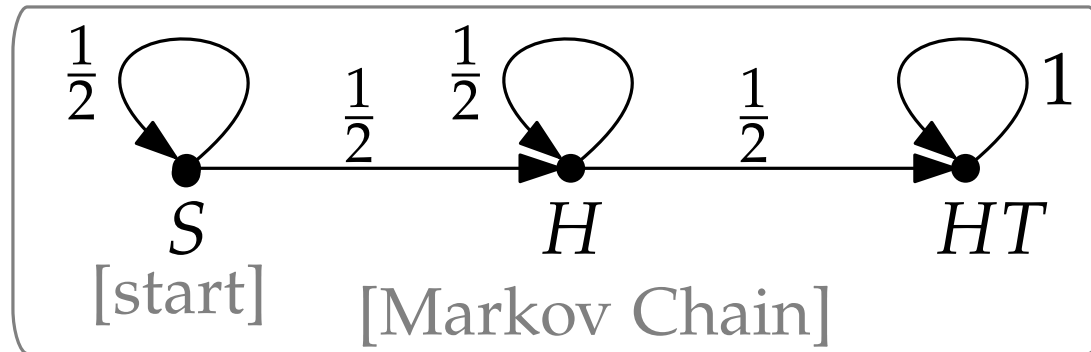
Coin Flipping (cont)

There is a better way!



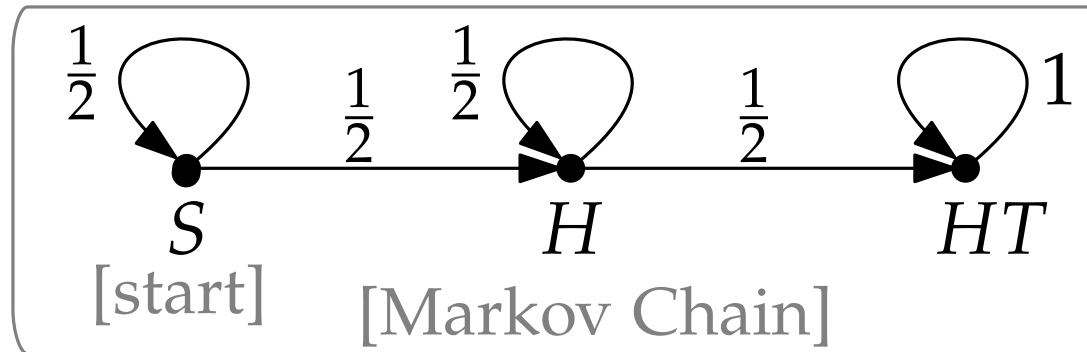
Coin Flipping (cont)

There is a better way!



Coin Flipping (cont)

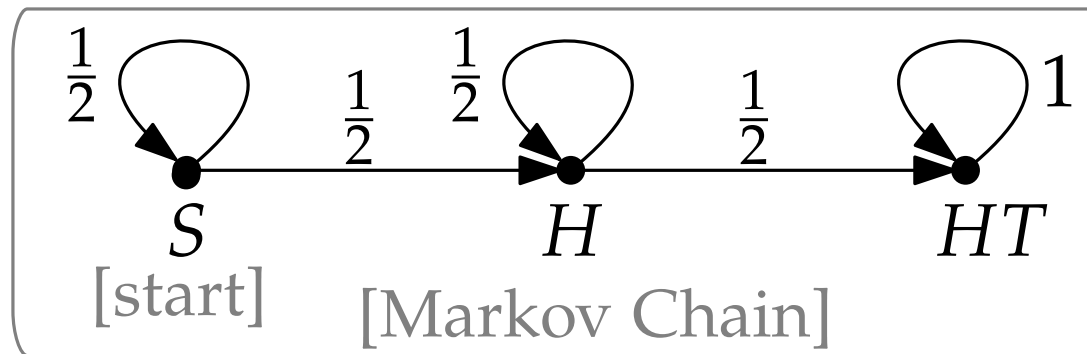
There is a better way!



Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

Coin Flipping (cont)

There is a better way!

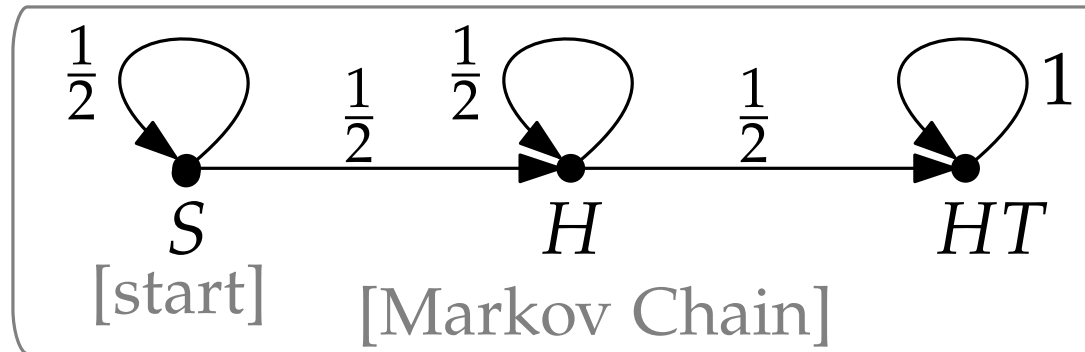


Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

Coin Flipping (cont)

There is a better way!



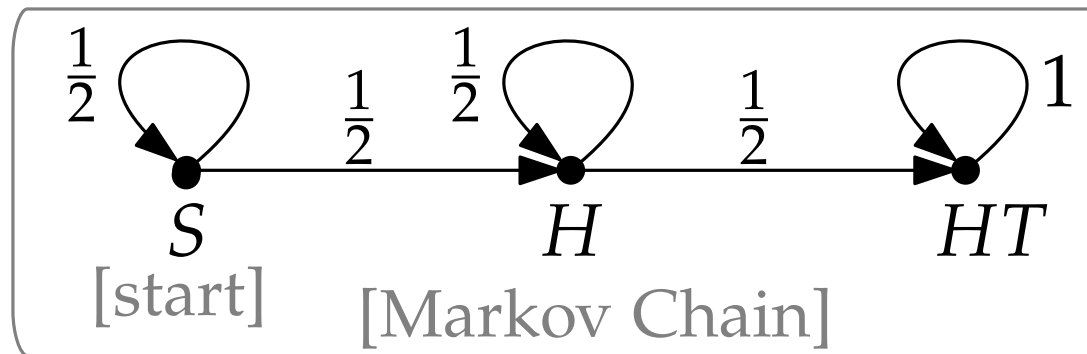
Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT}$$

Coin Flipping (cont)

There is a better way!



Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

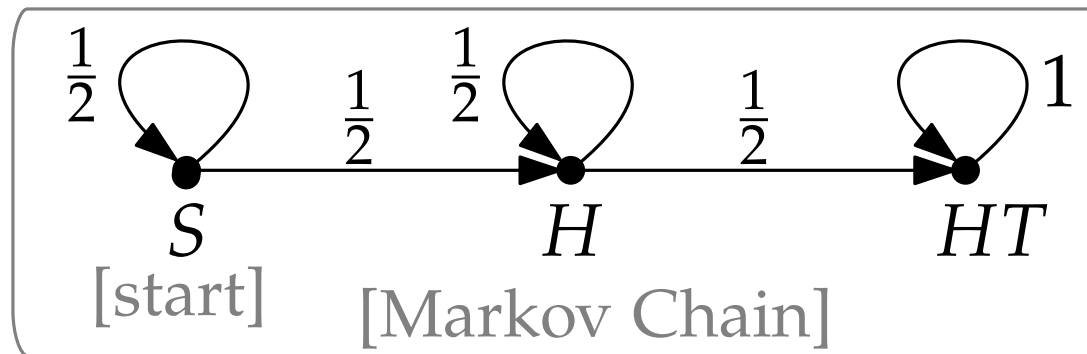
$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT}$$

$$E_{HT} = 0 \text{ [already done!]}$$

Coin Flipping (cont)

There is a better way!



Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

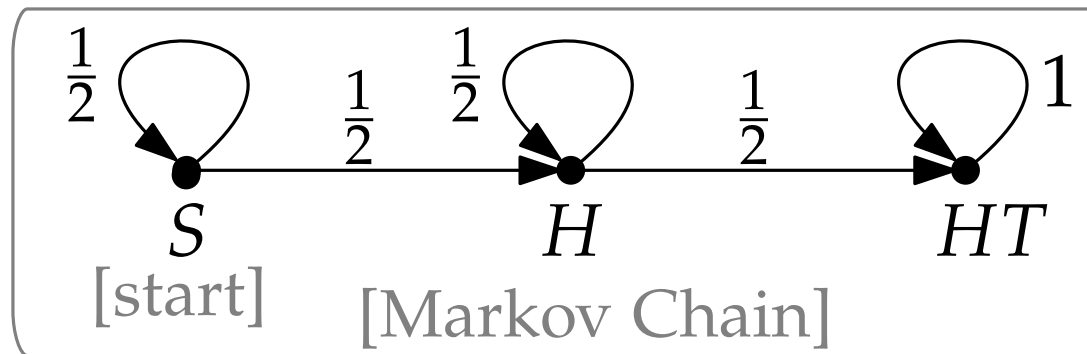
$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT} \rightsquigarrow E_H = 2 \text{ [re-arranging]}$$

$$E_{HT} = 0 \text{ [already done!]}$$

Coin Flipping (cont)

There is a better way!



Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

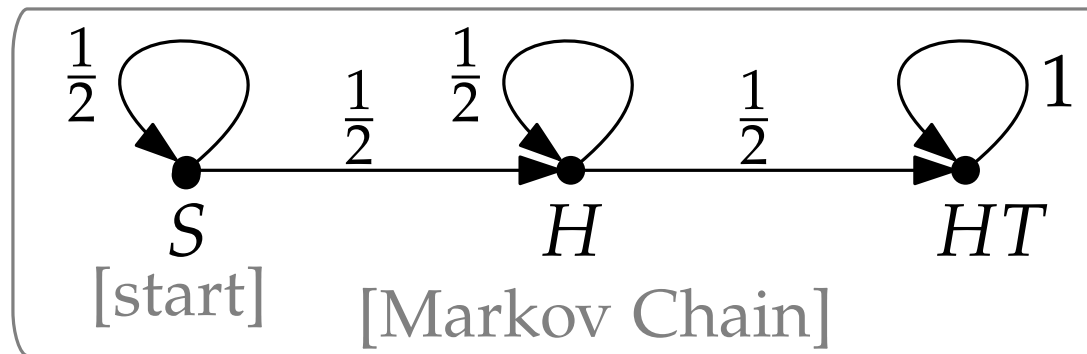
$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \quad \rightsquigarrow \quad E_S = 2 + \frac{1}{2}E_S$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT} \quad \rightsquigarrow \quad E_H = 2 \text{ [re-arranging]}$$

$$E_{HT} = 0 \text{ [already done!]}$$

Coin Flipping (cont)

There is a better way!



Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

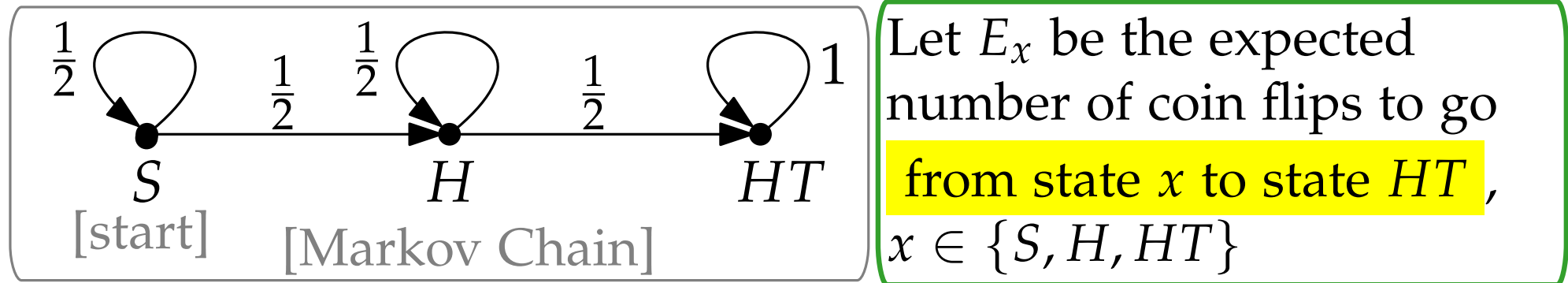
$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \quad \rightsquigarrow \quad E_S = 2 + \frac{1}{2}E_S \quad \rightsquigarrow \quad E_S = 4$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT} \quad \rightsquigarrow \quad E_H = 2 \text{ [re-arranging]}$$

$$E_{HT} = 0 \text{ [already done!]}$$

Coin Flipping (cont)

There is a better way!



$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \quad \rightsquigarrow \quad E_S = 2 + \frac{1}{2}E_S \quad \rightsquigarrow \quad E_S = 4$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT} \quad \rightsquigarrow \quad E_H = 2 \quad [\text{re-arranging}]$$

$$E_{HT} = 0 \quad [\text{already done!}]$$

- 4 is also the inverse of $\frac{1}{4}$ which is the probability of flipping HT...

Coin Flipping (cont)

What is the expected number of coin flips until we see HH .

Take a guess!

Coin Flipping (cont)

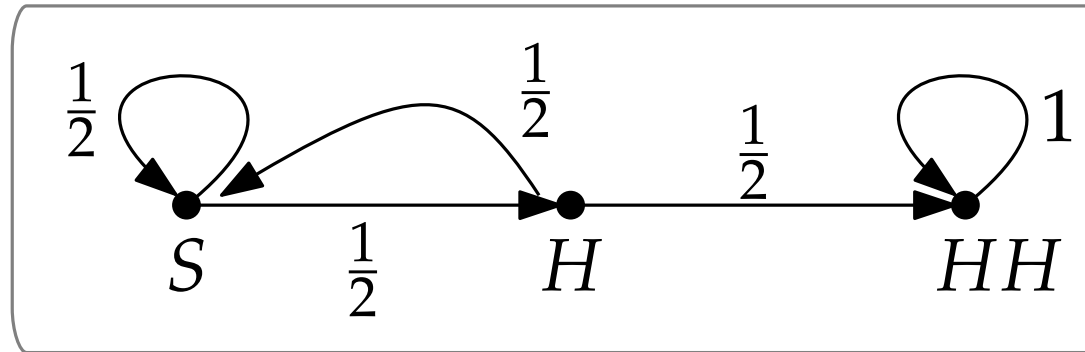
What is the expected number of coin flips until we see HH .

Take a guess!

Can you draw the Markov chain?

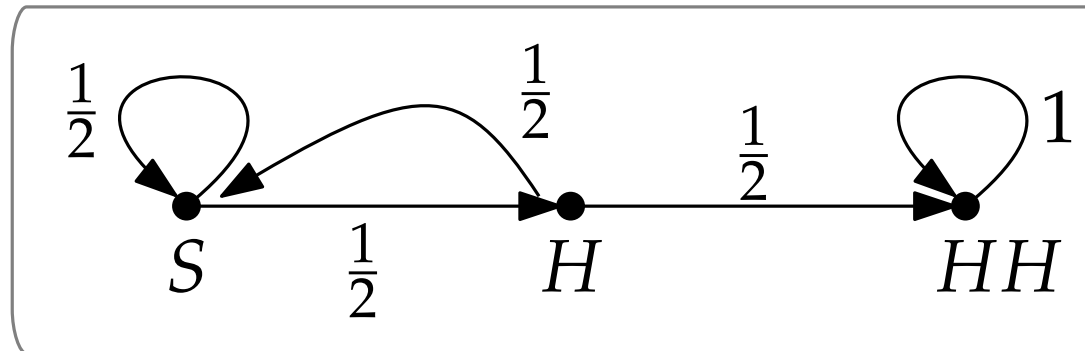
Coin Flipping (cont)

What is the expected number of coin flips until we see HH .



Coin Flipping (cont)

What is the expected number of coin flips until we see HH .



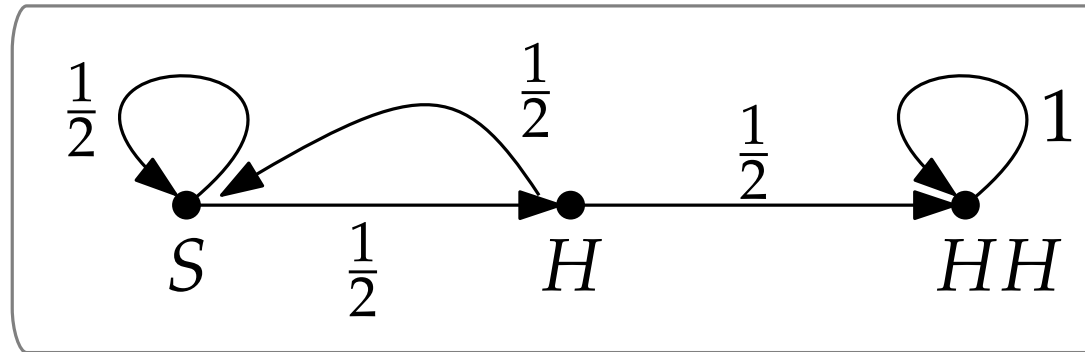
$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

$$E_H = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HH}$$

$$E_{HT} = 0 \text{ [already done!]}$$

Coin Flipping (cont)

What is the expected number of coin flips until we see HH .



$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

$$E_H = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HH}$$

$$E_{HT} = 0 \text{ [already done!]}$$

$$E_S = 6!$$

- 6 is not the inverse of $\frac{1}{4}$ which is the probability of flipping HH ...

Coin Flipping: systematically

Let X be the number of coin flips until we see HTH .

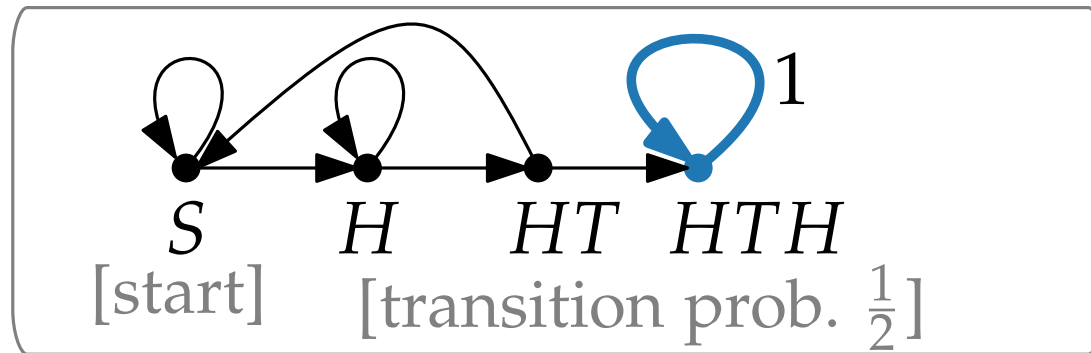
Coin Flipping: systematically

Let X be the number of coin flips until we see HTH .

Try Yourself!

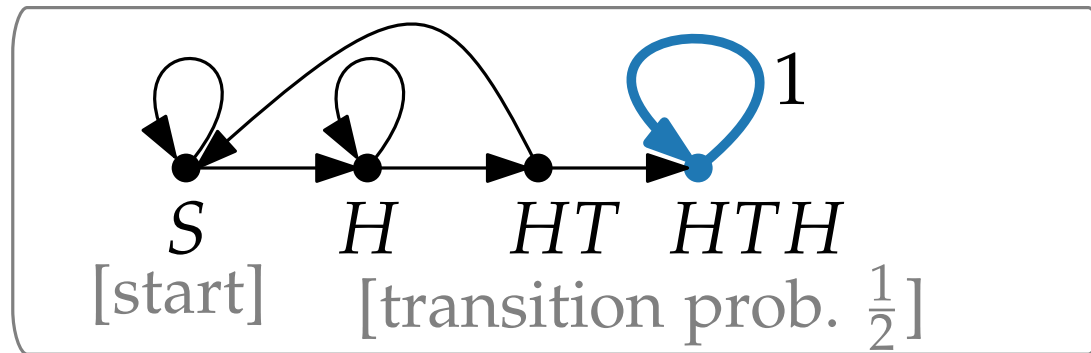
Coin Flipping: systematically

Let X be the number of coin flips until we see HTH .



Coin Flipping: systematically

Let X be the number of coin flips until we see HTH .



$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

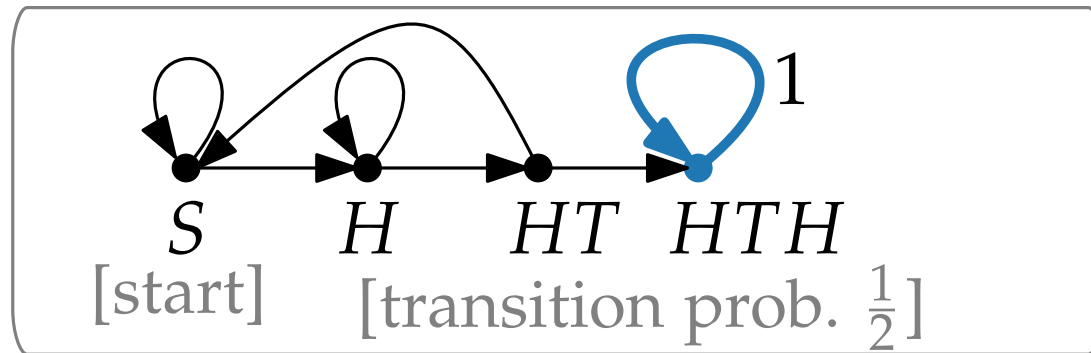
$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT}$$

$$E_{HT} = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HTH}$$

$$E_{HTH} = 0 \text{ [already done!]}$$

Coin Flipping: systematically

Let X be the number of coin flips until we see HTH .



$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT}$$

$$E_{HT} = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HTH}$$

$$E_{HTH} = 0 \text{ [already done!]}$$

$$E_S = 10!$$

- 10 is not the inverse of $\frac{1}{8}$ which is the probability of flipping HH...

Abacadabra Theorem

We actually measured two different things...

Abacadabra Theorem

We actually measured two different things...

- E_x (i.e., the system of equations) measure the number of coin flips before the we see a sequence for the *first* time.
- The inverse of the probability measures the number of coin flips before the process returns *again* to the state

Abracadabra Theorem

We actually measured two different things...

- E_x (i.e., the system of equations) measure the number of coin flips before the we see a sequence for the *first* time.

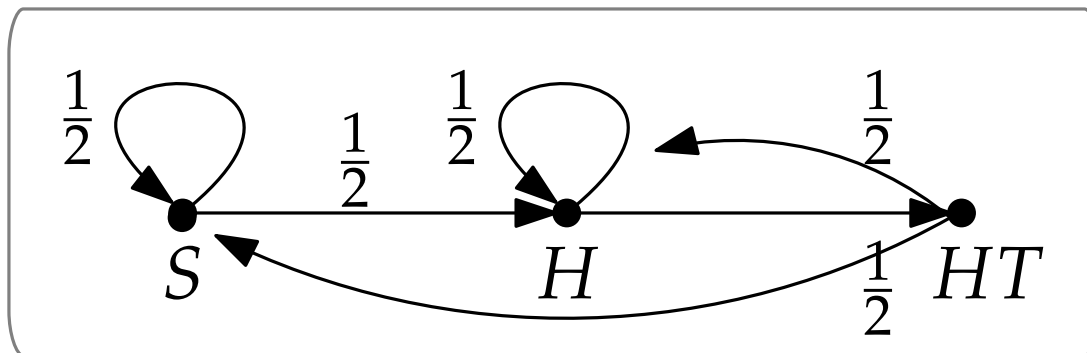
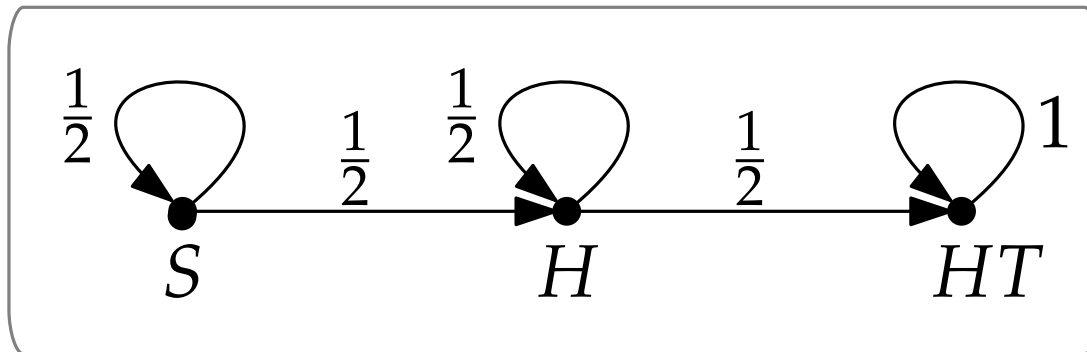
⇒ **First Passage Time**

- The inverse of the probability measures the number of coin flips before the process returns *again* to the state

⇒ **Recurrence Time**

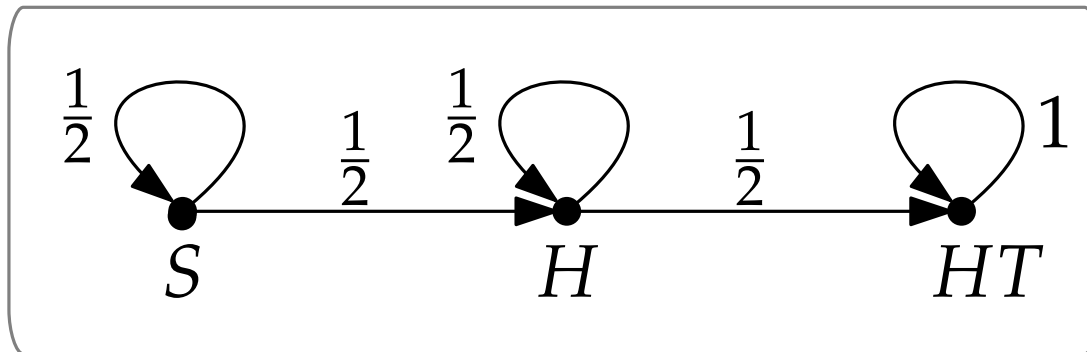
Abracadabra Theorem

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Abracadabra Theorem

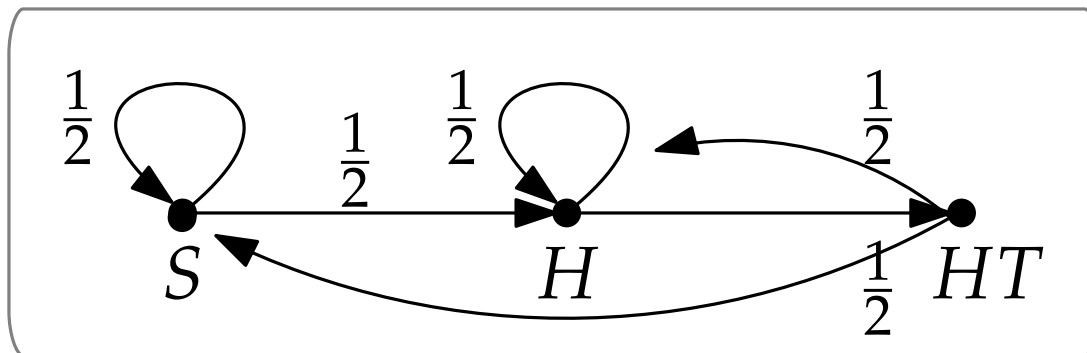
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First Passage Time

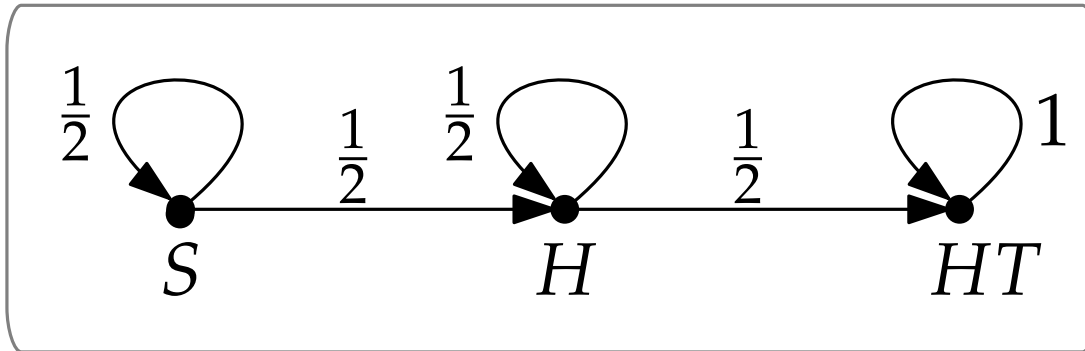
$$E_x = 1 + \sum_j P_{xj} E_j$$

[same as before]



Abracadabra Theorem

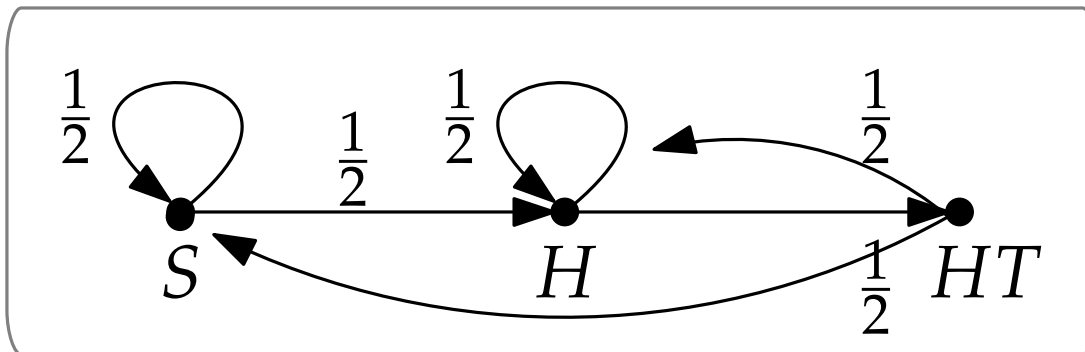
We actually measured two different things...



First Passage Time

$$E_x = 1 + \sum_j P_{xj} E_j$$

[same as before]



Recurrence Time

$$R_x = 1 + \sum_j P_{xj} E_j$$

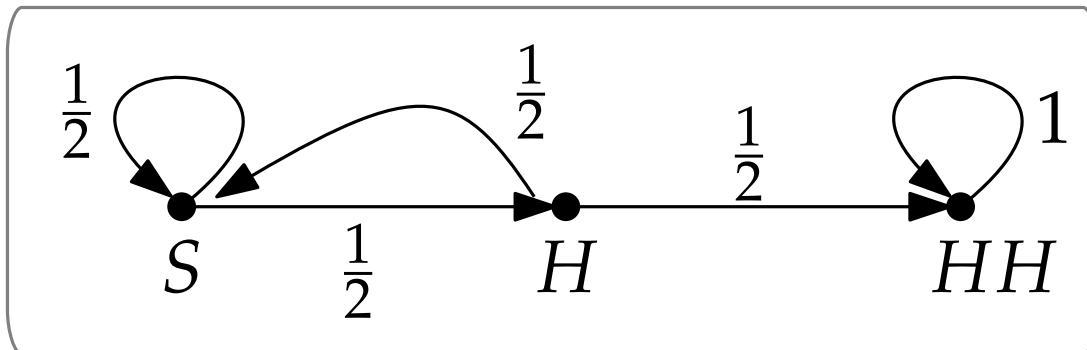
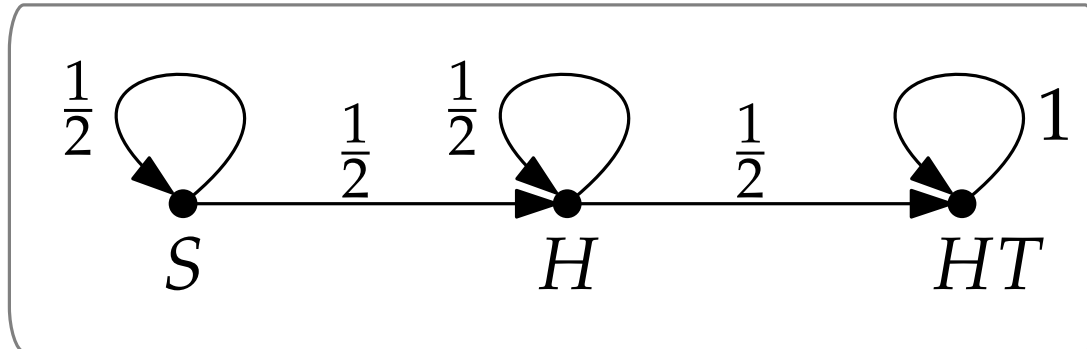
[different chain]

Recurrence time

Is there a faster way?

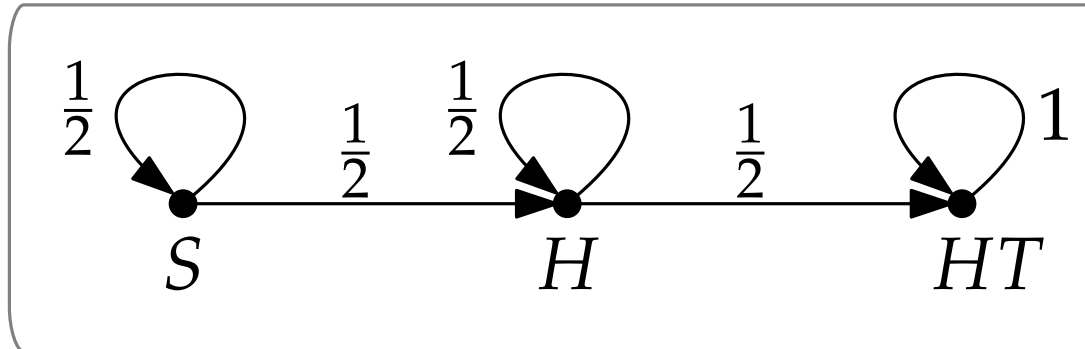
Recurrence time

Is there a faster way?

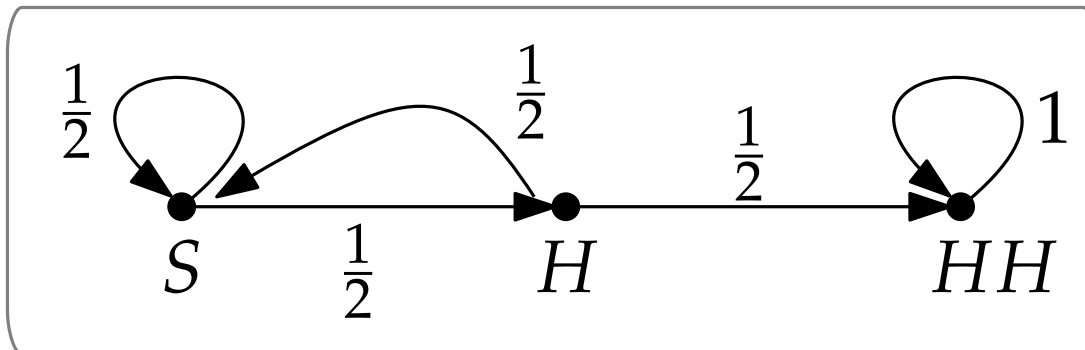


Recurrence time

Is there a faster way?



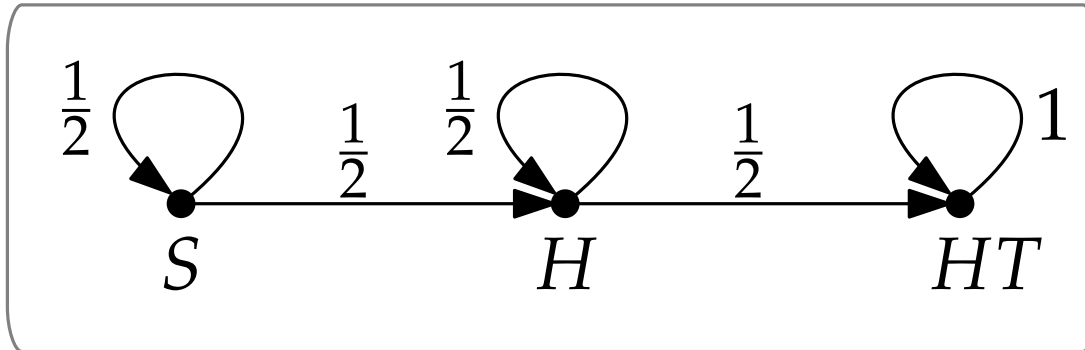
$S \longrightarrow HT$



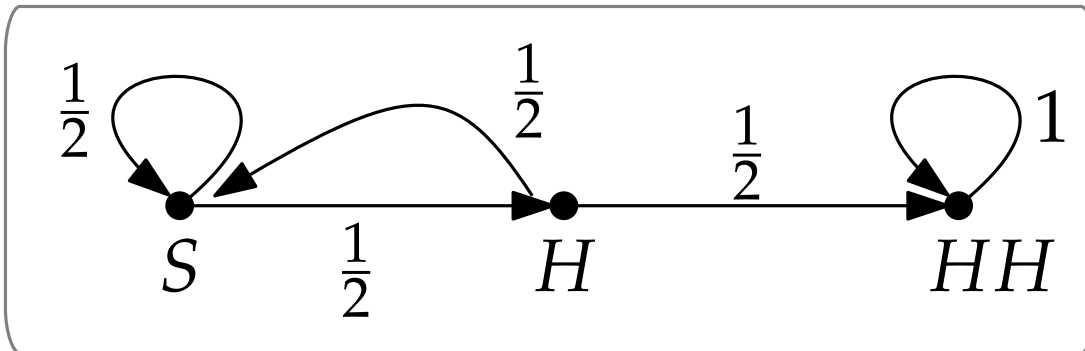
$S \longrightarrow H \longrightarrow HH$

Recurrence time

Is there a faster way?



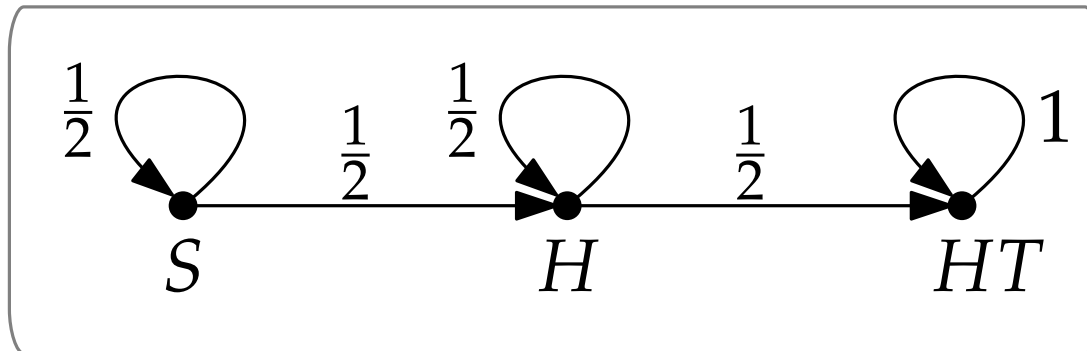
$$S \xrightarrow[4]{} HT$$



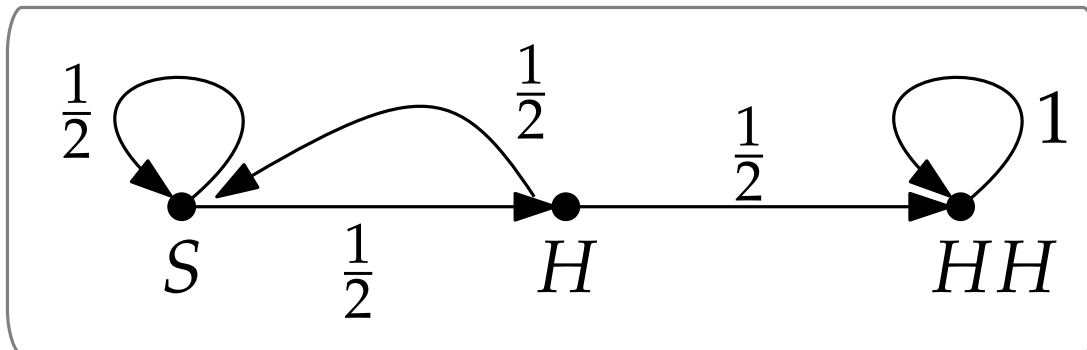
$$S \xrightarrow[2]{} H \xrightarrow[4]{} HH$$

Recurrence time

Is there a faster way?



$$S \xrightarrow[4]{} HT$$



$$S \xrightarrow[2]{} H \xrightarrow[4]{} HH$$

Since the Recurrence time is the time before a sequence *reappears again*, we need to check also that the subsequences *reappear again*.

Recurrence time

Consider the sequence $HTHTH$

- What is the probability that $HTHTH$ appears? $\frac{1}{32}$

Recurrence Time = 32

Expected Number of Flips = 42

Recurrence time

Consider the sequence $HTHTH$

- What is the probability that $HTHTH$ appears? $\frac{1}{32}$

Recurrence Time = 32

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The answer to the ultimate question of life, the universe and everything!



Recurrence time

Consider the sequence $HTHTH$

- What is the probability that $HTHTH$ appears? $\frac{1}{32}$

Recurrence Time = 32

Expected Number of Flips = 42

The answer to the ultimate question of life, the universe and everything!

No!

$$S \xrightarrow{2} H \xrightarrow{8} HT H \xrightarrow{32} HTHT H$$

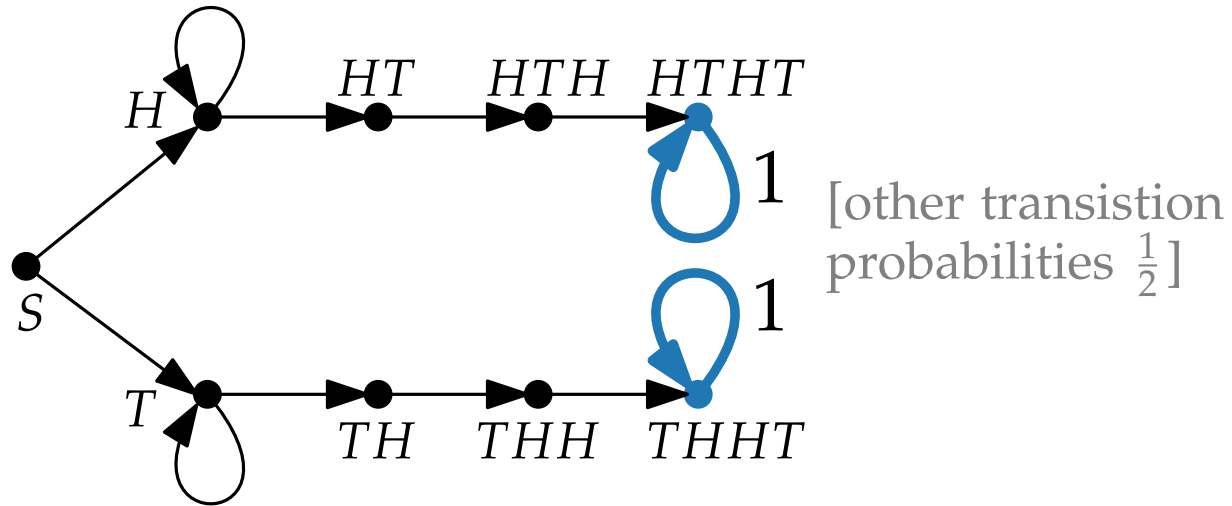


What's reached first?

Given two sequences of coin flips, which do you expect to reach first, e.g, between *HTHT* and *THHT*?

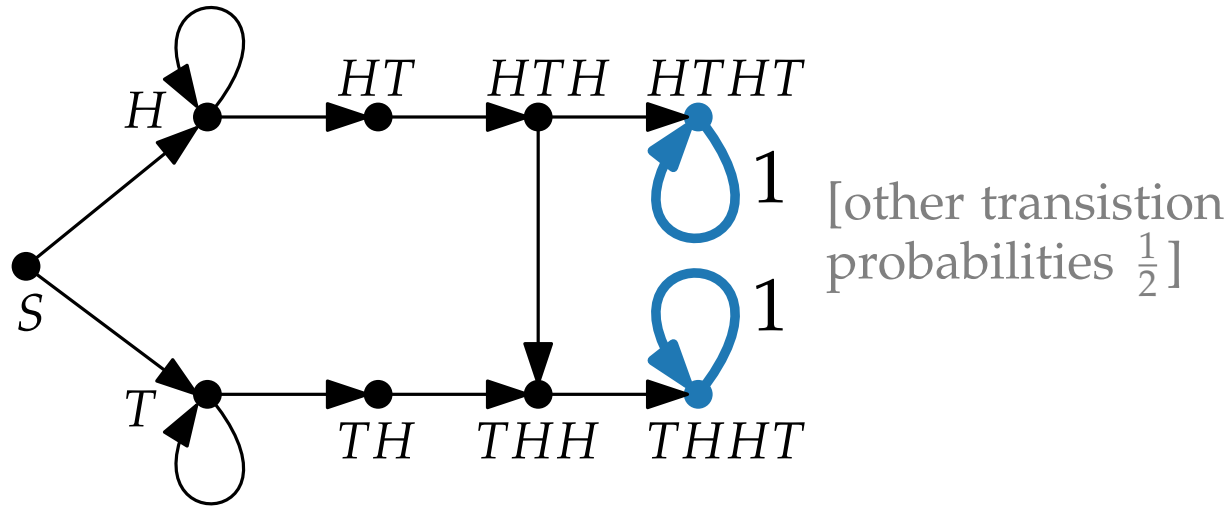
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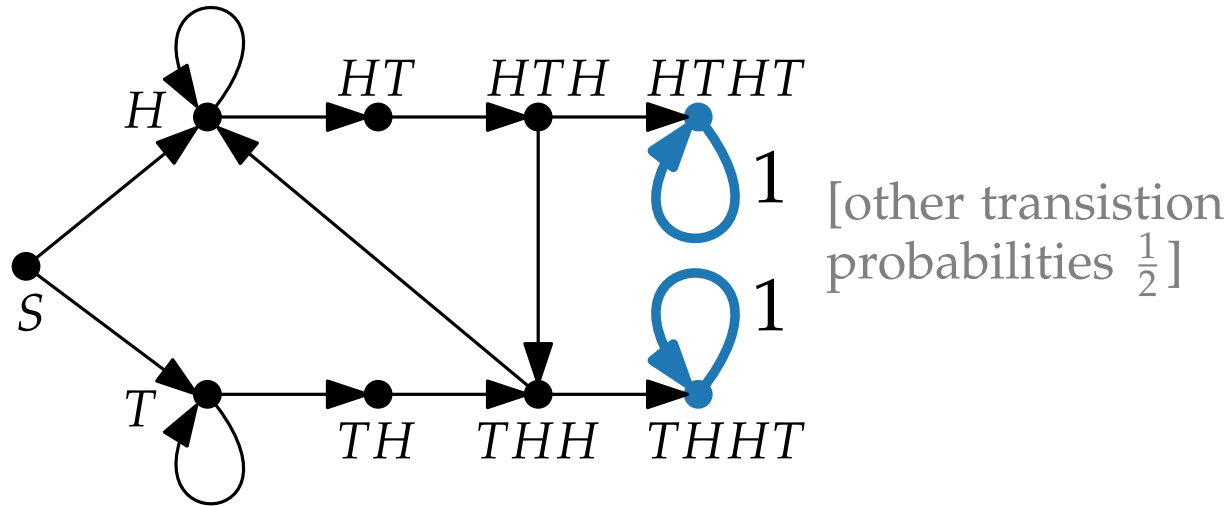
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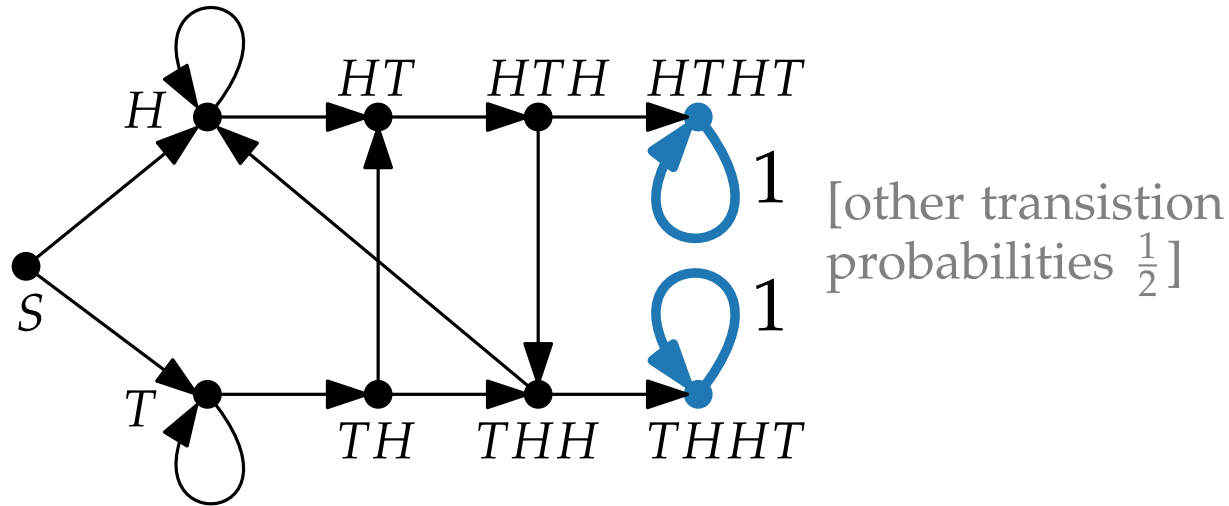
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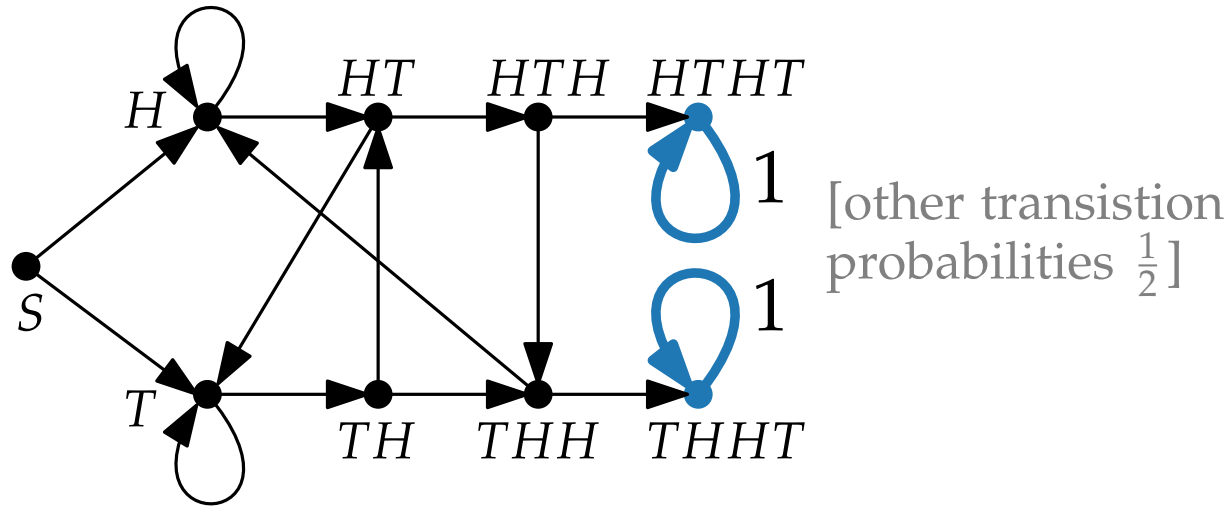
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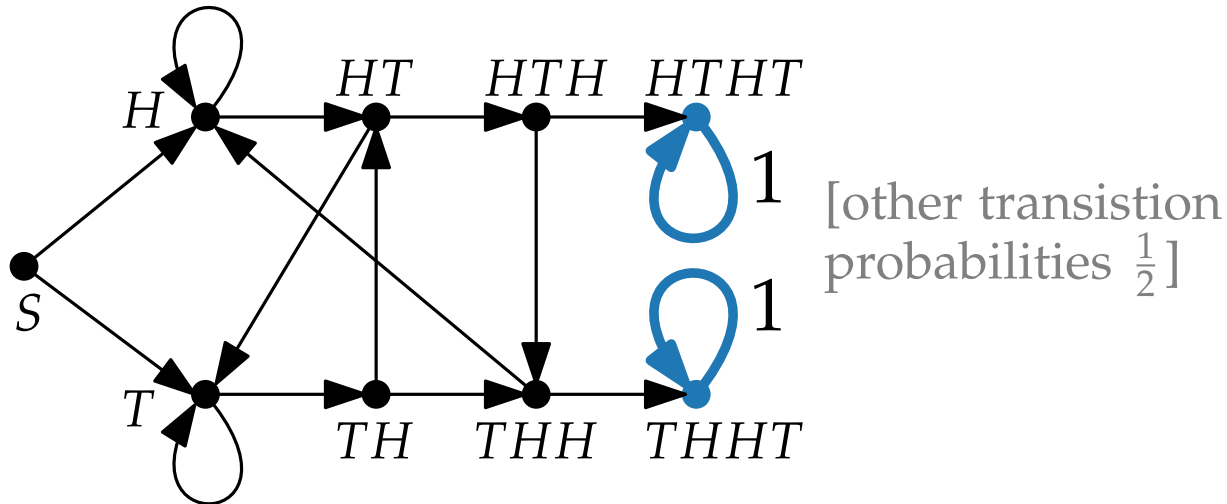
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What's reached first?

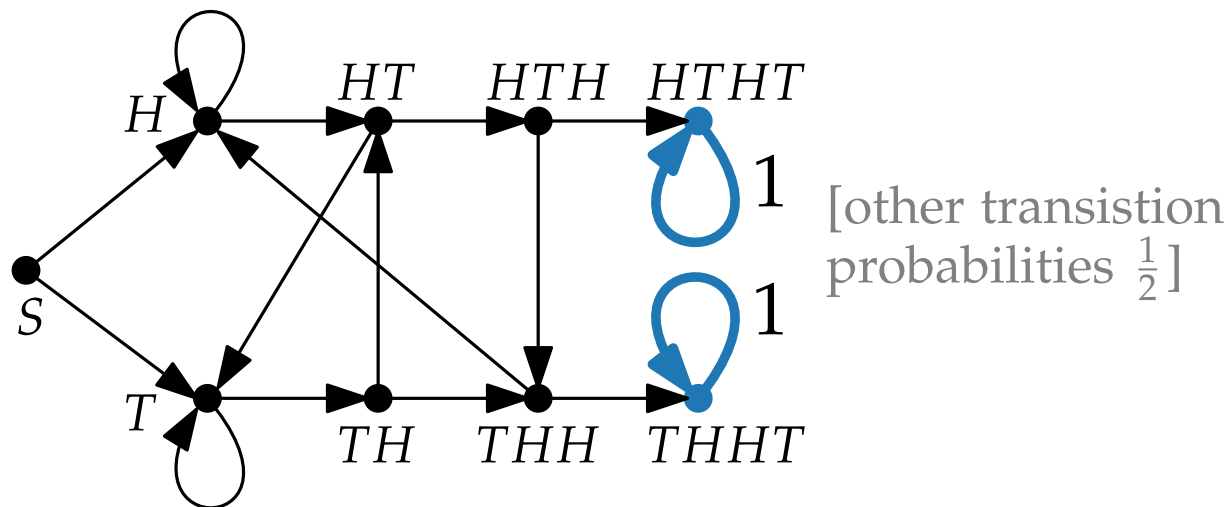
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Let P_x be the probability of reaching $HTHT$ before $THHT$ when starting from x

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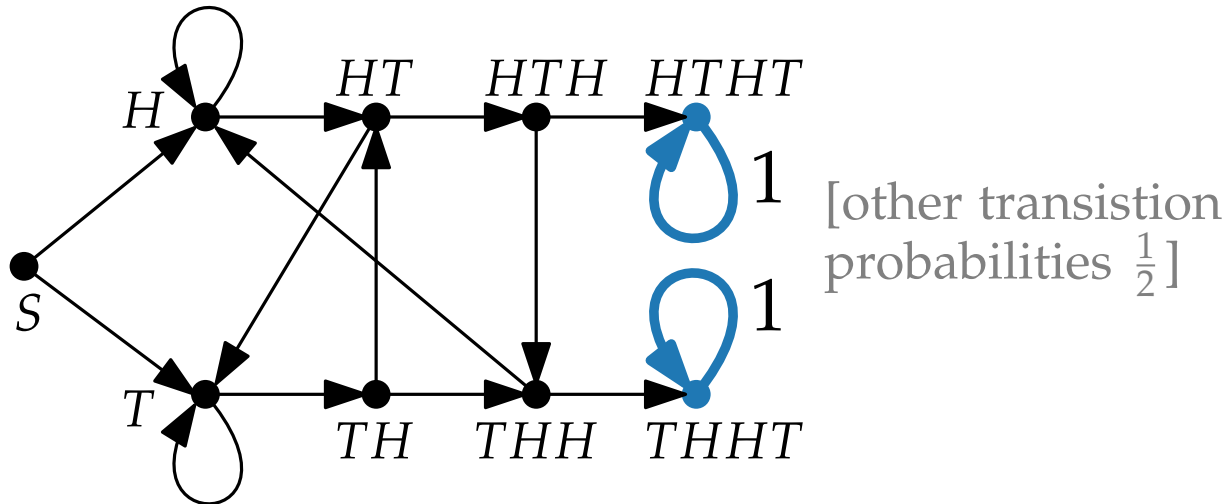
$$P_S = \frac{1}{2}P_H + \frac{1}{2}P_T$$

$$P_T = \frac{1}{2}P_T + \frac{1}{2}P_{TH}$$

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What's reached first?

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$$P_S = \frac{1}{2}P_H + \frac{1}{2}P_T$$

$$P_T = \frac{1}{2}P_T + \frac{1}{2}P_{TH}$$

$$P_H = \frac{1}{2}P_H + \frac{1}{2}P_{HT}$$

...and so on

$$P_{THHT} = 0$$

$$P_{HTHT} = 1$$

Markov Chains

Given the present, the future is independent of the past.



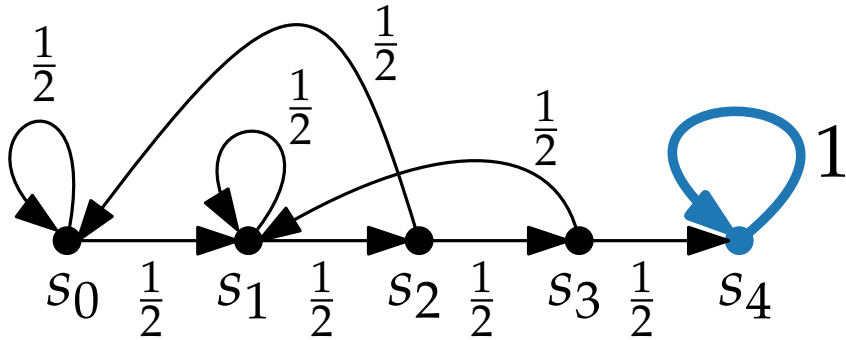
1856 –1922

Markov Chains

Given the present, the future is independent of the past.



1856 –1922



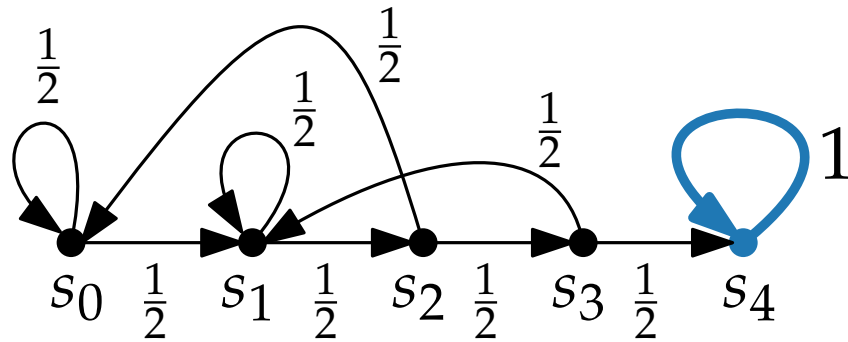
- *states:* s_0, s_1, s_2, s_3, s_4

Markov Chains

Given the present, the future is independent of the past.



1856 – 1922



• states: s_0, s_1, s_2, s_3, s_4

Transition Matrix

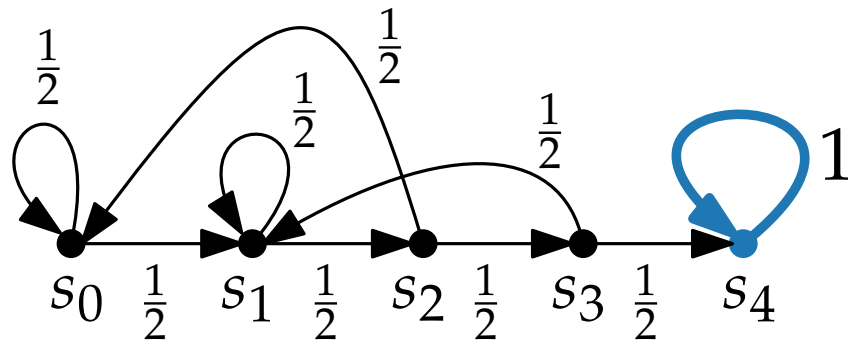
$$\begin{matrix} \text{from} \\ s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \end{matrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} \\ \\ \\ \\ \text{to} \end{matrix} \begin{matrix} s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \end{matrix} = P$$

Markov Chains

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1856 – 1922



Transition Matrix

$$\begin{array}{c} \text{from} \\ s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \end{array} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = P$$

to
 $s_0 \quad s_1 \quad s_2 \quad s_3 \quad s_4$

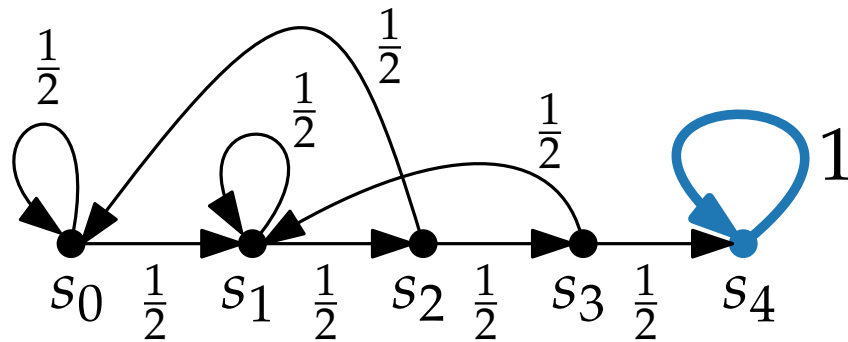
- *states*: s_0, s_1, s_2, s_3, s_4
- s_0, s_1, s_2, s_3 are *transient states* :
only visited **finitely often**

Markov Chains

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1856 – 1922



Transition Matrix

$$\begin{array}{c} \text{from} \\ \begin{matrix} s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \end{matrix} \end{array} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{array}{c} \\ \\ \\ \\ \text{to} \end{array} \begin{matrix} s_0 & s_1 & s_2 & s_3 & s_4 \end{matrix} = P$$

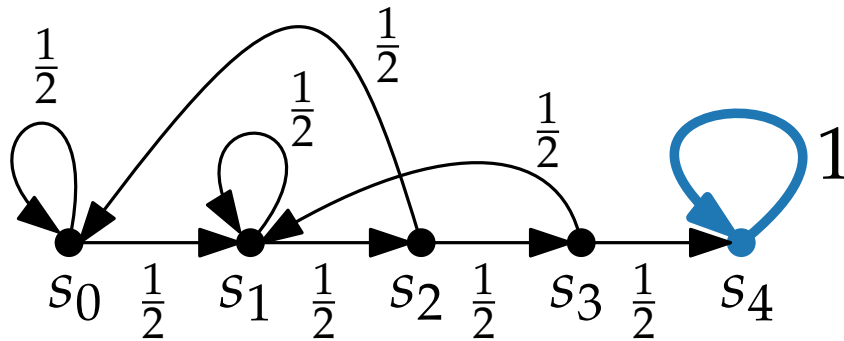
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Markov Chains

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1856 – 1922



Transition Matrix

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to
 $s_0 \quad s_1 \quad s_2 \quad s_3 \quad s_4$

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- s_4 is also an *absorbing state* : it is inescapable.



Maastricht University

Department of Data Science and Knowledge Engineering

MATSE: Operations Research

Lecture 6. Markov Chains

[based on notes by: Frank Thuijsman and Steven Chaplick]

1856 –1922

Markov Chains

Given the present, the future is independent of the past.



1856 –1922



Markov Chains

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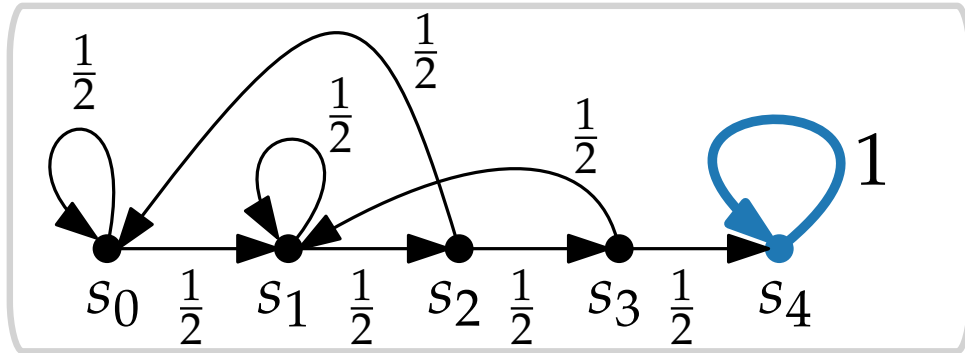


FIGURE 1.1. A randomly jumping frog. Whenever he tosses heads, he jumps to the other lily pad.

Levin, David A., and Yuval Peres. Markov chains and mixing times. Vol. 107. American Mathematical Soc., 2017.

Markov Chains

Given the present, the future is independent of the past.



- *states:* s_0, s_1, s_2, s_3, s_4

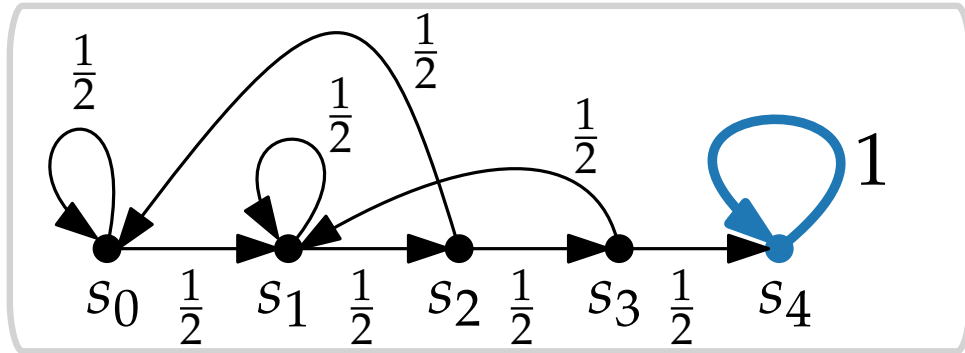
1856 – 1922

2 - 3



Markov Chains

Given the present, the future is independent of the past.



- states: s_0, s_1, s_2, s_3, s_4

1856 – 1922



from

Transition Matrix

$$\begin{matrix}
 s_0 \\
 s_1 \\
 s_2 \\
 s_3 \\
 s_4
 \end{matrix}
 \begin{bmatrix}
 \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\
 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\
 \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\
 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\
 0 & 0 & 0 & 0 & 1
 \end{bmatrix}
 = P$$

to

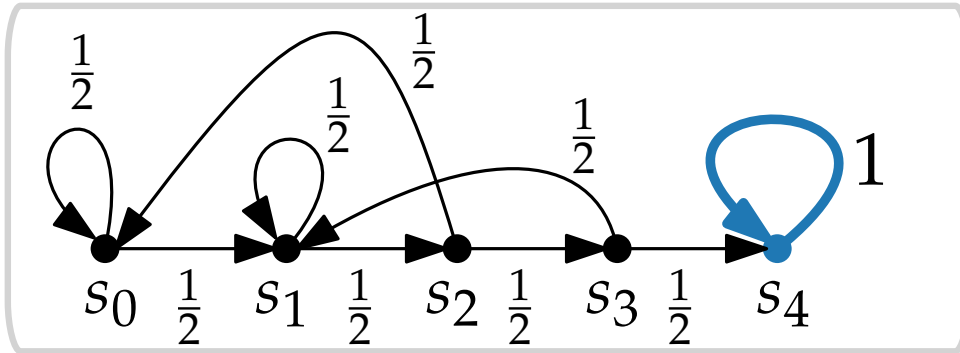
$s_0 \quad s_1 \quad s_2 \quad s_3 \quad s_4$

1856 – 1922



Markov Chains

Given the present, the future is independent of the past.



- *states:* s_0, s_1, s_2, s_3, s_4
- Note: sum of any row of $P = 1$

from

Transition Matrix

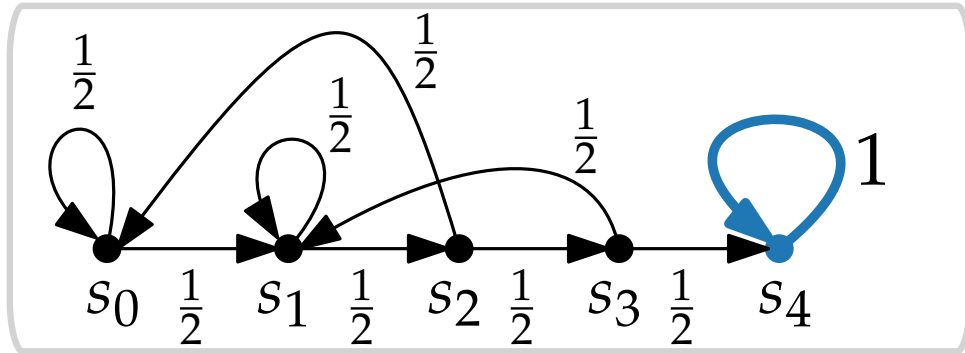
$$\begin{matrix} s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \end{matrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = P$$

to

$s_0 \quad s_1 \quad s_2 \quad s_3 \quad s_4$

Markov Chains

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only visited **finitely often**

1856 – 1922



from

Transition Matrix

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to

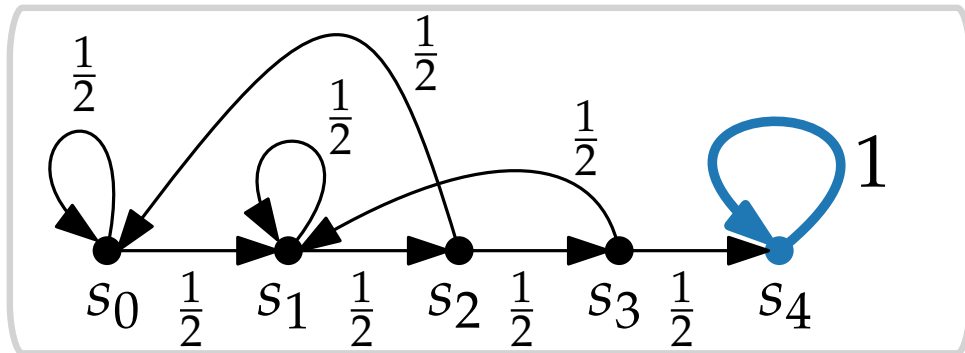
$s_0 \quad s_1 \quad s_2 \quad s_3 \quad s_4$

1856 – 1922



Markov Chains

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from

Transition Matrix

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 s_1 \\
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 \begin{bmatrix}
 \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\
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 \end{bmatrix}
 = P$$

to

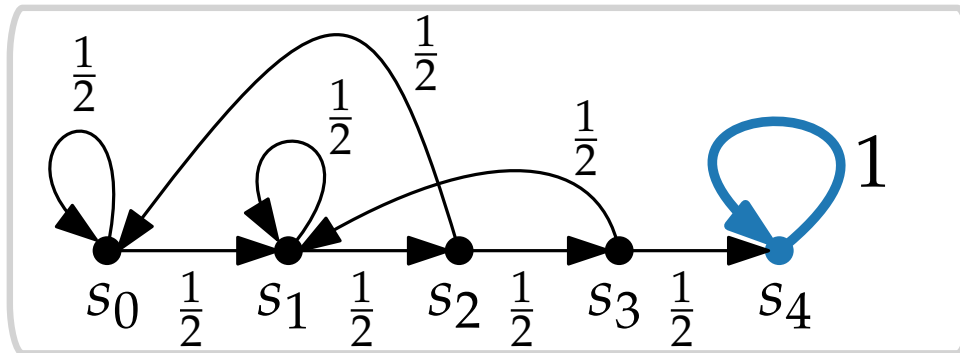
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1856 – 1922



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from

Transition Matrix

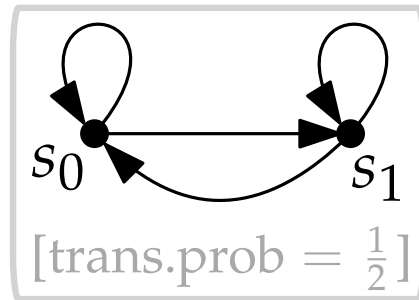
$$\begin{matrix}
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 \end{matrix}
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 = P$$

to

$s_0 \quad s_1 \quad s_2 \quad s_3 \quad s_4$

Recurrent vs Absorbing

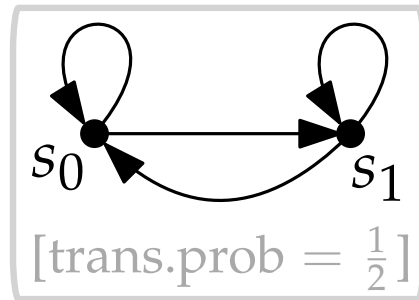
Consider the Markov chain



$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Recurrent vs Absorbing

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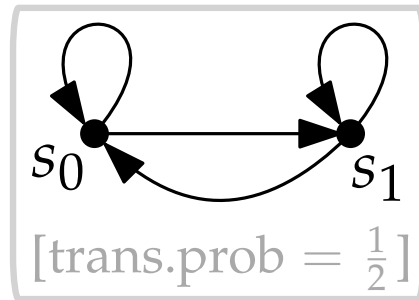


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- All states are recurrent
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Recurrent vs Absorbing

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- All states are recurrent
- None of them is absorbing
- Absorbing $\Rightarrow$ Recurrent
The vice-versa is not true.

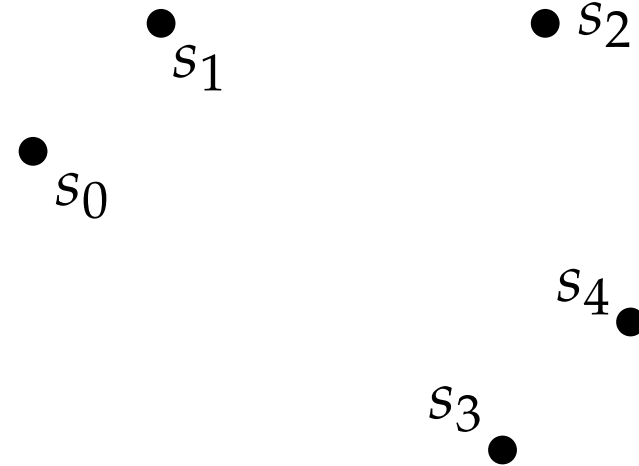
A Bigger Example

from \ to	s_0	s_1	s_2	s_3	s_4	
s_0	0.6	0.3	0	0.1	0	:
s_1	0.4	0.4	0.1	0	0.1	
s_2	0	0	1	0	0	
s_3	0	0	0	0.4	0.6	
s_4	0	0	0	0.3	0.7	

A Bigger Example

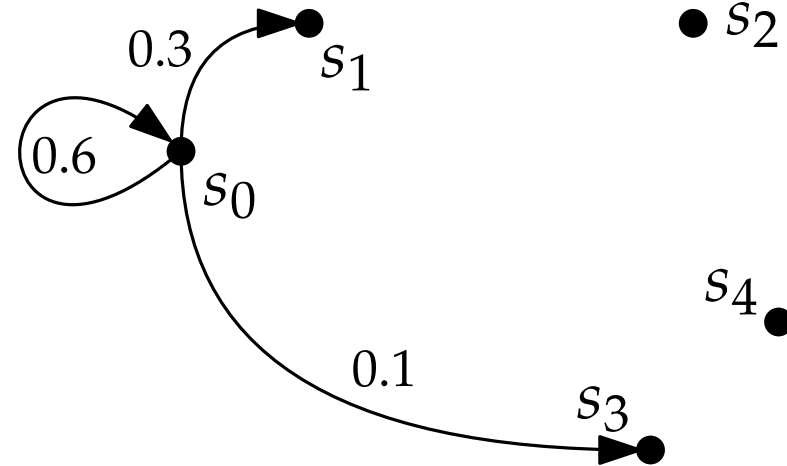
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s_4	0	0	0	0.3	0.7

:



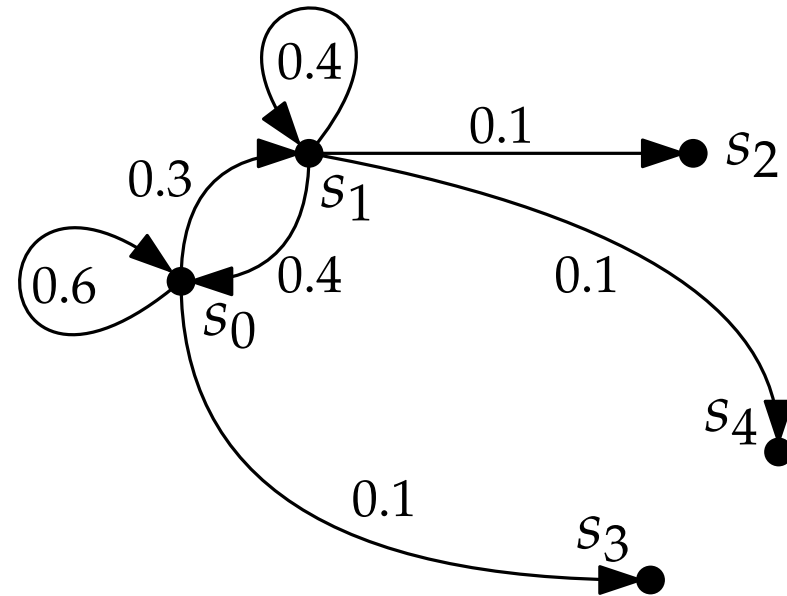
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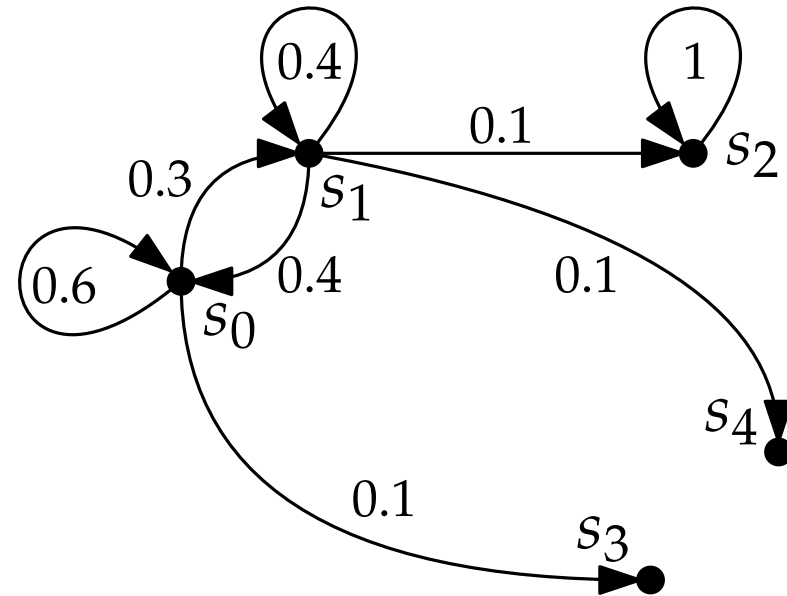
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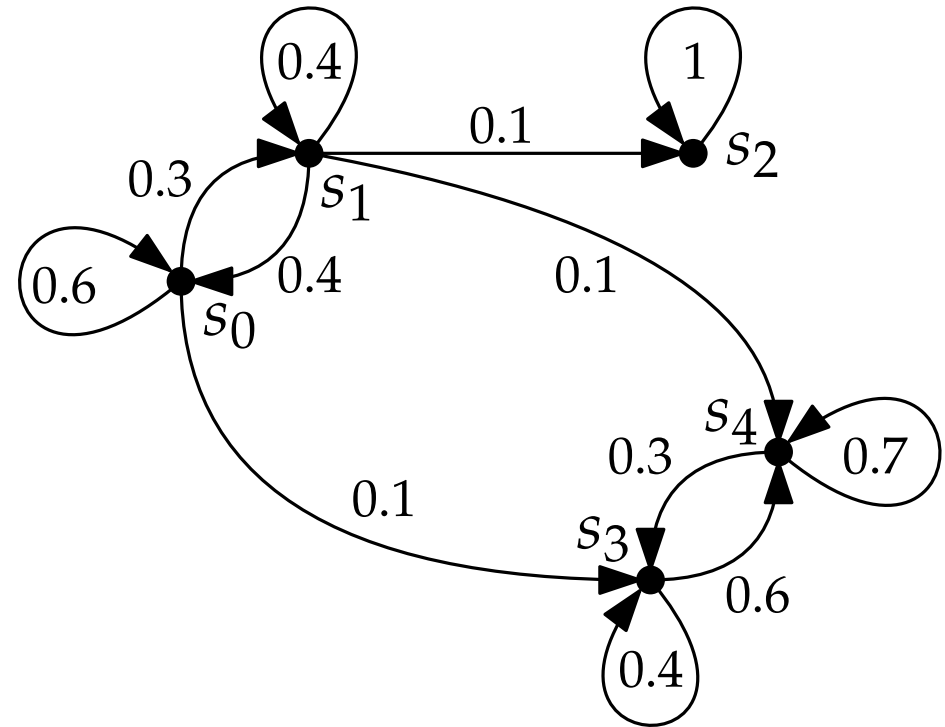
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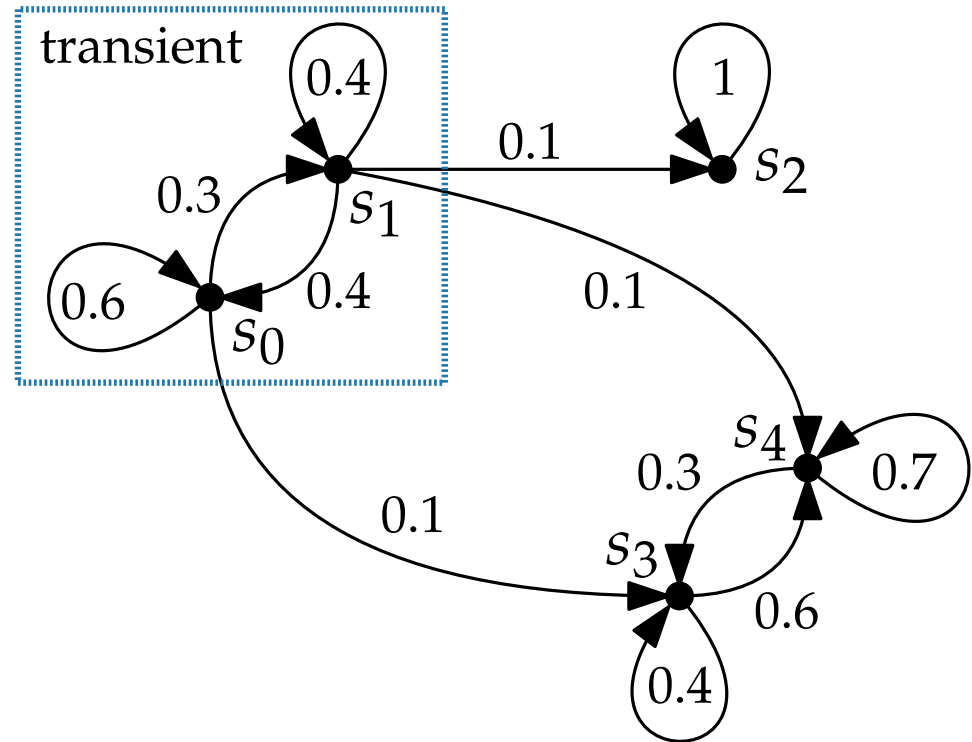
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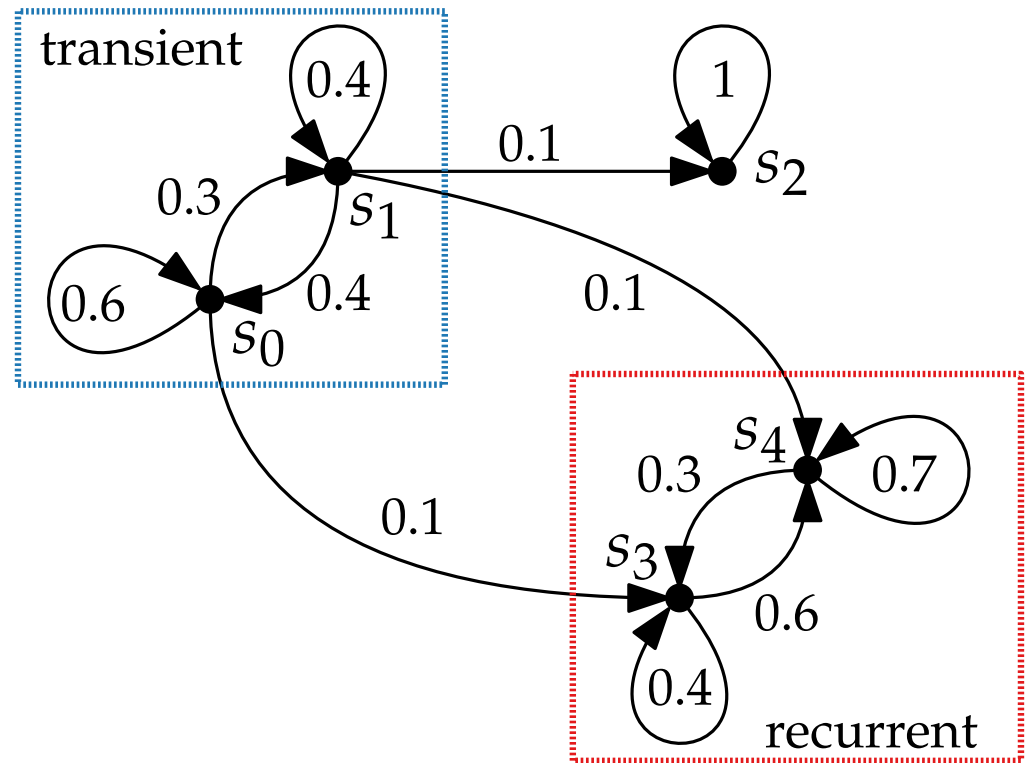
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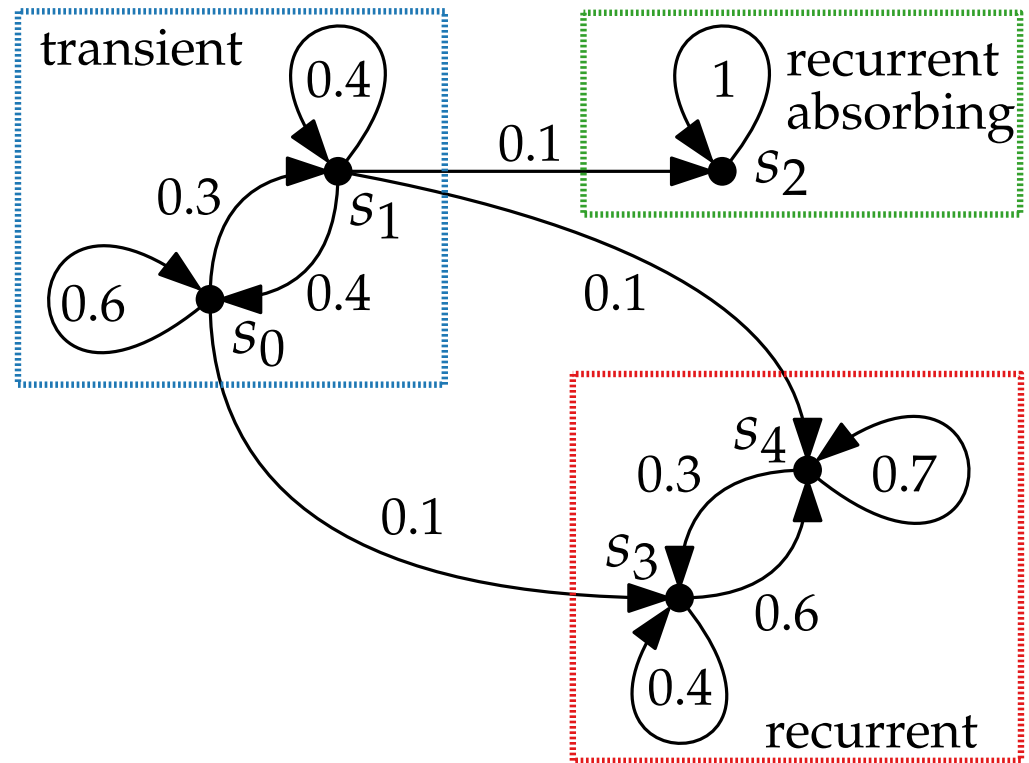
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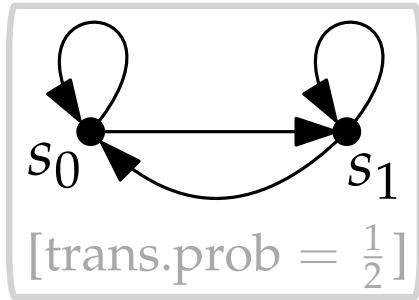


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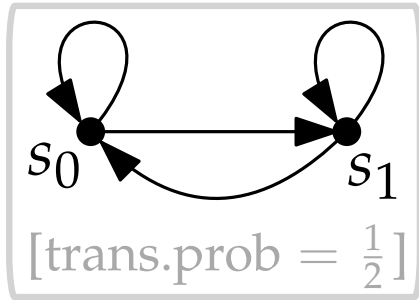
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Long Run Functions: π_i

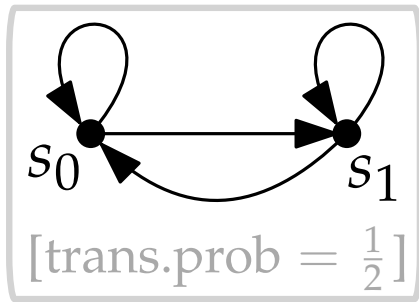


Long Run Functions: π_i



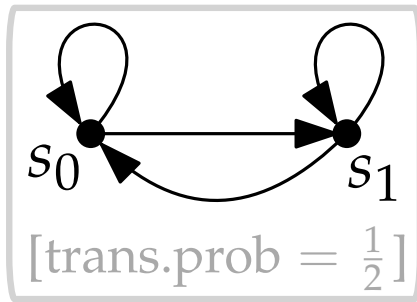
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Long Run Functions: π_i



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- *irreducible* $\coloneqq$ **all** pairs of states communicate.

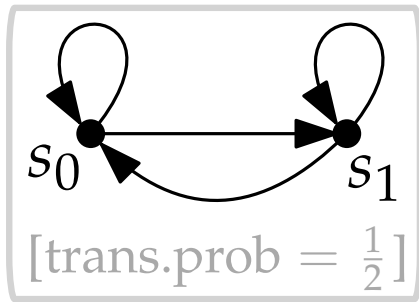
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Long Run Functions: π_i



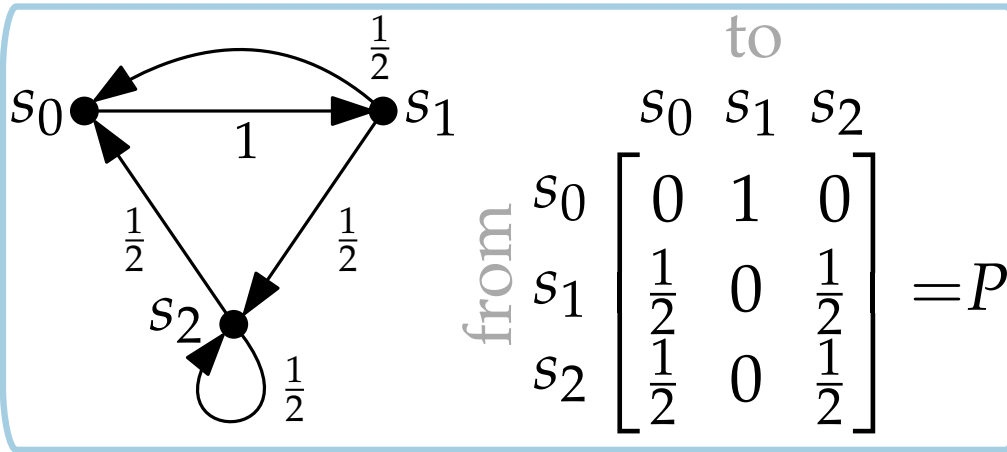
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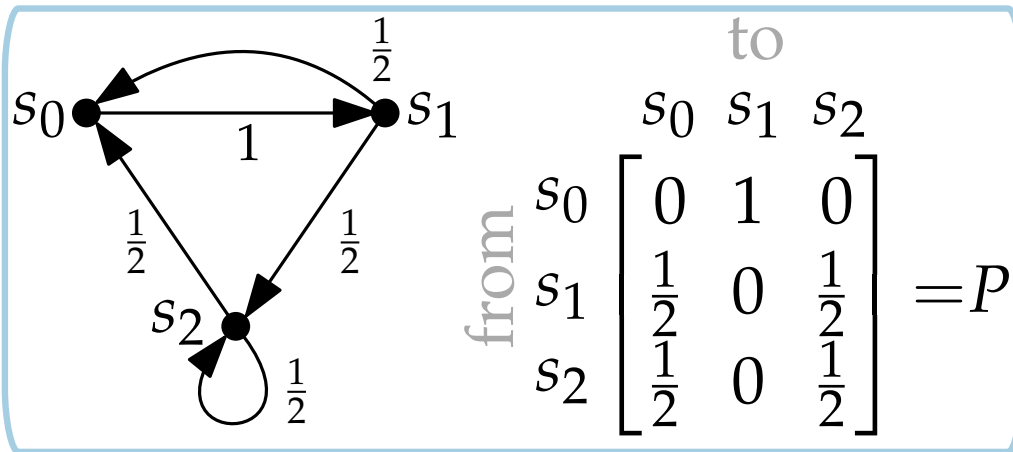
- *steady-state probabilities*: $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots, \pi_k)$

For a Markov chain, π_i is the fraction all time spent in state s_i .

Finding Steady-State Probabilities

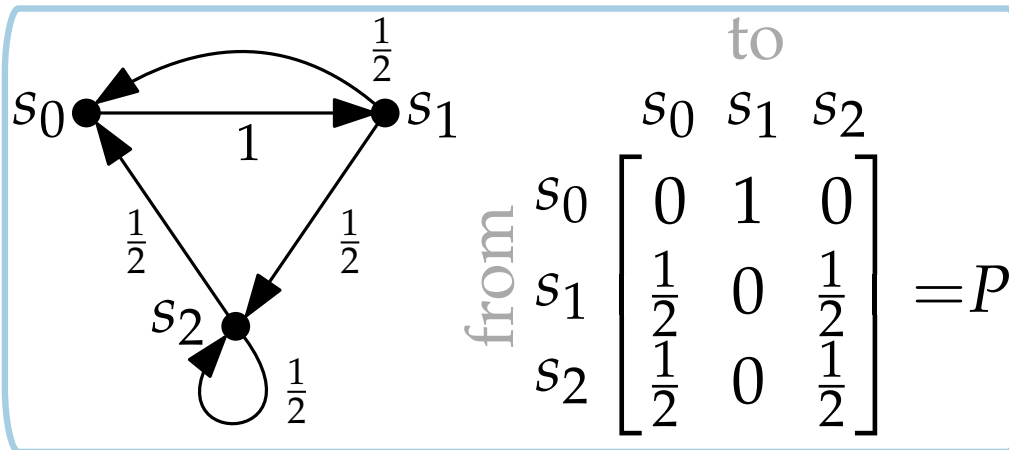


Finding Steady-State Probabilities



How to calculate $\pi = (\pi_0, \pi_1, \pi_2)$?

Finding Steady-State Probabilities



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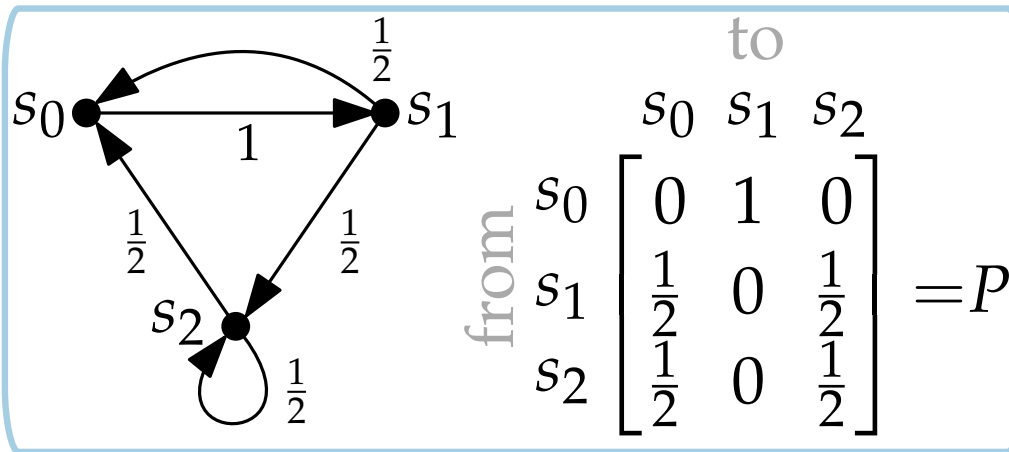
What properties hold for π ?

[when all pairs communicate]

$$\sum_i \pi_i = 1, \text{ and } \pi P = \pi$$

[Prob(being at s_i) = Prob(getting to s_i)
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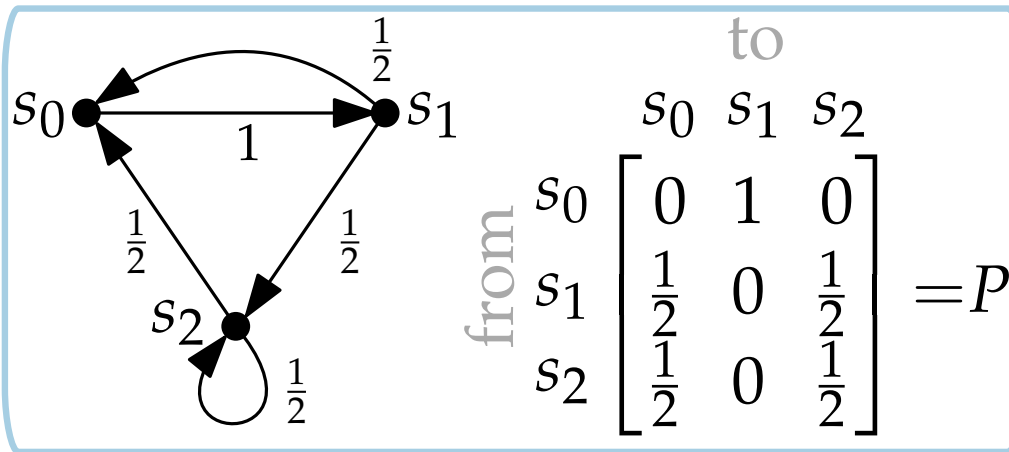
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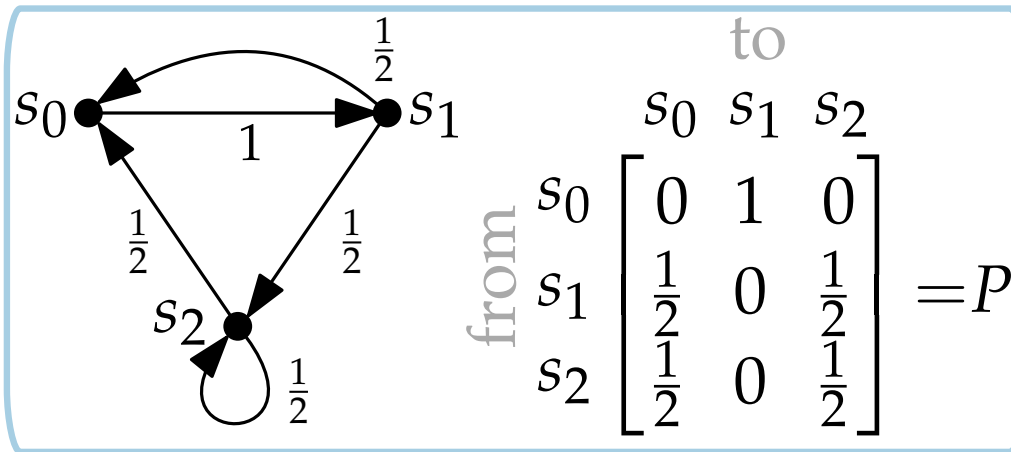
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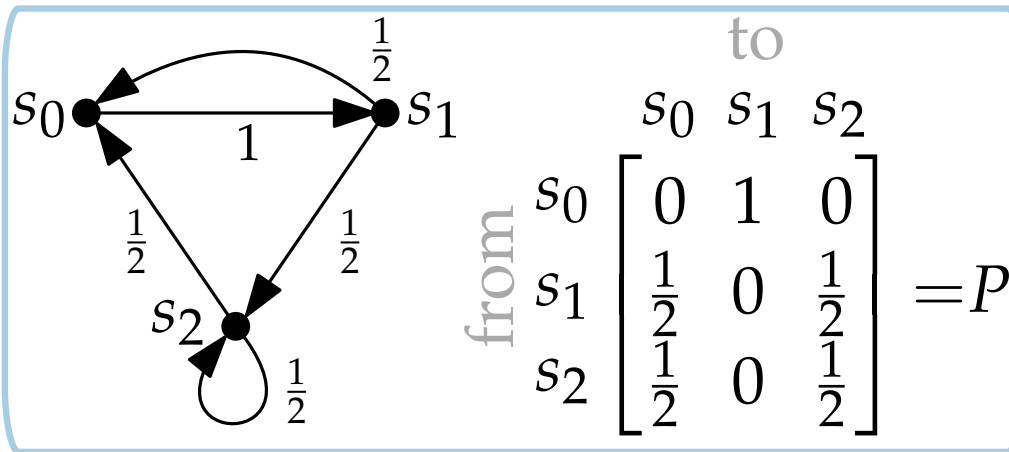
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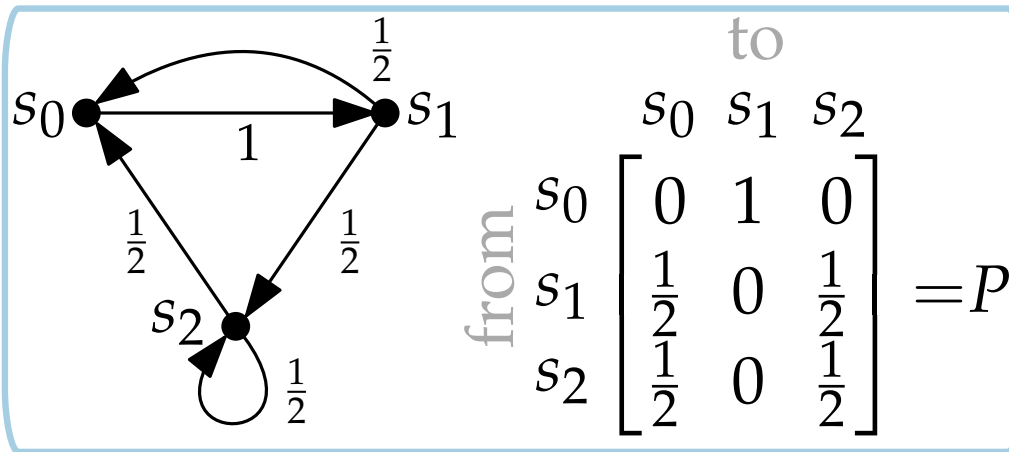
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Finding Steady-State Probabilities



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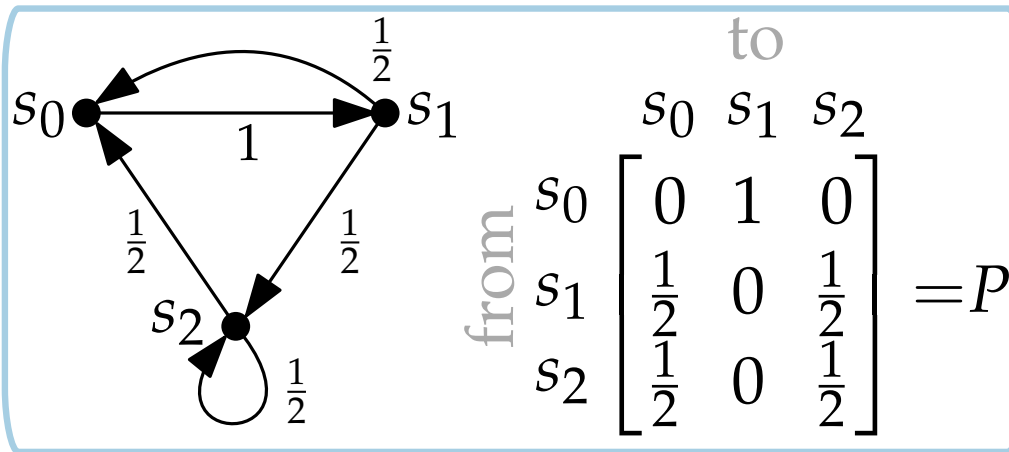
Fraction In = Fraction Out

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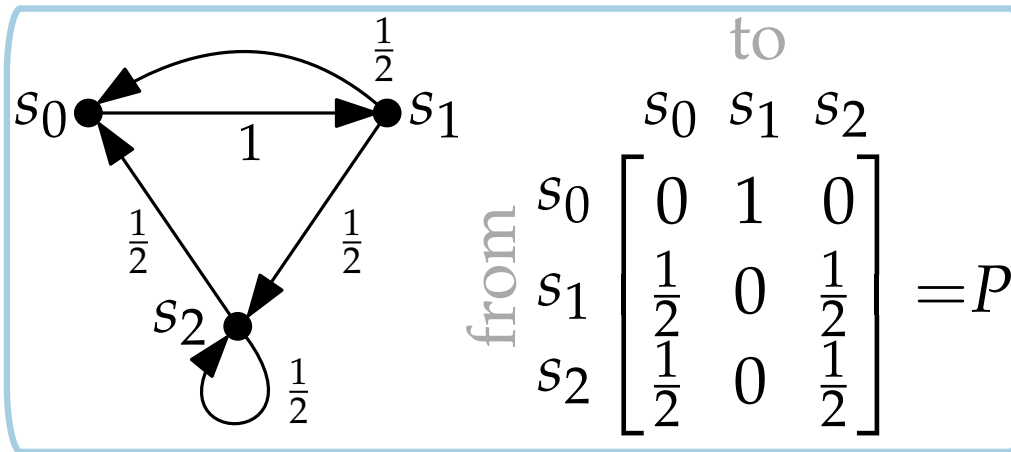
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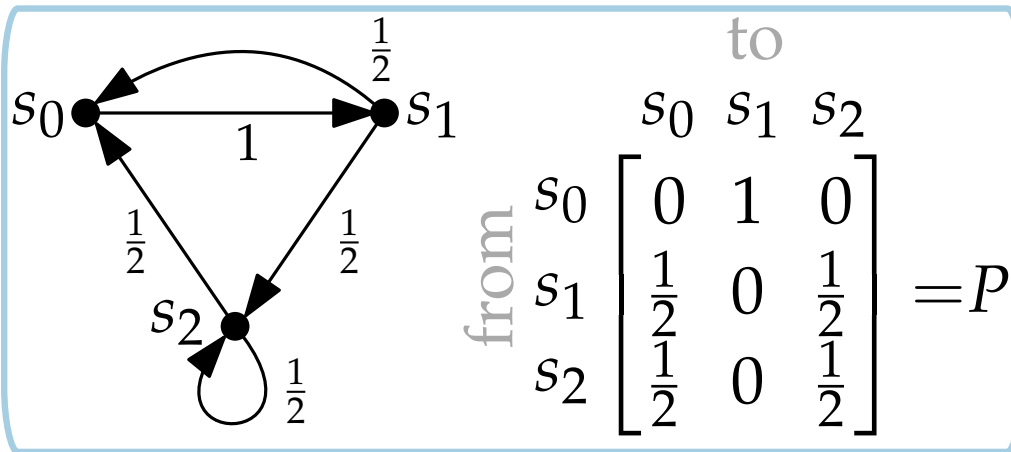
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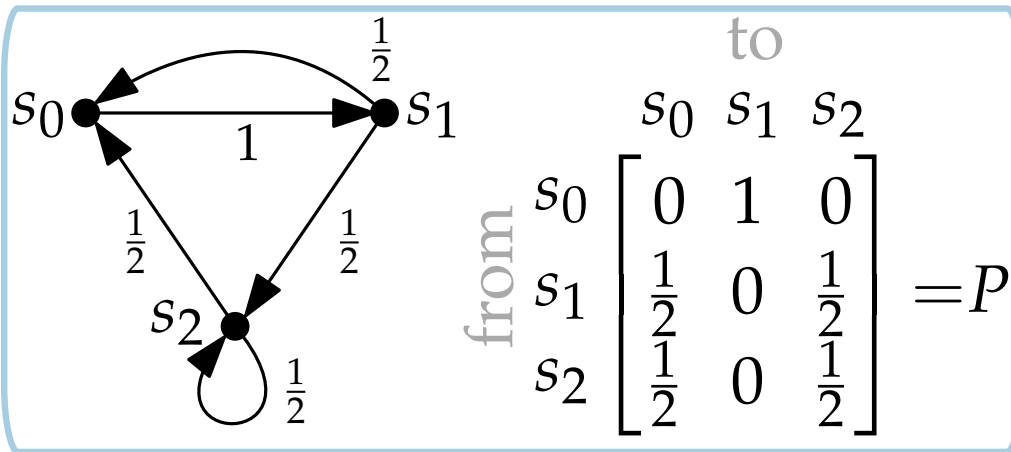
$$\pi = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

“Short” run?



$P :=$ transition matrix, **1-step**
transition probabilities

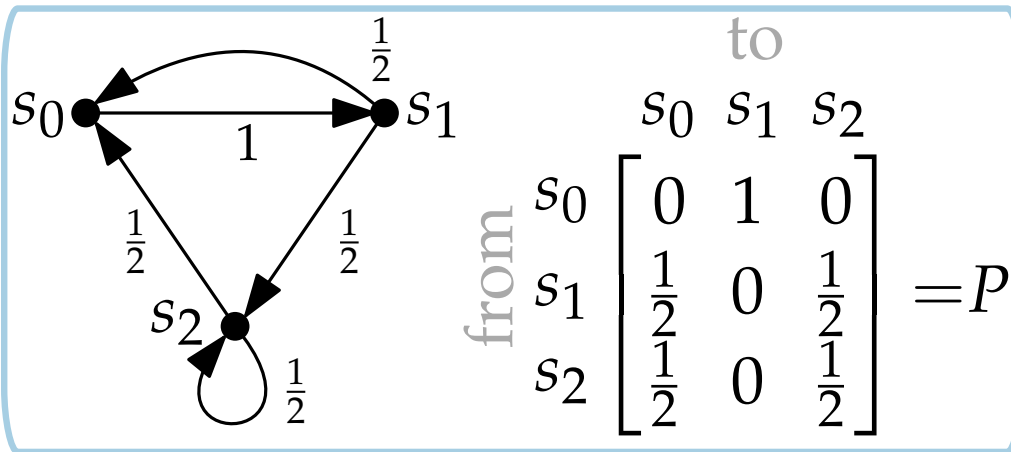
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Def: $P^{(t)}$ is the t -step transition matrix, i.e., entry ij of $P^{(t)}$,
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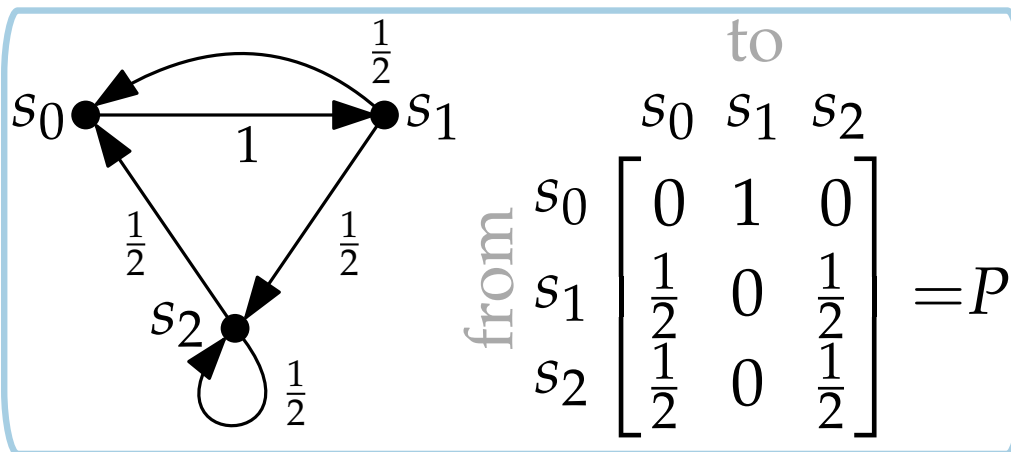


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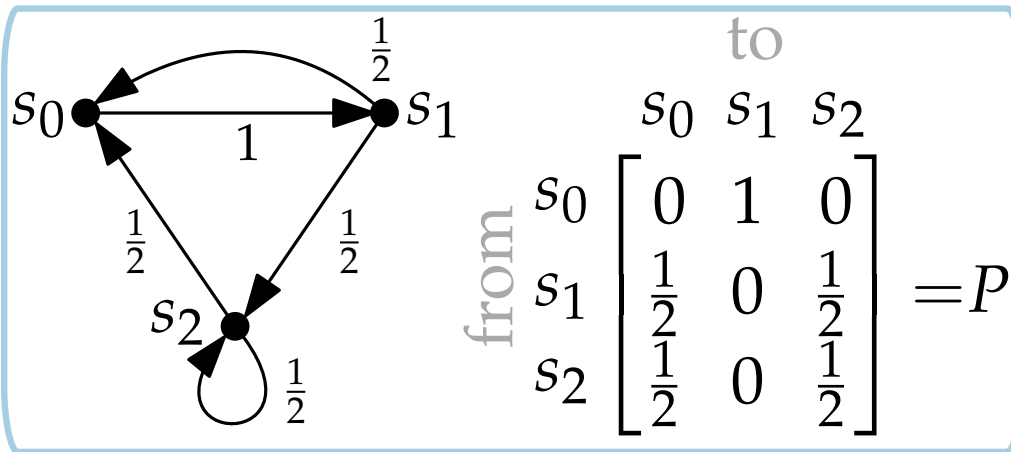
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$P_{ij}^{(2)}$ = probability we go from i to *somewhere* and then to j

$$= \sum_b \text{Prob}(i \rightarrow b \rightarrow j) = \sum_b P_{ib} P_{bj}$$

“Short” run?



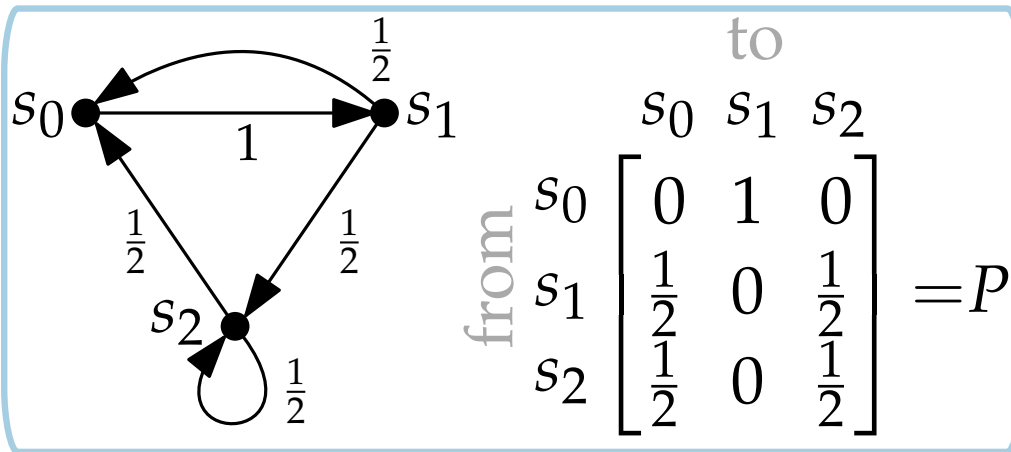
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Therefore, $P^{(2)} = PP = P^2 =$

$$\begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

"Longer" run

What about $P^{(t)}$?

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Therefore, $P^{(t)} = P^{(t-1)} P = P^{t-1} P = P^t$

"Longer" run

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Therefore, $P^{(t)} = P^{(t-1)} P = P^{t-1} P = P^t$

Can we relate long runs π to short runs $P^{(t)}$?

Maybe they relate in the **limit**?

“Long”run vs. “short” run

$P^t = P P P \dots P = t\text{-step transition matrix}$

“Long”run vs. “short” run

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$\infty\text{-step transition matrix } Q?$

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“Long”run vs. “short” run

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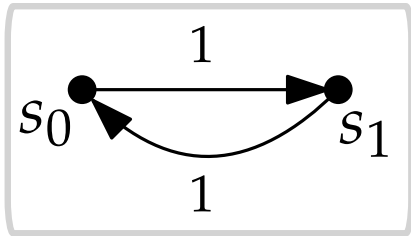
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“Long” run vs. “short” run

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$\infty\text{-step transition matrix } Q? \quad Q_{ij} = \lim_{t \rightarrow \infty} P_{ij}^t = \pi_j?$

Consider:

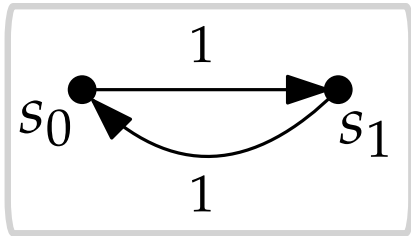


“Long” run vs. “short” run

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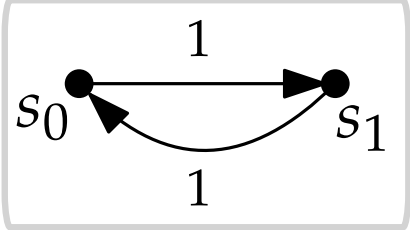
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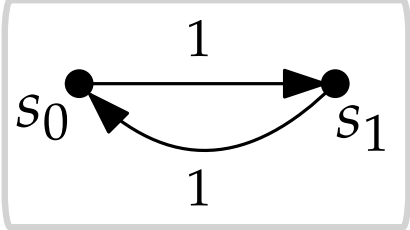
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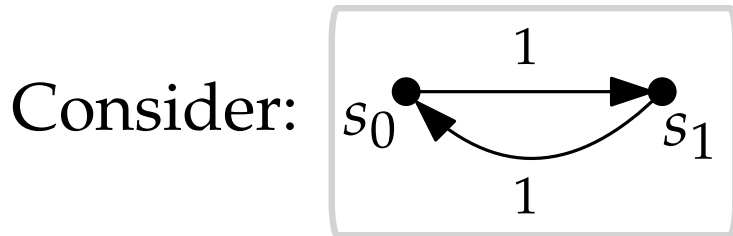
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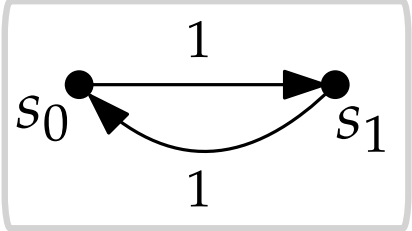
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$\rightsquigarrow$ No such limit for this *periodic* Markov chain!

Periodicity Properties

- **Period** of state i : the integer number t such that $p_{ii}^{(n)} = 0$ for all n but $t, 2t, 3t, \dots$

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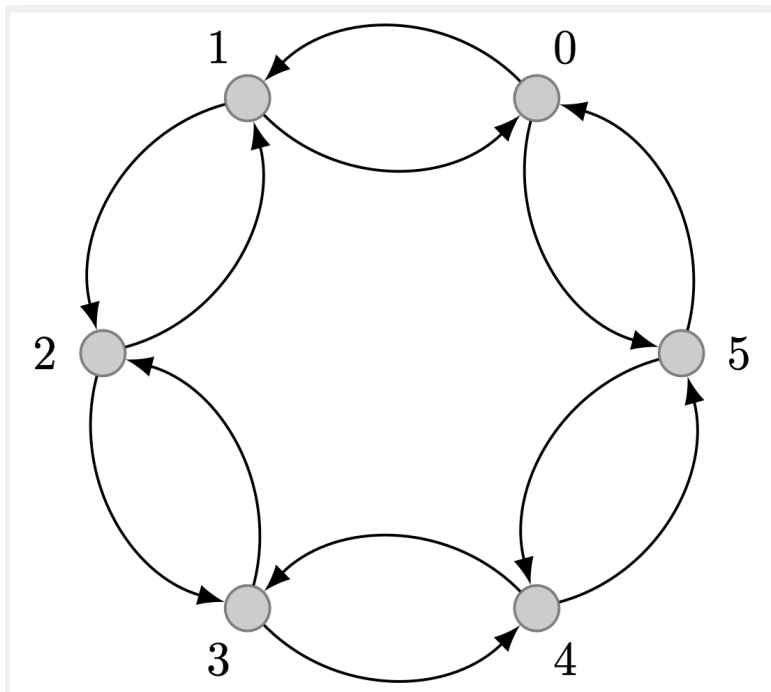
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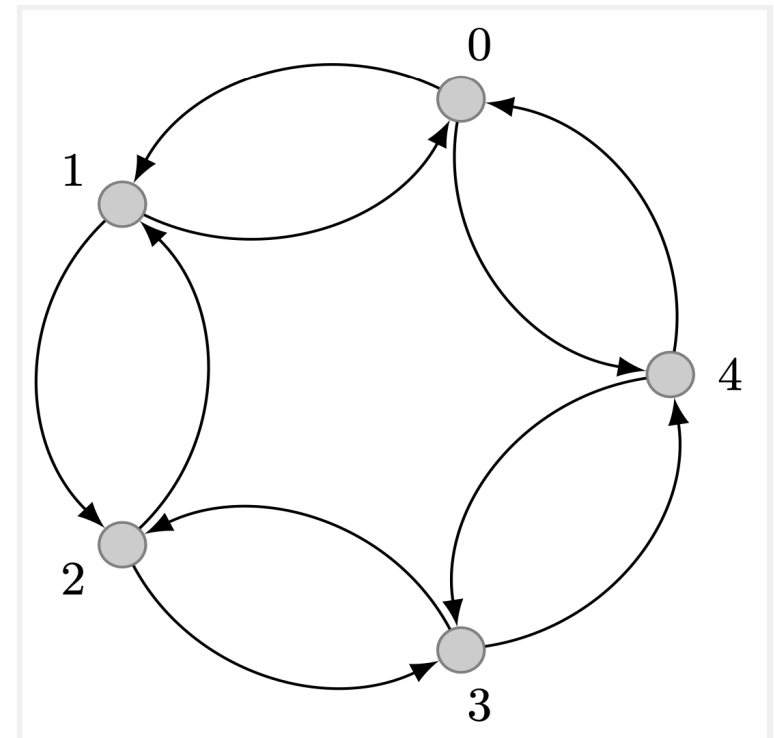
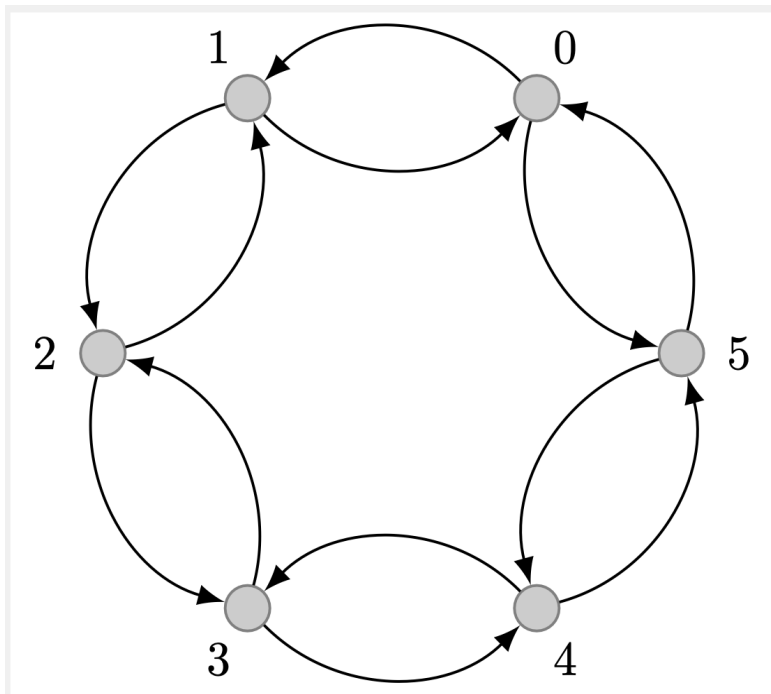


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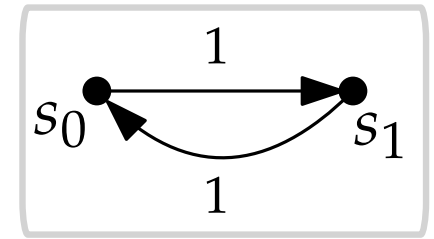
“Long”run vs. “short” run

$$Q_{ij} = \lim_{t \rightarrow \infty} P_{ij}^t = \pi_j$$

True if the Markov chain is *Aperiodic* and *Irreducible*

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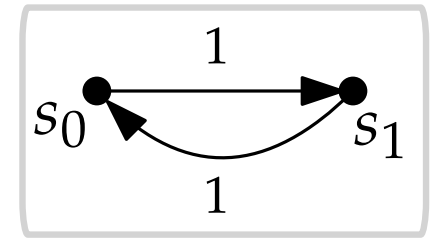
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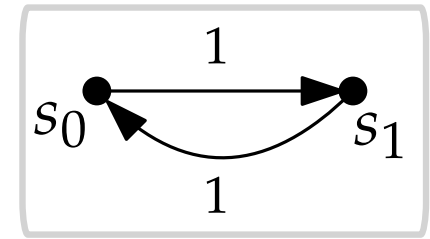
$Q_{ij}^{(t)} :=$ avg. time (over t -steps) in state j , starting in state i

$$Q^{(t)} = \frac{1}{t} \left(I + P^1 + P^2 + \dots + P^{t-1} \right) [I := \text{identity matrix}]$$

$$Q = \lim_{t \rightarrow \infty} \frac{1}{t} \left(I + P^1 + P^2 + \dots + P^t \right)$$

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row(i) of Q is π for starting state i .

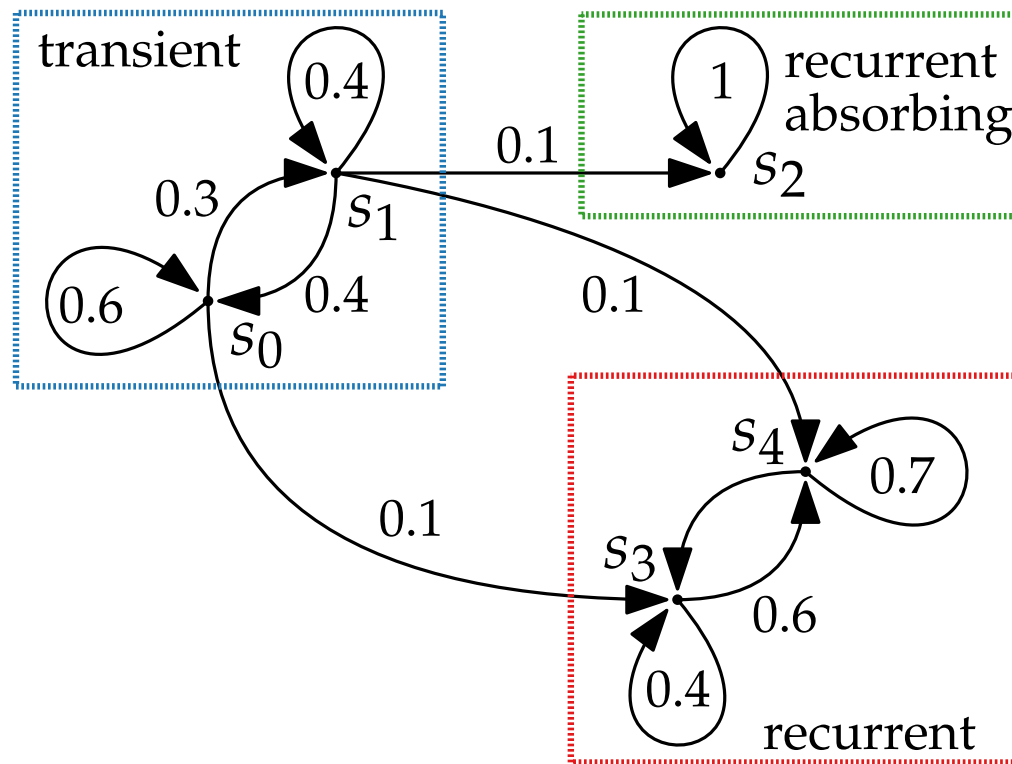
Irreducible $\Rightarrow$ each row of Q will be the same! e.g. $\pi_j = Q_{ij}$

Example

	from \ to	s_0	s_1	s_2	s_3	s_4
s_0		0.6	0.3	0	0.1	0
s_1		0.4	0.4	0.1	0	0.1
s_2		0	0	1	0	0
s_3		0	0	0	0.4	0.6
s_4		0	0	0	0.3	0.7

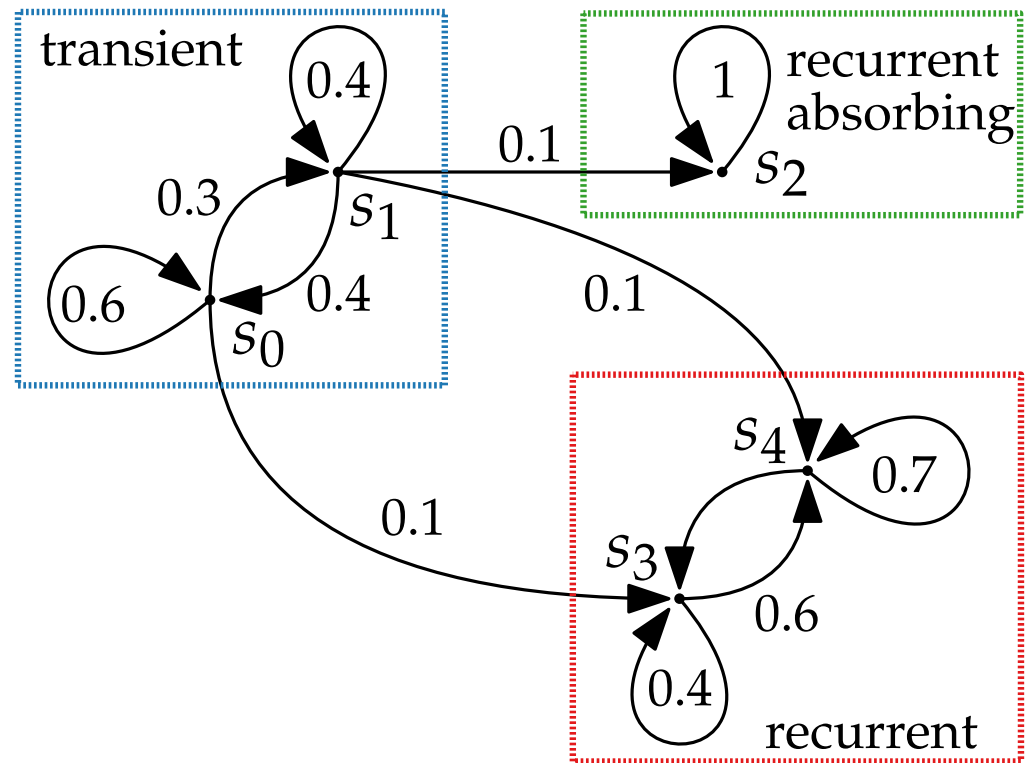
Starting in s_0 :

- Calculate the probability of reaching state (i) s_2 , (ii) s_3 , (iii) s_4 .
- Calculate $\pi = (\pi_0, \pi_1, \pi_2, \pi_3, \pi_4)$.
- On average (long run), how many **times** is (i) s_0 , (ii) s_1 visited?



Example

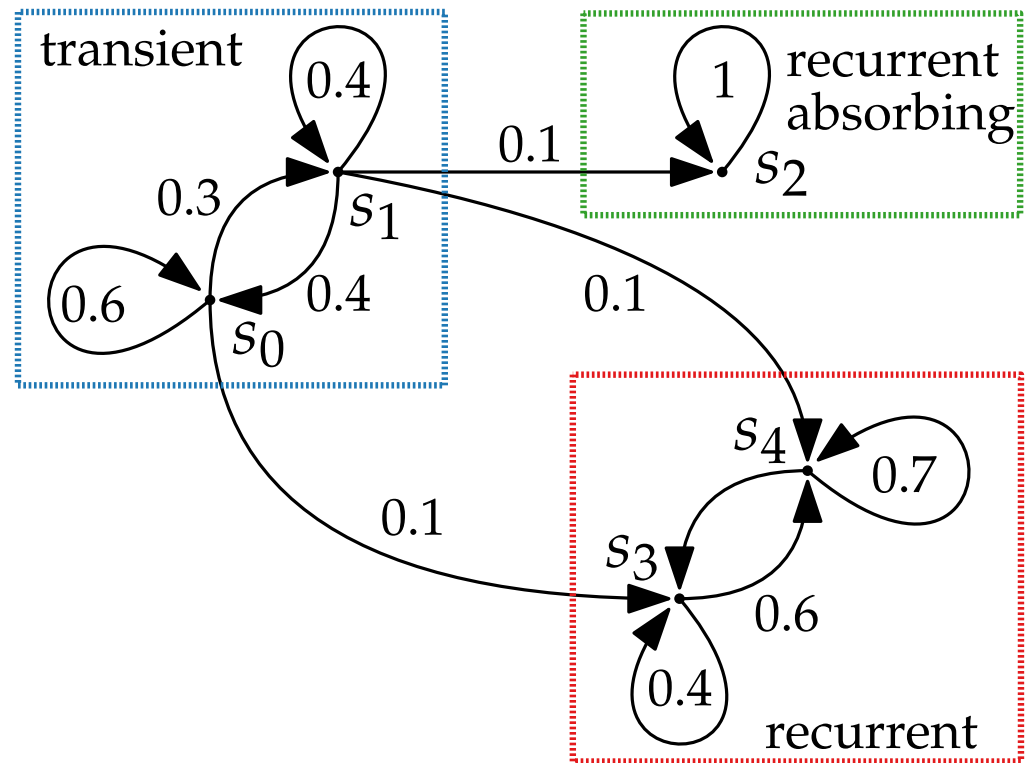
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Let P_x be the probability of reaching s_2 starting at x

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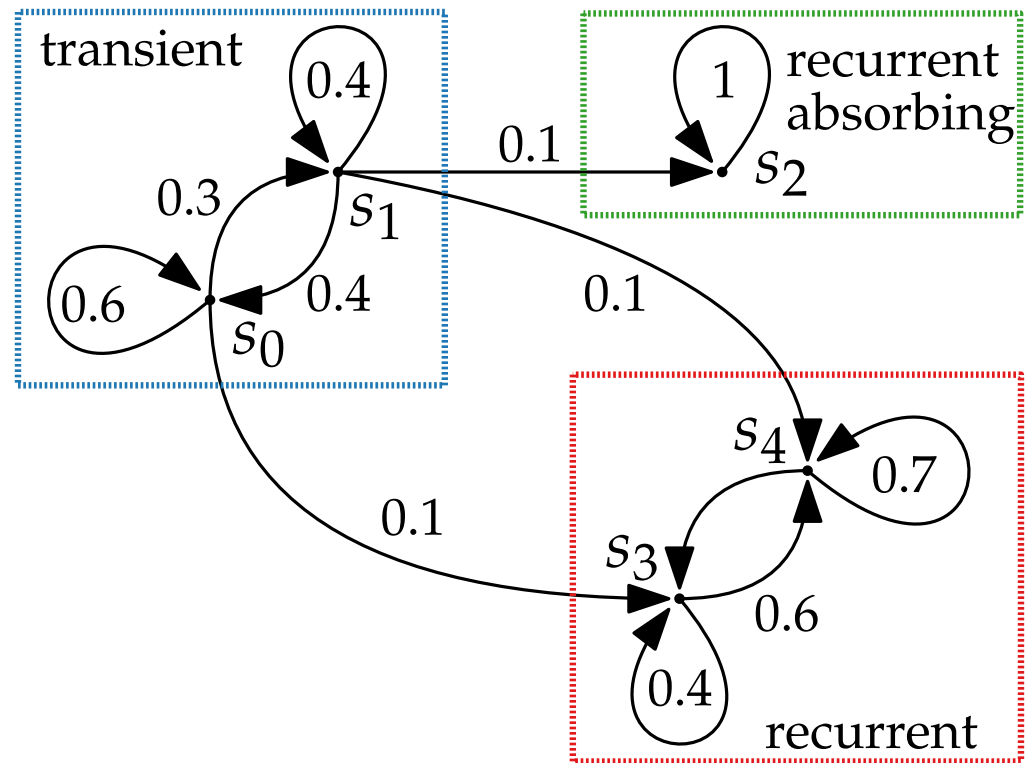


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$$\begin{cases} P_0 = 0.6P_0 + 0.3P_1 + 0.1P_3 \\ P_1 = 0.4P_0 + 0.4P_1 + 0.1P_2 + 0.1P_4 \\ P_2 = 1 \\ P_3 = P_4 = 0 \end{cases}$$

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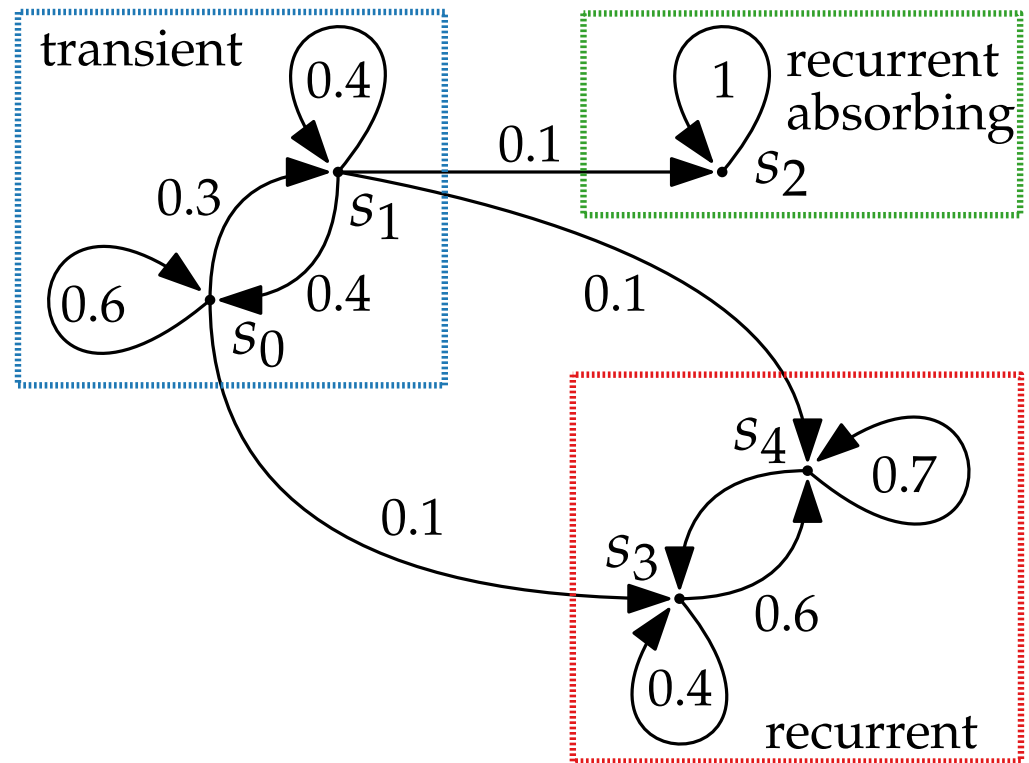


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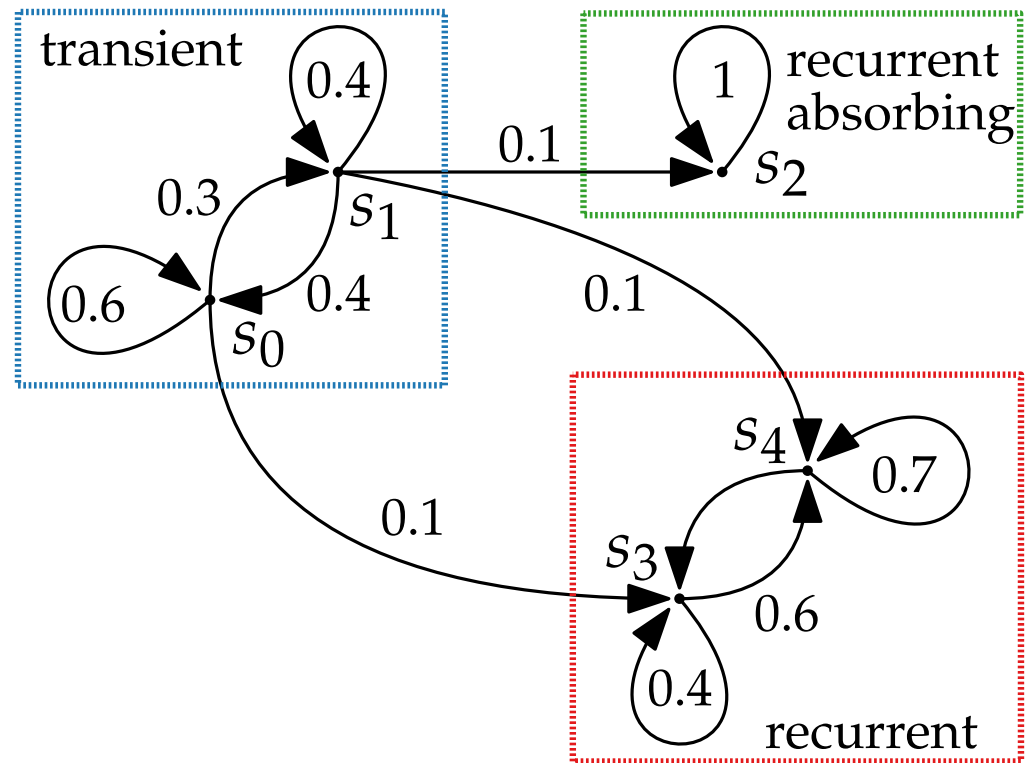
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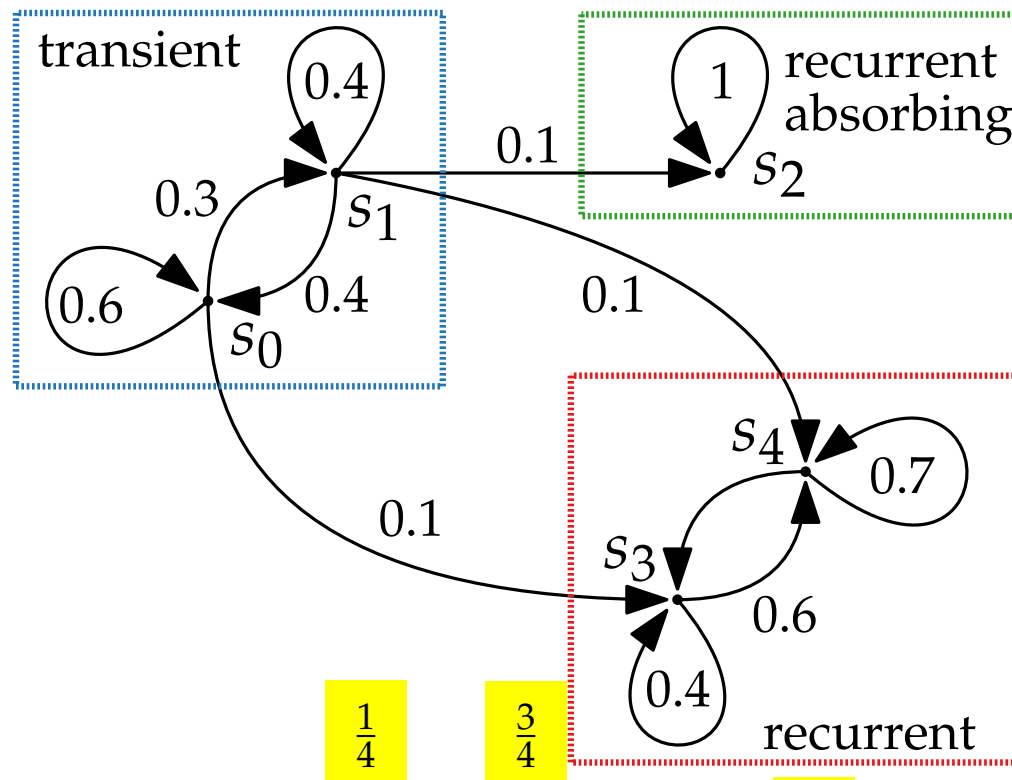
$$\Rightarrow P_{0,3} = P_{0,4} = 1 - P_{0,2} = 1 - \frac{1}{4} = \frac{3}{4}$$

Example

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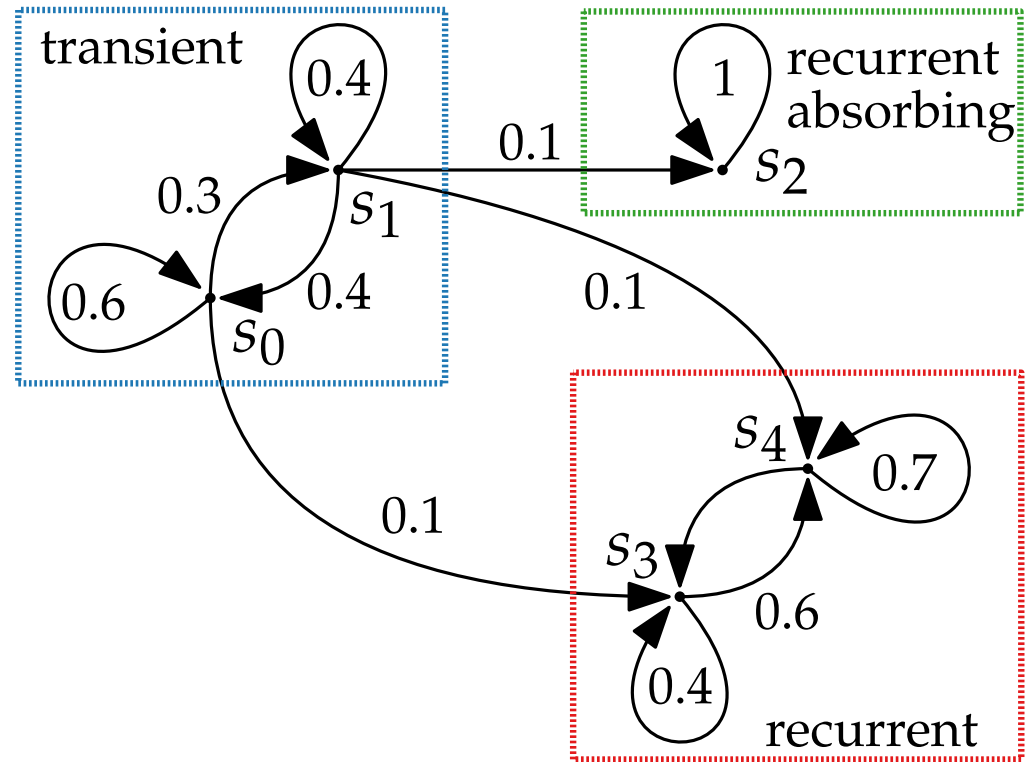
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to

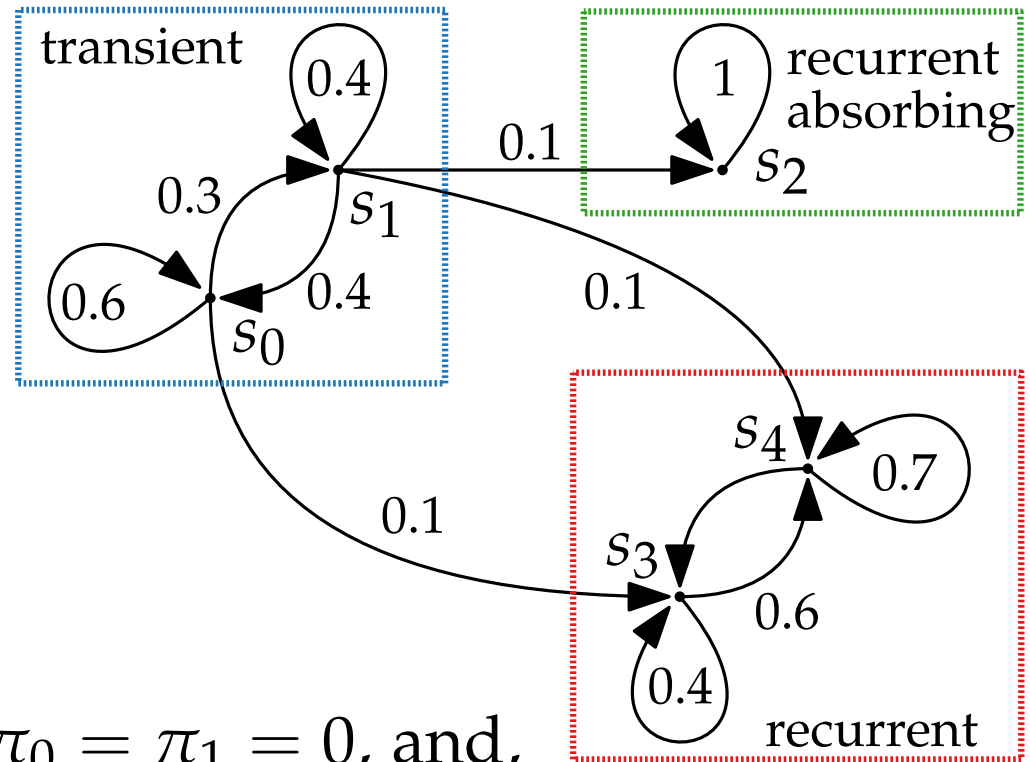
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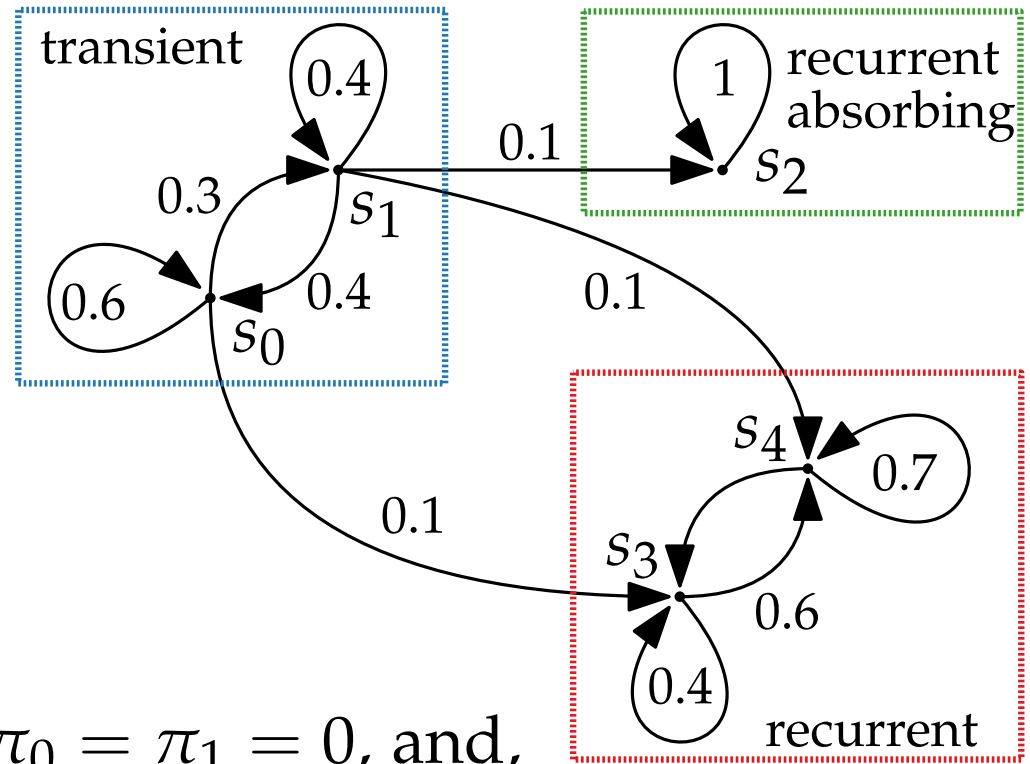


Since s_0 and s_1 are transient, $\pi_0 = \pi_1 = 0$, and, since s_2 is absorbing, from a)(i), $\pi_2 = \frac{1}{4}$.

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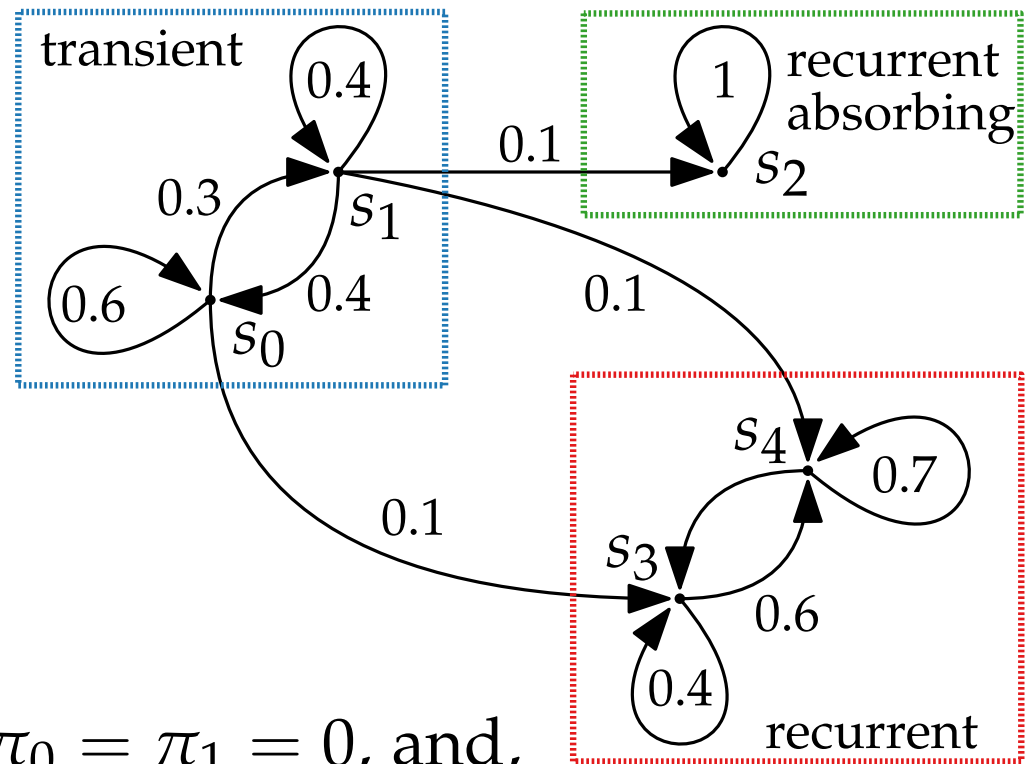
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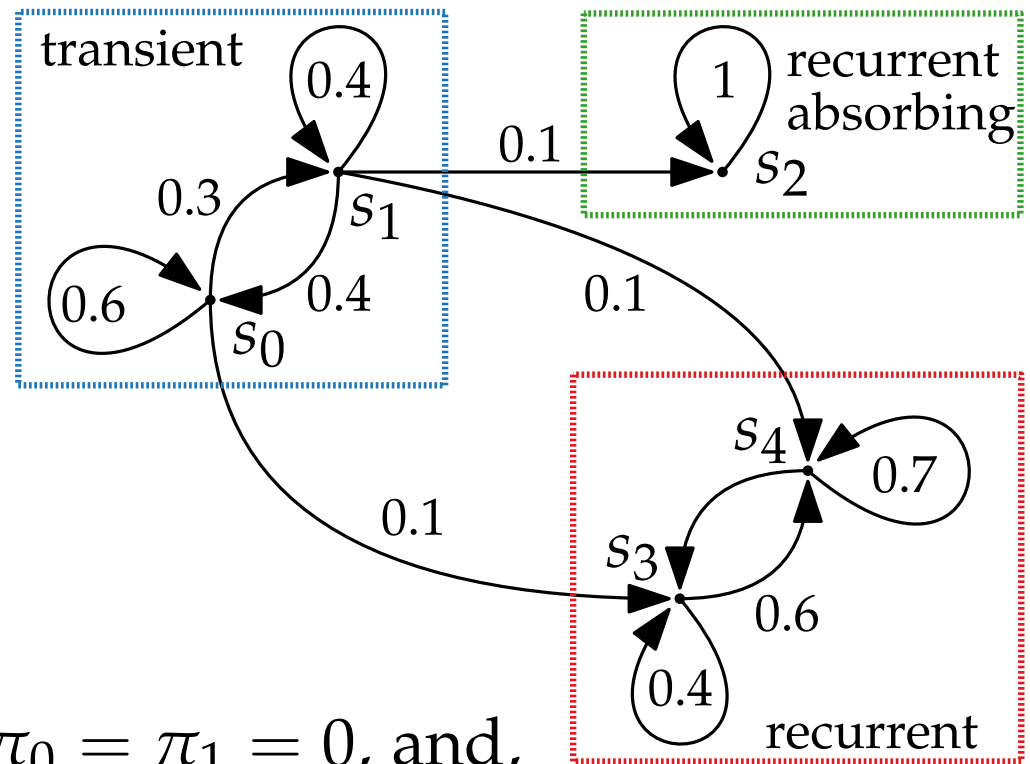
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Example

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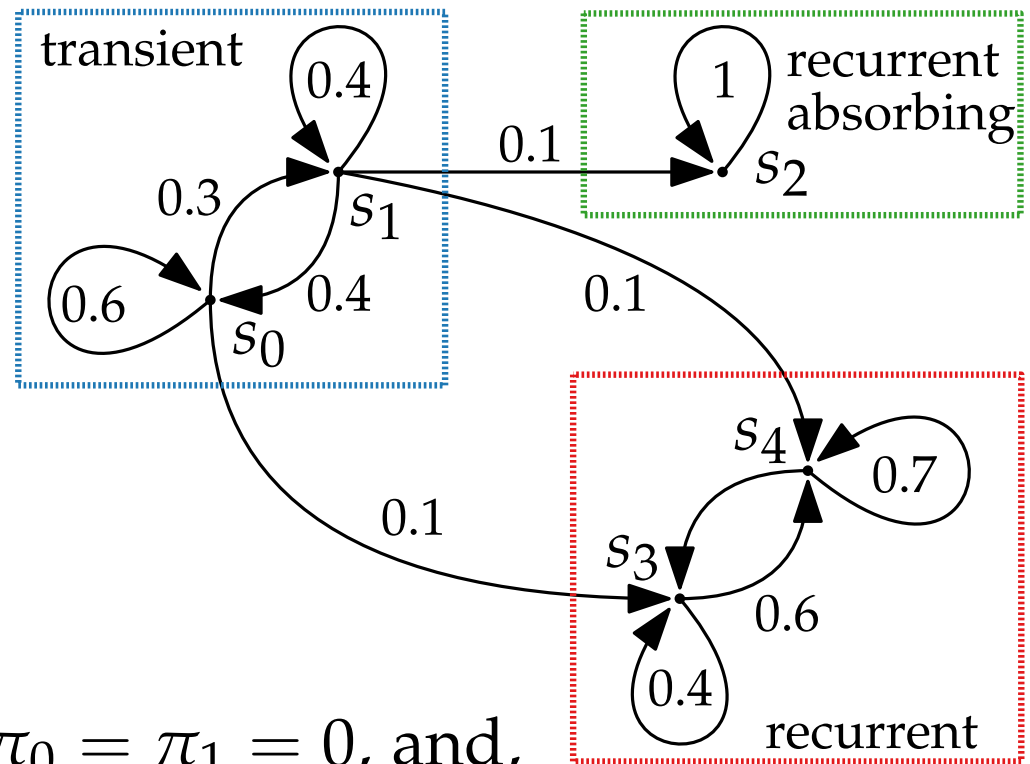
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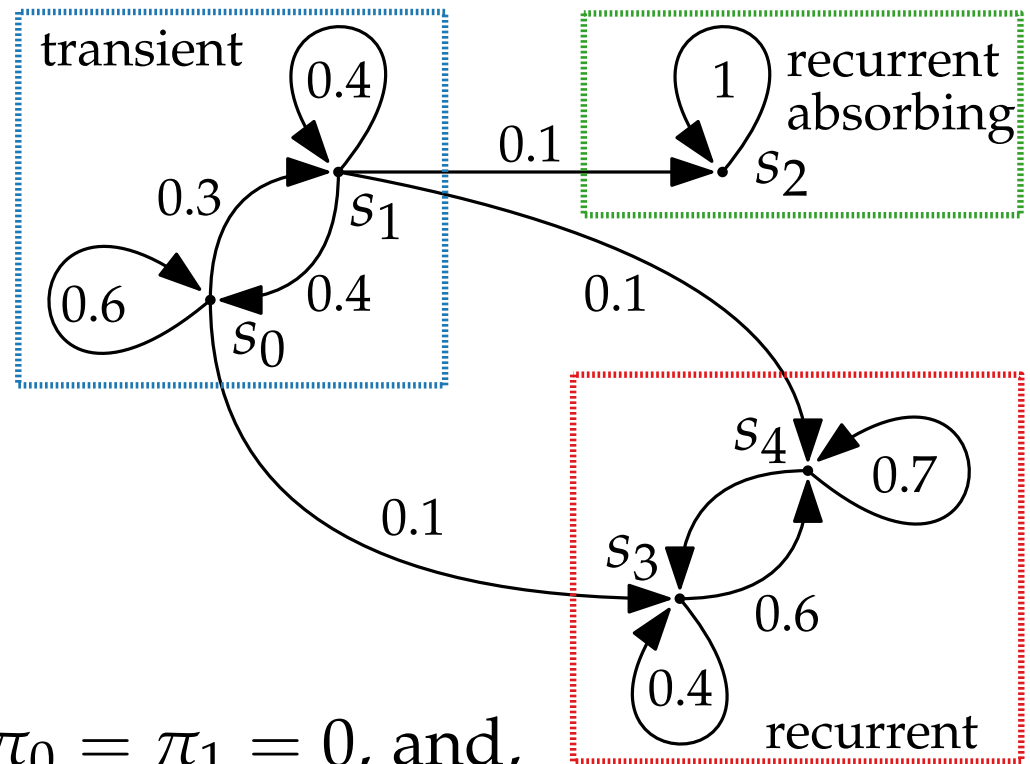
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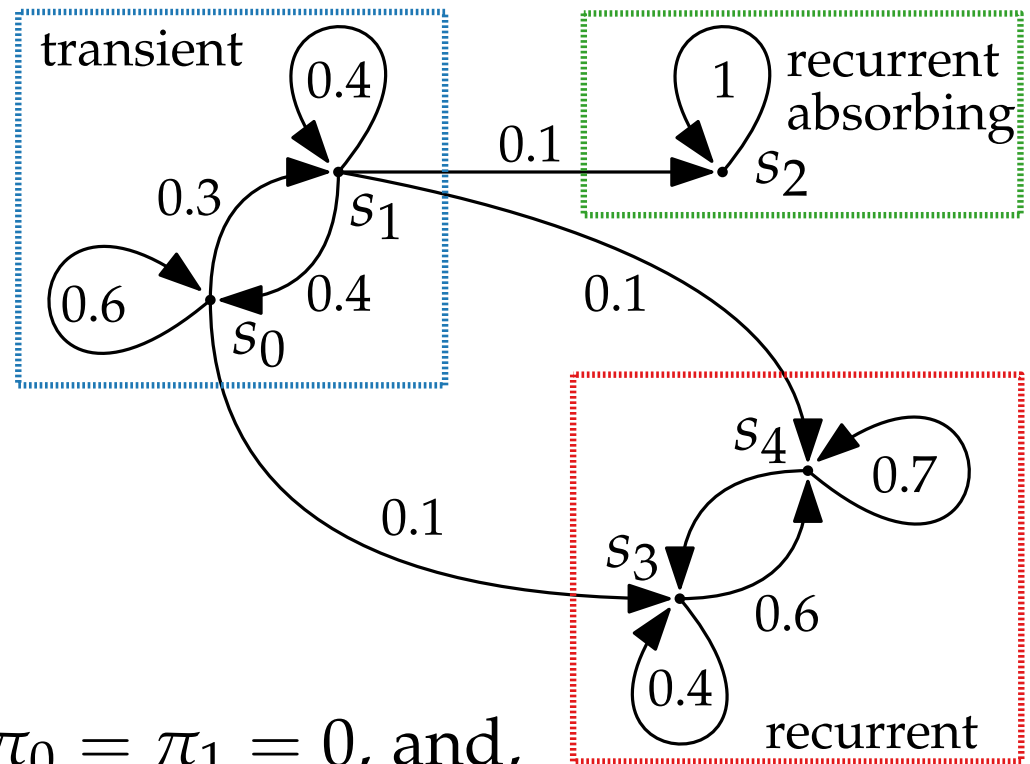
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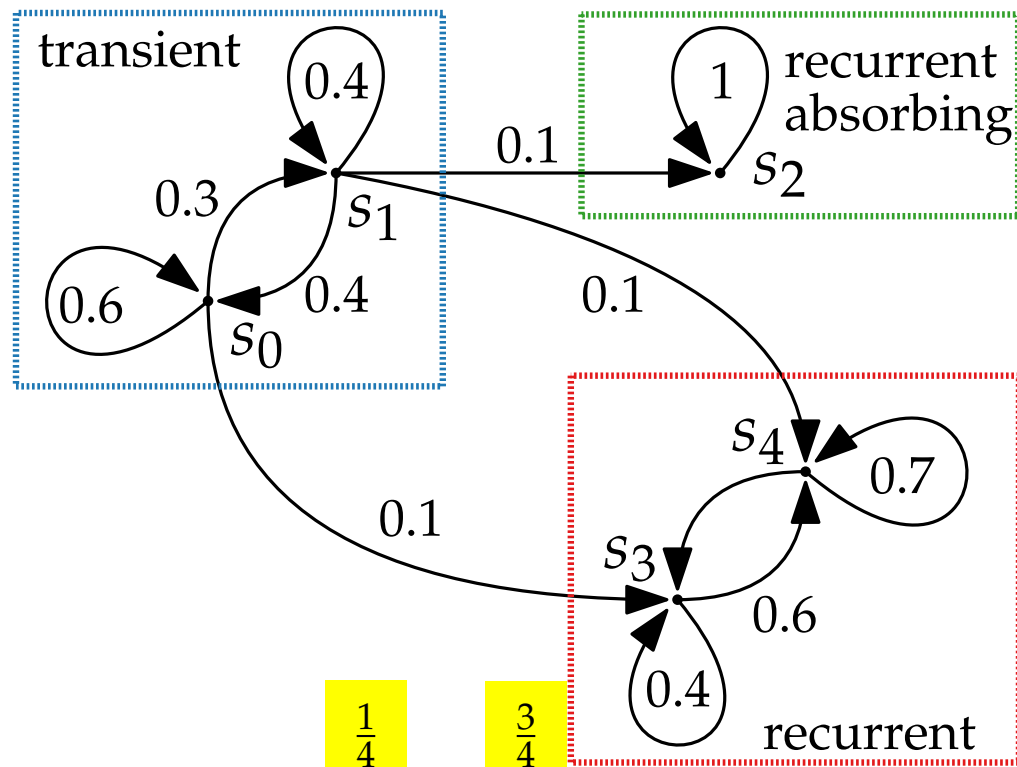
Thus, $\pi = (0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2})$

Example

	to	s_0	s_1	s_2	s_3	s_4	
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Starting in s_0 :

- Calculate the probability of reaching state (i) s_2 , (ii) s_3 , (iii) s_4 .
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Example

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$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

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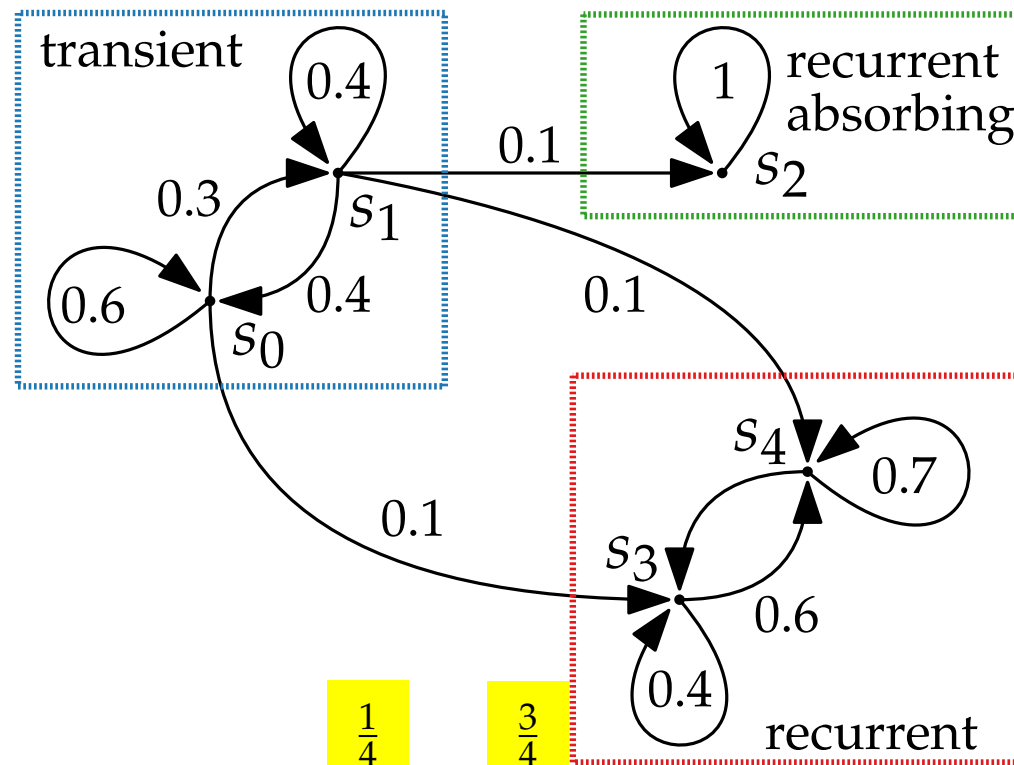
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Example

	from \ to	s_0	s_1	s_2	s_3	s_4
s_0		0.6	0.3	0	0.1	0
s_1		0.4	0.4	0.1	0	0.1
s_2		0	0	1	0	0
s_3		0	0	0	0.4	0.6
s_4		0	0	0	0.3	0.7



Starting in s_0 :

- Calculate the probability of reaching state (i) s_2 , (ii) s_3 , (iii) s_4 .
- Calculate $\pi = (\pi_0, \pi_1, \pi_2, \pi_3, \pi_4) = (0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2})$
- On average (long run), how many **times** is (i) s_0 , (ii) s_1 visited?

Starting in s_0 , we stay 5 times in state s_0 and $\frac{5}{2}$ times in state s_1 .

Queueing Theory

Barbara Franci

Queueing Processes

Call center:

- ▶ 30 calls per hour
- ▶ 3 agents reply and take 3 minutes per call (20 call per hour)
- ▶ If the agents are busy, costumers wait (up to 2)
- ▶ Any other call is lost, the costumers have to call another time

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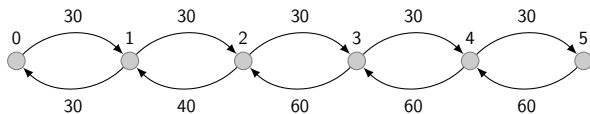
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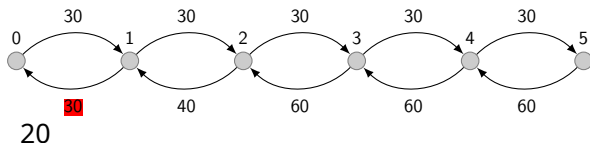


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This is a Markov Chain!

Queueing Processes

Interarrival time: time between calls

Service time: time to answer the call, i.e., how often a customer leaves

Queueing Processes

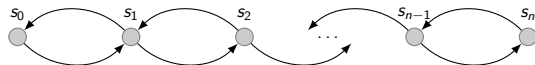
Interarrival time: time between calls

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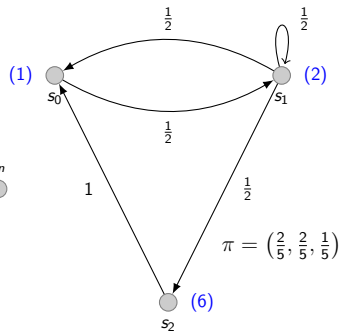
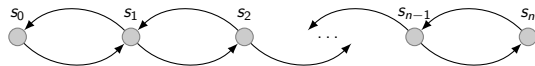
Goal:

- ▶ What fraction of time do the customers spend in each state?
- ▶ What is the average number of customers in the system?
- ▶ How much time does it take to go from 0 to 3?

Relation to Markov Chains



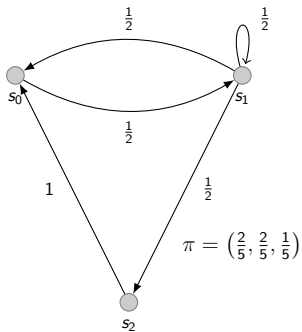
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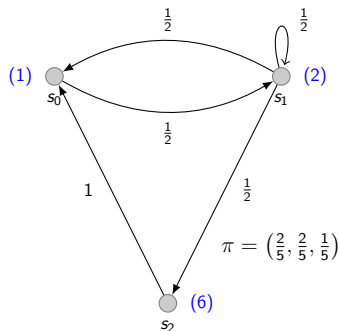
Problems

- Queues involve waiting...

Parenthesis



Parenthesis



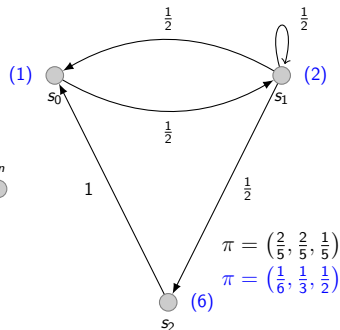
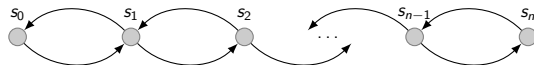
Let's consider the waiting time:

$$1 \cdot \frac{2}{5} + 2 \cdot \frac{2}{5} + 6 \cdot \frac{1}{5} = \frac{12}{5}$$

We have 12 time units to distribute among the states:

$$\Rightarrow \pi = \left(\frac{2}{12}, \frac{4}{12}, \frac{6}{12}\right) = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right)$$

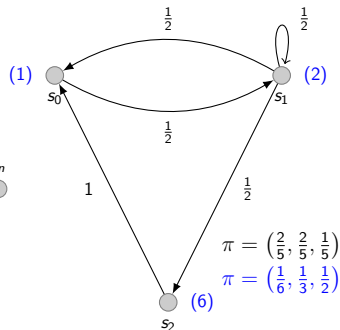
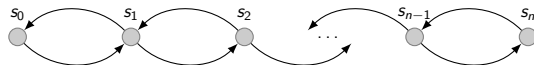
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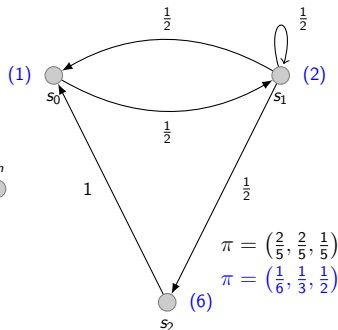
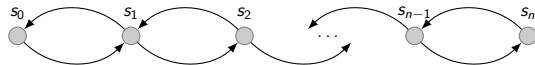
Relation to Markov Chains



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- ▶ Queues involve waiting...
- ▶ Customers do not arrive uniformly
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Relation to Markov Chains



Problems

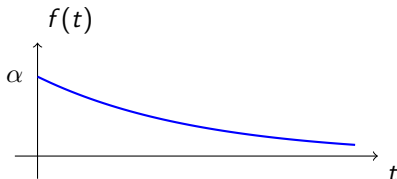
- ▶ Queues involve waiting...
 - ▶ Customers do not arrive uniformly
 - ▶ Customers are not processes uniformly
- ⇒ (something like) a Markov Chain with arrival/service times, according to a probability distribution
- But Markov Chains need a kind of memory-less distribution...

Exponential Distribution

Let X be a random variable, e.g., for interarrival or service time.

X has an exponential distribution with fixed parameter $\alpha > 0$ if the density function is

$$f(t) = \begin{cases} \alpha e^{-\alpha t} & t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

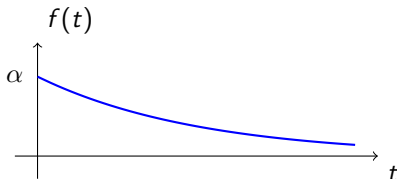


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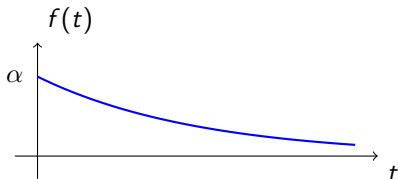
What is the expected time between arrivals?

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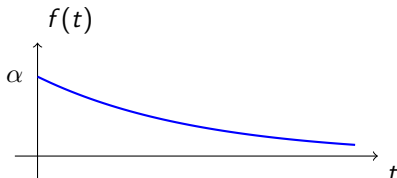
$$\begin{aligned} \mathbb{E}[X] &= \int_0^{\infty} t f(t) dt = \int_0^{\infty} t \alpha e^{-\alpha t} dt = -te^{-\alpha t} \Big|_0^{\infty} - \int_0^{\infty} -e^{-\alpha t} dt \\ &= -te^{-\alpha t} + \frac{1}{\alpha} e^{-\alpha t} \Big|_0^{\infty} = \frac{1}{\alpha} \end{aligned}$$

If α costumers arrive in 1 hour, then they arrive one every $\frac{1}{\alpha}$ time

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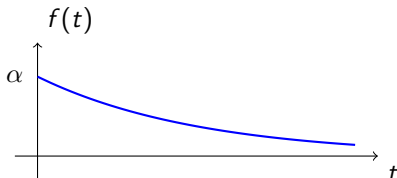
What is the probability of an event occurring after a certain time?

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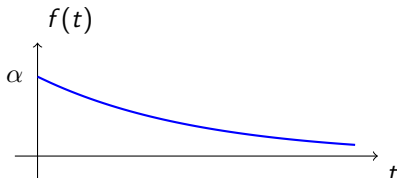
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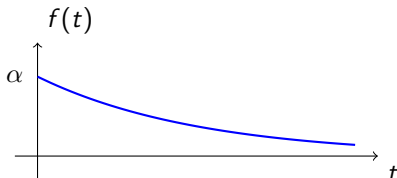
$$\mathbb{P}[a < X < b] = \int_a^b \alpha e^{-\alpha t} dt = -e^{-\alpha t} \Big|_a^b = e^{-\alpha a} - e^{-\alpha b}$$

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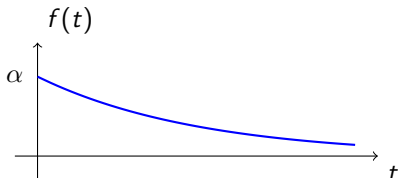
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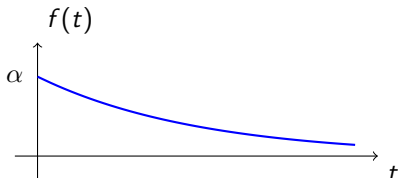
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$$\begin{aligned} \mathbb{P}[x > a + c \mid x > c] &= \frac{\mathbb{P}[x > a + c \text{ and } x > c]}{\mathbb{P}[x > c]} = \frac{\mathbb{P}[x > a + c]}{\mathbb{P}[x > c]} \\ &= \frac{e^{-\alpha(a+c)}}{e^{-\alpha c}} = e^{-\alpha a} = \mathbb{P}[x > a] \end{aligned}$$

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Lack of memory: the probability of a new arrival is always the same independently of how many events there have been before

More exponential distribution

X, Y : independent random vars. with $X \sim \exp(\alpha)$, $Y \sim \exp(\beta)$.

- ▶ Let $Z = \min\{X, Y\}$, i.e., Z is the first event among $\{X, Y\}$
Observe that $Z \sim \exp(\alpha + \beta)$

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$$\begin{aligned}\mathbb{P}(x < y) &= \int_0^\infty \mathbb{P}(x < y \mid x = t) f(t) dt = \int_0^\infty \mathbb{P}(y > t) f(t) dt \\ &= \int_0^\infty e^{-\beta t} \alpha e^{-\alpha t} dt = \int_0^\infty \alpha e^{-(\alpha+\beta)t} dt = \\ &= \left. -\frac{\alpha}{\alpha+\beta} e^{-(\alpha+\beta)t} \right|_0^\infty = \frac{\alpha}{\alpha+\beta}\end{aligned}$$

Poisson Distribution

Let X be a random variable for an arrival with $X \sim \exp(\alpha)$.

- ▶ Prob(n arrivals in next x time units) $\sim$ Poisson distribution
- ▶ Prob(n arrivals in next x time units) = Prob(n arrivals in the first x)
[memoryless]

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Let X be a random variable for an arrival with $X \sim \exp(\alpha)$.

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$$\begin{aligned}\mathbb{P}(1 \text{ event in } [0, x]) &= \int_0^x \mathbb{P}(0 \text{ events in } [0, x-t]) \alpha e^{-\alpha t} dt \\ &= \int_0^x e^{-\alpha(x-t)} \alpha e^{-\alpha t} dt \\ &= \int_0^x \alpha e^{-\alpha x} dt = \alpha t e^{-\alpha x} \Big|_0^x = \alpha x e^{-\alpha x}\end{aligned}$$

Poisson Processes

$$\mathbb{P}(0 \text{ events in } [0, x]) = \mathbb{P}(X > x) = e^{-\alpha x}$$

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$$\begin{aligned}\mathbb{P}(3 \text{ events in } [0, x]) &= \int_0^x \mathbb{P}(2 \text{ events in } [0, x-t]) \alpha e^{-\alpha t} dt \\&= \int_0^x \alpha^2 \frac{(x-t)^2}{2} e^{-\alpha(x-t)} \alpha e^{-\alpha t} dt \\&= \int_0^x \alpha^3 \frac{(x-t)^2}{2} e^{-\alpha x} dt \\&= \frac{\alpha^3}{6} (x-t)^3 \cdot e^{-\alpha x} \Big|_0^x = \frac{\alpha^3 x^3}{6} e^{-\alpha x}\end{aligned}$$

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This look very similar to the queueing process!

Recap

Goal:

- ▶ What fraction of time do the costumers spend in each state?
- ▶ What is the average number of costumers in the system?
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Goal:

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What we have is:

- ▶ Number of arrivals (via Poisson Distribution)
- ▶ Memory-Less property (via Exponential Distribution)
- ▶ Time before event happening (via $Z = \min\{X, Y\}$)
- ▶ Probability of event happening (via $\mathbb{P}(X < Y)$)

Is this enough?

An example

Medical Clinic with several types of patients, e.g., patients with flu symptoms arrive 10 /hour, and physical injury patients arrive 5/hour.
[both exp distributed]

- ▶ What are α, β , where $\text{flu} \sim \exp(\alpha)$ and $\text{injury} \sim \exp(\beta)$?
- ▶ How long until the first (i) flu; (ii) flu or injury patient arrives?
- ▶ What is $\mathbb{P}(\text{first patient has flu symptoms})$?

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 - (i) $\text{flu} \sim \exp(10)$, thus $\mathbb{E}(\text{first flu patient arrival}) = \frac{1}{10}$.
 - (ii) arrival of flu OR injury $\sim \min(\text{flu}, \text{injury}) \sim \exp(10 + 5)$, thus

$$\mathbb{E}(\text{first patient arrival}) = \frac{1}{15} \quad [\text{intuitively, 15 expected events}]$$

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- ▶ What is $\mathbb{P}(\text{first patient has flu symptoms})$?

$$\mathbb{P}(\text{first patient has flu symptoms}) = \frac{\alpha}{\alpha + \beta} = \frac{10}{15}$$

Example

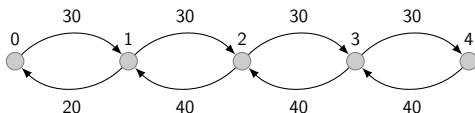
Call Center

- ▶ 30 calls per hour exponentially distributed, i.e., $\exp(30)$
- ▶ 2 agents reply and can serve 20 costumers per hour, i.e., $\exp(20)$
- ▶ If the agents are busy, costumers wait (up to 2)
- ▶ Any other call is lost, the costumers have to call another time

Example

Call Center

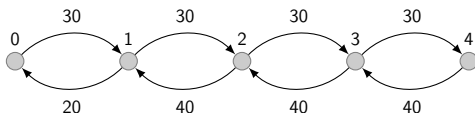
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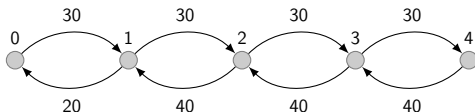


- ▶ What is the probability of leaving state s_3 for s_4 ?
- ▶ How much time do we spend in state s_2 ?

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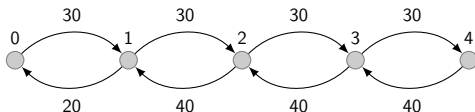


- ▶ What is the probability of leaving state s_3 for s_4 ? $\frac{30}{30+40}$
- ▶ How much time do we spend in state s_2 ? $\frac{1}{30+40}$

Example

Call Center

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How many events/per hour cause us to move from state s_i to state s_j ?

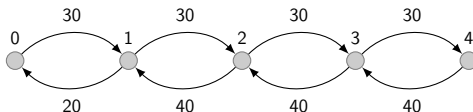
$s_i \rightarrow s_{i+1}$: a customer arrived.

- ▶ 30 customers/hour for all i

$s_{i+1} \rightarrow s_i$: a customer departed.

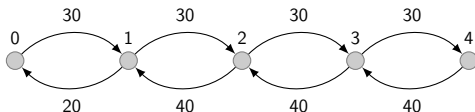
- ▶ 20 customers/hour ($i = 1$)
- ▶ 40 customers/hour ($i > 1$)

Example



What fraction of time do the costumers spend in each state?

Example



What fraction of time do the costumers spend in each state?

$$\text{In} = \text{Out}$$

$$20\pi_1 = 30\pi_0$$

$$30\pi_0 + 40\pi_2 = 20\pi_1 + 30\pi_1$$

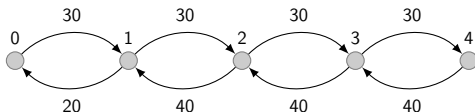
$$30\pi_1 + 40\pi_3 = 40\pi_2 + 30\pi_2$$

$$30\pi_2 + 40\pi_4 = 40\pi_3 + 30\pi_3$$

$$30\pi_3 = 40\pi_4$$

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$$

Example



What fraction of time do the costumers spend in each state?

$$\text{In} = \text{Out}$$

$$20\pi_1 = 30\pi_0$$

$$40\pi_2 = 30\pi_1$$

$$40\pi_3 = 30\pi_2$$

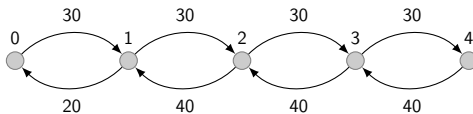
$$40\pi_4 = 30\pi_3$$

$$30\pi_3 = 40\pi_4$$

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$$

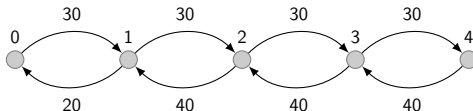
$$\pi = \frac{1}{653}(128, 192, 144, 108, 81)$$

Example



What is the average number of costumers in the system?

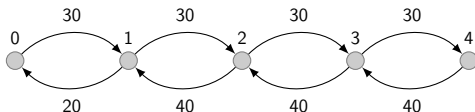
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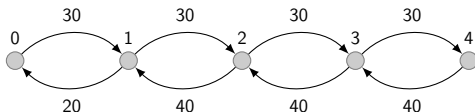


What is the average number of costumers in the system?

$$\pi = \frac{1}{653}(128, 192, 144, 108, 81)$$

$$\text{Avg. Num. Costumers} = 0 \cdot \pi_0 + 1 \cdot \pi_1 + 2 \cdot \pi_2 + 3 \cdot \pi_3 + 4 \cdot \pi_4$$

Example

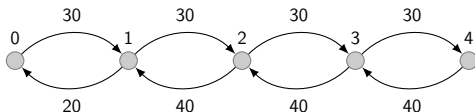


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$$\begin{aligned}\text{Avg. Num. Costumers} &= 0 \cdot \pi_0 + 1 \cdot \pi_1 + 2 \cdot \pi_2 + 3 \cdot \pi_3 + 4 \cdot \pi_4 \\ &= \left(0 \cdot \frac{128}{653} + 1 \cdot \frac{192}{653} + 2 \cdot \frac{144}{653} + 3 \cdot \frac{108}{653} + 4 \cdot \frac{81}{653} \right) \\ &= \frac{1128}{653} \sim 1.7\end{aligned}$$

Example

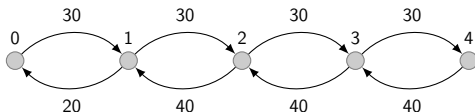


How much time does it take to go from 0 to 3?

$$\begin{aligned} E_x &= \text{expected time until there are 3 customers when starting in } x \\ &= \text{expected time to go from } s_x \text{ to } s_3 \\ &= \text{Time in } x + \sum P_{xi} E_i \end{aligned}$$

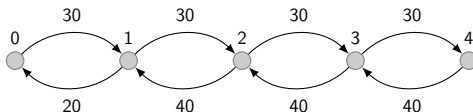
- ▶ How much time in x ?
- ▶ What are the transition probabilities P_{xi} ?

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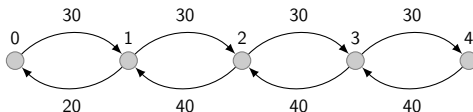
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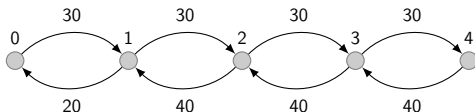
- ▶ How much time in x ?
 - ▶ 30 events/hour leave $s_0 \Rightarrow \frac{1}{30}$ hours/visit to s_0
 - ▶ $20 + 30$ events/hour leave $s_1 \Rightarrow \frac{1}{50}$ hours/visit to s_1
 - ▶ 70 events/hour leave $s_i, i \in \{2, \dots, w+1\} \Rightarrow \frac{1}{70}$ hours/visit to s_i
 - ▶ 40 events/hour leave $s_{w+2} \Rightarrow \frac{1}{40}$ hours/visit to s_{w+2}
- ▶ What are the transition probabilities P_{xi} ?

Example



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- ▶ What are the transition probabilities P_{xi} ?
 - ▶ in s_2 , leaving to s_1 occurs with probability $\frac{40}{40+30}$
 - ▶ in s_2 , leaving to s_3 occurs with probability $\frac{30}{40+30}$

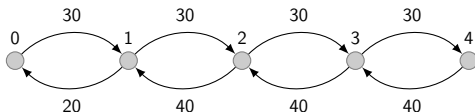
Example



How much time does it take to go from 0 to 3?

⇒ Set up the system and solve:

Example



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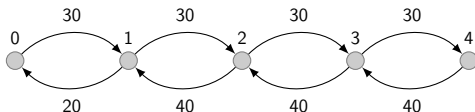
$$E_0 = \frac{1}{30} + E_1$$

$$E_1 = \frac{1}{50} + \frac{2}{5}E_0 + \frac{3}{5}E_2$$

$$E_2 = \frac{1}{70} + \frac{4}{7}E_1 + \frac{3}{7}E_3$$

$$E_3 = 0$$

Example



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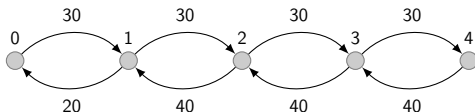
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$$E_0 = \frac{53}{270} \text{ hours}$$

Example



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$$E_0 = 2 + E_1$$

$$E_1 = \frac{6}{5} + \frac{2}{5}E_0 + \frac{3}{5}E_2$$

$$E_2 = \frac{6}{7} + \frac{4}{7}E_1 + \frac{3}{7}E_3$$

$$E_3 = 0$$

$$E_0 = \frac{53}{270} \text{ hours}$$

$$E_0 = \frac{106}{9} \sim 11,8 \text{ minutes}$$



MATSE: Operations Research

Lecture 8. Game Theory

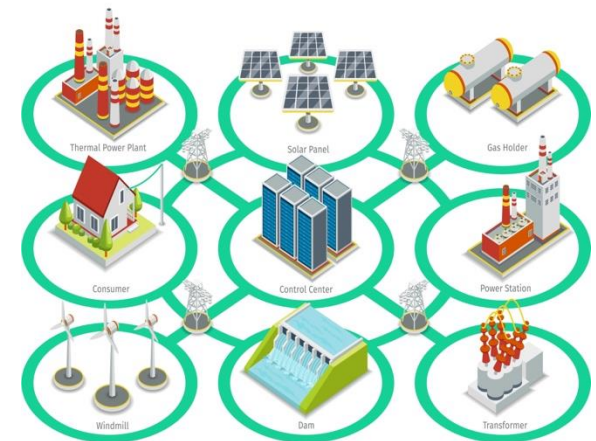
[based on slides and notes by: Bob Thuijsman, Steven Chaplick, Monica Salvioli]

What is Game Theory?

- Mathematical model for decision makers who interact
- Each of them tries to maximize its profit while influencing the others
- The choice made by one affects also the others interests

⇒ Cooperation, Competition, Conflicts

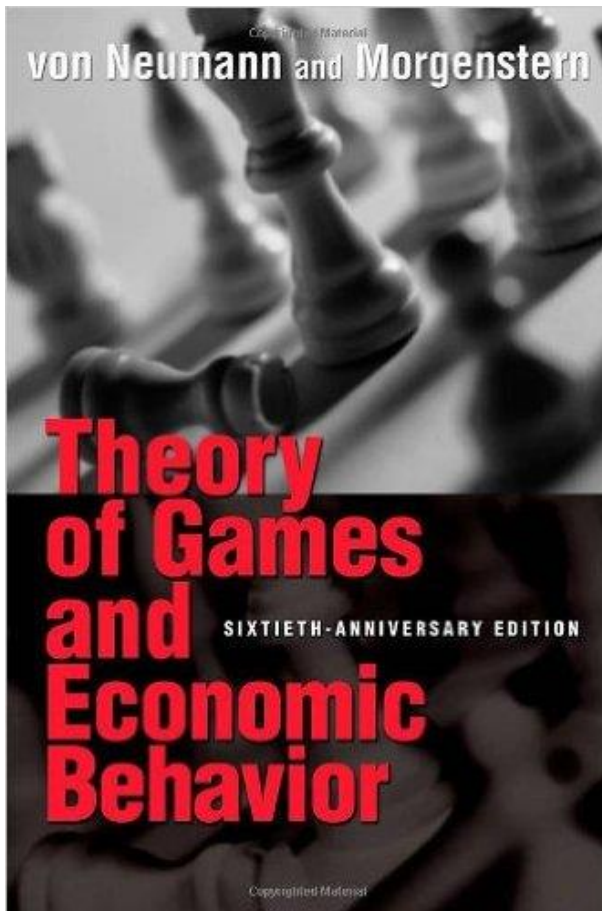
- Many applications



Images from images.google.com and freepik.com

Milestones

1944



- Formalization of the concept of game
- Winning strategies are independent of the others behaviour
- It was born during the cold war...

Milestones

Nash Jr, J. F. (1950). Equilibrium points in n -person games. *Proceedings of the national academy of sciences*, 36(1), 48-49.

This follows from the arguments used in a forthcoming paper.¹² It is proved by constructing an "abstract" mapping cylinder of λ and transcribing into algebraic terms the proof of the analogous theorem on CW-complexes.

* This note arose from consultations during the tenure of a John Simon Guggenheim Memorial Fellowship by MacLane.

¹ Whitehead, J. H. C., "Combinatorial Homotopy I and II," *Bull. A.M.S.*, 55, 214-245 and 453-496 (1949). We refer to these papers as CH I and CH II, respectively.

² By a complex we shall mean a connected CW complex, as defined in §5 of CH I. We do not restrict ourselves to finite complexes. A fixed 0-cell $e^0 \in K^0$ will be the base point for all the homotopy groups in K .

³ MacLane, S., "Cohomology Theory in Abstract Groups III," *Ann. Math.*, 50, 736-761 (1949), referred to as CT III.

⁴ An (unpublished) result like Theorem 1 for the homotopy type was obtained prior to these results by J. A. Zilber.

⁵ CT III uses in place of equation (2.4) the stronger hypothesis that λB contains the center of A , but all the relevant developments there apply under the weaker assumption (2.4).

⁶ Eilenberg, S., and MacLane, S., "Cohomology Theory in Abstract Groups II," *Ann. Math.*, 48, 326-341 (1947).

⁷ Eilenberg, S., and MacLane, S., "Determination of the Second Homology . . . by Means of Homotopy Invariants," these *PROCEEDINGS*, 32, 277-280 (1946).

⁸ Blikers, A. L., "Some Relations Between Homology and Homotopy Groups," *Ann. Math.*, 49, 428-461 (1948), §12.

⁹ The hypothesis of Theorem C, requiring that $\nu^{-1}(1)$ not be cyclic, can be readily realized by suitable choice of the free group X , but this hypothesis is not needed here (cf. *).

¹⁰ Eilenberg, S., and MacLane, S., "Homology of Spaces with Operators II," *Trans. A.M.S.*, 65, 49-99 (1949); referred to as HSO II.

¹¹ $C(K)$ here is the $C(K)$ of CH II. Note that $\bar{K}$ exists and is a CW complex by (N) of p. 231 of CH I and that $p^{-1}K^0 = \bar{K}^0$, where p is the projection $p: \bar{K} \rightarrow K$.

¹² Whitehead, J. H. C., "Simple Homotopy Types." If $W = 1$, Theorem 5 follows from (17:3) on p. 155 of S. Lefschetz, *Algebraic Topology*, (New York, 1942) and arguments in §6 of J. H. C. Whitehead, "On Simply Connected 4-Dimensional Polyhedra" (*Comm. Math. Helv.*, 22, 48-92 (1949)). However this proof cannot be generalized to the case $W \neq 1$.

EQUILIBRIUM POINTS IN N -PERSON GAMES

By JOHN F. NASH, JR.*

PRINCETON UNIVERSITY

Communicated by S. Lefschetz, November 16, 1949

One may define a concept of an n -person game in which each player has a finite set of pure strategies and in which a definite set of payments to the n players corresponds to each n -tuple of pure strategies, one strategy being taken for each player. For mixed strategies, which are probability

distributions over the pure strategies, the pay-off functions are the expectations of the players, thus becoming polylinear forms in the probabilities with which the various players play their various pure strategies.

Any n -tuple of strategies, one for each player, may be regarded as a point in the product space obtained by multiplying the n strategy spaces of the players. One such n -tuple counters another if the strategy of each player in the countering n -tuple yields the highest obtainable expectation for its player against the $n - 1$ strategies of the other players in the countered n -tuple. A self-countering n -tuple is called an equilibrium point.

The correspondence of each n -tuple with its set of countering n -tuples gives a one-to-many mapping of the product space into itself. From the definition of countering we see that the set of countering points of a point is convex. By using the continuity of the pay-off functions we see that the graph of the mapping is closed. The closedness is equivalent to saying: if $P_1, P_2, \dots$ and $Q_1, Q_2, \dots, Q_n, \dots$ are sequences of points in the product space where $Q_n \rightarrow Q$, $P_n \rightarrow P$ and Q_n counters P_n then Q counters P .

Since the graph is closed and since the image of each point under the mapping is convex, we infer from Kakutani's theorem¹ that the mapping has a fixed point (i.e., point contained in its image). Hence there is an equilibrium point.

In the two-person zero-sum case the "main theorem"² and the existence of an equilibrium point are equivalent. In this case any two equilibrium points lead to the same expectations for the players, but this need not occur in general.

* The author is indebted to Dr. David Gale for suggesting the use of Kakutani's theorem to simplify the proof and to the A. E. C. for financial support.

¹ Kakutani, S., *Duke Math. J.*, 8, 457-459 (1941).

² Von Neumann, J., and Morgenstern, O., *The Theory of Games and Economic Behaviour*, Chap. 3, Princeton University Press, Princeton, 1947.

REMARK ON WEYL'S NOTE "INEQUALITIES BETWEEN THE TWO KINDS OF EIGENVALUES OF A LINEAR TRANSFORMATION"*

By GEORGE POLYA

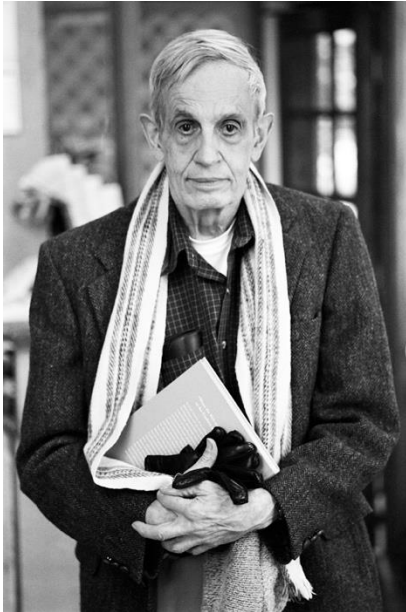
DEPARTMENT OF MATHEMATICS, STANFORD UNIVERSITY

Communicated by H. Weyl, November 25, 1949

In the note quoted above H. Weyl proved a Theorem involving a function $\varphi(\lambda)$ and concerning the eigenvalues α_i of a linear transformation A and those, κ_i , of A^*A . If the κ_i and $\lambda_i = |\alpha_i|^2$ are arranged in descending order,

Milestones

Nash Jr, J. F. (1950). Equilibrium points in n-person games. *Proceedings of the national academy of sciences*, 36(1), 48-49.



- Proposed a fundamental concept that will be known as Nash equilibrium
- Harsanyi-Nash-Salten Nobel 1994

Two other Nobel:

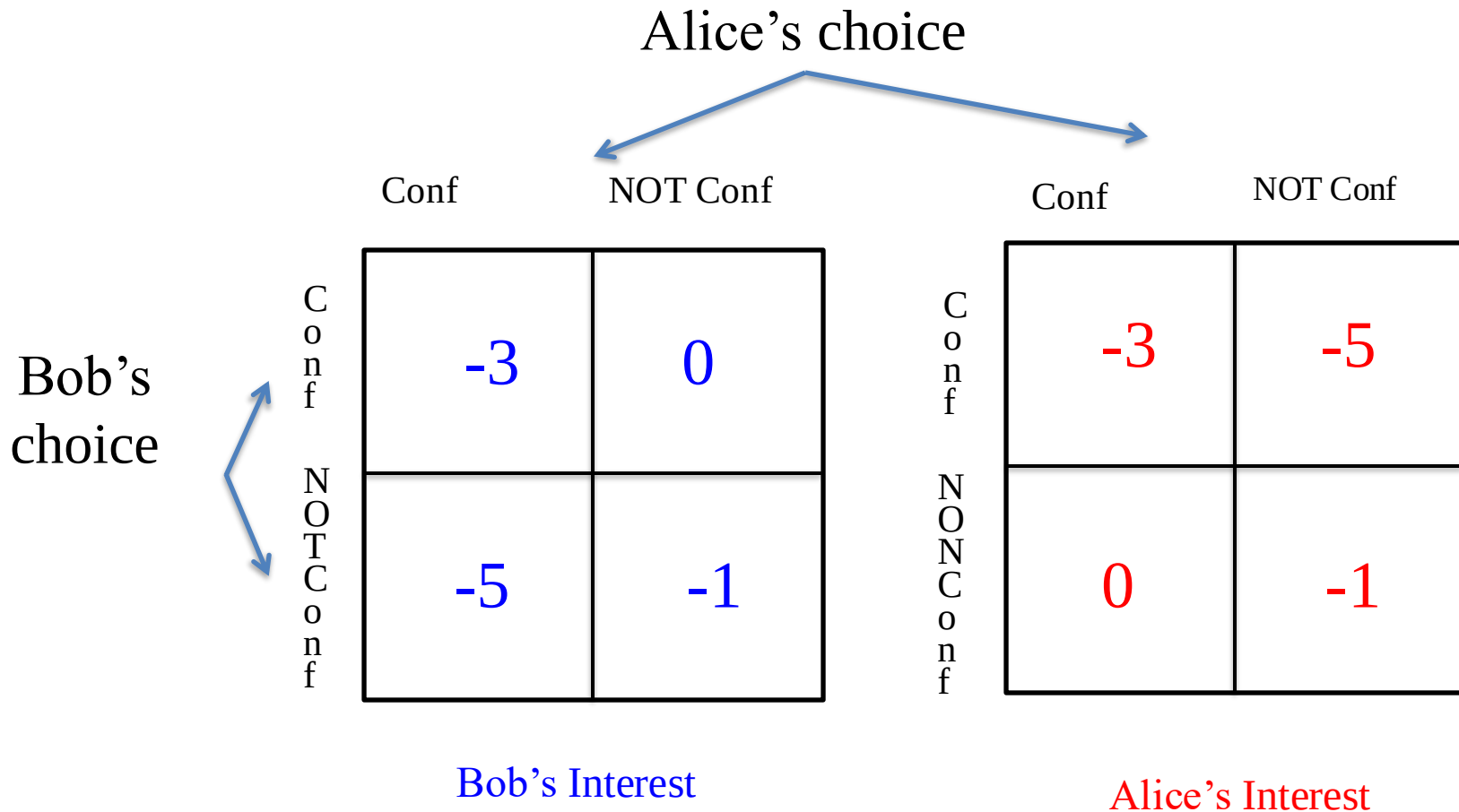
- 2005 Aumann-Shelling - Conflicts and Cooperation
- 2012 Roth-Shapley - Applications on Martkes

Prisoner's Dilemma

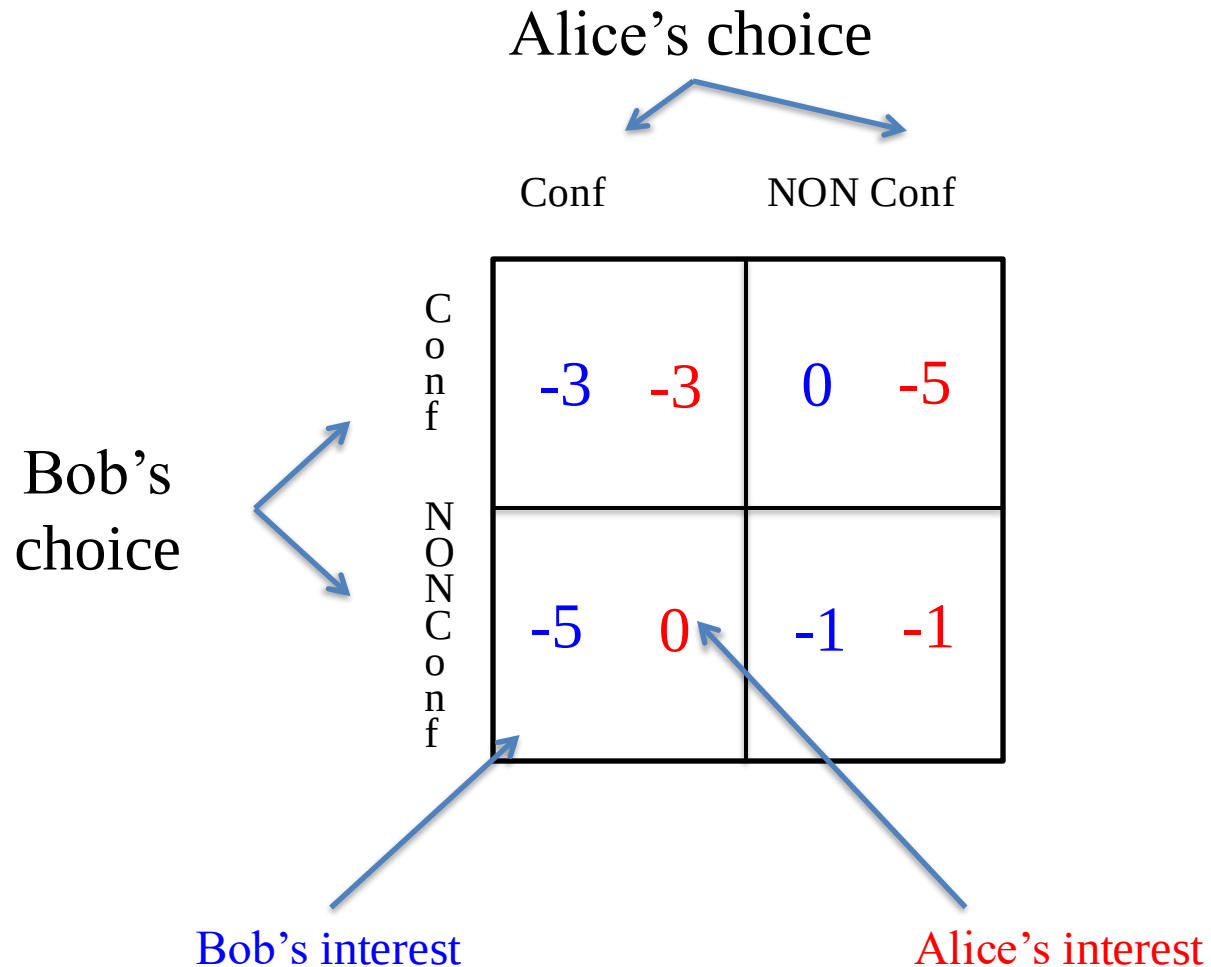
Bob and Alice have been arrested and accused of a crime. They are questioned separately (with no possibility of talking with each other) and they are proposed to Confess.

- If both Confess, both serve 3 years
- If one betrays the other, the first is free while the second serves 5 years
- If both don't Confess, they serve one year

Formalizing the game



Formalizing the game



Formalizing the game

Alice			
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

If Alice Confess, it is better for Bob to Confess.

Formalizing the game

		Alice	
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

If Alice doesn't Confess, it is better for Bob to Confess.

Formalizing the game

		Alice	
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

It is always better for Bob to Confess!

Formalizing the game

		Alice	
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

It is always better for Bob to Confess!
Conf is a **dominant strategy** for Bob.

Formalizing the game

		Alice	
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

If Bob doesn't Confess, it is better for Alice to Confess.

Formalizing the game

Alice			
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

If Bob doesn't Confess, it is better for Alice to Confess.

Conf is a **dominant strategy** for Alice.

Formalizing the game

Alice			
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

Rational Choice: both Confess

Formalizing the game

Alice			
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

Rational Choice: both Confess

But it would be much better if they didn't!

Prisoner's Dilemma

Conf is a dominant strategy: rational choice

Implicit underlying assumptions:

- Bob and Alice cannot talk with each other
- We play the game only once
- The interest is determined only in the years they have to serve

Other possibilities:

- Repeated games
- Different interests for the players (altruism, fear...)

Altruistic Prisoner's Dilemma

Alice's choice

Conf NON Conf

Bob's choice

Conf	-3 -3	-2 -5
NON Conf	-5 -2	-1 -1

Conf

NON Conf

Conf

NON Conf

-3 -3

-2 -5

-5 -2

-1 -1

Altruistic Prisoner's Dilemma

Alice's choice

Conf NON Conf

Bob's choice

Conf

NON Conf

Conf	-3 -3	-2 -5
NON Conf	-5 -2	-1 -1

The matrix shows the following payoffs (Alice, Bob):

- If Alice Confesses and Bob Confesses: (-3, -3)
- If Alice Confesses and Bob does not Confess: (-2, -5)
- If Alice does not Confess and Bob Confesses: (-5, -2)
- If Alice does not Confess and Bob does not Confess: (-1, -1)

If Alice Confess, it is better for Bob to Confess.

If Alice doesn't Confess, it is better for Bob to not Confess.

Altruistic Prisoner's Dilemma

Alice's choice

Conf NON Conf

Bob's choice

Conf	-3 -3	-2 -5
NON Conf	-5 -2	-1 -1

If Alice Confess, it is better for Bob to Confess.

If Alice doesn't Confess, it is better for Bob to not Confess.

Altruistic Prisoner's Dilemma

Alice's choice

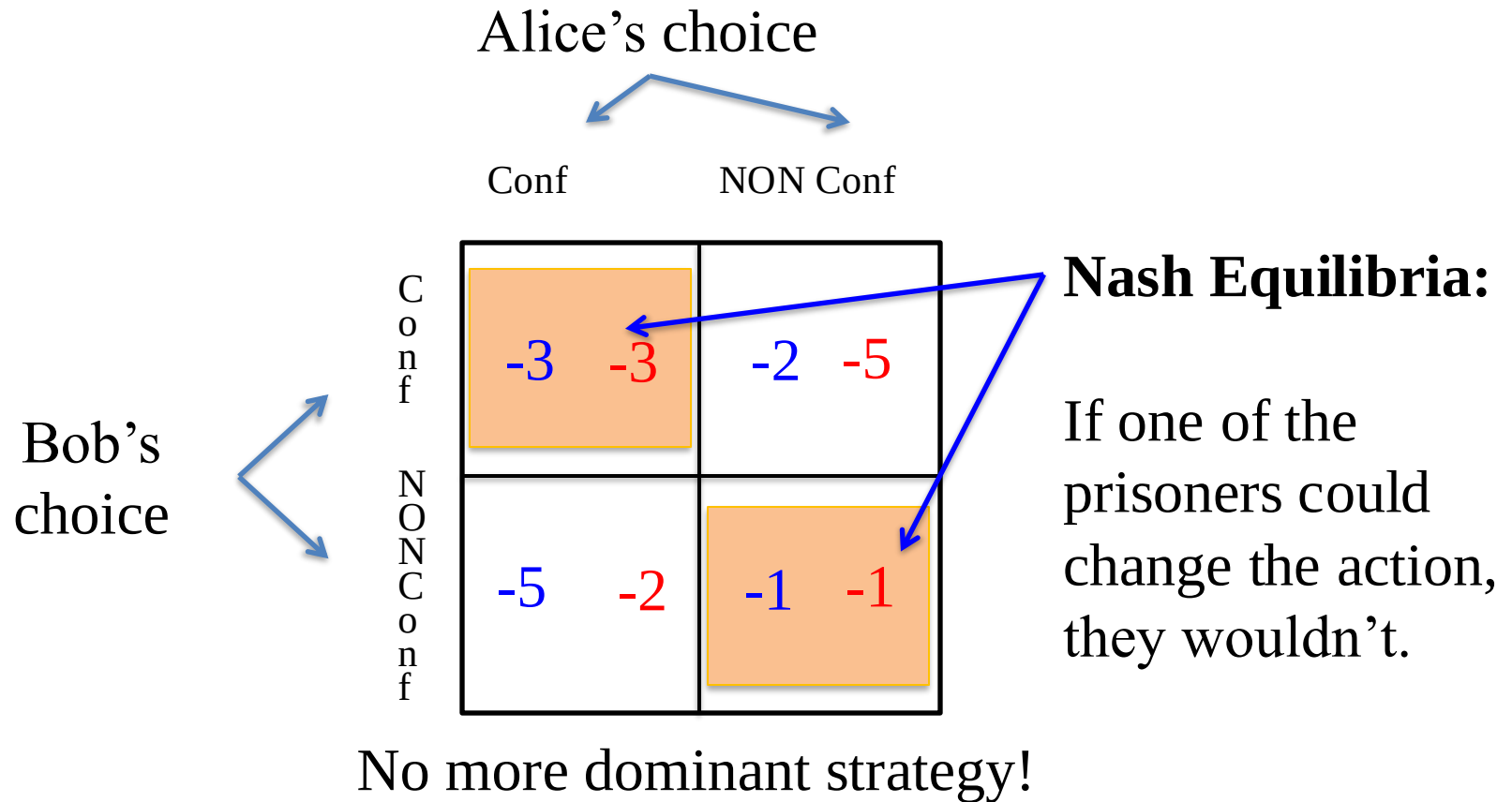
Conf NON Conf

Bob's choice

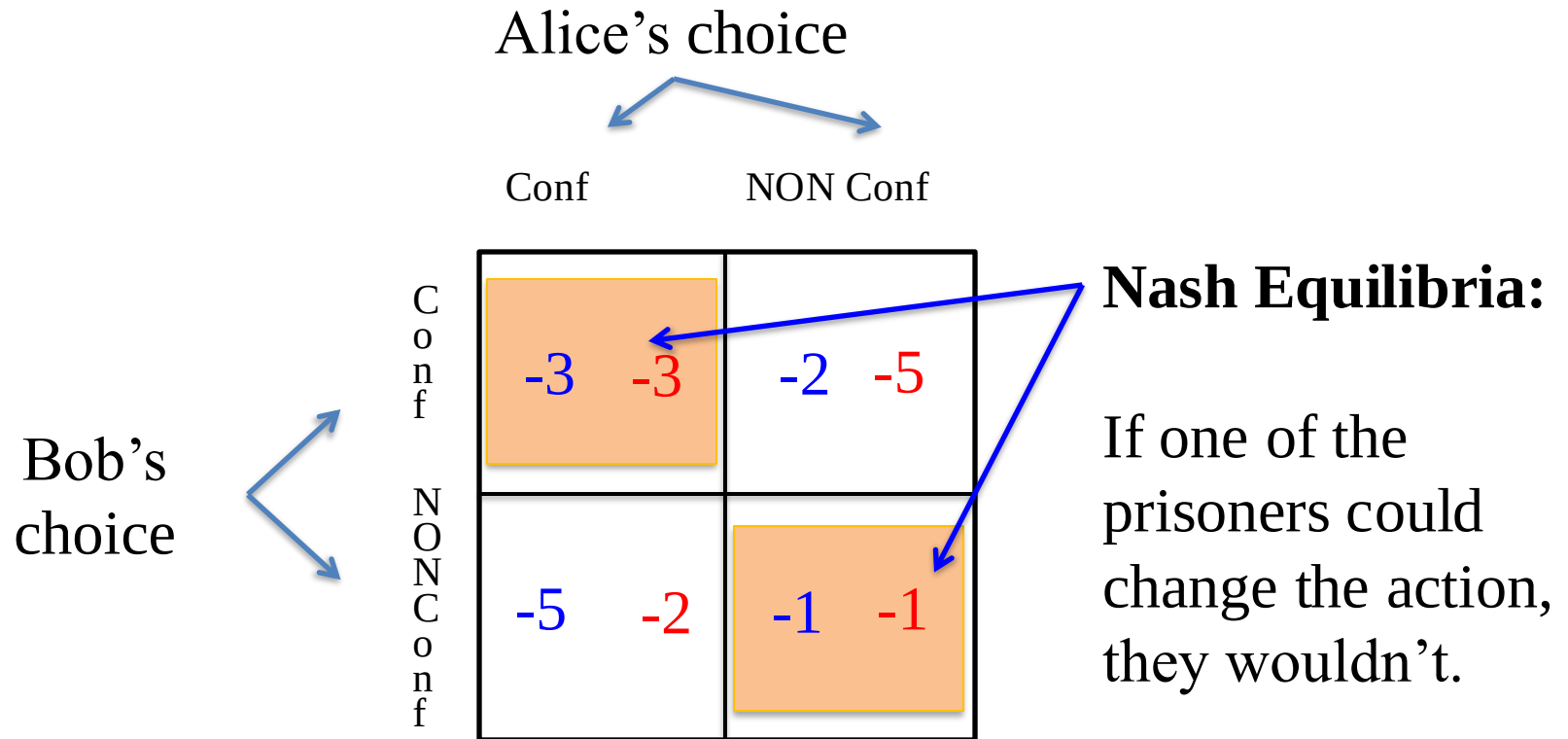
Conf	-3 -3	-2 -5
NON Conf	-5 -2	-1 -1

No more dominant strategy!

Altruistic Prisoner's Dilemma

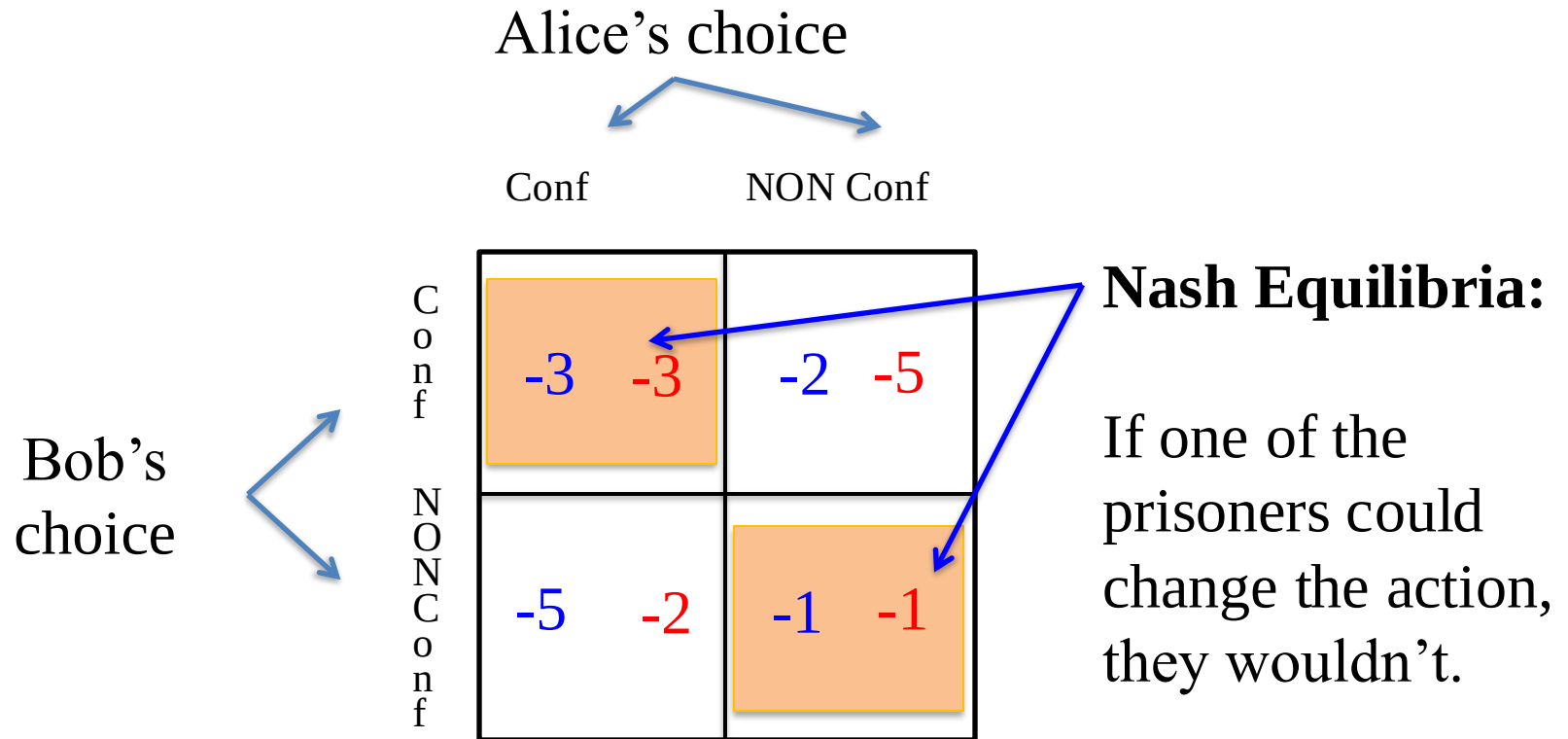


Altruistic Prisoner's Dilemma



However, one of the equilibria is better...

Altruistic Prisoner's Dilemma



However, one of the equilibria is better...

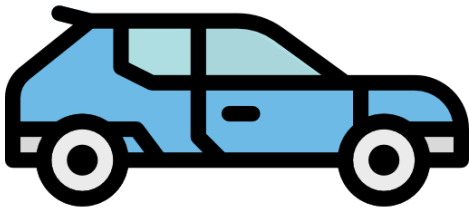
Coordination game

Chicken's game

Two cars drive towards each other.

Every driver can decide whether to swerve or keep going.

- If one swerve while the other doesn't, the first is considered a 'chicken'
- If both don't swerve, they crash
- If both swerve, they'll have a bad reputation among their peers



Chicken's game

- Crash: -10
- Chicken: -1
- Indifference: 0
- Brave: 1

	Swerve	Straight
Swerve	0 0	-1 1
Straight	1 -1	-10 -10

Chicken's game

- Crash: -10
- Chicken: -1
- Indifference: 0
- Brave: 1

	Swerve	Straight
Swerve	0 0	-1 1
Straight	1 -1	-10 -10

Can you compute the equilibria?

Chicken's game

- Crash: -10
- Chicken: -1
- Indifference: 0
- Brave: 1

	Swerve	Straight
Swerve	0 0	-1 1
Straight	1 -1	-10 -10

Can you compute the equilibria?

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Anti-Coordination game: each player has a preferred equilibrium

Chicken's game

- Crash: -10
- Chicken: -1
- Indifference: 0
- Brave: 1

	Swerve	Straight
Swerve	0 0	-1 1
Straight	1 -1	-10 -10

Anti-Coordination game: each player has a preferred equilibrium

Best strategy: convince the other you are not going to swerve!

Introducing some math...

Focus of game theory:

Situations (**games**) in which **players** (agents, actors, individuals, decision makers) **interact** with one another, and in which the **outcome** (profit, loss, costs, utility) of each player may depend not just on their own decisions but also on the decisions of the other participants.

How do we make this formal?

Bimatrix games

Prisoner's Dilemma

$$\begin{array}{c}
 P_1 \\
 \begin{array}{c} C \\ NC \end{array}
 \end{array}
 \begin{array}{c}
 P_2 \\
 \begin{array}{c} C \\ NC \end{array}
 \end{array}
 \begin{bmatrix}
 -3, -3 & 0, -5 \\
 -5, 0 & -1, -1
 \end{bmatrix}$$

Bob

		Alice	
		Conf	NON Conf
Bob	Conf	-3 -3	0 -5
	NON Conf	-5 0	-1 -1

Bimatrix games

Prisoner's Dilemma

$$\begin{array}{c}
 P_1 \\
 \begin{array}{c} C \\ NC \end{array}
 \end{array}
 \begin{array}{cc}
 & \begin{array}{c} P_2 \\ C \quad NC \end{array} \\
 \left[\begin{array}{cc} -3, -3 & 0, -5 \\ -5, 0 & -1, -1 \end{array} \right]
 \end{array}$$

Normal-form description of the game:

{players, their possible actions, outcomes per player for all possible actions}

$$P_1, P_2 \quad \Gamma_1 = \Gamma_2 = \{C, NC\} \quad J_1(C, C) = -3, \dots, J_2(NC, NC) = -1$$

Bimatrix games

Prisoner's Dilemma

		P_2	
		C	NC
P_1	C	$[-3, -3]$	$[0, -5]$
	NC	$[-5, 0]$	$[-1, -1]$

Normal-form description of the game:

{**players**, **their possible actions**, **outcomes per player for all possible actions**}

P_1, P_2 $\Gamma_1 = \Gamma_2 = \{C, NC\}$ $J_1(C, C) = -3, \dots, J_2(NC, NC) = -1$

Strictly dominant strategy: a strategy that is strictly better than others, independently of what the other player does.

When a player has a strictly dominant strategy, we assume that they will play it.

Indeed, C(onfess) is a strictly dominant strategy.

Best response (reply)

Best response (set) R_1 for player P_1 to any action $\gamma_2 \in \Gamma_2$ of player P_2 is defined as:

$$R_1(\gamma_2) = \{\rho \in \Gamma_1 : J_1(\rho, \gamma_2) \geq J_1(\gamma_1, \gamma_2) \forall \gamma_1 \in \Gamma_1\}$$

if each player i maximizes their objective J_i .

If this set a singleton, we Conf about *strict best response*

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		P_2	
		L	R
P_1	T	(0,0)	(3,1)
	B	(1,3)	(4,2)

(Profit)

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if each player i maximizes their objective J_i .

If this set is a singleton, we can talk about *strict best response*

		P_2	
		L	R
P_1	T	0,0	3,1
	B	1,3	4,2

(Profit)

Mathematically, denoting the best response of player i by R_i , we can write:

$$R_1(L) = B$$

$$R_1(R) = B$$

$$R_2(T) = R$$

$$R_2(B) = L$$

Best response (reply)

Best response (set) R_1 for player P_1 to any action $\gamma_2 \in \Gamma_2$ of player P_2 is defined as:

$$R_1(\gamma_2) = \{\rho \in \Gamma_1 : J_1(\rho, \gamma_2) \geq J_1(\gamma_1, \gamma_2) \forall \gamma_1 \in \Gamma_1\}$$

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		P_2	
		L	R
P_1	T	(0,0)	(3,1)
	B	(1,3)	(4,2)

(Profit)

Mathematically, denoting the best response of player i by R_i , we can write:

$$R_1(L) = B$$

$$R_1(R) = B$$

$$R_2(T) = R$$

$$R_2(B) = L$$

It appears that the pair of strategies (B, L) is special:
 B is the best response to L and L is the best response to B .

Nash Equilibrium

Even when there are no dominant strategies, we should expect players to use strategies that are best responses to each other.

Suppose that P_1 chooses a strategy γ_1 and P_2 chooses a strategy γ_2 .

This pair of strategies (γ_1, γ_2) is a *Nash equilibrium* if γ_1 is a best response to γ_2 and γ_2 is a best response to γ_1 .

Nash Equilibrium

		P_2	
		L	R
P_1	T	(0,0)	(3,1)
	B	(1,3)	(4,2)

(Profit)

Note: At Nash equilibrium, there is no unilateral profitable deviation from any of the players involved.

In other words, no player in the game would take a different action as long as every other player remains the same.

Summary so far

- Dominant Strategy
- Best Response
- Nash equilibrium
 - Games with multiple equilibria
 - Choice depends on the strategy and on the ‘temporal dimension’
 - Games without equilibria?

Zero-Sum games

Matching Pennies

- Two Players (A and B)
decide for H or T
- If they get HH or TT,
player A wins
- Otherwise B wins

Zero-Sum games

Matching Pennies

- Two Players (A and B) decide for H or T
- If they get HH or TT, player A wins
- Otherwise B wins

		B		H	T
A	H	1	-1	-1	1
	T	-1	1	1	-1

Zero-Sum games

Matching Pennies

- Two Players (A and B) decide for H or T
- If they get HH or TT, player A wins
- Otherwise B wins

		B	
		H	T
A	H	1 -1	-1 1
	T	-1 1	1 -1

No Nash equilibrium!

Typical in zero-sum games...

Nash's Nobel

How is it that Nash received a Nobel with the following result?

Theorem: *Any game with a finite number of players and a finite number of actions has a mixed-strategy Nash equilibrium.*

Nash's Nobel

How is it that Nash received a Nobel with the following result?

Theorem: *Any game with a finite number of players and a finite number of actions has a mixed-strategy Nash equilibrium.*

- Repeated probabilistic strategies
- In the coin flipping example the equilibrium will be to play H 50% of the time and T 50% of the time

Symmetric games

In a *symmetric game*, every player is identical with respect to the game rules

			
	0,0	-1,1	1,-1
	1,-1	0,0	-1,1
	-1,1	1,-1	0,0

Mixed strategy Nash equilibrium

				
	0	-1	1	(profit)
	1	0	-1	
	-1	1	0	

Mixed strategy Nash equilibrium

				
	0	-1	1	(profit)
	1	0	-1	
	-1	1	0	

A pair of strategies (now in probabilities) so that each is a best response to the other.

Nash prove that a mixed strategy Nash equilibria exists for any game with a finite number of players and a finite number of strategies.

But how do we find it?

Mixed strategy Nash equilibrium

	p 	q 	$1 - p - q$ 	
	$\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$			(profit)
				
				

Let us add probability p to playing rock, probability q to playing paper, and probability $1-p-q$ to play scissors.

Mixed strategy Nash equilibrium

	p 	q 	$1 - p - q$ 	
	$\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$			(profit)
				
				

Let us add probability p to playing rock, probability q to playing paper, and probability $1-p-q$ to play scissors.

- Expected payoff from playing rock is
- Expected payoff from playing paper is
- Expected payoff from playing scissors is

Mixed strategy Nash equilibrium

	p 	q 	$1 - p - q$ 	
	$\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$			(profit)
				
				

Let us add probability p to playing rock, probability q to playing paper, and probability $1-p-q$ to play scissors.

- Expected payoff from playing rock is $0p - 1q + 1(1 - p - q)$
- Expected payoff from playing paper is $1p - 0q - 1(1 - p - q)$
- Expected payoff from playing scissors is $-1p - 1q + 0(1 - p - q)$

If we are at Nash equilibrium, all of these have to be the same.

Mixed strategy Nash equilibrium

	p 	q 	$1 - p - q$ 
	$\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$		
			
			

(profit)

$$\begin{aligned}
 0p - 1q + 1(1 - p - q) &= 1p - 0q - 1(1 - p - q) \\
 1p - 0q - 1(1 - p - q) &= -1p + 1q + 0(1 - p - q)
 \end{aligned}$$

Linear System!
(2 equations in 2 variables)

Mixed strategy Nash equilibrium

	p 	q 	$1 - p - q$ 	
	$\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$			(profit)
				
				

$$\begin{aligned}
 0p - 1q + 1(1 - p - q) &= 1p - 0q - 1(1 - p - q) \\
 1p - 0q - 1(1 - p - q) &= -1p + 1q + 0(1 - p - q)
 \end{aligned}$$

Linear System!
(2 equations in 2 variables)

$$p = q = \frac{1}{3}$$

Another bimatrix game

$$\begin{pmatrix} 1,3 & 0,2 \\ 2,-1 & 0,1 \end{pmatrix} \quad (\text{costs})$$

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}$$

Another bimatrix game

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}$$

(costs)

- Imagine the row player plays
top row with probability x_1 ,
bottom row with probability $x_2 = 1 - x_1$
- Imagine the column player plays
left column with probability y_1
right column with probability $y_2 = 1 - y_1$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Another bimatrix game

$$A = \begin{matrix} & \overset{x_1}{\underset{x_2}{}} & \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix} \\ & \underset{y_1}{\underset{y_2}} & \end{matrix} \quad \text{(costs)} \quad B = \begin{matrix} & \overset{x_1}{\underset{x_2}{}} & \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix} \\ & \underset{y_1}{\underset{y_2}} & \end{matrix}$$

- The expected outcome of Player A is

$$1x_1y_1 + 0x_1y_2 + 2x_2y_1 + 0x_2y_2 = x^T A y$$

- The expected outcome of Player B is

$$3x_1y_1 + 2x_1y_2 + (-1)x_2y_1 + 1x_2y_2 = x^T B y$$

Another bimatrix game

$$A = \begin{matrix} & \begin{matrix} \textcolor{red}{x}_1 & \textcolor{red}{x}_2 \end{matrix} \\ \begin{matrix} \textcolor{green}{y}_1 \\ \textcolor{green}{y}_2 \end{matrix} & \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix} \end{matrix} \quad \text{(costs)} \quad B = \begin{matrix} & \begin{matrix} \textcolor{red}{x}_1 & \textcolor{red}{x}_2 \end{matrix} \\ \begin{matrix} \textcolor{green}{y}_1 \\ \textcolor{green}{y}_2 \end{matrix} & \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix} \end{matrix}$$

- The expected outcome of Player A is

$$1x_1y_1 + 0x_1y_2 + 2x_2y_1 + 0x_2y_2 = x^T A y$$

- The expected outcome of Player B is

$$3x_1y_1 + 2x_1y_2 + (-1)x_2y_1 + 1x_2y_2 = x^T B y$$

Def. A pair $\textcolor{red}{x}^*, \textcolor{green}{y}^*$ is said to constitute a **(mixed-strategy) Nash equilibrium** solution to a bimatrix game A, B where both players minimize their costs, if

$$\begin{aligned} \textcolor{red}{x}^{*T} A \textcolor{green}{y}^* &\leq \textcolor{red}{x}^T A \textcolor{green}{y}^* \text{ for all possible } \textcolor{red}{x} \\ \textcolor{red}{x}^{*T} B \textcolor{green}{y}^* &\leq \textcolor{red}{x}^{*T} B \textcolor{green}{y} \text{ for all possible } \textcolor{green}{y} \end{aligned}$$

Nash's Nobel

How is it that Nash received a Nobel with the following result?

Theorem: *Any game with a finite number of players and a finite number of actions has a mixed-strategy Nash equilibrium.*

Graphical Method

Bob's loss matrix

Alice's
payoff
matrix

		B plays	
		b_1	b_2
A plays	a_1	1	3
	a_2	4	2

Graphical Method

Alice's payoff matrix

Bob's loss matrix

B plays

A plays

	b_1	b_2	
a_1	1	3	$1-p$
a_2	4	2	p

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.

Graphical Method

Alice's payoff matrix

Bob's loss matrix

B plays

A plays

	b_1	b_2	
a_1	1	3	$1-p$
a_2	4	2	p

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?

Graphical Method

Alice's payoff matrix

		Bob's loss matrix	
		B plays	
A plays	a_1	b_1	b_2
	a_2	b_1	b_2
		1	3
		4	2
		$1-p$	p

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?
best guaranteed payoff

Graphical Method

Alice's payoff matrix

		Bob's loss matrix	
		B plays	
A plays	a_1	b_1	b_2
	a_2	b_1	b_2
		1	3
		4	2
		$1-p$	p

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?
best guaranteed payoff
 pick p to max the min payoff.

Graphical Method

Alice's payoff matrix

Bob's loss matrix

A plays

B plays

		b_1	b_2	
a_1		1	3	$1-p$
a_2		4	2	p

$1+3p$ $3-p$ summarized payoff

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?
best guaranteed payoff
 pick p to max the min payoff.

Graphical Method

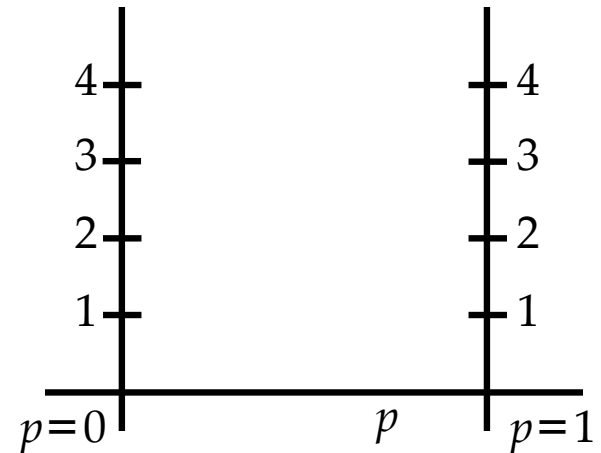
Alice's payoff matrix

Bob's loss matrix

		B plays		
		a_1	a_2	
A plays	b_1	1	4	$1+3p$
	b_2	3	2	$3-p$

summarized payoff

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?
best guaranteed payoff
 pick p to max the min payoff.



Graphical Method

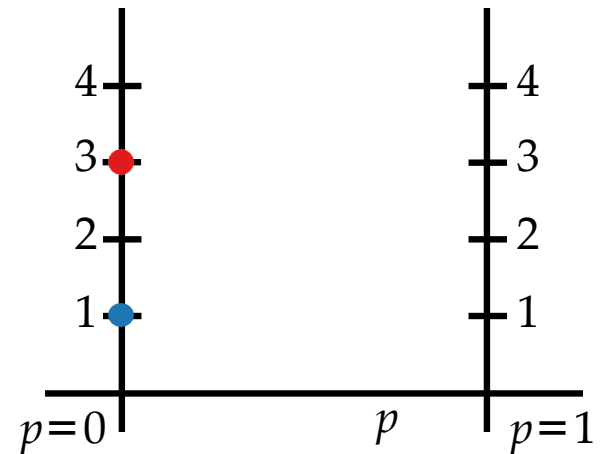
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$1+3p$ $3-p$ summarized payoff

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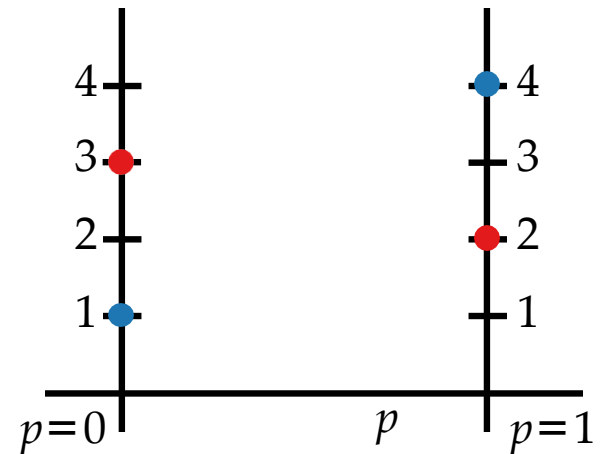
Graphical Method

Alice's payoff matrix

		Bob's loss matrix		
		b_1	b_2	
A plays	a_1	1	3	$1-p$
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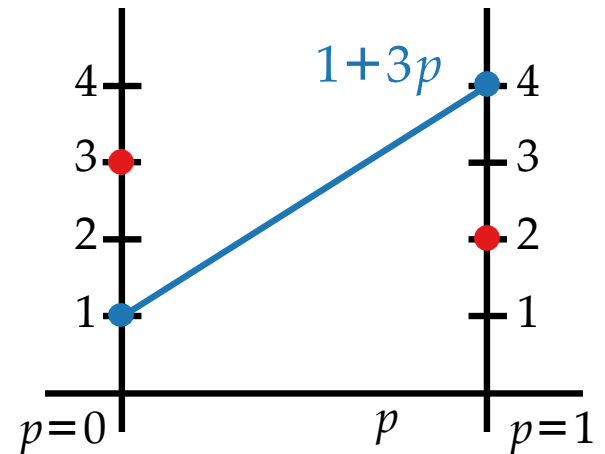
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Graphical Method

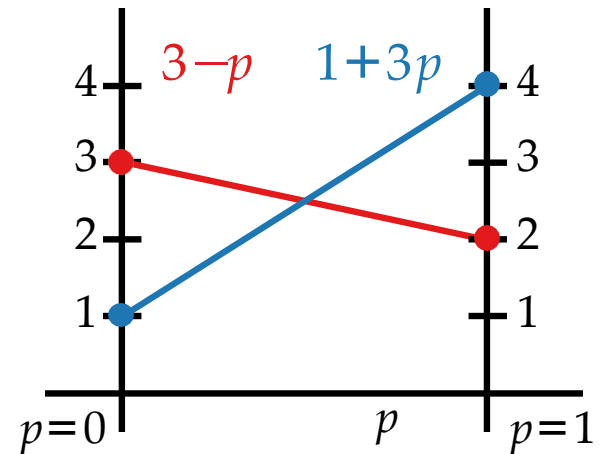
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Bob's loss matrix

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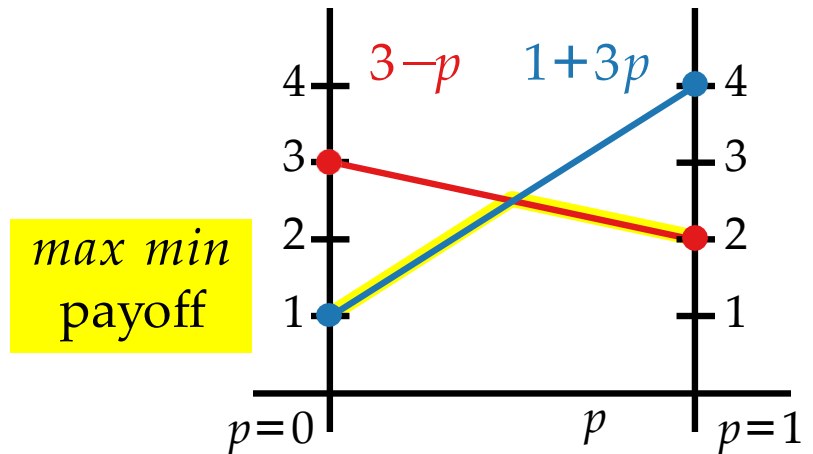
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Alice's payoff matrix

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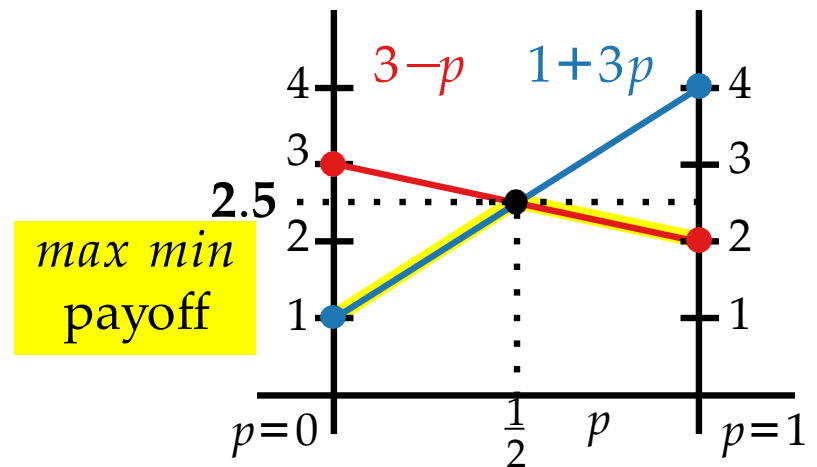
Graphical Method

Alice's payoff matrix

		Bob's loss matrix B plays		
		A plays	b_1	b_2
	a_1	1	3	$1-p$
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$1+3p$ $3-p$ summarized payoff

- Alice plays a_2 with prob. p , and a_1 with prob. $1-p$.
- What p should Alice pick?
best guaranteed payoff
 pick p to max the min payoff.



max min payoff is 2.5 and occurs when $p = \frac{1}{2}$

Two-Player, Zero-Sum Games

- What q should Bob pick?

		B	
		b_1	b_2
A	a_1	1	3
	a_2	4	2
		$1-q$	q

Two-Player, Zero-Sum Games

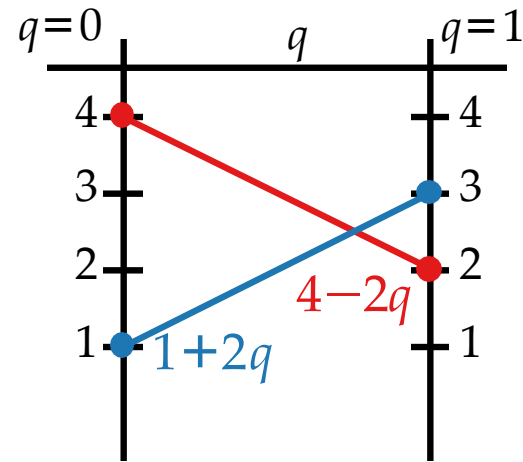
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Two-Player, Zero-Sum Games

- What q should Bob pick?

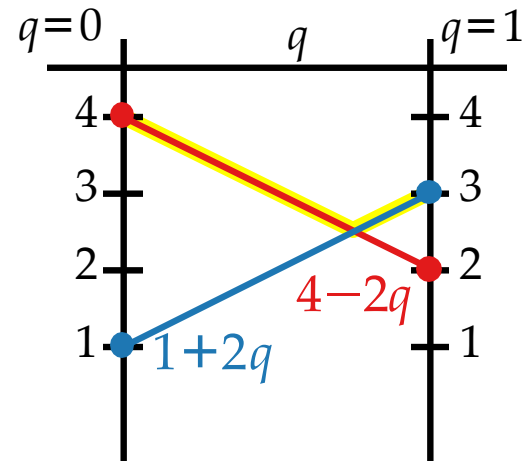
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Two-Player, Zero-Sum Games

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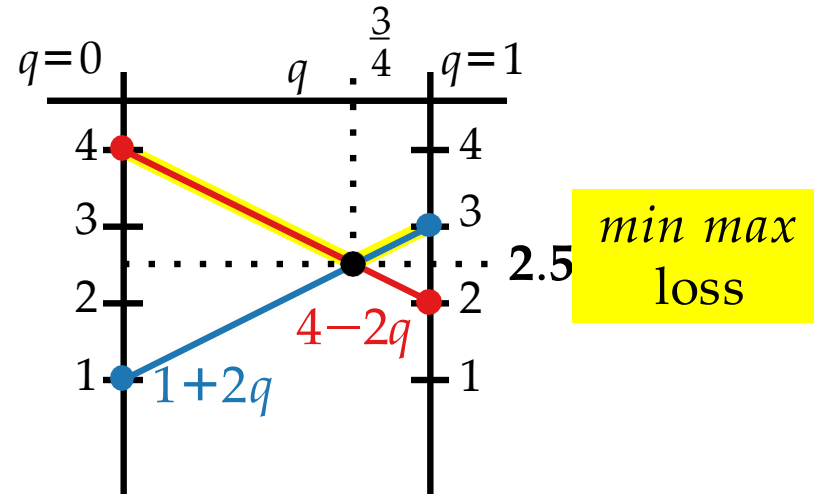


min max
loss

Two-Player, Zero-Sum Games

- What q should Bob pick?

		B		
		b_1	b_2	
A	a_1	1	3	$1+2q$
	a_2	4	2	$4-2q$
		$1-q$	q	



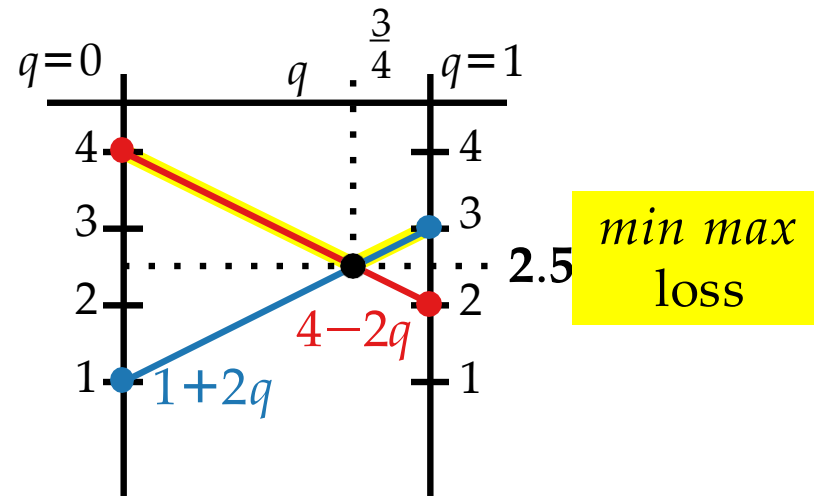
min max loss is 2.5 and
occurs when $q = \frac{3}{4}$

Two-Player, Zero-Sum Games

- What q should Bob pick?

		B		
		b_1	b_2	
A	a_1	1	3	$1+2q$
	a_2	4	2	$4-2q$
		$1-q$	q	

The value is the same!
coincidence?



min max loss is 2.5 and
occurs when $q = \frac{3}{4}$

Two-Player, Zero-Sum Games

Alice's payoff matrix

		Bob's loss matrix	
		B plays	
A plays	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$
	$\frac{1}{2}$	1	3
	$\frac{1}{2}$	4	2

Strategies $(\frac{1}{2}, \frac{1}{2})$ for A, $(\frac{1}{4}, \frac{3}{4})$ for B are

optimal since they both ensure to **maximize the minimum payoff** and (symmetrically) both **minimize the maximum loss**.

The game's **value** is this max min payoff for the first player, here **2.5**.

Two-Player, Zero-Sum Games

Alice's payoff matrix

		Bob's loss matrix	
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John von Neumann

[1928] The MiniMax Theorem

Two-Player, Zero-Sum Games

Alice's payoff matrix

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A plays	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$
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The game's **value** is this max min payoff for the first player, here **2.5**.



John von Neumann

[1928] The MiniMax Theorem

Every $(n \times m)$ matrix S has a **value**, i.e., there is a unique number v with

$$\max_x \min_y xSy^T = v = \min_y \max_x xSy^T$$

row vectors $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$

Non-zero Sum Games?

Alice's
payoff
matrix

		B plays	
		b_1	b_2
A plays	a_1	1	3
	a_2	4	2

(max)

Bob's
payoff
matrix

		B plays	
		b_1	b_2
A plays	a_1	6	-4
	a_2	0	5

Non-zero Sum Games?

(A,B) payoff matrix	B plays			
	A plays	b_1	b_2	
	a_1	1, 6	3, -4	
	a_2	4, 0	2, 5	

Non-zero Sum Games?

		B plays	
		b_1	b_2
(A,B) payoff matrix	A plays a_1	1, 6	3, -4
	a_2	4, 0	2, 5

- Is there a Pure-Strategy Nash equilibrium?

Non-zero Sum Games?

		B plays	
(A,B) payoff matrix	A plays	b_1	b_2
	a_1	1, 6	3, -4
	a_2	4, 0	2, 5

- Is there a Pure-Strategy Nash equilibrium?

$$R_A(b_1) = a_2$$

$$R_A(b_2) = a_1$$

$$R_B(a_1) = b_1$$

$$R_B(a_2) = b_2$$

No Pure Equilibrium

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1, 6$	$3, -4$	$1-p$
		$4, 0$	$2, 5$	p
		$1-q$	q	

- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1, 6$	$3, -4$	$1-p$
		$4, 0$	$2, 5$	p
		$1-q$	q	

- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?

$$A : 1 - q + 3q = 1 + 2q \quad a_1$$

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1, 6$	$3, -4$	$1-p$
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- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?

$$A : \quad 1 - q + 3q = 1 + 2q \quad a_1$$

$$4(1 - q) + 2q = 4 - 2q \quad a_2$$

$B :$

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1, 6$	$3, -4$	$1-p$
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$$B : \quad 6(1 - p) \quad b_1$$

$$-4(1 - p) + 5p = -4 + 9p \quad b_2$$

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1-q$	q	
	a_1	1, 6	3, -4	$1-p$
	a_2	4, 0	2, 5	p

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$$\begin{array}{ll}
 A : & 1 - q + 3q = 1 + 2q \quad a_1 \\
 & 4(1 - q) + 2q = 4 - 2q \quad a_2 \\
 B : & 6(1 - p) \quad b_1 \\
 & -4(1 - p) + 5p = -4 + 9p \quad b_2
 \end{array}
 \qquad
 \begin{array}{l}
 1 + 2q = 4 - 2q \\
 6(1 - p) = -4 + 9p
 \end{array}$$

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1, 6$	$3, -4$	$1-p$
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 \begin{array}{ll}
 a_1 & \\
 a_2 & \\
 b_1 & \\
 b_2 &
 \end{array}
 \begin{array}{l}
 q = \frac{3}{4} \\
 p = \frac{2}{3}
 \end{array}$$

Non-zero Sum Games?

		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1-q$	q	
	a_1	1, 6	3, -4	$1-p$
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 a_1 & \\
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 b_2 &
 \end{array}
 \begin{array}{l}
 q = \frac{3}{4} \\
 p = \frac{2}{3}
 \end{array}
 \begin{array}{l}
 v_A = 2.5 \\
 v_B = 2
 \end{array}$$

Non-zero Sum Games?

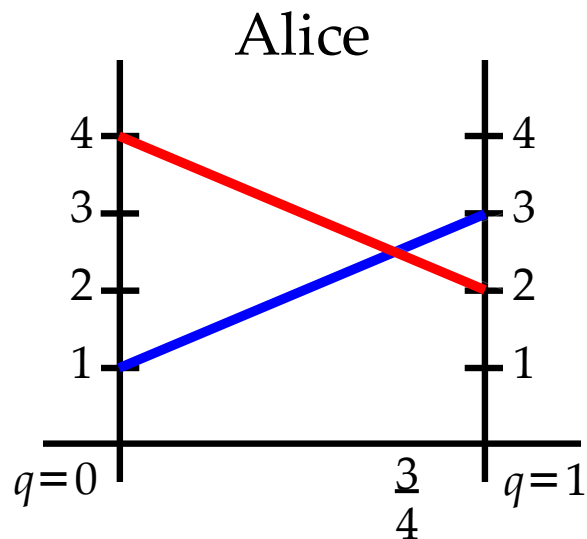
		B plays		
		a_1	a_2	
(A,B) payoff matrix	A plays	b_1	b_2	
		$1, 6$	$3, -4$	$1-p$
		$4, 0$	$2, 5$	p
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Non-zero Sum Games?

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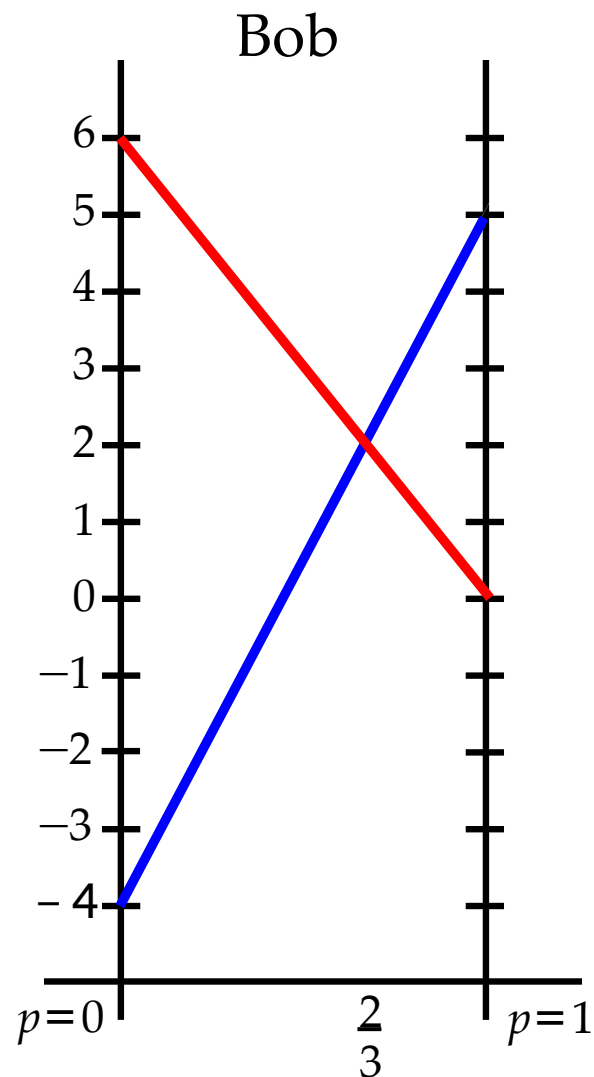
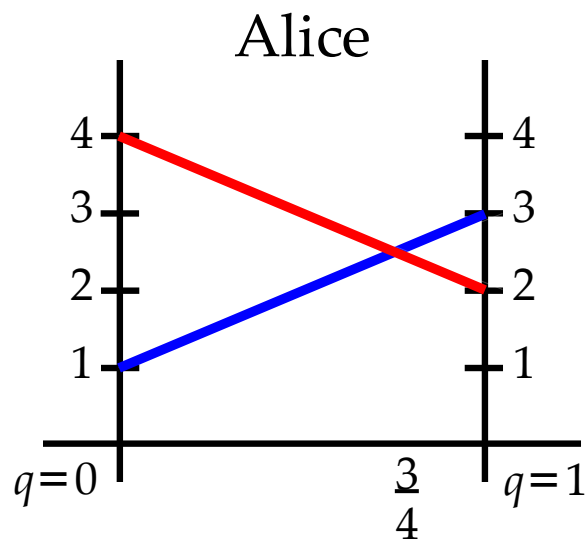


Non-zero Sum Games?

(A,B)
payoff
matrix

		B plays		
		A plays	b_1	b_2
	a_1		1, 6	3, -4
	a_2		4, 0	2, 5
			$1-q$	q
			$1-p$	
			p	

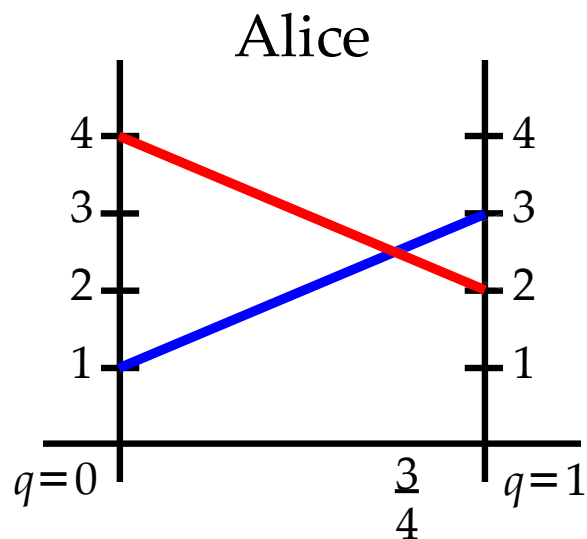
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Non-zero Sum Games?

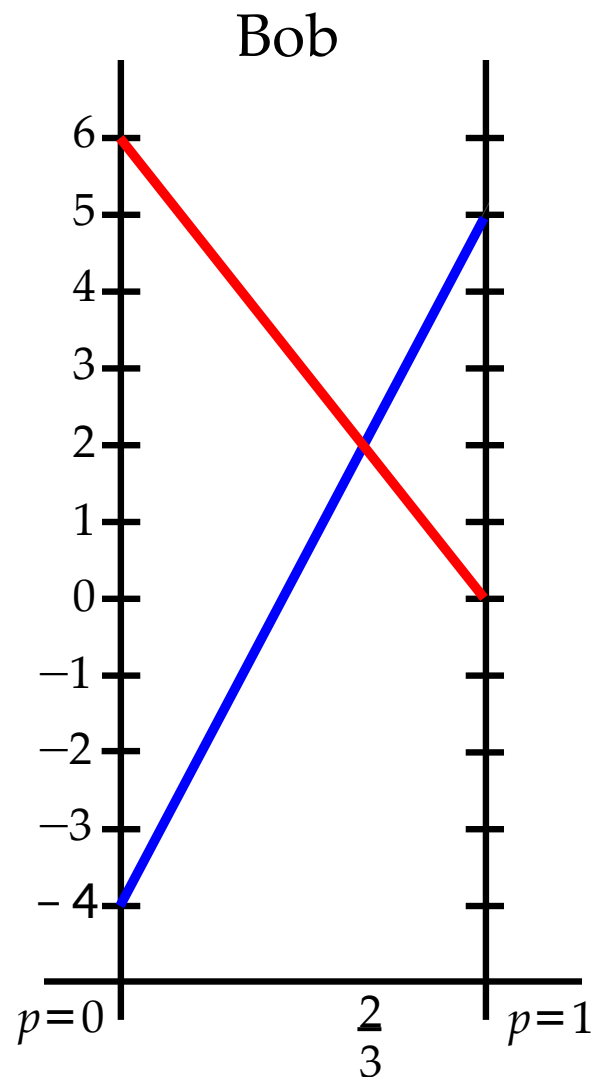
		B plays			
		A plays	b_1	b_2	
(A,B) payoff matrix	a_1		1, 6	3, -4	$1-p$
	a_2		4, 0	2, 5	p
			$1-q$	q	

- Is there a Pure-Strategy Nash equilibrium?
- Is there a Mixed-Strategy Nash equilibrium?



$$v_A = 2.5$$

$$v_B = 2$$



MATSE: Operations Research

Lecture 9. Game Theory

[based on slides and notes by: Frank Thuijsman, Steven Chaplick]

Two-Player, Zero-Sum Games

Setup: Two players Alice and Bob act **simultaneously**
 (Alice uses strategy a_i ; Bob uses b_j)
 Alice's score is s_{ij} and Bob's is $-s_{ij}$

Alice's payoff matrix	B plays	
	A plays	$b_1 \quad b_2$
	a_1	$\begin{bmatrix} 1 & 3 \end{bmatrix}$
	a_2	$\begin{bmatrix} 4 & 2 \end{bmatrix}$

Bob's payoff matrix	B plays	
	A plays	$b_1 \quad b_2$
	a_1	$\begin{bmatrix} -1 & -3 \end{bmatrix}$
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Alice's payoff matrix	B plays		Bob's loss matrix
	A plays	b_1	b_2
		a_1	a_2
		$\begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$	

Bob's payoff matrix	B plays		Bob's loss matrix
	A plays	b_1	b_2
		a_1	a_2
		$\begin{bmatrix} -1 & -3 \\ -4 & -2 \end{bmatrix}$	

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Bob's payoff matrix	B plays		Bob's loss matrix
	A plays	b_1	b_2
		a_1	a_2
		$\begin{bmatrix} -1 & -3 \\ -4 & -2 \end{bmatrix}$	

Called **Zero-Sum** because each play (a_i, b_j) leads to a total score of **zero**.
 To analyze such games, we must assume *rational* players.

Two-Player, Zero-Sum Games

Setup: Two players Alice and Bob act **simultaneously**
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Alice's payoff matrix	B plays		Bob's loss matrix
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|

Bob's payoff matrix	B plays		Bob's loss matrix
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- If we know Alice's play, then Bob can plan a "best" counter strategy.

Two-Player, Zero-Sum Games

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 Alice's score is s_{ij} and Bob's is $-s_{ij}$

		B plays				B plays			
Alice's payoff matrix	A plays	b_1	b_2	Bob's loss matrix		Bob's payoff matrix	A plays	b_1	b_2
	a_1	$\begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$					a_1	$\begin{bmatrix} -1 & -3 \\ -4 & -2 \end{bmatrix}$	
	a_2						a_2		

Called **Zero-Sum** because each play (a_i, b_j) leads to a total score of **zero**.
 To analyze such games, we must assume *rational* players.

- If we know Alice's play, then Bob can plan a "best" counter strategy.
- But, players act simultaneously, maybe one strategy is more likely than another?
 $\leadsto$ Analyze such games by probabilistic play!

Two-Player, Zero-Sum

Some games have redundant options , or a deterministic strategy.

Two-Player, Zero-Sum

Some games have redundant options , or a deterministic strategy.

- Maybe some rows (cols.) would never be played ...

Two-Player, Zero-Sum

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- Maybe some rows (cols.) would never be played ...

		B plays			
		b_1	b_2		
A plays	a_1	1	3	Would Alice every play the new row a_3 ?	
	a_2	4	2		
	a_3	3	1		

Two-Player, Zero-Sum

Some games have redundant options , or a deterministic strategy.

- Maybe some rows (cols.) would never be played ...

		B plays		Would Alice every play the new row a_3 ? No, because ...
A plays				
	a_1	a_2	a_3	
	b_1	b_2		
	1	3		
	4	2		
	3	1		

Two-Player, Zero-Sum

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Would Alice every play the new row a_3 ?

No, because ...

For each col. j , the entry $(a_2)_j \geq (a_3)_j$.

Two-Player, Zero-Sum

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$\rightsquigarrow a_3$ is **dominated** by a_2 .

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Thus, always better to use a_2 than a_3 !

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Thus, always better to use a_2 than a_3 !

Two-Player, Zero-Sum

Some games have redundant options , or a deterministic strategy.

- Maybe some rows (cols.) would never be played ...

		B plays	
		b_1	b_2
A plays	a_1	1	3
	a_2	4	2
	a_3	3	1

Would Alice every play the new row a_3 ?

No, because ...

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But Bob playing $b_1 \rightsquigarrow$ Alice playing a_2

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But Alice playing $a_2 \rightsquigarrow$ Bob playing b_2

But Bob playing $b_1 \rightsquigarrow$ Alice playing a_2

But Bob playing $b_2 \rightsquigarrow$ Alice playing a_1

unstable! :(

Saddle points

When does a game have a **pure** strategy, i.e., where Alice picks a_i and Bob picks b_j ?

		B plays		
		b_1	b_2	b_3
A plays	a_1	1	3	0
	a_2	4	2	-2
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- (i, j) is a **saddle point**, when S_{ij} is both the min entry in row i and the max entry in col j .

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Alice picking $a_1 \rightsquigarrow$ Bob picking b_3
 Bob picking $b_3 \rightsquigarrow$ Alice picking a_1

stable!

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 Bob picking $b_3 \rightsquigarrow$ Alice picking a_1

stable!

- (i, j) is a **saddle point**, when S_{ij} is both the min entry in row i and the max entry in col j .
 $\rightsquigarrow$ A saddle point (i, j) provides an *optimal* solution, the *value* being S_{ij} for Alice and $-S_{ij}$ for Bob.

Multiple Saddle Points

What if there are multiple saddle points?

Suppose that an $(n \times m)$ matrix S has two saddle points (i, j) and (i', j') . Show that

(a) $S_{ij} = S_{i'j'}$, and (b) (i', j) and (i, j') are also saddle points.

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Suppose that an $(n \times m)$ matrix S has two saddle points (i, j) and (i', j') . Show that

(a) $S_{ij} = S_{i'j'}$, and (b) (i', j) and (i, j') are also saddle points.

Consider (i', j) and (i, j') :

- $S_{i'j}$ is in the same column j of S_{ij}

$\Rightarrow S_{i'j} \leq S_{ij}$ because (i, j) is a saddle point
It is the max entry in the column

- $S_{i'j}$ is in the same row i' of $S_{i'j'}$

$\Rightarrow S_{i'j} \geq S_{i'j'}$ because (i', j') is a saddle point
It is the min entry in the row

- $S_{ij'}$ is in the same column j' of $S_{i'j'}$

$\Rightarrow S_{ij'} \leq S_{i'j'}$

- $S_{ij'}$ is in the same row i of S_{ij}

$\Rightarrow S_{ij'} \geq S_{ij}$

$$\begin{bmatrix} S_{ij} & S_{ij'} \\ S_{i'j} & S_{i'j'} \end{bmatrix}$$

Two-Player, Zero-Sum Games

Probabilistic play.

Alice's payoff matrix

Bob's loss matrix

B plays

A plays

	b_1	b_2
a_1	1	3
a_2	4	2

Two-Player, Zero-Sum Games

Probabilistic play.

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.

Alice's
payoff
matrix

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	b_1	b_2	
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- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?

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Bob's loss matrix

		B plays	
		b_1	b_2
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	a_2	4	2

$1-p$ p

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Probabilistic play.

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?
best guaranteed payoff

Alice's
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Bob's loss matrix

		B plays	
		b_1	b_2
A plays	a_1	1	3
	a_2	4	2

$1-p$ p

Two-Player, Zero-Sum Games

Probabilistic play.

- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?
best guaranteed payoff
 pick p to max the min payoff.

Alice's
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Bob's loss matrix

		B plays	
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$1-p$
 p

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- Alice plays a_2 with prob. p , and a_1 with prob. $1 - p$.
- What p should Alice pick?
best guaranteed payoff
 pick p to max the min payoff.
- What q should Bob pick?

$\Rightarrow$ Expected Payoff

$\Rightarrow$ Graphical Method

Alice's
payoff
matrix

Bob's loss matrix

B plays

A plays

	b_1	b_2	
a_1	1	3	$1-p$
a_2	4	2	p
	$1-q$	q	

Two-Player, Zero-Sum Games

Alice's payoff matrix

Bob's loss matrix

		B plays	
A plays	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$
	$\frac{1}{2}$	1	3
	$\frac{1}{2}$	4	2

Strategies $(\frac{1}{2}, \frac{1}{2})$ for A, $(\frac{1}{4}, \frac{3}{4})$ for B are **optimal** since they both ensure to **maximize the minimum payoff** and (symmetrically) both **minimize the maximum loss**.

The game's **value** is this max min payoff for the first player, here **2.5**.

Two-Player, Zero-Sum Games

Alice's
payoff
matrix

Bob's loss matrix

		B plays	
A plays	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$
	$\frac{1}{2}$	1	3
	$\frac{1}{2}$	4	2
	$\frac{1}{2}$		

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John von Neumann

[1928] The MiniMax Theorem

Every $(n \times m)$ matrix S has a **value**, i.e., there is a unique number v with

$$\max_x \min_y xSy^T = v = \min_y \max_x xSy^T$$

row vectors $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$

A bigger example

		B plays		
		b_1	b_2	b_3
A plays	a_1	0	-2	5
	a_2	4	10	-1
	a_3	3	2	-3

A bigger example

		B plays		
A plays		b_1	b_2	b_3
	a_1	0	-2	5
	a_2	4	10	-1
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- Check for dominated rows/cols

A bigger example

B plays

A plays

	b_1	b_2	b_3
a_1	0	-2	5
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		B plays		
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- Check for dominated rows/cols

A bigger example

		B plays		
A plays		b_1	b_2	b_3
		a_1	$\begin{bmatrix} 0 & -2 & 5 \end{bmatrix}$	
	a_2	$\begin{bmatrix} 4 & 10 & -1 \end{bmatrix}$		

- Check for dominated rows/cols any more?

A bigger example

		B plays		
A plays		b_1	b_2	b_3
		a_1	$\begin{bmatrix} 0 & -2 & 5 \end{bmatrix}$	
	a_2	$\begin{bmatrix} 4 & 10 & -1 \end{bmatrix}$		

- Check for dominated rows/cols
any more? No!
- Check for saddle points

A bigger example

		B plays		
A plays		b_1	b_2	b_3
		a_1	$\begin{bmatrix} 0 & -2 & 5 \end{bmatrix}$	
	a_2	$\begin{bmatrix} 4 & 10 & -1 \end{bmatrix}$		

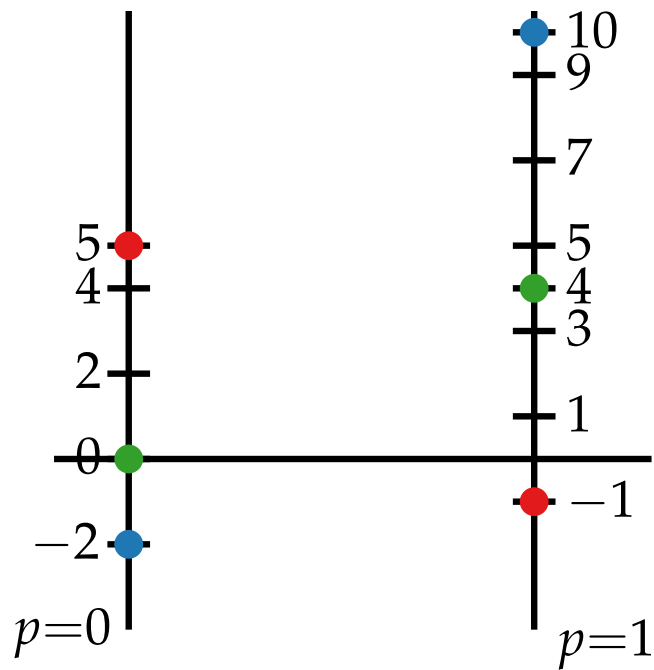
- Check for dominated rows/cols
any more? No!
- Check for saddle points
no saddle points $\rightsquigarrow$ consider a mixed strategy

A bigger example

		B plays		
		q_1	q_2	q_3
A plays	$1-p$	a_1	$\begin{bmatrix} 0 & -2 & 5 \end{bmatrix}$	
	p	a_2	$\begin{bmatrix} 4 & 10 & -1 \end{bmatrix}$	

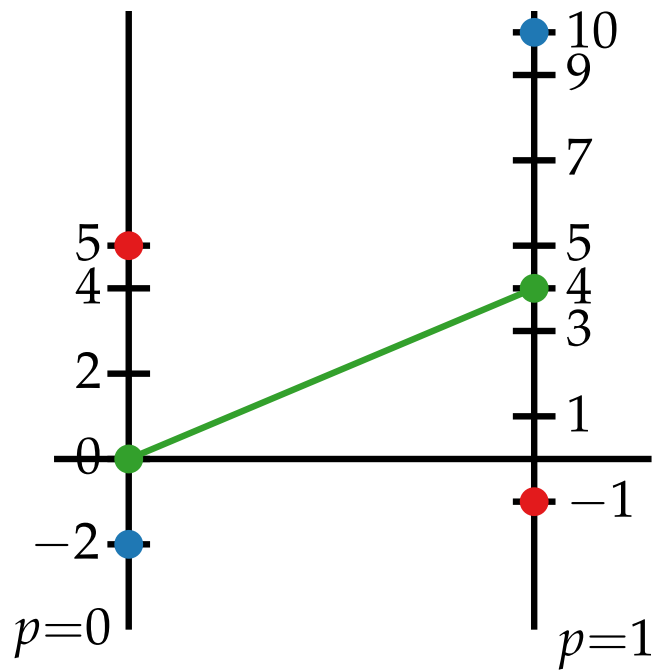
A bigger example

		B plays		
		q_1	q_2	q_3
A plays	a_1	b_1	b_2	b_3
	a_2	b_1	b_2	b_3
$1-p$		0	-2	5
p		4	10	-1



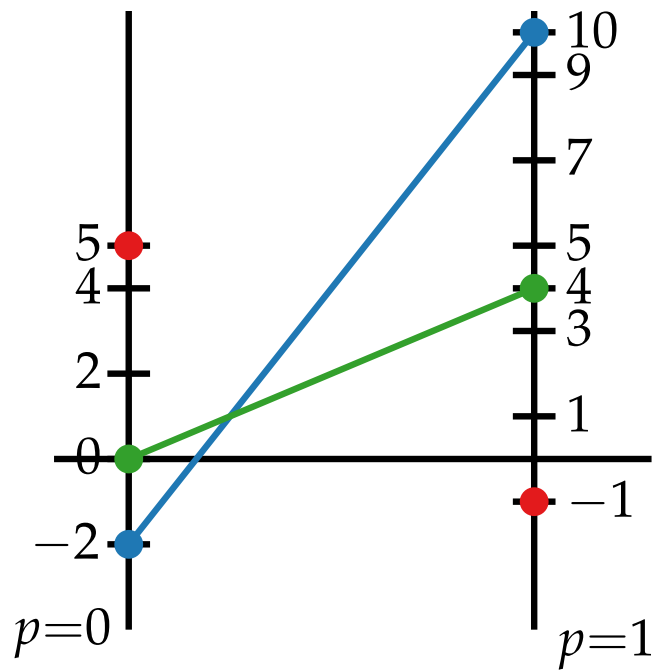
A bigger example

		B plays		
		q_1	q_2	q_3
A plays	b_1	b_1	b_2	b_3
	$1-p$ a_1	0	-2	5
	p a_2	4	10	-1
		$4p$		



A bigger example

		B plays		
		q_1	q_2	q_3
A plays	b_1	b_1	b_2	b_3
	$1-p$ a_1	0	-2	5
	p a_2	4	10	-1
		$4p$	$12p-2$	

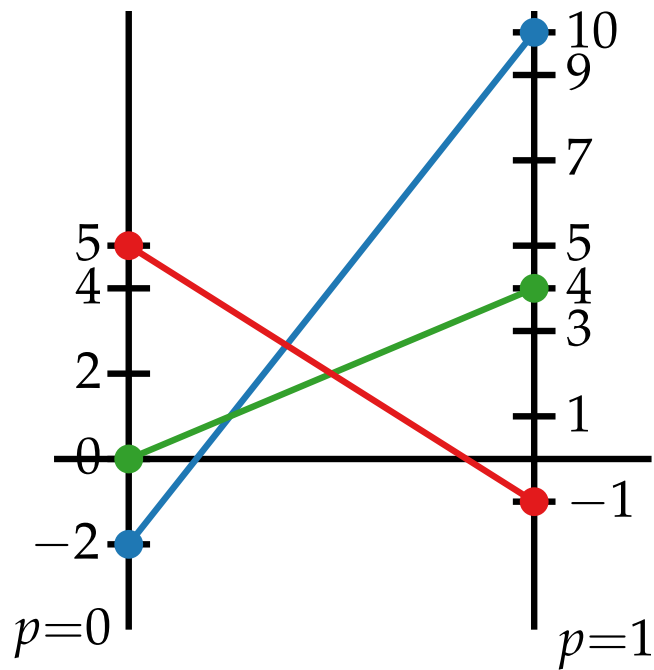


A bigger example

B plays q_1 q_2 q_3

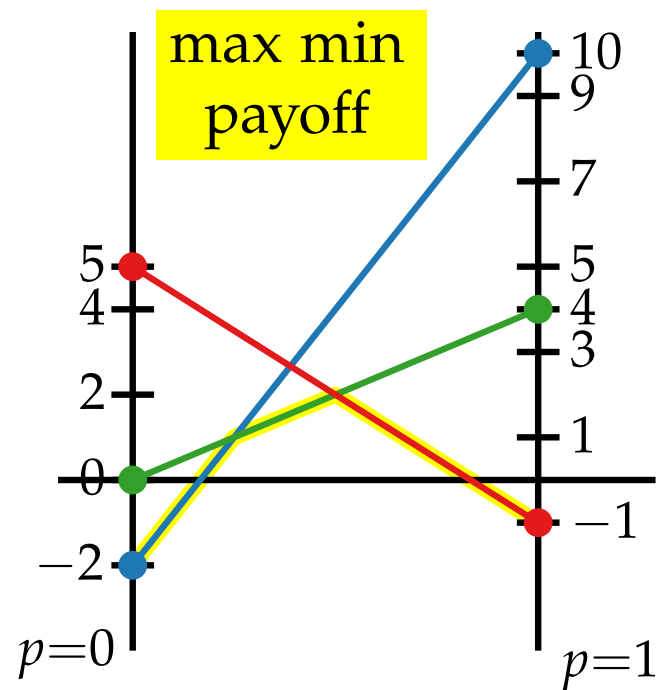
A plays b_1 b_2 b_3

$1-p$ a_1	0	-2	5
p a_2	4	10	-1
	$4p$	$12p-2$	$5-6p$



A bigger example

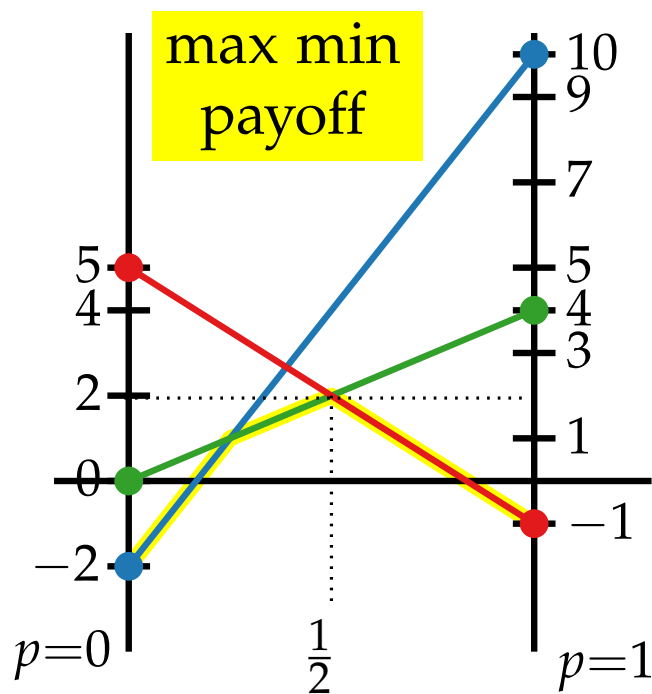
		B plays		
		q_1	q_2	q_3
A plays	b_1	0	-2	5
	b_2	4	10	-1
$1-p$ a_1		$4p$	$12p-2$	$5-6p$
p a_2				



A bigger example

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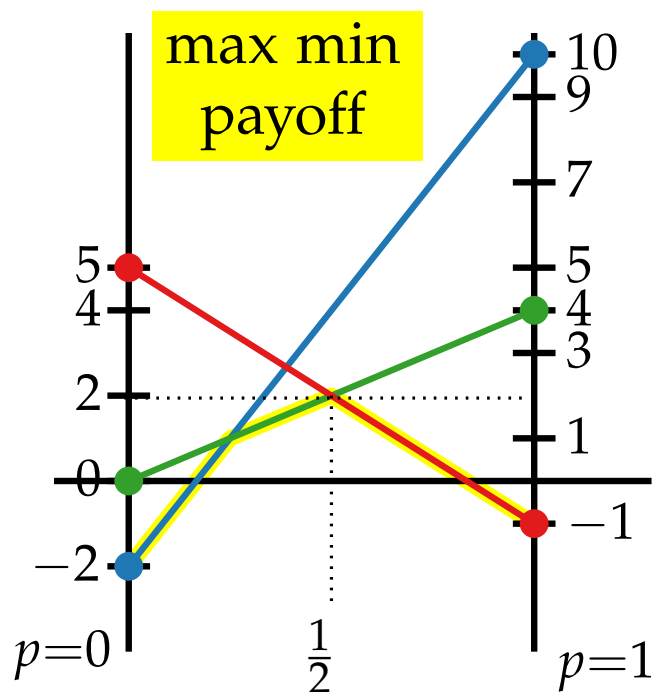
Thus, $p = \frac{1}{2}$, and the **value** v^* of the game is 2.



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		q_1	q_2	q_3
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	a_1	0	-2	5
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		$4p$	$12p-2$	$5-6p$

Thus, $p = \frac{1}{2}$, and the **value** v^* of the game is 2.
 But, we didn't find $\mathbf{q} = (q_1, q_2, q_3) \dots$



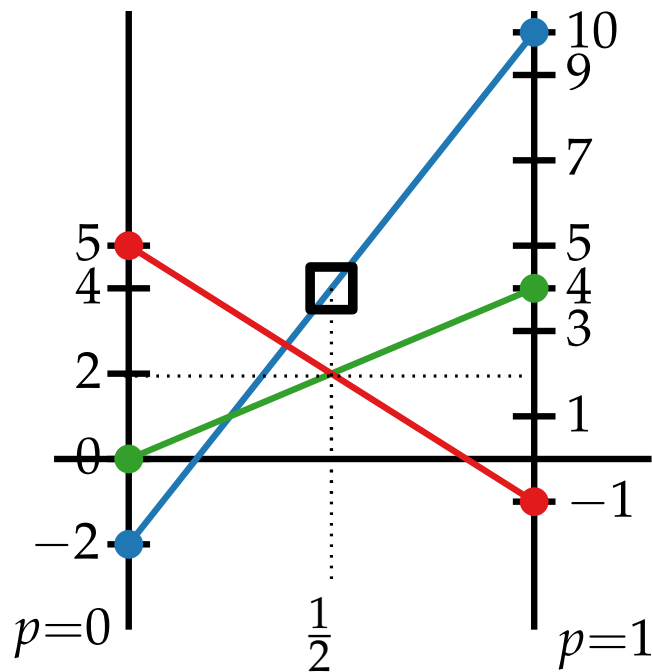
A bigger example

		B plays		
		q_1	q_2	q_3
A plays	$1-p$ a_1	b_1	b_2	b_3
	p a_2	b_1	b_2	b_3
		$4p$	$12p-2$	$5-6p$

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$$\Rightarrow q_2 = 0$$



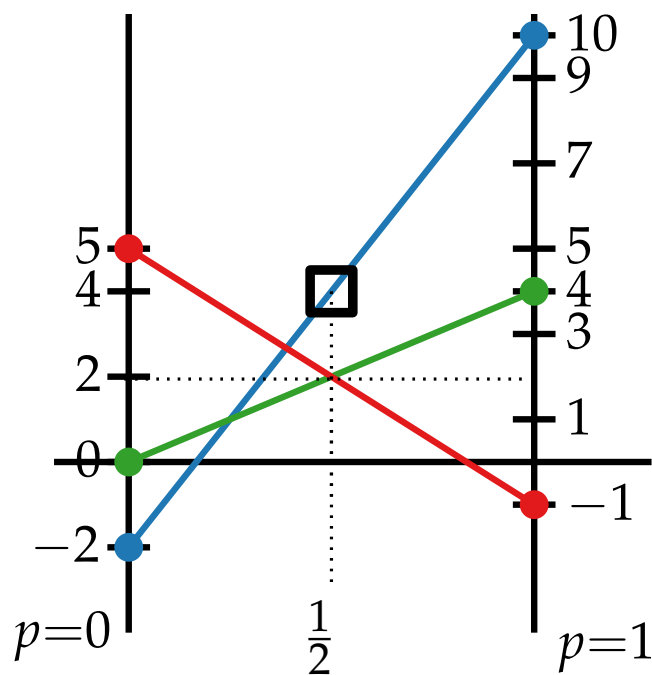
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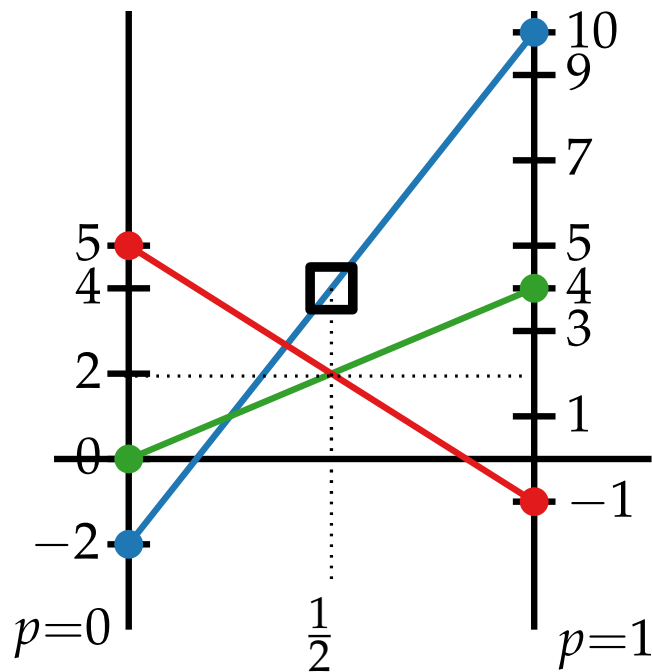
A plays a_1 a_2

$$\begin{array}{c}
 1-p \\
 p
 \end{array}
 \begin{bmatrix}
 0 & -2 & 5 \\
 4 & 10 & -1
 \end{bmatrix}
 \begin{array}{c}
 -2q_2 + 5q_3 \\
 4p \quad 12p-2 \quad 5-6p
 \end{array}$$

Thus, $p = \frac{1}{2}$, and the **value** v^* of the game is 2.

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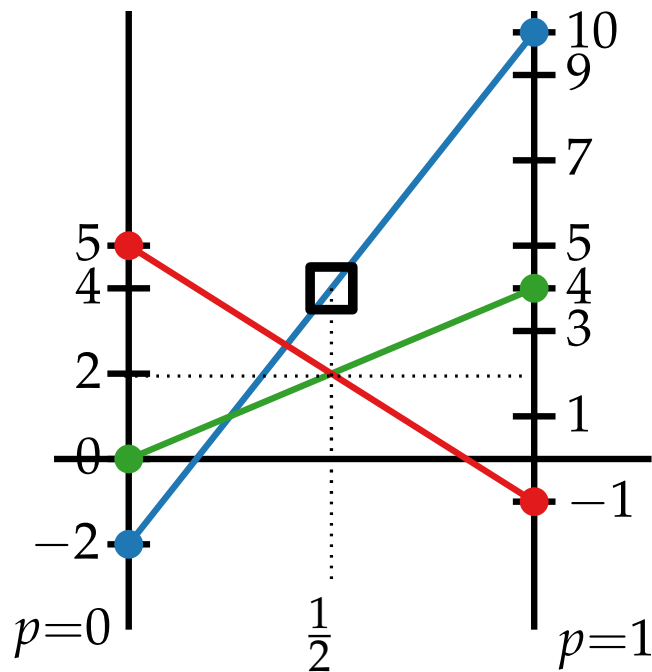
A bigger example

		B plays			
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A plays	b_1	b_1	b_2	b_3	
	a_1	0	-2	5	$-2q_2 + 5q_3$
$1-p$	a_2	4	10	-1	$4q_1 + 10q_2 - 1q_3$
p		$4p$	$12p-2$	$5-6p$	

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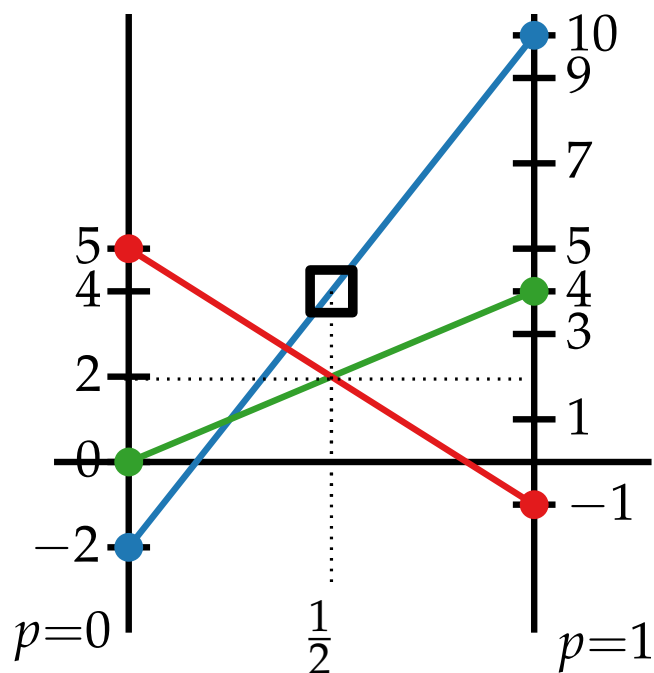
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A bigger example

		B plays			
		q_1	q_2	q_3	
A plays	b_1	b_1	b_2	b_3	
	a_1	0	-2	5	$-2q_2 + 5q_3$
	a_2	4	10	-1	$4q_1 + 10q_2 - 1q_3$
		$4p$	$12p - 2$	$5 - 6p$	



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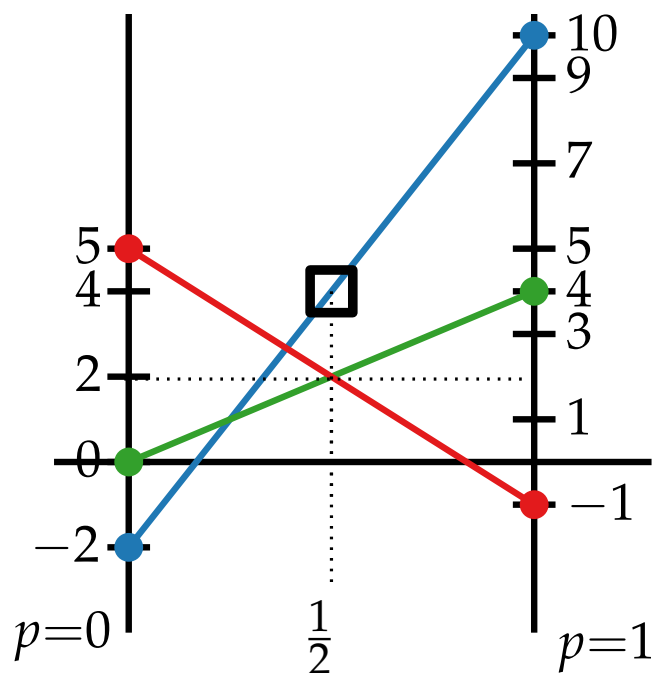
[Minimax thm. $\mathbf{pS}\mathbf{q}^T = v^*$]

$$(1-p)(-2q_2 + 5q_3) + p(4q_1 + 10q_2 - 1q_3) = v^*$$

$$q_1(4p) + q_2(12p - 2) + q_3(5 - 6p) = v^*$$

A bigger example

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		q_1	q_2	q_3
A plays	b_1	b_2	b_3	
	a_1	a_2		
$1-p$	a_1	0	-2	5
p	a_2	4	10	-1
		$4p$	$12p-2$	$5-6p$
		$-2q_2 + 5q_3$		
		$4q_1 + 10q_2 - 1q_3$		



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[Minimax thm. $\mathbf{pS}\mathbf{q}^T = v^*$]

$$(1-p)(-2q_2 + 5q_3) + p(4q_1 + 10q_2 - 1q_3) = v^*$$

$$q_1(4p) + q_2(12p - 2) + q_3(5 - 6p) = v^*$$

$$p = 0 \rightsquigarrow -2q_2 + 5q_3 = 2$$

A bigger example

		B plays		
		q_1	q_2	q_3
A plays	b_1	b_2	b_3	
	$1-p$ a_1	0	-2	5
	p a_2	4	10	-1
		$4p$	$12p-2$	$5-6p$

$-2q_2 + 5q_3$
 $4q_1 + 10q_2 - 1q_3$

Thus, $p = \frac{1}{2}$, and the **value** v^* of the game is 2.

But, we didn't find $\mathbf{q} = (q_1, q_2, q_3) \dots$

$$\Rightarrow q_2 = 0$$

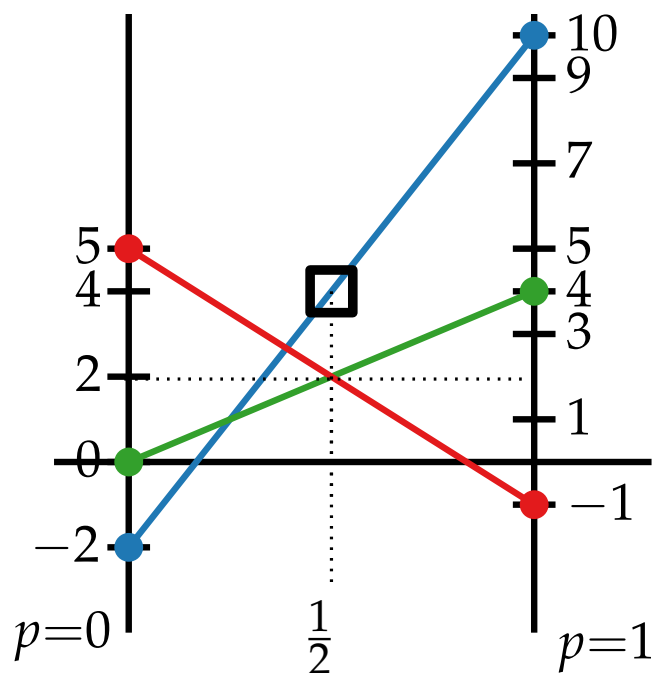
[Minimax thm. $\mathbf{pS}\mathbf{q}^T = v^*$]

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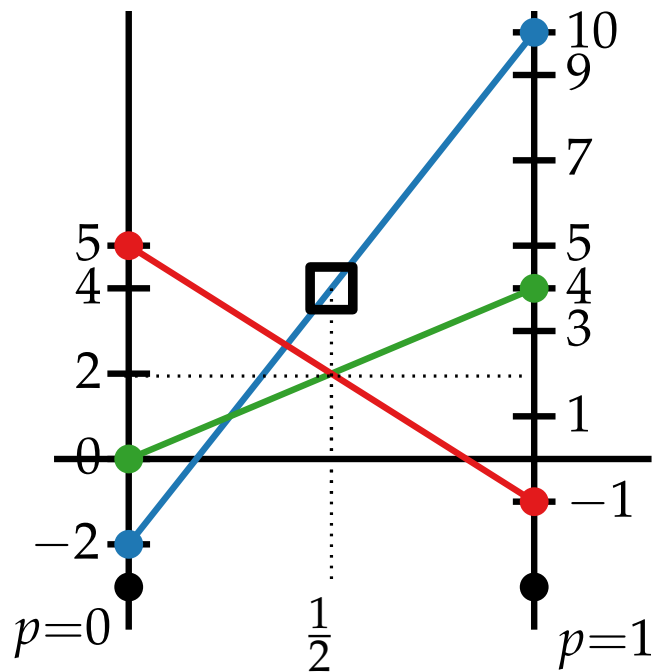
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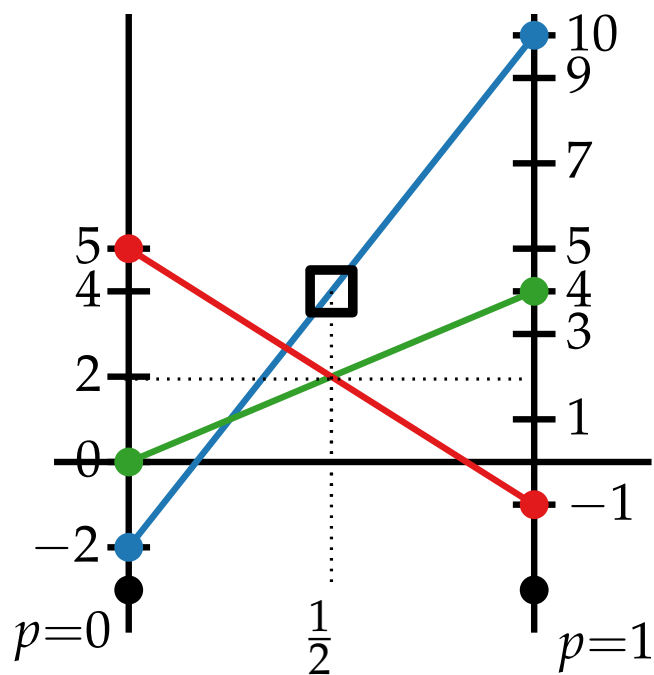
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A bigger example

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		q_1	q_2	q_3	
A plays	b_1	b_1	b_2	b_3	
	a_1	0	-2	5	$-2q_2 + 5q_3$
	a_2	4	10	-1	$4q_1 + 10q_2 - 1q_3$
		$4p$	$12p - 2$	$5 - 6p$	



Thus, $p = \frac{1}{2}$, and the **value** v^* of the game is 2.

But, we didn't find $\mathbf{q} = (q_1, q_2, q_3) \dots$

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$$\mathbf{q} = \left(\frac{3}{5}, 0, \frac{2}{5}\right)$$

Flip the Matrix? [Let S be the (simplified) game of prev. slide]

B plays

A plays

$$\begin{matrix} & b_1 & b_2 \\ a_1 & 0 & 4 \\ a_2 & -2 & 10 \\ a_3 & 5 & -1 \end{matrix} = S^T$$

- Is this the same game?

Flip the Matrix? [Let S be the (simplified) game of prev. slide]

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No, payoff/loss is reversed!

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